

National Innovation Systems

Chapter 14

Lecture Guide

CLO	Weekly Learning Outcomes	Asynchronous WLOs	Weeks
K 3 Critically evaluate important sources of innovations and what affects the innovativeness of individuals S1- Apply key concepts to identify factors that facilitate and hinder the creation and adoption of innovations	1. Distinguish those nations with a stronger record of innovation; 2. Distinguish the institutions and policies that influence the rate of innovation; 3. Analyse the nature of the institutions that contribute to innovation; 4. Describe the policies that contribute to innovation in UAE	4	14

Topics to be covered

During the self-paced session, you will cover the following WLO:

- Evaluate the policies that contribute to innovation in UAE.

Advanced Organizer

- Some countries are known to be more innovative than others. One characteristic of innovative countries is that they have developed and implemented national innovation policies that create a conducive support system for Innovation to thrive. In this session, you will learn about different major national innovation policies in UAE.

GII rank	Economy	Score	Income group rank	Region rank
1	Switzerland	67.6	1	1
2	Sweden	64.2	2	2
3	United States	63.5	3	1
4	United Kingdom	62.4	4	3
5	Singapore	61.5	5	1
6	Finland	61.2	6	4
7	Netherlands (Kingdom of the)	60.4	7	5
8	Germany	58.8	8	6
9	Denmark	58.7	9	7
10	Republic of Korea	58.6	10	2
11	France	56.0	11	8
12	China	55.3	1	3
13	Japan	54.6	12	4
14	Israel	54.3	13	1
15	Canada	53.8	14	2
16	Estonia	53.4	15	9
17	Hong Kong, China	53.3	16	5
18	Austria	53.2	17	10
19	Norway	50.7	18	11
20	Iceland	50.7	19	12
21	Luxembourg	50.6	20	13
22	Ireland	50.4	21	14
23	Belgium	49.9	22	15
24	Australia	49.7	23	6
25	Malta	49.1	24	16
26	Italy	46.6	25	17
27	New Zealand	46.6	26	7
28	Cyprus	46.3	27	2
29	Spain	45.9	28	18
30	Portugal	44.9	29	19
31	Czech Republic	44.8	30	20
32	United Arab Emirates	43.2	31	3
33	Slovenia	42.2	32	21
34	Lithuania	42.0	33	22
35	Hungary	41.3	34	23
36	Malaysia	40.9	2	8
37	Latvia	39.7	35	24
38	Bulgaria	39.0	3	25
39	Türkiye	38.6	4	4
40	India	38.1	1	1
41	Poland	37.7	36	26
42	Greece	37.5	37	27
43	Thailand	37.1	5	9
44	Croatia	37.1	38	28
45	Slovakia	36.2	39	29
46	Viet Nam	36.0	2	10
47	Romania	34.7	40	30
48	Saudi Arabia	34.5	41	5
49	Brazil	33.6	6	1
50	Qatar	33.4	42	6

Global Innovation Index 2023 Rankings

National/ Regional Policies Supporting Innovation in UAE

There are a number of policies in UAE that support Innovation at national, regional and sectoral levels. The intention of this document is not to make a comprehensive overview of all these innovation policies, but instead to present the details of four major policies:

1. UAE National Innovation Policy
2. Science, technology, & Innovation Policy
3. UAE Vision 2021 & UAE Centennial 2071
4. Dubai 10X Initiative
5. 'UAE Strategy for Artificial Intelligence (AI), 2017
6. Green Economy for Sustainable Development (<https://u.ae/en/about-the-uae/economy/green-economy-for-sustainable-development>)
7. UAE Innovates (<https://uaeinnovates.gov.ae/>) & Government Innovation Lab (<https://www.mbrcgi.gov.ae/en/experiment/government-innovation-labs>)



1. UAE National Innovation Policy

UAE NATIONAL INNOVATION STRATEGY



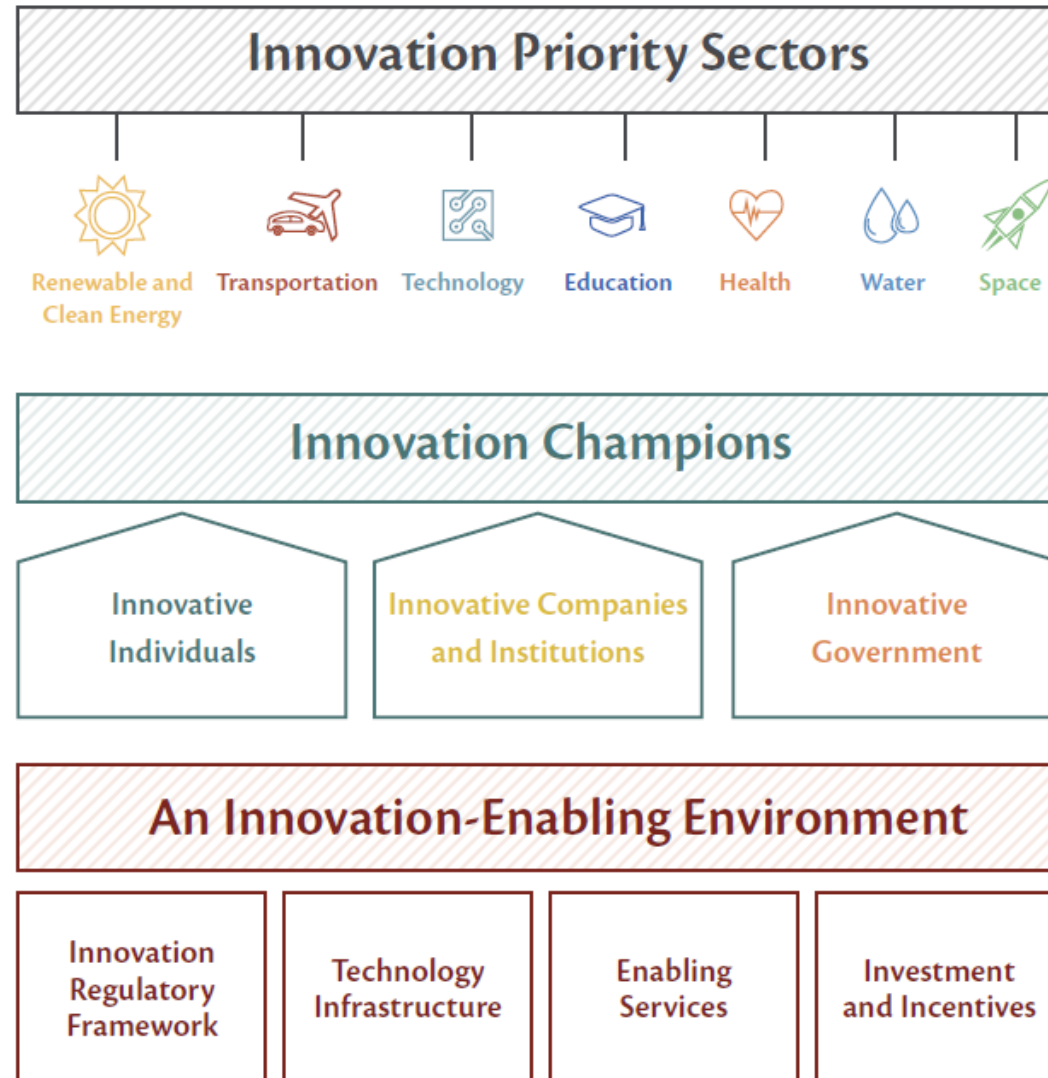
Source: UAE National Innovation Policy, Prime Minister's Office at the UAE Ministry of Cabinet Affairs 2015

UAE National Innovation Strategy (Source: NIS (2015))

- The policy was launched by HH Shaikh Mohammed bin Rashid Al Maktoum, Vice-President and Prime Minister of the UAE and Ruler of Dubai on 19 October 2014.
- Innovation is defined as
 - *“the aspiration of individuals, private institutions and governments to achieve development by generating creative ideas and introducing new products, services and operations that improve the overall quality of life. Innovation is key to promoting economic growth, increasing competitiveness and providing new job opportunities.”*
- The National Innovation Strategy (NIS) aims to take innovation in the UAE to new levels, where a culture of innovation is embedded amongst individuals, companies and governments. It primarily focuses on identified seven priority sectors that will drive future innovation.
- The NIS considers that Innovation is essential to:
 - Improving the quality of life
 - Increasing economic diversification
 - Enhancing knowledge economy
 - Improving competitiveness
 - Creating high skilled jobs
 - Promoting entrepreneurship
- The NIS Framework is structured around three key pillars (see figure below):
 - 1. An Innovation-Enabling Environment
 - 2. Innovation Champions
 - 3. Innovation Priority Sector

UAE National Innovation Strategy (Source: NIS (2015))

Figure: Innovation Framework in UAE



An Innovation-Enabling Environment (Source: NIS (2015))

Creating the right environment for innovation is key to the success of innovation efforts. It defines the environment in which individuals, firms and governments work, and demonstrates a nation's ability to attract investments and innovative ideas. The NIS seeks to enabling environment for innovation by a four pronged strategy: 1. developing the right regulatory framework, 2. providing comprehensive enabling services, 3. enhancing the technology infrastructure and 4. ensuring the availability of investments and incentives.

1. Innovation Regulatory Framework

An ideal environment for innovation is underpinned by supporting laws and regulatory frameworks. The NIS seeks to establish the innovation regulatory framework by developing rules and regulations that promote innovation, allowing for the rapid enactment of relevant legislations. It aims as well to design efficient and effective patent registration procedures, while raising awareness of the importance of intellectual property rights and the best methods to protect those rights.

2. Enabling Services

The NIS is committed to develop a first-rate education system as a base for innovation. It will promote research and development across universities, besides encouraging incubators to sponsor and assist innovators in transforming their ideas into leading projects. The incubators will provide innovators with supportive counseling, training, and administrative support services, while promoting and funding their innovations.

3. Technology Infrastructure

Innovation is enabled by a technology infrastructure that supports the generation, dissemination and retention of knowledge needed by innovators. Hence, the NIS aims to ensure the best technology infrastructure for effective communication and to promotes innovation amongst individuals, companies and governments. The infrastructure will support the optimal use of information and communications technology in acquiring and sharing knowledge amongst universities, research institutions and innovation incubators.

4. Investment and Incentives

The NIS seeks to introduce new innovative funding options that meet the needs of both individuals and companies. While an advanced financial system is one of the key elements of an innovative culture, financial rules and regulations will be revisited to ensure their effectiveness in securing various funding options and promoting innovative ideas.

Innovation Champions (Source: NIS (2015))

1. Innovative Individuals

Human Capital is a cornerstone of innovation. Given the importance of the citizen in the UAE's development, the NIS focuses on developing individuals and entrepreneurs who exemplify a spirit of innovation. It aspires to empower the nation to drive local innovation by developing innovative national talents and capabilities in science, technology, engineering, mathematics and entrepreneurship. Schools and universities will play a bigger role in promoting innovation by collaborating with top global academic institutions to introduce new specialized educational materials, while launching innovation challenges and recognizing national innovators. The NIS will thus develop structured mechanisms to recognize and sponsor creative and talented individuals at an early age, and help develop their capabilities and harness their full potential. It also aims to embed a national culture of innovation and entrepreneurship that recognizes innovators and promotes collaboration between the government, private sector and media agencies. Furthermore, the NIS emphasizes the importance of attracting best global talented scientists and innovators, and infuses a shift towards multi-cultural work teams to share knowledge and experience that help in building strong innovation skills.

2. Innovative Companies and Institutions

Most businesses around the world build their competitiveness around innovation. Companies with a sustainable approach to innovation achieve higher gross value added per employee, and higher wages and profits. These companies help in enhancing the competitiveness and standard of living of the countries where they operate. The NIS seeks to promote a culture of innovation generally across businesses, and specifically in small and medium enterprises. It aims to support national companies in developing, promoting and commercializing their innovations worldwide, thus enhancing their global market accessibility. In addition, the NIS aspires to build innovation and scientific research centers, adopt cutting-edge technologies, sponsor innovative nationals, and launch a national corporate innovation award. Furthermore, The NIS aims to attract the world's leading innovative companies, while promoting the role and reputation of the UAE as a global hub for testing innovations across identified innovation priority sectors. It also calls for the development of specialized zones for the identified sectors in line with the Internet, Media, Renewable Energy, Industry and other specialized zones currently present in the UAE. Social Innovation represents as well one of the NIS pillars, hence the call for innovating services and products that help address social challenges in the UAE and around the world. The NIS also aims to drive research institutions toward applied research in the identified innovation priority sectors.

3. Innovative Government

Government innovation plays an essential role in enabling public sector entities to provide customers and businesses with highly efficient services at a lower cost. Emphasizing the significance of innovation in enhancing public sector performance, the NIS aspires to embed a culture of innovation across government entities. It aims to promote the UAE as a global innovation hub, making it a leader of innovation in government services worldwide, and driving innovation in government policies, processes and procedures to increase overall efficiency. The NIS aims to introduce a comprehensive set of advanced tools, training programs and incentive schemes to promote innovation within government entities, and maintain an updated database of various national, regional and international innovative practices that can be shared across public sector entities within the UAE.

7 Priority Areas identified in NIS (Source: NIS (2015))



2. Science, Technology, & Innovation Policy



"Innovation, scientific research and a knowledge economy will create sustainable opportunity for our future generations. The new Science, Technology and Innovation Policy is a turning point in our march for development and progress."

His Highness Sheikh Khalifa bin Zayed Al Nahyan,
President of the UAE

"The Science, Technology and Innovation Policy is our roadmap to building a better future for generations to come. We have the human capital, effective governance and financial resources to accomplish a transformation of scientific progress in the UAE."

His Highness Sheikh Mohammed bin Rashid Al Maktoum,
UAE Vice President, Prime Minister, and Ruler of Dubai

Source: STI report; UAE Government
2015

2. Science, Technology and Innovation (STI) policy

Source: STI report; UAE Government 2015

1. Innovative Individuals

This Science, Technology and Innovation (STI) policy, adopted by His Highness Sheikh Khalifa bin Zayed Al Nahyan, President of the UAE, aims mainly to prepare the UAE for a post oil world. The timing of the policy coincides with the announcement of 2015 as the Year of Innovation in the UAE, as per the directives of His Highness the President of the UAE. It is also in line with the overall guidance of His Highness Sheikh Mohammed bin Rashid Al Maktoum, UAE Vice President, Prime Minister, and Ruler of Dubai, to launch the National Innovation Strategy and to form the National Science, Technology and Innovation Committee.

The STI policy starts includes a brief review of the UAE Vision 2021 and the National Innovation Strategy that form the umbrella and the main purpose for setting this policy; it then highlights the importance of science, technology and innovation as well as the current assets of the UAE. Finally, it details the policy's general framework that includes the UAE's ambition in the fields of science, technology and innovation, the focus areas for this policy and its enablers. The UAE government will work on implementing this policy and achieving its ambition, in order to make the UAE among the most innovative countries in the world, owing to advanced and innovative talent, resources, legislations and infrastructure.

The Importance Of A Science, Technology And Innovation Policy

Source: STI report; UAE Government 2015

Governments all over the world understand the importance of investing in Science, Technology and Innovation (STI) in order to achieve socio-economic development, as STI opens up opportunities for faster economic growth and creates sustainable wealth that independent of natural and non-renewable resources.

Moreover, a focus on STI fosters investment in talent and human capital required for development, and provides innovative solutions for a number of challenges in health, security, environment and society.

International Comparisons show a strong linkage between the Research and Development (R&D) activities carried out by countries and their level of economic development, as the countries focused on STI through spending on R&D have higher per capita income levels.

Further, public investment in R&D contributed to achieving big leaps in innovation and development in many fields such as the Internet, space and public health, while various evidence point to high returns on investment for the firms in the private sector that invest in R&D.

In this regards, in its attempt to remain one of the most prosperous countries in the long run, the UAE is set on transitioning into a knowledge and innovation based economy independent of oil resources, and on enhancing scientific and technological innovations in its public, private and academic sectors. That is why the UAE is in a unique position to invest and create the right environment and enhance the required culture to successfully achieve innovation based on science and technology.

National Science, Technology and Innovation Committee

Source: STI report; UAE Government 2015

In November 2014, the National Science, Technology and Innovation Committee was formed to bring under its umbrella a number of Ministers and Chairmen of federal government entities leading the innovation sectors.

The Committee will monitor the implementation of the National Innovation Strategy and the policies and initiatives that emanate from it; enhance coordination, cooperation and exchange of expertise among federal and local entities; follow up progress of innovation initiatives and its related indices nationwide; and engage the private sector and ensure that its social and economic contributions stimulate innovation. The National Science, Technology and Innovation Committee and the different working teams emanating from it worked on the development of the STI policy, which will be updated on an annual basis.

The Essential Role Of Science & Technology In Innovation

There are three main types of innovation:(**Source: STI report; UAE Government 2015**)

1. Business innovation,
2. Technology-based innovation, and
3. Science-based innovation.

The goal of the UAE's Science, Technology & Innovation policy is to achieve robust science and technology-based innovation.

1. Business Innovation

The most important input to business innovation is good and innovative ideas. Business innovation does not require formal scientific or technological expertise, and it often requires modest investments to commercialize. Examples of such innovation include: Shared economy applications (e.g. transportation and hospitality); and • Internet-based crowdfunding platforms

2. Technology-based Innovation

Technology-based innovation requires subject matter expertise, and may also require specialized infrastructure. Software development, for example, requires coding expertise and some computing equipment. Development of the touchscreen computer interface, on the other hand, required expertise in several areas including computer science, electrical engineering, and mechanical engineering. Mastering these various fields of knowledge requires more time and infrastructure than simply learning to code.

Examples of Technology-based innovation include: Self-driving vehicles; and Intelligent and virtual personal assistants.

3. Science-based innovation

This type of innovation may usually be undertaken only by those with highest levels of knowledge and experience, and with access to specialized and usually expensive laboratory equipment and facilities. It also typically requires cooperation of teams of scientists and other technical experts, as is the case for the development of new pharmaceuticals. It is worth noting that Science-based innovation often provides higher return on investment compared to the other two types.

Examples of Science-based innovation include: Genomics and biotechnology; and Solar cells.

National Science, Technology and Innovation Framework



Source: STI report; UAE Government 2015

3. THE UAE VISION 2021

Source: STI report; UAE Government 2015

The UAE and its prudent leadership recognition of the importance of innovation and the primary role it plays in economic progress is reflected in the UAE Vision 2021 that highlights science, technology and innovation as main drivers of growth and progress. UAE Vision 2021 emphasizes their role in the UAE's transition towards a knowledge based economy and in ensuring sustainable development for the country.

The UAE's National Agenda 2021 identified many indicators in science, technology and innovation and set ambitious targets for them. These include the UAE becoming one of the top ten countries in the world in the Global Innovation Index, and increasing Research and Development expenditure three folds by 2021. Because of the great role of human capital in enhancing innovation, the UAE is also seeking to increase the share of knowledge workers to 40% of total workforce, and to advance the rank of its students in mathematics, science and reading to become among the 20 highest ranked countries by 2021.

“Innovation, research, science and technology will form the pillars of a knowledge-based, highly productive and competitive economy, driven by entrepreneurs in a business-friendly environment where public and private sectors form effective partnerships”

UAE Vision 2021

We want to be among the
best countries in the world by **2021**

Source: UAE Vision 2021 document, released in 2010

UAE Vision 2021



Launched in 2010

UNITED IN AMBITION
AND DETERMINATION



Source: UAE Vision 2021 document,
released in 2010

“ Work is a true criterion of **citizenship**. It is evidence of **sincerity** and **loyalty**. We all share the **responsibility** of building this country, protecting its **sovereignty** and **safeguarding** the gains”

His Highness Sheikh Khalifa bin Zayed Al Nahyan
President of the United Arab Emirates



“ The word 'impossible' is not in **leaders'** dictionaries. No matter how big the **challenges**, strong **faith**, **determination** and **resolve** will overcome them”

HH Sheikh Mohammed Bin Rashid Al Maktoum
UAE Vice President, Prime Minister and Ruler of Dubai



UAE Vision 2021: Some excerpts

Source: UAE Vision 2021 document, released in 2010

UAE aspires to be among the best countries in the world by 2021

In a **strong** and **safe** Union, **Knowledgeable** and **innovative** Emiratis will confidently build a **competitive** and **resilient** economy. They will thrive as a **cohesive society** bonded to its identity, and enjoy highest standards of living within a **nurturing** and **sustainable** environment.

We want the UAE to transform its economy into a model where growth is driven by knowledge and innovation. Productivity and competitiveness will come to rival the best in the world, as a result of investment in science, technology , research and development throughout the fabric of UAE economy.

An ambitious and confident nation grounded in its heritage.

Ambitious and responsible Emiratis will successfully carve out their future, actively engaging in an evolving socio-economic environment and drawing on their strong family and communities, moderate Islamic values, and deep-rooted heritage to build a vibrant and well-knit society.

Knowledge-based and highly productive economy

Innovation, research, science, and technology will form the pillars of a knowledge-based highly productive and competitive economy, driven by entrepreneurs in a business-friendly environment where public and private sectors form effective partnerships.

UAE CENTENNIAL PLAN 2071

- The UAE Centennial 2071, launched by H.H. Sheikh Mohammed bin Rashid Al Maktoum, Vice President and Prime Minister of the UAE and Ruler of Dubai, is a long-term vision that spans five decades and represents a clear roadmap for government work to promote the UAE's reputation and soft power.
- The UAE Centennial 2071 Plan aims to invest in future generations and equip them with the skills and knowledge needed to keep pace with the global rapid changes in order to make the UAE the best country in the world by its centennial in 2071.
- The UAE Centennial 2071 plan was based on the historic lecture given by H. H. Sheikh Mohammed bin Zayed Al Nahyan, President of the UAE, in which he spoke about the strategies to prepare the UAE for future generations, ensuring sustainable development and embracing happiness and positivity.

UAE CENTENNIAL PLAN 2071

Pillars of UAE Centennial Plan 2071

1. Future-focused government.
2. Excellent education
3. A diversified knowledge economy
4. A happy and cohesive society

The UAE Centennial 2071 plan includes

- Promoting the UAE's reputation.
- Diversifying non-oil imports and exports.
- Investing in advanced technology-based education.
- Building Emirati values and ethics for the future generations.
- Raising productivity of the national economy.
- Enhancing social cohesion.

UAE CENTENNIAL 2071

Pillars of UAE Centennial 2071

The UAE Centennial 2071 is based on four pillars as follows:

Future-focused government: The objectives of the government under UAE Centennial 2071 include establishing the government of the UAE as the best government in the world, with a long-term vision and inspirational leadership that anticipates and prepares for the future. Other objectives include achieving happiness in society and spreading positive messages internally and to the world and developing mechanisms for monitoring long term variables in various sectors.

Excellent education: Regarding education, UAE Centennial 2071 highlights the importance of excellent quality of education. Certain areas of focus in education include advanced science and technology, space science, engineering, innovation and health sciences. Other educational measures include teaching students, mechanisms for discovering their individual talents early. On the institutional level, educational institutions are encouraged to be incubators of entrepreneurship and innovation and to be international research centres.

A diversified knowledge economy: The UAE's economy is aimed to be competitive and one of the best economies worldwide. This can be achieved by increasing productivity of national economy, support of national companies, investment in scientific research and promising sectors, focus on innovation, entrepreneurship and advanced industries, development of a national strategy to shape the future of the UAE's economy and industry, and place the UAE among international important economies. Knowledge economy can be achieved by a generation of UAE inventors and scientists and supporting them in technical sciences.

A happy and cohesive society: Community development is an integral part of UAE Centennial 2071. Some objectives in this regard include establishing a secure, tolerant, cohesive and ethical society that embraces happiness and a positive lifestyle and a high quality of life. The pillar also focuses on developing programmes to prepare future generations to serve as the UAE's goodwill ambassadors, as well as promoting women's participation in all sectors, making the UAE one of the best places to live in.

Dubai 10X Initiative: Launched Feb 2017



Dubai 10X Initiative: Launched Feb 2017

(Source : <http://dubai10x.ae/about-dubai-10x/>)

This initiative launched by His Highness Sheikh Mohammed bin Rashid Al Maktoum, UAE Vice President, Prime Minister and Ruler of Dubai, contributes to the implementation of the vision of Dubai as a city of the future, through new mechanisms of action mimic the future, and contribute to the sustainability of the competitive Dubai.

The government action is not limited to the provision of services and development, but also includes change in the work and mechanisms of systems, to make Dubai the largest laboratory for future government experiences in the world.

All Dubai Government entities will therefore embrace disruptive innovation, and develop services and innovations today that other world cities will introduce after ten years.

Disruptive innovation refers to exploiting available technologies to deliver new or existing services in radically different ways that are design-thinking-based and customer-focused, in contrast to incremental innovation, which focuses on making good services better for existing customers. Instead, disruptive innovation will create new operating and business models that replace traditional services and provide multiple times the value for end users and customers.

X-Units for each government entity

Every government entity will set up (with staff with adequate resources) – an independent X-Unit mandated with the objective of finding ways to disrupt their own organization's practices and develop their own moonshot solutions to deliver beyond their mission.

UAE AI Strategy

- The UAE sets a clear vision through its AI Strategy, to become the world leader in AI by 2031. Implementing this vision on the ground requires rigorous dedication and clear steps that outline the path for success.
- The AI Strategy aligns with the UAE Centennial 2071, to make the UAE the best country in the world by 2071.
- The AI Strategy will contribute significantly in education, economy, government development, and community happiness through various AI technologies implementations in different sectors to include energy, tourism, and education, to list few.

UAE AI Strategy

The UAE AI and Blockchain Council will overlook the implementation of the AI Strategy throughout all emirates but ultimately, the implementation will be a multi-stakeholders effort and cooperation from different local and federal entities in the UAE. There are eight strategic objectives outlined in the AI Strategy, namely:

- Build a reputation as an AI destination.
- Increase the UAE competitive assets in priority sectors through deployment of AI.
- Develop a fertile ecosystem for AI.
- Adopt AI across customer services to improve lives and government.
- Attract and train talent for future jobs enabled by AI.
- Bring world-leading research capability to work with target industries.
- Provide the data and supporting infrastructure essential to become a test bed for AI.
- Ensure strong governance and effective regulation.

UAE Strategy for Artificial Intelligence (AI)

In October 2017, the UAE Government launched '**UAE Strategy for Artificial Intelligence (AI)**'. This marks the post-mobile government phase which will rely on various future services, sectors and infrastructure projects.

The strategy aims to:

- Achieve the objectives of UAE Centennial 2071
- Boost government performance at all levels
- Use an integrated smart digital system that can overcome challenges and provide quick efficient solutions
- Make the UAE the first in the field of AI investments in various sectors
- Create new vital market with high economic value.

Sectors

- Transport – to reduce accidents and cut operational costs
- Health – to minimise chronic and dangerous diseases
- Space – to help conduct accurate experiments, reduce rate of costly mistakes
- Renewable energy – to manage facilities
- Water – to conduct analysis and studies to provide water sources
- Technology – to increase productivity and help with general spending
- Education – to cut costs and enhance desire for education
- Environment – to increase forestation rate
- Traffic – to reduce accidents and traffic jams and draw more effective traffic policies.

UAE Strategy for Artificial Intelligence (AI)

Themes

The AI strategy has five themes:

- the formation of the UAE AI Council
- workshops, programmes, initiatives and field visits to government bodies
- develop capabilities and skills of all staff operating in the field of technology and organise training courses for government officials
- provide all services via AI and the full integration of AI into medical and security services
- launch leadership strategy and issue a government law on the safe use of AI.

UAE Green Economy for Sustainable Development

- Under the Green Economy for Sustainable Development initiative, the UAE seeks to become a global hub and a successful model of the new green economy.
- The UAE aims to become one of the world leaders in this area as well as a center for the export and re-export of green products and technologies, and to maintain a sustainable environment to support long-term economic growth while protecting the environment.
- By virtue of Vision 2021, the UAE is striving to diversify its income resources by moving away from oil.
- In January 2012, Sheikh Mohammed launched the Green Economy initiative under the slogan: A green economy for sustainable development. Under this initiative, the UAE seeks to become a global hub and a successful model of the new green economy, to enhance the country's competitiveness and sustainability and preserve its environment for future generations.
- Through this initiative, the UAE aims to become one of the world leaders in this area as well as a center for the export and re-export of green products and technologies, and to maintain a sustainable environment to support long-term economic growth while protecting the environment.

UAE Green Economy for Sustainable Development

The initiative includes a range of programmes and policies in the areas of energy, agriculture, investment and sustainable transport in addition to new environmental and constructional policies.

- The first field of green energy aims to promote the production and use of renewable energy.
- The second field includes government policies aimed to encourage investments in green economy and to facilitate the production, import, export and re-export of green products and technologies.
- The third field relates to developing urban planning policies that preserve the environment and to raise the efficiency of housing and buildings environmentally.
- The fourth field consists of means for dealing with the effects of climate change, promote organic agriculture, maintain biodiversity and protect the ecological balance.
- The fifth field aims at rationalising the use of water resources, electricity and natural resources and recycle waste.
- The sixth field includes development and promotion of green technology.

UAE Innovates

UAE Innovates is a nation-wide celebration and one of the largest innovation festivals in the world. It was launched in 2015 and has become a yearly flagship event in the United Arab Emirates. It is a consolidated effort by the government, the private sector, and individuals to help create a widespread culture of innovation

- UAE Innovates' activities aim to:
 - Celebrate innovation in the UAE and support the implementation of the advanced national innovation strategy
 - Enhance the development of innovative ideas and capabilities, and create a widespread culture of innovation
 - Contribute to strengthening the country's position as a global hub for innovation
- In its yearly edition of the UAE Innovates Award, the Mohammed bin Rashid Center for Government Innovation invite you to participate in the UAE Innovates Award by nominating an innovative project your organization worked on.

Supporting Materials & Extra Reading

Read the following policy documents:

- 1.UAE National Innovation Policy
- 2.Science, technology, & Innovation Policy
- 3.UAE Vision 2021
- 4.Dubai 10X Initiative
- 5.UAE Strategy for Artificial Intelligence

Summary

During the session, you learnt the major thrusts of four major policy documents that affect innovation in UAE.

1. UAE National Innovation Policy
2. Science, technology, & Innovation Policy
3. UAE Vision 2021 & UAE Centennial 2071
4. Dubai 10X Initiative
5. 'UAE Strategy for Artificial Intelligence (AI), 2017
6. Green Economy for Sustainable Development
7. UAE Innovates & Government Innovation Lab

Formative Assessment

1. What is the definition of Innovation in the National Innovation System?

Feedback:

Innovation is defined as

“the aspiration of individuals, private institutions and governments to achieve development by generating creative ideas and introducing new products, services and operations that improve the overall quality of life. Innovation is key to promoting economic growth, increasing competitiveness and providing new job opportunities.”

2. What is the objective of the National Innovation Strategy (NIS)?

Feedback:

The National Innovation Strategy (NIS) aims to take innovation in the UAE to new levels, where a culture of innovation is embedded amongst individuals, companies and governments. It primarily focuses on seven identified priority sectors that will drive future innovation.

3. What is UAE’s Science, Technology and Innovation (STI) policy?

Feedback:

The Science, Technology and Innovation (STI) policy, adopted by His Highness Sheikh Khalifa bin Zayed Al Nahyan, President of the UAE, aims mainly to prepare the UAE for a post oil world. The timing of the policy coincides with the announcement of 2015 as the Year of Innovation in the UAE, as per the directives of His Highness the President of the UAE. It is also in line with the overall guidance of His Highness Sheikh Mohammed bin Rashid Al Maktoum, UAE Vice President, Prime Minister, and Ruler of Dubai, to launch the National Innovation Strategy and to form the National Science, Technology and Innovation Committee

Formative Assessment

4. What is the relationship of UAE Vision 2021 with Innovation?

Feedback:

The UAE and its prudent leadership recognition of the importance of innovation and the primary role it plays in economic progress is reflected in the UAE Vision 2021 that highlights science, technology and innovation as main drivers of growth and progress. UAE Vision 2021 emphasizes their role in the UAE's transition towards a knowledge based economy and in ensuring sustainable development for the country.

The UAE's National Agenda 2021 identified many indicators in science, technology and innovation and set ambitious targets for them. These include the UAE becoming one of the top ten countries in the world in the Global Innovation Index, and increasing Research and Development expenditure three folds by 2021. Because of the great role of human capital in enhancing innovation, the UAE is also seeking to increase the share of knowledge workers to 40% of total workforce, and to advance the rank of its students in mathematics, science and reading to become among the 20 highest ranked countries by 2021.

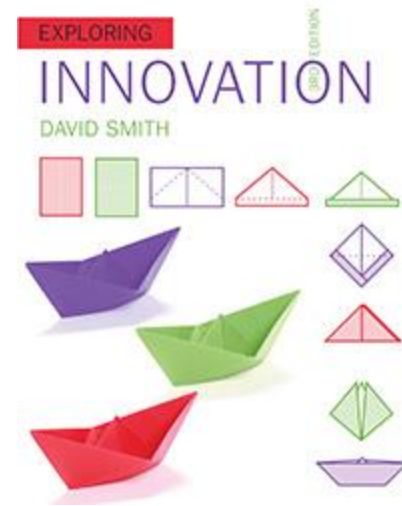
5. What is the Dubai 10X initiative?

Feedback:

This initiative contributes to the implementation of the vision of Dubai as a city of the future, through new mechanisms of action mimic the future, and contribute to the sustainability of the competitive Dubai. The government action is not limited to the provision of services and development, but also includes change in the work and mechanisms of systems, to make Dubai the largest laboratory for future government experiences in the world. All Dubai Government entities will therefore embrace disruptive innovation, and develop services and innovations today that other world cities will introduce after ten years. Disruptive innovation refers to exploiting available technologies to deliver new or existing services in radically different ways that are design-thinking-based and customer-focused, in contrast to incremental innovation, which focuses on making good services better for existing customers. Instead, disruptive innovation will create new operating and business models that replace traditional services and provide multiple times the value for end users and customers.

Green Innovation

Chapter 13



Objectives

When you have completed this chapter you should be able to:

- Describe the UAE Green Economy for Sustainable Development policy
- Understand the nature of Circular economy;
- Analyse the origins of the concept of green innovation;
- Categorise the various forms that green innovations can take;
- Analyse the factors that drive green innovation;
- Analyse the barriers that deter the adoption and diffusion of green innovation;
- Evaluate the business strategies that can be employed when launching green innovations.



Green innovation

- The excessive use of non-renewable resources and fast economic development has resulted in an adverse effect on the environment
- Resource limitations and environmental degradation emphasized sustainable operations and safeguarding of the environment a major global issue.
- Government institutions globally are enacting and consolidating environmental laws and regulations to address degradation
- Green innovation comprises all type of innovations that contribute to the creation of key products, services, or processes to reduce the harm, impact, and deterioration of the environment while optimizing the use of natural resources. It is an innovation that emphasizes waste reduction, pollution prevention and implementation of an environmental management system (Eiadat et al., 2008)

Terminology of Green Innovation

- Eco-innovation/Ecological innovation
- Environmental innovation
- Sustainable innovation
- Green innovation

Rationale for Green Innovation

- The excessive use of non-renewable resources due to rapid economic development has hurt the atmosphere and elevated various environmental worries (Atlin and Gibson, 2017).
- Human activities is affecting the planet in alarming ways:
 - Sea levels are rising.
 - Global temperatures are slowly increasing.
 - The oceans are getting incrementally warmer.
 - Ice sheets around the world are shrinking.
 - Glaciers are retreating, extreme weather events are on the rise, there is less snow cover and overall pollution levels keep increasing.
- At the same time, the global population is also increasing, at a rate of approximately 1.1 percent every year, with about 80 million people being added.
- Therefore, we need to drastically change our lifestyles to preserve our planet.

Source: The Marttime Eexeutive, July 2020



Rationale for Green Innovation

- Innovation has long been seen as one of the primary sources of economic growth and prosperity. Consequently, society in general and governments in particular have sought to foster advances in science and technology, since they generally underpin innovation.
- As Dodgson et al. (2008: 323) observe, ‘as long as new technologies were being developed and applied they were generally thought to be beneficial’.
- Recently, it was recognized that there may be limits to this process, because innovations in certain circumstances can lead to serious environmental problems.
- Rachel Carson, in her book *Silent Spring* (1962) highlighted for the first time the negative impact on the environment of chemical treatments designed to eradicate pests on an industrial scale. This marked the start of the environmental movement.



Rationale for Green Innovation

- In the 1970s successive energy crises led to recognition for the first time that natural resources were finite and exhaustable.
- In the 1980s climate change was added to the agenda with the discovery of a hole in the ozone layer above the Antarctic resulting from the industrial use of CFC gases. Since then freak weather conditions and a series of authoritative reports such as the Stern Review (2006) have highlighted the seriousness of climate issues in particular and environmental issues in general.
- Increased customer environmental consciousness has supported demand for environmentally friendly products leading firms to adopt green innovation. Green innovation became an increasingly important topic both in industry and in academia, and one that is only likely to increase in importance in the future.

Rationale for Green Innovation

- Environmental legislations has frequently induced a somewhat negative reaction from the business community. Regulatory controls were grudgingly accepted as a trade-off between environmental and economic performance.
- An important turning point occurred when Porter & van der Linde (1995) suggested that environmental regulation represented a positive opportunity. Firms would innovate to reduce the costs of compliance and production. The resulting innovations in the form of new products and processes would improve firms' competitiveness leading to a positive relationship between environmental and economic performance.

UAE Green Economy for Sustainable Development

- Under the Green Economy for Sustainable Development initiative, the UAE seeks to become a global hub and a successful model of the new green economy. The UAE aims to become one of the world leaders in this area as well as a centre for the export and re-export of green products and technologies, and to maintain a sustainable environment to support long-term economic growth while protecting the environment.
- By virtue of Vision 2021, the UAE is striving to diversify its income resources by moving away from oil. In January 2012, Sheikh Mohammed launched the Green Economy initiative under the slogan: A green economy for sustainable development.
- Under this initiative, the UAE seeks to enhance the country's competitiveness and sustainability and preserve its environment for future generations.
- Through this initiative, the UAE aims to become one of the world leaders in this area as well as a centre for the export and re-export of green products and technologies, and to maintain a sustainable environment to support long-term economic growth while protecting the environment.



UAE Green Economy for Sustainable Development

The initiative includes a range of programmes and policies in the areas of energy, agriculture, investment and sustainable transport in addition to new environmental and constructional policies.

- 1.The first field of green energy aims to promote the production and use of renewable energy.
- 2.The second field includes government policies aimed to encourage investments in green economy and to facilitate the production, import, export and re-export of green products and technologies.
- 3.The third field relates to developing urban planning policies that preserve the environment and to raise the efficiency of housing and buildings environmentally.
- 4.The fourth field consists of means for dealing with the effects of climate change, promote organic agriculture, maintain biodiversity and protect the ecological balance.
- 5.The fifth field aims at rationalising the use of water resources, electricity and natural resources and recycle waste.
- 6.The sixth field includes development and promotion of green technology.



UAE Hosted COP 28 in November 2023 @ Expo City

The 2023 United Nations Climate Change Conference named COP28 UN Climate Change Conference was held from 30 November to 13 December at Expo City, Dubai, the United Arab Emirates.

It was the biggest of its kind with 85 000 participants, including more than 150 Heads of State and Government, civil society, business, Indigenous Peoples, youth, philanthropy, and international organizations in attendance.



Circular Economy

In recent years the concept of Circular Economy(CE) has received growing attention from civil society, businesses and policy makers. It has emerged as an alternative to the linear model ("take-make-dispose" model), which is based on continuous growth and conversion of natural resources into products/services for human consumption.

The circular economy is defined as a model of production and consumption, which involves sharing, leasing, reusing, repairing, refurbishing, and recycling existing materials and products as long as possible. In this way, the life cycle of products is extended. In practice, it implies reducing waste to a minimum (European Parliament, 2015). In practice, when a product reaches the end of its life, its materials are kept within the economy wherever possible thanks to recycling thereby keeping waste to a minimum. The goods may then be productively used again and again, thereby creating further value.

Traditionally, companies have adopted a linear economic model, based on "take, make, and dispose" (Murray et al., 2017). In this model, goods are manufactured by consuming raw materials, and the goods are then purchased by consumers, used and ultimately disposed of as waste (European Parliament, 2015).

The 'take-make-dispose' or linear model is now recognized to be unsustainable for the planet. The importance of closing material cycles (Aguilar Esteva et al., 2021), has led to moving from a "take and discard" logic (Hopkinson et al., 2018), associated with a linear economy, to "reuse, recycle, and recover" (David et al., 2019; Harrold, 2009), typical of the Circular Economy.

UAE Circular Economy 2021-31

- The objectives of UAE Circular Economy Policy are to:

Achieve sustainable management of the economy and efficient use of natural resources

Promote circular economy and Sustainable Consumption Production patterns that reduce environmental stress and meet basic needs.

Encourage the private sector to shift to cleaner industrial production methods and techniques including the use of Artificial Intelligence.

Priority Sectors



Sustainable

Manufacturing



Green

Infrastructure



Sustainable

Transportation

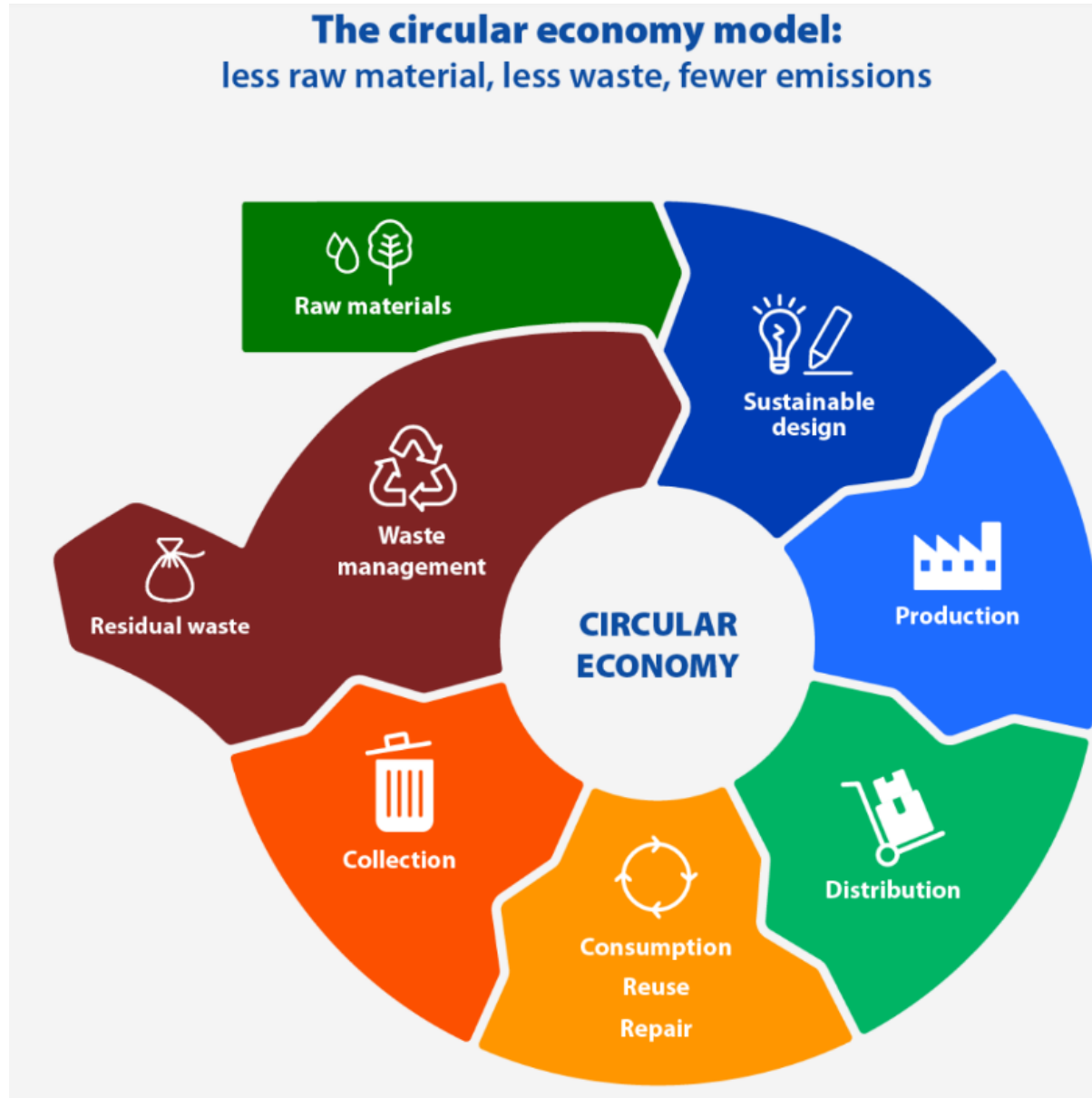


Sustainable food production

and consumption

European Parliament Model

(<https://www.europarl.europa.eu/topics/en/article/20151201STO05603/circular-economy-definition-importance-and-benefits>)





Circular Economy

Conversely, the CE model plays a fundamental positive role in the environment, and its interactions with the economic system. It is poised to transform economic systems and contribute to sustainable development by developing innovative business models, processes and products that close material loops.

Therefore, a circular economy may be considered as an economic system designed to minimize waste and optimise resource use by ensuring that materials and products are:

- Reused,
- Repaired,
- Reduced
- Recycled,
- Refused

for as long as possible.

In a circular economy, the traditional linear "take-make-dispose" model is replaced with one that prioritizes sustainability, resource efficiency, and environmental stewardship.



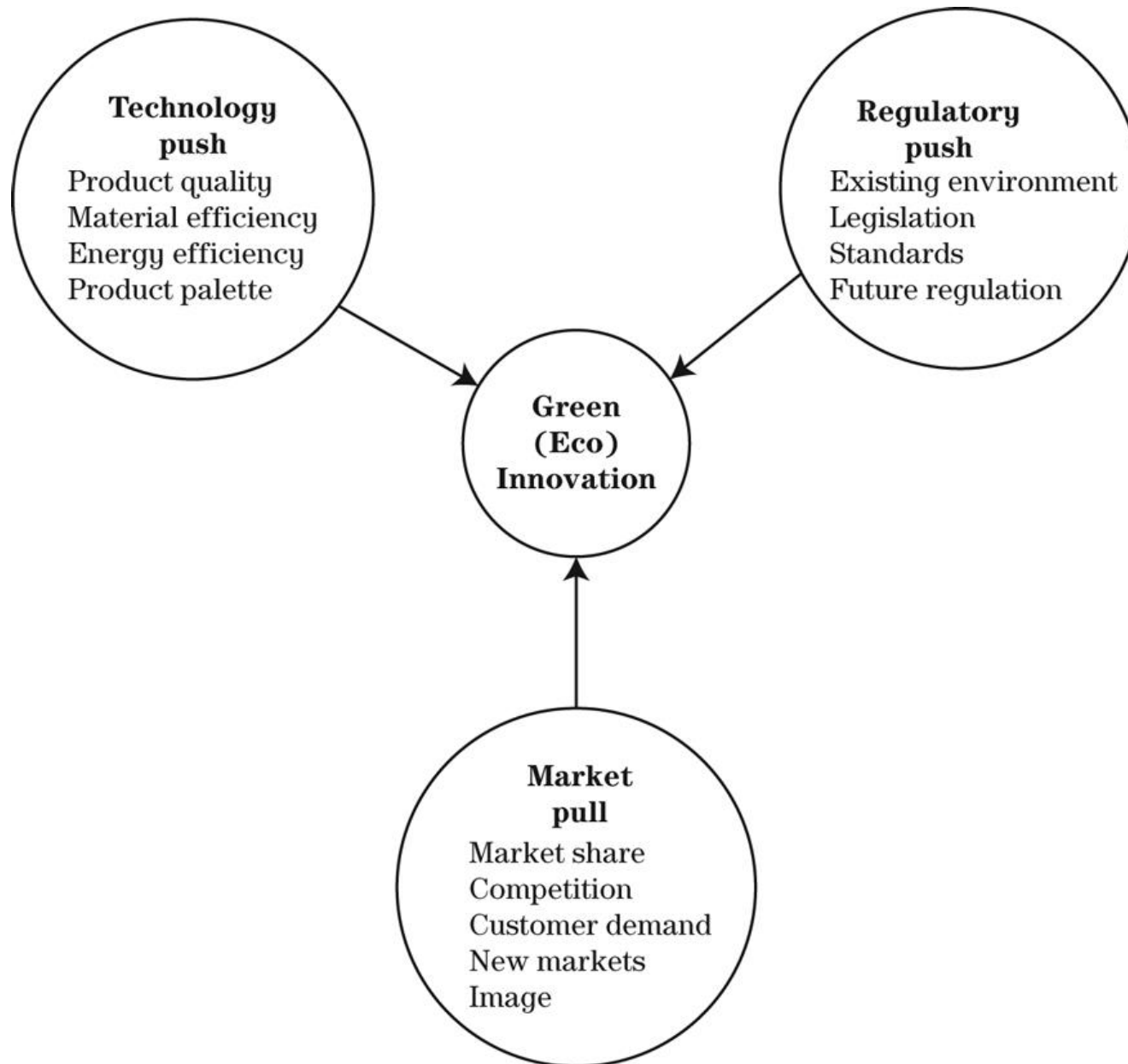
Circular Economy

The transition to a circular economy requires collaboration and innovation across sectors, including industry, government, academia, and civil society. By transitioning to a circular economy, societies can reduce their ecological footprint, mitigate resource scarcity, create new economic opportunities, and build more resilient and sustainable communities for the future.

Key strategies for implementing a circular economy include:

- **Design for Longevity**: Products are designed to be durable, repairable, recyclable, and easily disassembled so that their components can be reused or recycled at the end of their life cycle.
- **Resource Efficiency**: The focus is on using resources efficiently and minimizing waste generation throughout the entire value chain, from extraction and production to consumption and disposal.
- **Regenerative Practices**: The circular economy aims to regenerate natural systems by minimizing environmental impact, preserving ecosystems, and restoring natural capital.
- **Supply Chain Optimization**: Streamlining supply chains to minimize waste and resource consumption.
- **Business Model Innovation**: Adopting new business models such as product-as-a-service, sharing platforms, and leasing to encourage reuse and prolong product lifespans.
- **Policy Support**: Implementing policies and regulations that incentivize circular practices, such as extended producer responsibility (EPR) schemes, eco-design standards, and waste reduction targets.
- **Consumer Awareness**: Educating consumers about the benefits of a circular economy and promoting sustainable consumption patterns.

Drivers of Green Innovation



Technology push

Technological advances associated with:

- Material efficiency
 - Energy efficiency
 - Improved product quality
- Innovation in technologies and how they are applied are key to enabling industry to create new business values while also benefiting people and the planet.
 - In recent years, manufacturing companies have been upgrading their efforts towards sustainable manufacturing from pollution prevention to integrated approaches that take into account product lifecycles and wider impacts.
 - Eco-innovation enables this evolution through a combination of technological and non-technological changes that can yield substantial environmental improvements.



Market pull associated with groups of consumers such as:

- 'Green idealists'
 - i.e. moral, philosophical interests who are keen to utilize green innovations
- 'Trend setters'
 - i.e. 'image', association interests
 - interested in green innovations by virtue of their 'image. The latter are typically consumers keen to be seen by others as using or associated with green innovations.
 - Toyota Prius, the first mass-produced hybrid electric car sold unexpectedly well in the US, in part because many Hollywood stars including Cameron Diaz, Demi Moore, Dustin Hoffman and Matt Damon bought and used the car apparently because it was seen to be a green innovation.

Regulatory push

Command and control instruments

- Mandatory requirements regarding environmental standards
- e.g. outlawing the use of materials considered to be toxic. For instance, in the 1980s chloroflourocarbons (CFCs) used as refrigerants, propellants (i.e. for aerosols) and solvents were banned because their use had led to the depletion of the ozone layer in the Earth's upper atmosphere.
- E.g. Customers may buy recyclable bags and they have to pay for Plastic bags in supermarkets. Absence of drinking straws and use of forks made from bamboo fast foods such as Freedom Pizza.

Market-based instruments

- Financial incentives/penalties e.g. taxes, emission permits, subsidies, take-back schemes, etc.
- e.g. the system of Vehicle Excise Duty introduced in the UK in 2001 that linked the tax paid on vehicles to their CO2 emissions



Barriers to green innovation

The 'double' hurdle:

- Technological externalities
 - i.e. like most innovations there is the problem of appropriability where others get the benefits of the innovator's creative endeavour
- Environmental externalities
 - i.e. unattributed environmental externalities where the polluter does not pay for the adverse negative environmental impact of his or her activities



Types of barrier

- In addition to this ‘double hurdle’ brought on by market failure, there are also a number of other factors that can act as significant barriers to the successful diffusion of green innovations.
 - Economic barriers
 - Technological barriers
 - The notion of ‘technological lock-in’. This centres on the cost of transition to new cleaner technologies. The actual phenomenon of technological lock-in is associated with the development of standards (i.e. norms or requirements) that may be less than optimal. E.g. Internal combustion engines in cars and VHS tapes.
 - Institutional barriers



- For more than 125 years the Coca-Cola Company has been using a distinctive shaped bottle. But in 2009 the company announced a new innovation affecting the iconic Coke bottle.
- Henceforth plastic Coke bottles would be made not from conventional oil-based plastic but from a new form of plastic derived from plants.
- Key features of the PlantBottle™'s breakthrough technology are that it:
 - looks, feels and functions like traditional PET plastic
 - is fully recyclable, and
 - is made from up to 30 per cent renewable, sustainably sourced plant-based material.

Case Study: New Coke Bottle

Case Study: New Coke Bottle

- The plant-based material used in the PlantBottle™ is made through a process that converts natural sugars found in plants into a key component of PET (polyethylene terephthalate) plastic. The sugar itself is sourced from Brazil as sugarcane ethanol.
- An initial life-cycle analysis indicated that the introduction of PlantBottle™ packaging resulted in a 12–19 per cent reduction in carbon impact. According to Coca-Cola, in 2010 the use of the new PlantBottle™ technology helped it reduce its carbon dioxide emissions by 30 000 tons annually.
- Beverage brands ranging from DASANI to Coca-Cola to Gold Peak have used PlantBottle packaging, which today accounts for 30% of the company's packaging volume in North America and 7% globally.
- By replacing up to 30% of the petroleum used to make PET plastic bottles with material from sugar cane and other plant matter, PlantBottle has avoided the CO2 emissions equivalent of taking nearly 1 million vehicles off the road since 2009.
- Since introducing PlantBottle, Coca-Cola has allowed non-competitive companies to use the technology and brand in their products –from Heinz Tomato Ketchup, to the fabric interior in certain Ford Fusion hybrid sedans.

The Porter hypothesis

“environmental regulation represents a positive opportunity for innovation because it would induce firms to innovate in order to reduce the costs of compliance and production.”



Formula 1 Makes Switch
leantechnica.com

Green Innovation: Lessons from Formula One

Three years ago, the Formula One authorities announced big changes to the rules governing engine size and fuel capacity. 2.4 litre normally aspirated V8 engines had to be replaced by 1.6 litre turbo-charged engines. Simultaneously, unlimited fuel consumption had to be reduced from 160 to 100 kg.

- Lots of smart technologists and engineers work for the F1 industry and the new regulations forced them to find ways of making cars more fuel efficient without loss of power. They recognised that in an internal combustion engine only around one-third of the fuel used actually propels the car, and went about recovering some of the lost energy.
- As a result, this year's F1 cars have two new energy-recovery systems: kinetic energy released when driver slams on the brakes is converted into electrical energy; and energy formerly lost through the exhaust is turned into electrical energy. The old V8 engines formerly used in F1 produced more than 750 brake horsepower, but the one-third smaller V6 engines produce 600 bhp (only 20 per cent less). In addition, the energy recovery systems are designed to provide an additional 160 bhp for 33 seconds every lap. So the new engines will produce as much power as the old engines, using 40 per cent less fuel.

Green Innovation: Lessons from Formula One

- This is not just a matter that should interest petrol heads. John Llewellyn says applying this technology to everyday motor vehicles could cut global oil consumption by 2 per cent or more a year.
- This F1 experience has a deeper significance: it shows what clever people can do when motivated. The same approach and commitment used by governments and other stakeholders will definitely help in the fight against global warming.

Seven Examples of Green Innovations

Some major Green projects are on the forefront of green-friendly solutions and are already changing things

1. 3D Printing

- 3D printing is being used to preserve the environment. Plastic currently makes about 13 percent of municipal waste in the U.S. and often ends up in landfills. The Perpetual Plastic Project is seeking to turn recycled plastic products into 3D plastic printing materials. On another front, Pembient is utilizing 3D printing to create artificial rhino horns and elephant ivory, which are then used to keep the animals from going extinct. Their hope is that the artificial products will undercut poachers, eventually driving them out of business.

(Source: Maritime Executive- <https://www.maritime-executive.com/editorials/seven-green-innovations-that-are-changing-the-world>):

Seven Examples of Green Innovations

2. Battery Storage Capacity

Elon Musk has created batteries which can store large amounts of energy and then be used to power everything from cars to houses. Musk's company Tesla is well known for producing electric cars, but this could have an even bigger impact.

3. Vertical Farming

As the population grows, space declines and the environment becomes increasingly unfriendly toward plants, new methods of farming will be required. Vertical farming, which involves going upward rather than outward, offers great promise in that area.



Seven exceptional green innovations

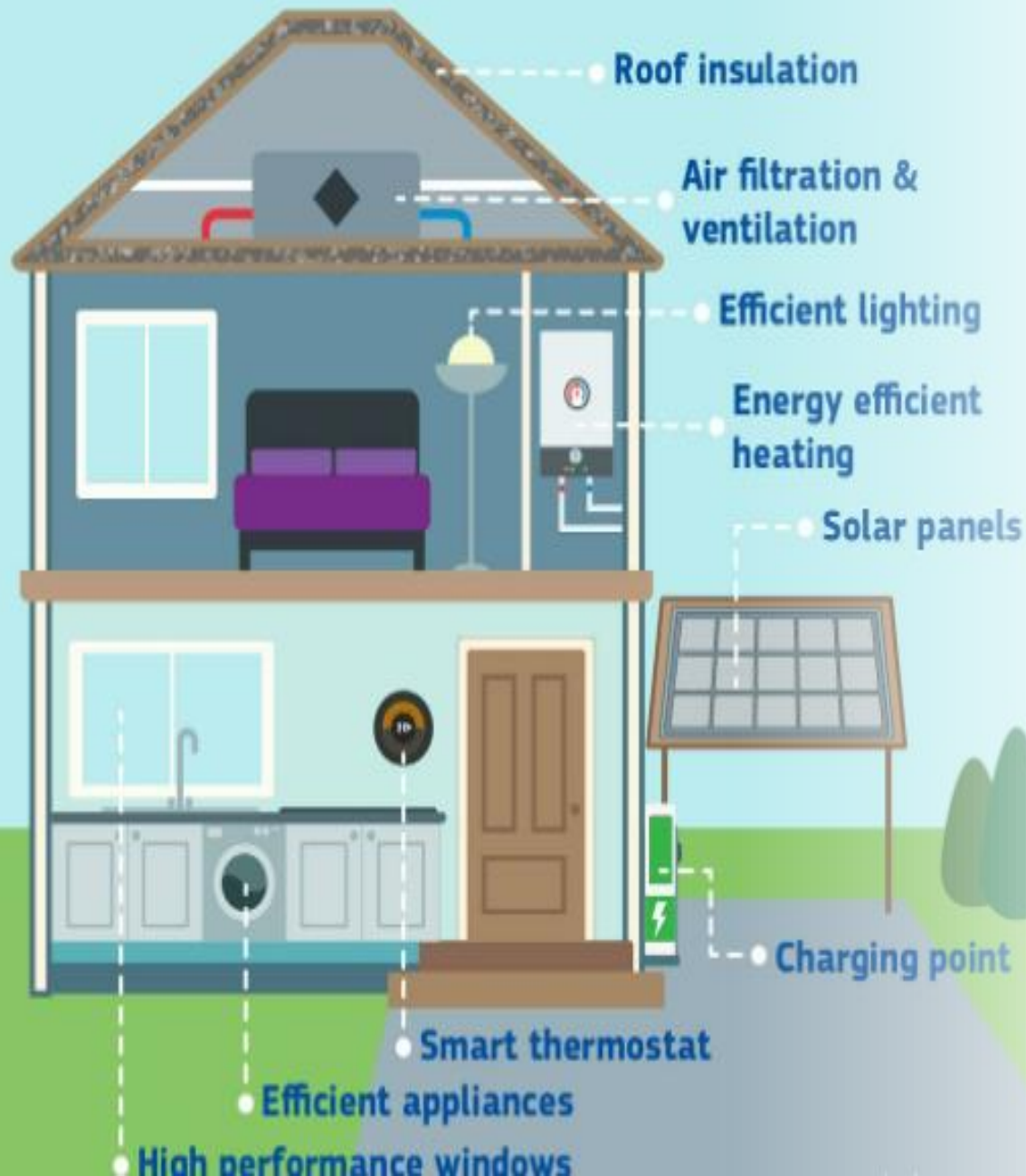
4. Desalination through renewable energy

- Typically, cleaning and desalination of water requires energy, which ties them closely to fossil fuels. However, in recent years, scientists have begun creating water cleaning solutions that depend on renewable energy. A team from Jain Irrigation and MIT recently devised a way to turn brackish water into 100 percent clean water. Instead of needing fossil fuels, this process relies completely on renewable, solar energy. It works by pulling salt out of the water and then using ultraviolet light to disinfect the water.

5. Emissions-Free (Electric) Vehicles

- vehicles are a huge source of harmful emissions.. The Tesla Model S being a completely Zero Emissions Vehicle. Unfortunately, Tesla is expensive, although automakers such as Toyota and Honda entering the market. In the future, there will likely be emissions-free vehicles including boats and planes. The greatest challenge will be producing these vehicles inexpensively enough.

Nearly Zero-Energy Buildings (NZEB)



6. Zero-Emissions Buildings

- About 40 percent of U.S. energy consumption comes from residential and commercial buildings. The increasing movement toward green buildings, particularly school houses that add zero emissions to the environment is encouraging. Astonishingly, the Australian architecture firm Archiblox recently revealed the world's first carbon-positive home, which actually adds energy back into the grid. These types of innovations are the future of environmental preservation.

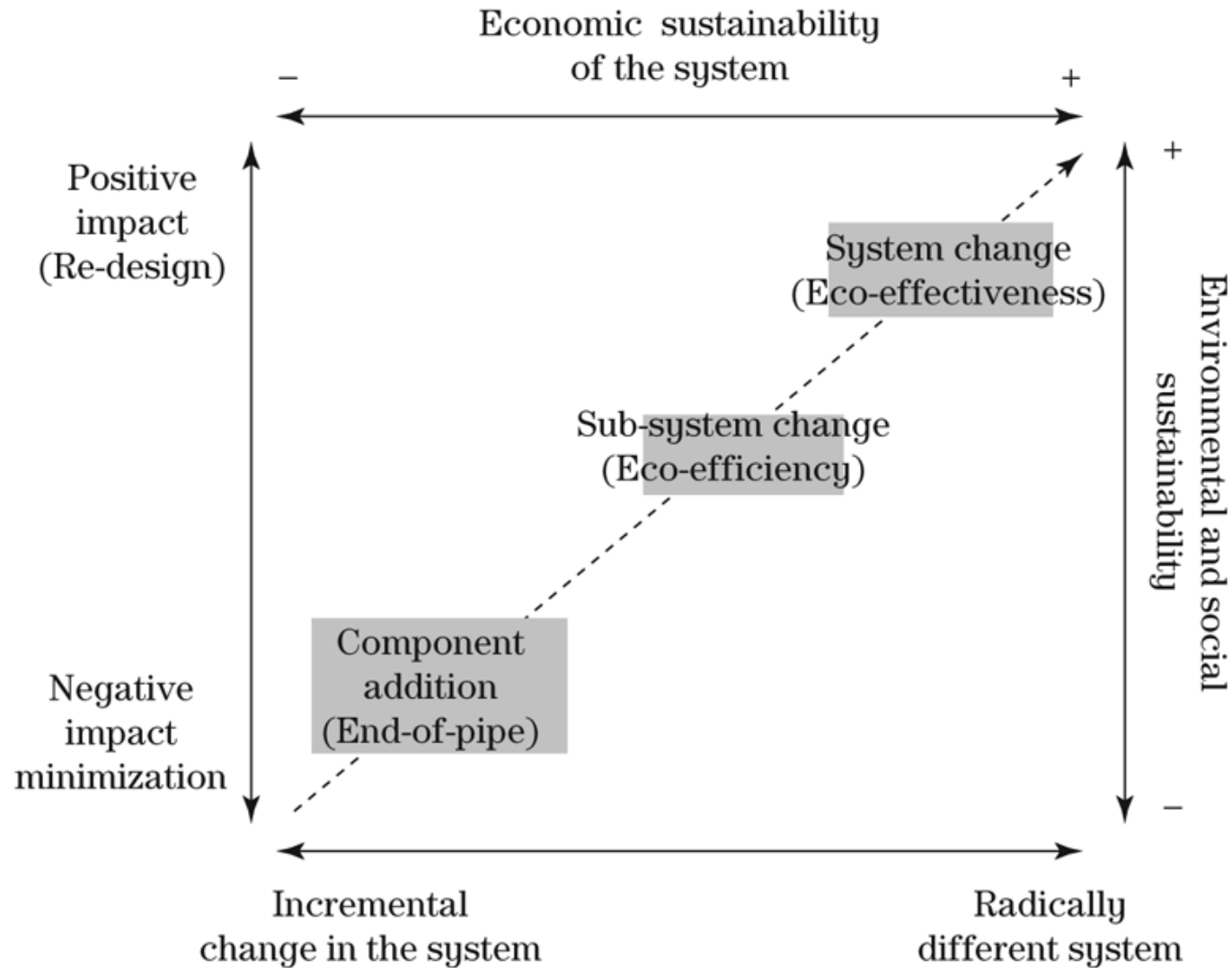
7. Removing Plastic From The Ocean

- The Ocean Cleanup Project, an ambitious effort to remove half the trash in the Great Pacific Garbage Patch. Adidas is phasing out plastic bags from their stores. The company has also introduced the first footwear created using plastic bottles.

Types of Green Innovation

- Component
- Sub-system/modular
- System

Types of Green Innovation



Catalytic Converters: What type of Green Innovation?

- Catalytic converters, or 'cats', were introduced in 1975 as a means to reduce the amount of pollution in car exhaust emissions. It converts some of the engine's dangerous pollutants into carbon dioxide and water vapor.
- Since 1993, all new cars incorporate a catalytic converter as well as a specific filter to comply with the latest Euro 6 emissions standards.
- A metal exterior hides a delicate honeycomb structure that's coated in precious metals such as palladium, rhodium and platinum. When heated, these materials cause a chemical reaction that purifies the exhaust gases. Because they need to be hot to work best, catalytic converters are usually placed near the engine where the temperature is highest.



Stack Chimneys: What type of Green Innovation?





Hybrid Cars

- A hybrid combines at least one electric motor with a gasoline engine to move the car, and its system recaptures energy via regenerative braking. Sometimes the electric motor does all the work, sometimes it's the gas engine, and sometimes they work together. The result is less gasoline burned and, therefore, better fuel economy. Adding electric power can even boost performance in certain instances.
- Electricity comes from a high-voltage battery pack that's replenished by capturing energy from deceleration that's typically lost to heat generated by the brakes in conventional cars. Hybrids also use the gas engine to charge and maintain the battery. Car companies use different hybrid designs to accomplish different missions, ranging from maximum fuel savings to keeping the vehicle's cost as low as possible.

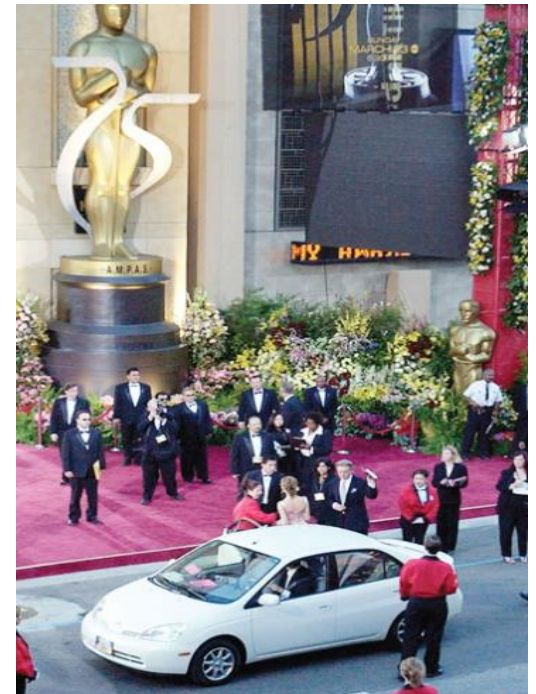


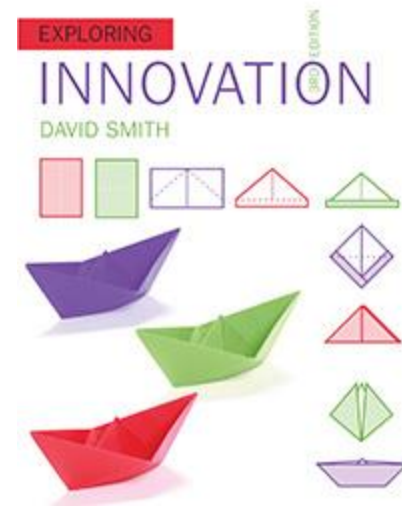
Electric Cars



Strategies for Green Innovation

- Niche strategy
 - e.g. Volvo Bus. Electric vehicles (EVs) which suffer the weakness of limited range are particularly appropriate in environments where most journeys are relatively short range, such as major cities.
- Endorsement strategy
 - e.g. Hollywood and the Toyota Prius
 - Chinese PC manufacturer Lenovo, gained valuable endorsement for green aspects of its products when in 2007 Greenpeace ranked the company in first place among electronics manufacturers for its recycling and toxic content policies
- Partnership strategy
 - Coca-Cola and Heinz and the development of the PlantBottle™. In 2011 Heinz signed a strategic partnership agreement Coca-Cola to develop and commercialize a new bio-based plastic technology for its bottles. This enabled Heinz to launch the bottle more rapidly than if it had tried to develop the technology on its own.





Innovation Offshoring

Chapter 12

Objectives

At the end of this chapter you will be able to:

- Evaluate the impact of globalisation on transaction costs;
- Differentiate and distinguish between outsourcing and offshoring;
- Appreciate and understand the concept of innovation offshoring;
- Analyse the factors that have led to the increased mobility of research and development (R&D);
- Distinguish between global innovation networks and global production networks;
- Analyse the impact of innovation offshoring on the process of innovation.



What is Innovation Offshoring?

- Innovation offshoring involves the practice of relocating a company's research and development (R&D) and other innovation-related activities to foreign countries.
- Represents the act of transferring to the offshore supplier any element of the innovation process, including collaborative research and development (R&D), supplier involvement in new product development (NPD) and the commercialization of innovations (Gusenbauer et al., 2015, Quinn, 2000).
- Driven by the globalization of markets for technology and knowledge workers, as well as changes in corporate innovation management.
- This may be typically illustrated by the significant impact on new product and process innovations in manufacturing industries due to the rapid advancement in electronics and software technologies (Nieto and Santamaria, 2007, Quinn, 2000).
- Companies in Western countries are at the forefront of this trend, but Asian governments and firms are also playing an increasingly active role as promoters and new sources of innovation.

Innovation Offshoring Vs Outsourcing

- Domestic outsourcing involves contracting a firm's activities to be conducted by other firms **locally**.
- Offshoring is a transaction that **crosses borders**.
- Offshoring is associated with the following benefits:
 1. Outsourcing allows firms to reduce factor costs and restructure their operations towards higher value-added activities such as R&D and innovation (e.g., Glass and Saggi 2001).
 2. If Offshoring takes place to technologically advanced countries, it may provide access to higher quality inputs (e.g., Griffith et al. 2006). This allows the firm to learn new technologies and expand its technological frontier.



Growth & Development of Offshoring

- Innovation offshoring began with the movement of production activities overseas, in search of lower costs, e.g., textiles and electronics.
- Led to creation of Global Production Networks. Driven by the search for new knowledge and greater efficiency.
- In the past few decades have therefore seen an increase in internationalization of research and development (R&D) by firms from advanced economies (Bertrand and Mol, 2013; Tojeiro-Rivero and Moreno, 2019).
- Technological complexity and shorter product life cycles in advanced economies combined by access to cheaper skilled labor and new markets in emerging economies has led to an increase in R&D internationalization (Cantwell, 1995).
- Moreover, the continuous rise in Innovation Offshoring (IO) relationships since early 2000s has proven that IO allows buying firms to access and benefit from offshore suppliers and country-specific resources, such as specialized talents, technological advances, and lower innovation costs than in developed countries (Lewin et al., 2009, Spithoven and Teirlinck, 2015, Van Beers and Zand, 2014).

Enablers facilitating Innovation Offshoring

- Changes in Corporate Innovation Management
 - i.e. shorter lead times; increased development of multi-technology products
- Increased specialisation through modularisation
- Increased availability and mobility of knowledge
- Globalisation of markets for technology

Factors encouraging Innovation Offshoring

- Improved technological linkages
 - i.e. *Electronification*
- Growth of markets for technology
- Growth of high tech clusters in Asia
- Growth of expertise and capability overseas
 - i.e. through technological catch-up and upgrading

THE NEW MOBILITY OF INNOVATION

Innovation offshoring runs counter to established wisdom. It is widely assumed that for a firm to grow, it must control resources that are valuable, rare, and to a significant extent immobile, and that “a firm’s rate of growth is limited by the growth of knowledge within it.”²¹ A related assumption is that innovation, in contrast to most other stages of the value chain, is highly immobile: it remains tied to specific locations, despite a rapid geographic dispersion of markets, finance, and production.²²

Only a decade ago, research on the geographical distribution of patents demonstrated that innovative activities of the world’s largest firms were among the least internationalized of their functions.²³ Experts assumed that innovation within the firm was and would always be highly localized because it usually requires dense exchange of knowledge (much of it tacit) between the users and producers of the resultant new technologies. Attempts to explain such spatial stickiness of innovation have focused on the *dynamics* of spatial agglomeration within localized innovation clusters.²⁴

There is no question that the demanding requirements of managing complex innovation projects tend to concentrate innovation in the home country. However, research on globalization has clearly established that the center of gravity has shifted beyond the national economy.²⁵ International linkages proliferate as markets for capital, goods, services, technology, and knowledge workers are integrated across borders.²⁶ While integration is far from perfect,

Source: Ernst(2006) Asia role in GIN

Why is Innovation Offshoring Important for Businesses?

- **Cost Efficiency:** By offshoring innovation activities to countries with lower labor costs, companies can significantly reduce their R&D expenses. This allows them to allocate more resources to other critical areas of their business.
- **Access to Specialized Skills:** Offshoring enables companies to tap into a global talent pool, accessing specialized skills and expertise that may not be readily available in their home country. This is particularly important for high-tech industries where specific technical skills are in high demand.
- **Faster Time-to-Market:** By leveraging time zone differences, companies can achieve round-the-clock development cycles, speeding up the innovation process and reducing time-to-market for new products and services.
- **Market Expansion:** Offshoring innovation activities can help companies better understand and cater to local markets, leading to more effective product localization and increased market penetration.
- **Risk Diversification:** By spreading innovation activities across multiple locations, companies can mitigate risks associated with political instability, economic downturns, or other local disruptions.

Innovation Offshoring (Source: Ernst, 2006) Asia role in GIN)

Critical Factors supporting economic growth and competitiveness in the USA and other developed countries have been policies that encourage strong investment in research and development (R&D) and innovation.

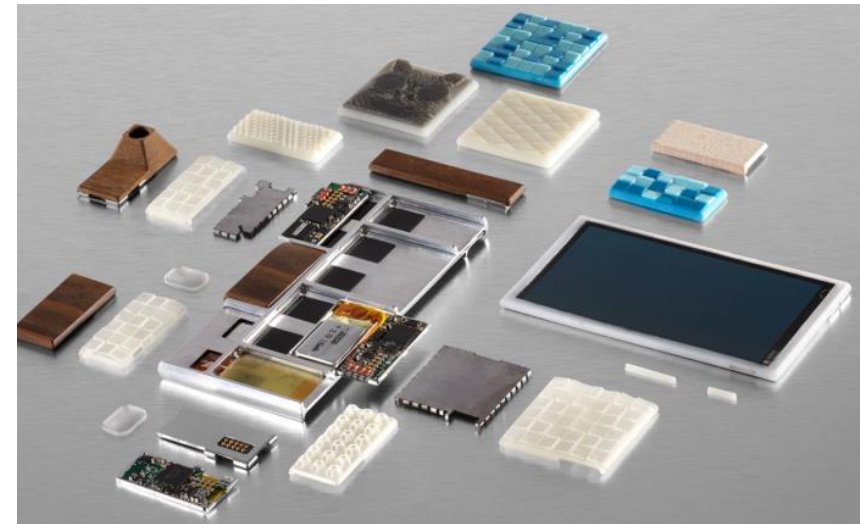
The traditional perception was that technological innovation must be based in the country to remain a pillar of its economy.

In the new millennium, Asia has risen as an important location for "innovation offshoring"

Innovation offshoring is driven by profound changes in corporate innovation management as well as by the globalization of markets for technology and knowledge workers. Western companies are at the forefront of this trend, but Asian governments and firms are playing an increasingly active role as promoters and new sources of innovation.

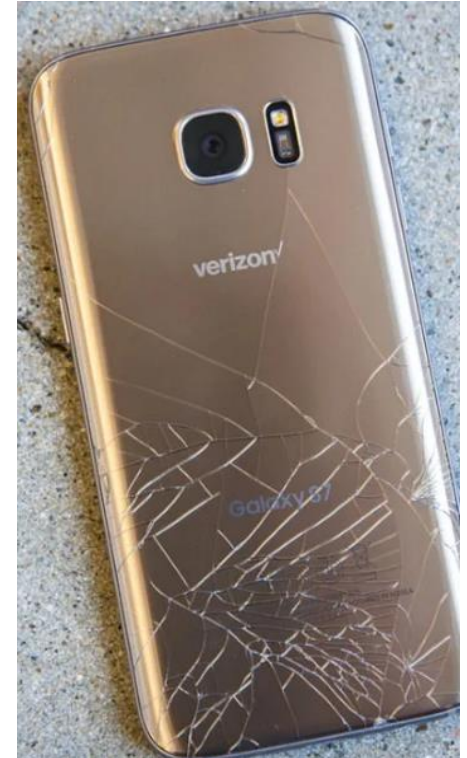
Modularisation

- Modular product architecture
- Use of “de-composable building blocks” Ernst (2005)
- Results in *interchangeable* modules
- Facilitates ‘de-coupling’ in the production of different modules
- Allows outside firms to be given responsibility for design/development of complete modules



Electronics right to repair

- The right to repair is a new legislation that allows consumers of electronic products to potentially repair and modify their own consumer electronic devices. Previously, the manufacturer of electronic devices required the consumer to use only their offered services
- The **UK has introduced** right-to-repair rules that legally require manufacturers to make spare parts available to people buying electrical appliances.
- The European Commission has announced plans for right-to-repair rules for smartphones, tablets and laptops.
- From June 2022, the right to repair law, called the Digital Fair Repair Act, will mean customers no longer have to take their damaged or broken electronic products back to the original makers to get an authorized repair, and can instead fix things like phones or laptops themselves, or take them to lower-cost independent repair shops





Electronics right to repair

- Experts expect the rules to apply to consumer electronic devices such as phones or game consoles.
- Tractor manufacturer John Deere opposed the idea, saying it posed a safety risk.
- Technology giants such as Amazon, Apple and Microsoft also opposed the legislation and say independent repair could affect the security and safety of devices.
- Self-repair groups like iFixit supply parts, tools and instructions to fix thousands of items. iFixit hailed the legislation calling it "one giant leap for repairkind".
- YouTubers and Reddit communities help others get the most out of their devices by answering questions or giving advice on repairs.

Offshoring & IPR

- Fritsch and Görg 2013) found that outsourcing is associated at the firm level with:
 - Spending on research and development.
 - The introduction of new products.
 - Upgrading existing products.
- The benefits of Offshoring depend on the level of protection of intellectual property rights in the economy. Firms will increase their R&D outsourcing expenditure if their business environment intensively protects intellectual property rights.

Innovation Offshoring as a Win-Win Partnership

Source: Ernst(2006). Asia role in GIN

- Some analysts fear that one country's gain will lead to another country's fall. Yet, Innovation offshoring does not have to be a zero-sum game.
- However, the globalization of markets for technology and knowledge workers and the growth of knowledge diffusion through Global Innovation Networks create benefits for all parties. The Western economy may also enhance its capacity to produce better innovations and market-defining standards.

Innovation Offshoring (Source: Lewin et al, 2008)

Source: Ernst(2006) Asia role in GIN

- In the early 1980s several leading-edge companies such as Texas Instruments, Motorola, and General Electric established technology centers in India and China to secure strategic advantages such as favorable political treatment and access to talent.
- Twenty years later, small entrepreneurial firms are increasingly offshoring new product development because their ability to grow, need to increase speed to market, or simply their survival depend on it.
- Asia, in particular, is playing a central role in the growing global innovation networks, as indicated, for example, by the growth in US patents granted to companies in Asia between 1986 and 2003. Major Asian countries in that respect include South Korea, Singapore, China, Taiwan and India.

Global Innovation Networks

- “Collaborations and interactions between different firms and organisations producing and exploiting new knowledge in a global scale for purposes of fostering innovation activities” (EU, 2010)
- The scale of the move towards GINs can be gauged from work by Contractor (2011: 12) who shows that multinational companies have moved about a quarter of their R&D outside the home country.
- India in 2005, for example, had 100 multinational companies with R&D facilities located in the country.



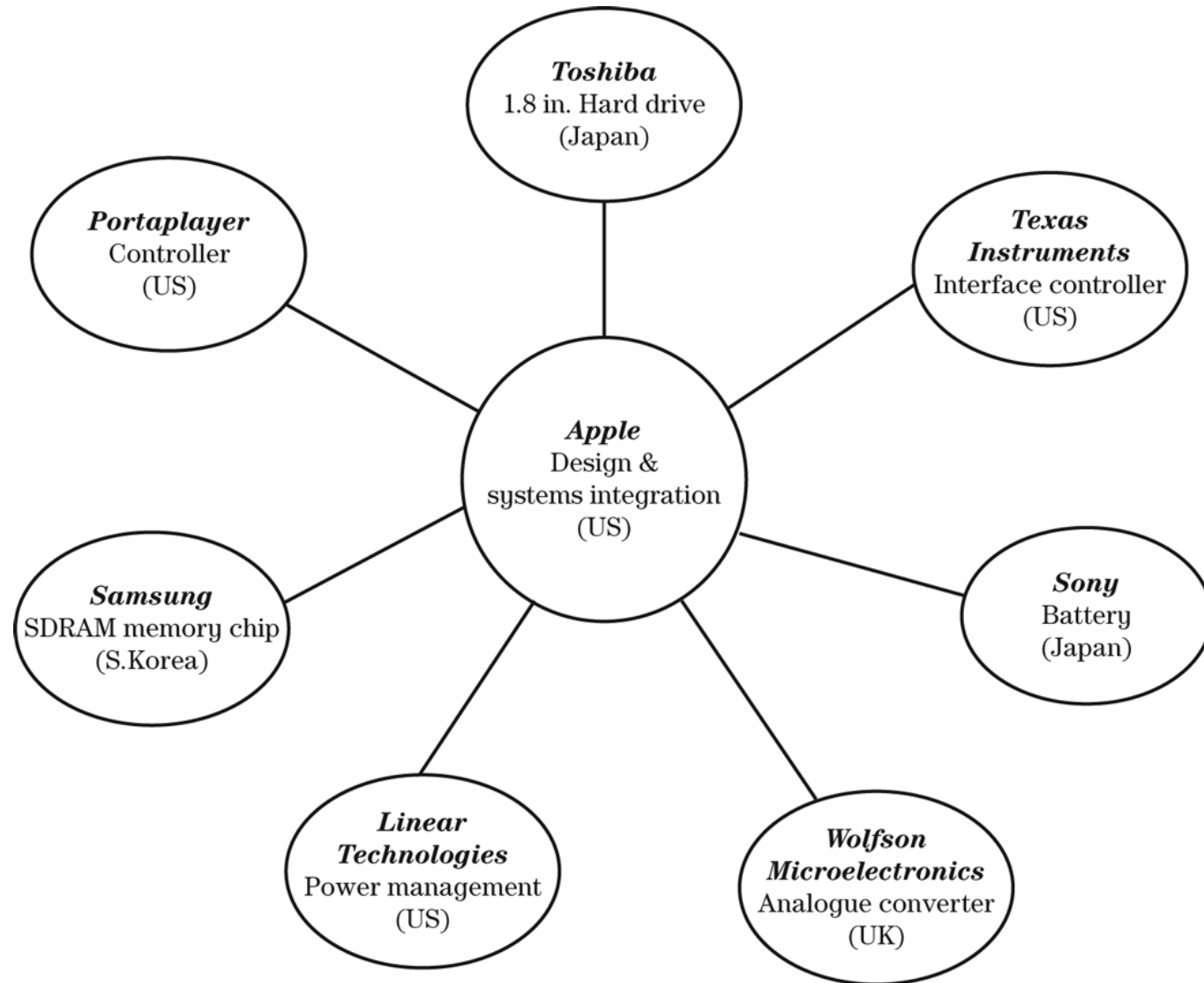
Practical Examples of Innovation Offshoring

- In the early 1980s several leading-edge companies such as Texas Instruments, Motorola, and General Electric established technology overseas to secure strategic advantages such as favorable political treatment and access to talent.
- General Electric (GE): GE has established multiple R&D centers in countries like India and China. These centers focus on developing products tailored to local markets, such as low-cost medical devices for emerging economies. By offshoring innovation, GE has been able to reduce costs and accelerate product development cycles¹.
- IBM: IBM has a significant presence in India, where it conducts a substantial portion of its software development and R&D activities. This offshoring strategy has allowed IBM to access a large pool of highly skilled software engineers and reduce development costs, while also fostering innovation through collaboration with local universities and research institutions¹.
- Microsoft: Microsoft has offshored various innovation activities to countries like India. Microsoft has established an R&D center overseas that focuses on cybersecurity and artificial intelligence. This center has contributed to the development of cutting-edge technologies that are now integral to Microsoft's product offerings.

Case Study: GIN at Apple

- In order to introduce the iPod within a period of 6-months, Apple selected firms that had the specialist expertise to rapidly develop the various systems and parts required.
- These companies were from Asia and Europe as well as America. They included Sony and Toshiba from Japan, Samsung from South Korea; Wolfson Microelectronics from Scotland; and Texas Instruments, Portalplayer and Linear Technologies Inc from the US.
- All were involved in the new product development process. Portalplayer, for example, had some 200 staff working on the project at one time.
- Overall design and systems integration was in the hands of Apple. Design itself was the responsibility of Apple's Jonathan Ive.

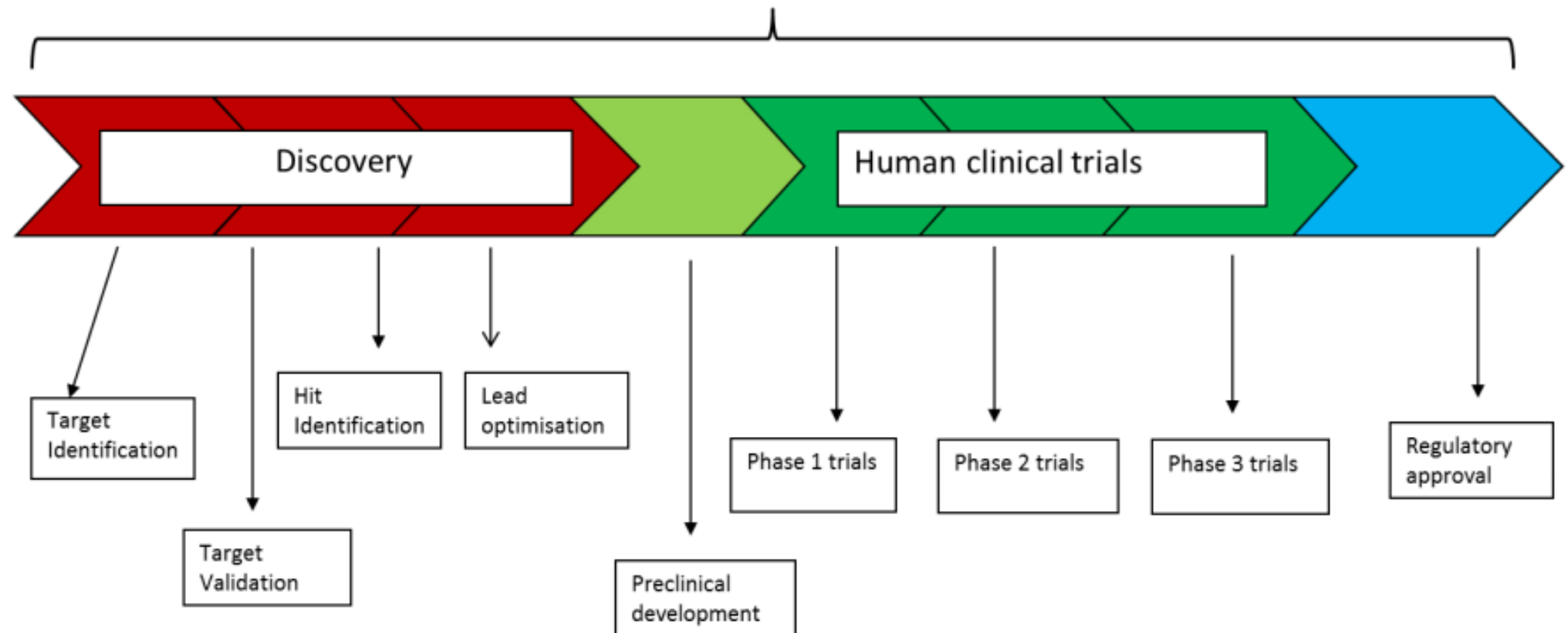
Global Innovation Network for the Apple iPod



Pharmaceutical Sector

Response: Externalisation of R&D

AZ, GSK, Pfizer, Merck: Target is to externalise 40%-50% of whole R&D pipeline (2012)





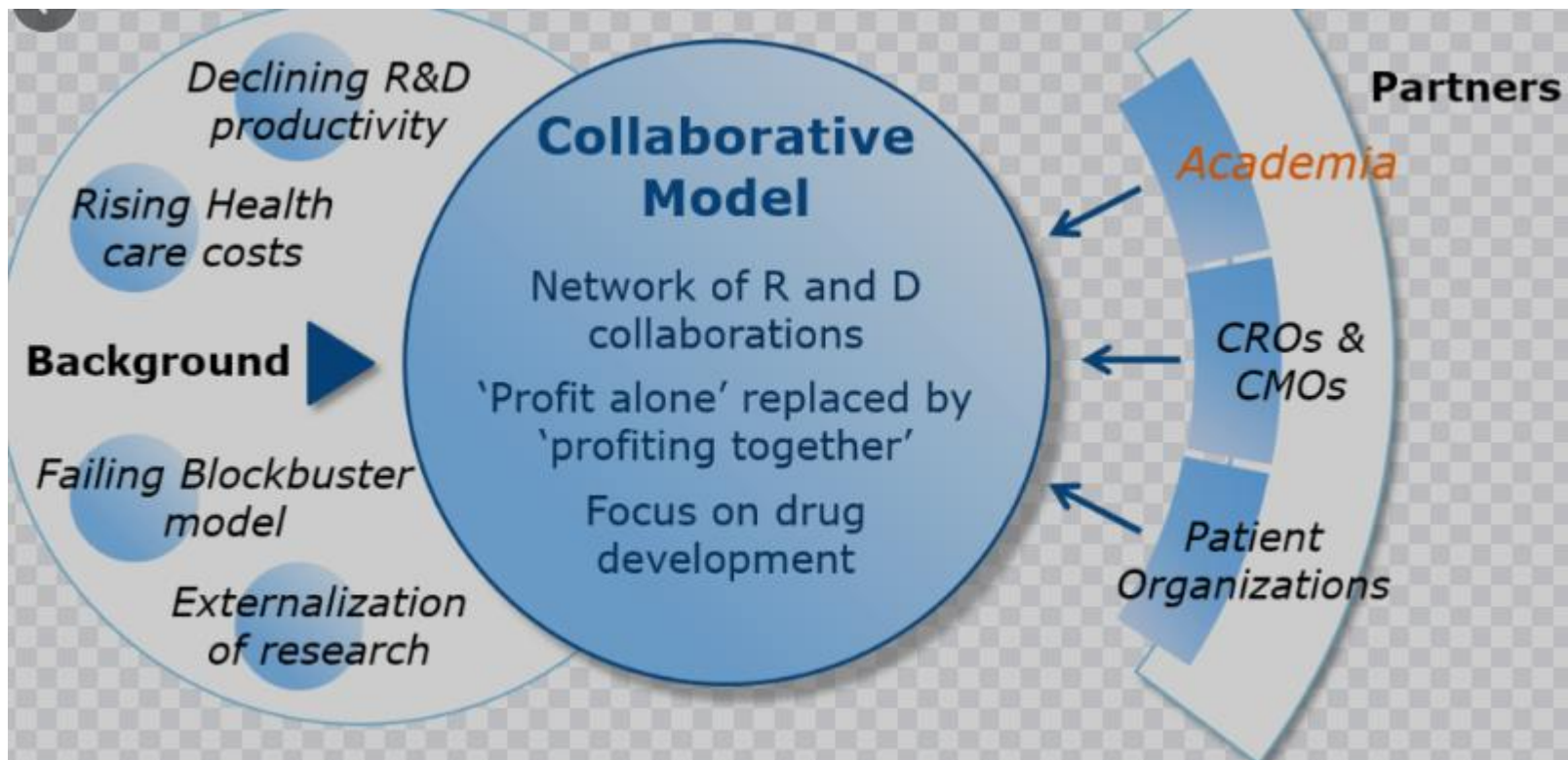
Pharmaceutical Sector

Source: Steven Hutchins, 2011

- Launching new drugs is a high-risk, high-cost, and only rarely high-reward enterprise. For every 10,000 new potential drugs investigated, only 1 will make it to market, and the process from concept to commercialization lasts over 10 years. In fact, very few marketed drugs ever recover the important R&D investments incurred in their development, which is estimated to exceed \$800 million.
- These issues, along with a general decline in R&D productivity (as defined by the number of NDAs [new drug applications] granted annually versus industry expenses) and the loss of exclusivity on high-revenue marketed products, explain the difficulty currently experienced the pharmaceutical industry. The consequences are— megamergers, rationalization, and resulting major reductions in R&D.
- Nonetheless, global R&D budgets remain unaffected, and the percentage of global R&D expenses for outsourcing has increased, specifically in the exploratory phase of drug discovery, which remains relatively low when compared to preclinical and clinical outsourcing.
- Therefore, pharmaceutical and biotech organizations increasingly use external research options earlier in the drug discovery process. The relationship between the sponsor and the external R & D firm is evolving from functional to integrated collaborations.

Open Innovation in Healthcare

Source: Center for Open Innovation and Lead Discovery



CRO=Contract Research Organizations
CMO=Contract Manufacturing Organizations.

Market Factors: Opening of R&D facilities in China-Pharmaceuticals

Roche	The R&D Centre in Zhangjiang Hi-Tech Park, Shanghai	2004
Sanofi-Aventis	Chinese R&D Centre	2005
Pfizer	Pfizer China Research and Development Center, Shanghai	2005
AstraZeneca	AstraZeneca Innovation Center China, in Shanghai.	2006
GSK	Established global R&D Centre on neurodegeneration in Shanghai. Established Clinical Research Centers.	2007
Novartis	Novartis Institute for BioMedical Research , Shanghai.	2007
Bayer	Announcement that R&D centre to be established in Beijing	2009
Merck & Co	Asian R&D headquarters, Beijing	2011

Example of Country Competitive Advantage (Source: Deepak R Gourad)

Competitive advantage of india

Country Attractiveness Index Criteria and Weighting [pwc report]

Patient Pool (30%)

- Size and availability of suitable patient pool

Cost Efficiency (20%)

- Cost efficiency of labor
- Cost efficiency of facilities and travel



Regulatory Conditions (20%)

- Food and Drug Administration visibility
- Country's regulatory laws
- Strength of intellectual protection

Relevant Expertise (15%)

- Number of clinical research organizations
- Number of clinical trials
- Size and availability of labor force with relevant skills

Infrastructure and Environment (15%)

- Protection of intellectual property
- Health-care infrastructure
- Country infrastructure
- Country risk factors

Features of Global Innovation Networks

- Structure
- Linkages
- Geography
- Hi-tech sectors
- Governance
- Collaborative nature

Recommendations to Improve GIN

1. Improve access to and collection of innovation-related data to inform the national policy debate;
2. Address “home-made” causes of innovation offshoring by sustaining and building upon existing strengths of the U.S. innovation system;
3. Support corporate innovation by (1) providing tax incentives to spur early-stage investments in innovative start-ups and (2) reforming the U.S. patent system so it is more accessible to smaller inventors and innovators; and
4. Upgrade the U.S. talent pool of knowledge workers by (1) providing incentives to study science and engineering; (2) encouraging the development of management, interpretive, cross-cultural, and other “soft” capabilities; and (3) encouraging immigration of highly skilled workers.

Source: Ernst(2006) Asia role in GIN

Globalisation & Transaction costs

- Globalisation is a key factor behind the growth of innovation offshoring
- Globalisation has come about because of the lowering of transaction costs
- Transaction costs are the costs of:
 - searching
 - negotiating
 - enforcing
 - agreements and contracts in a world of imperfect information

Two types of transaction costs

- Coordination costs
- Opportunism costs

In recent years both of these elements have shown very marked reductions in cost caused largely by advances in *technology* and trade *liberalisation*



Impact of Globalisation and Transaction costs

Reductions in transaction costs have encouraged:

1. Outsourcing

- i.e. organisation re-structuring where increasingly supply/value chain activities are carried out by third parties

2. Offshoring

- i.e. the geographical re-location of the supply/value chain overseas

Among the industry sectors that have led this process are textiles and electronics

Offshoring and MNE

- Innovation is increasingly due to global networks that connect geographically dispersed knowledge centers.
- The multinational enterprise (MNE) generates value by integrating knowledge across national borders.
- Three megatrends shape the strategy of the modern MNE:
- The disaggregation of the value chain into ever narrower activities, the migration of value to knowledge-intensive intangibles and the rise of huge emerging markets like China and India, whose domestic firms can grow to enormous size before venturing abroad.
- These megatrends are linked to falling spatial transaction costs, enabled in the main, by digitalization.
- The literature has focused on organizational pipelines, not on personal relationships, which are particularly important for both entrepreneurship and radical innovation

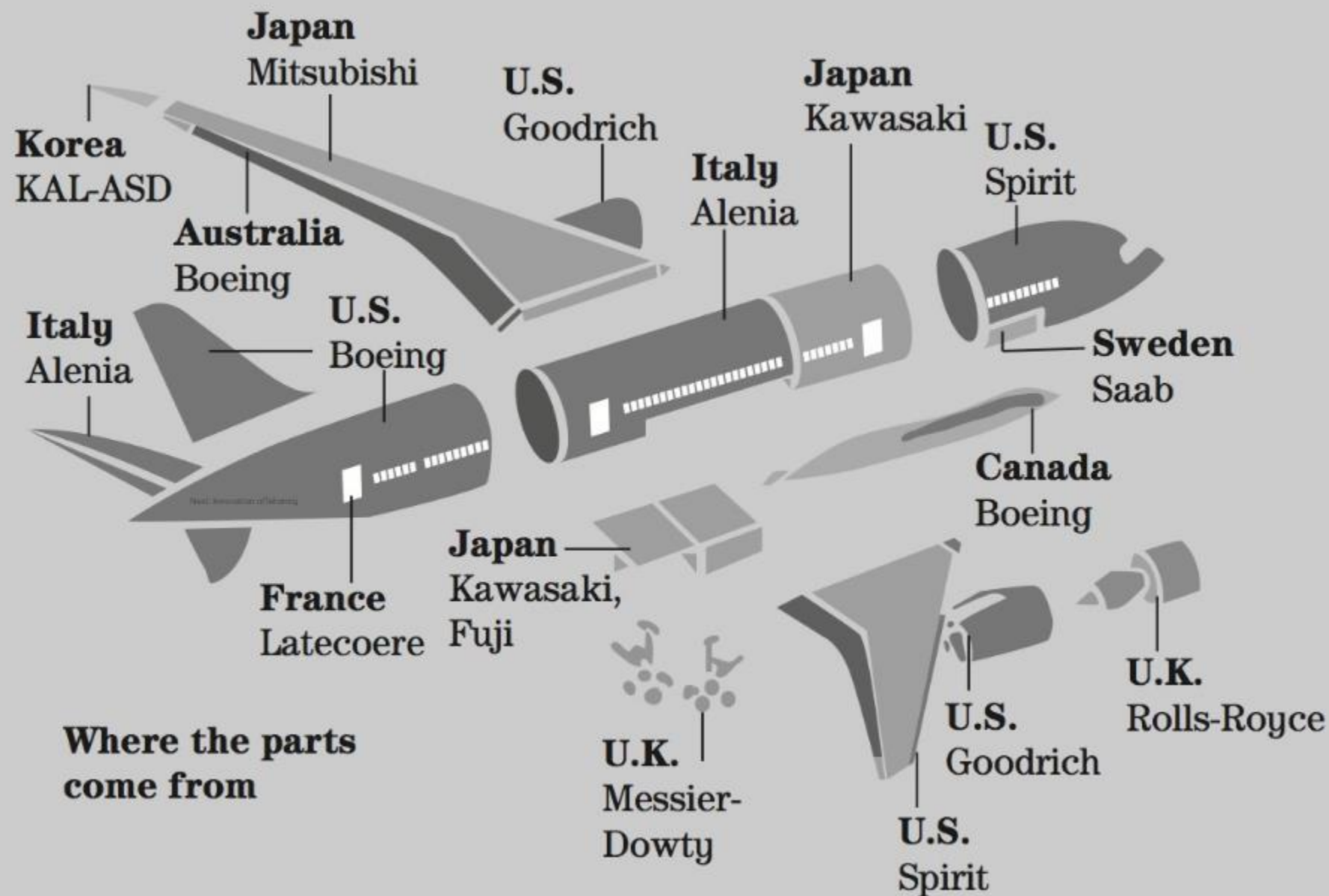
Source: Cano-Kollmann, Hannigan, and Ram Mudambi (2018)

Organisational and Spatial options

		Geographic relocation →	
↓ Organisational restructuring		<i>Home</i>	<i>Offshore</i>
	<i>In-house</i>	A In-house	B Multinational (wholly-owned subsidiaries)
	<i>Cooperative</i>	C Partnership (i.e. networked innovation)	D Global innovation network (i.e. innovation offshoring)
	<i>Outsourcing</i>	E Outsourced (i.e. arm's length contracting)	F Global production network (i.e. production offshoring)

Case Study: Offshoring at Boeing

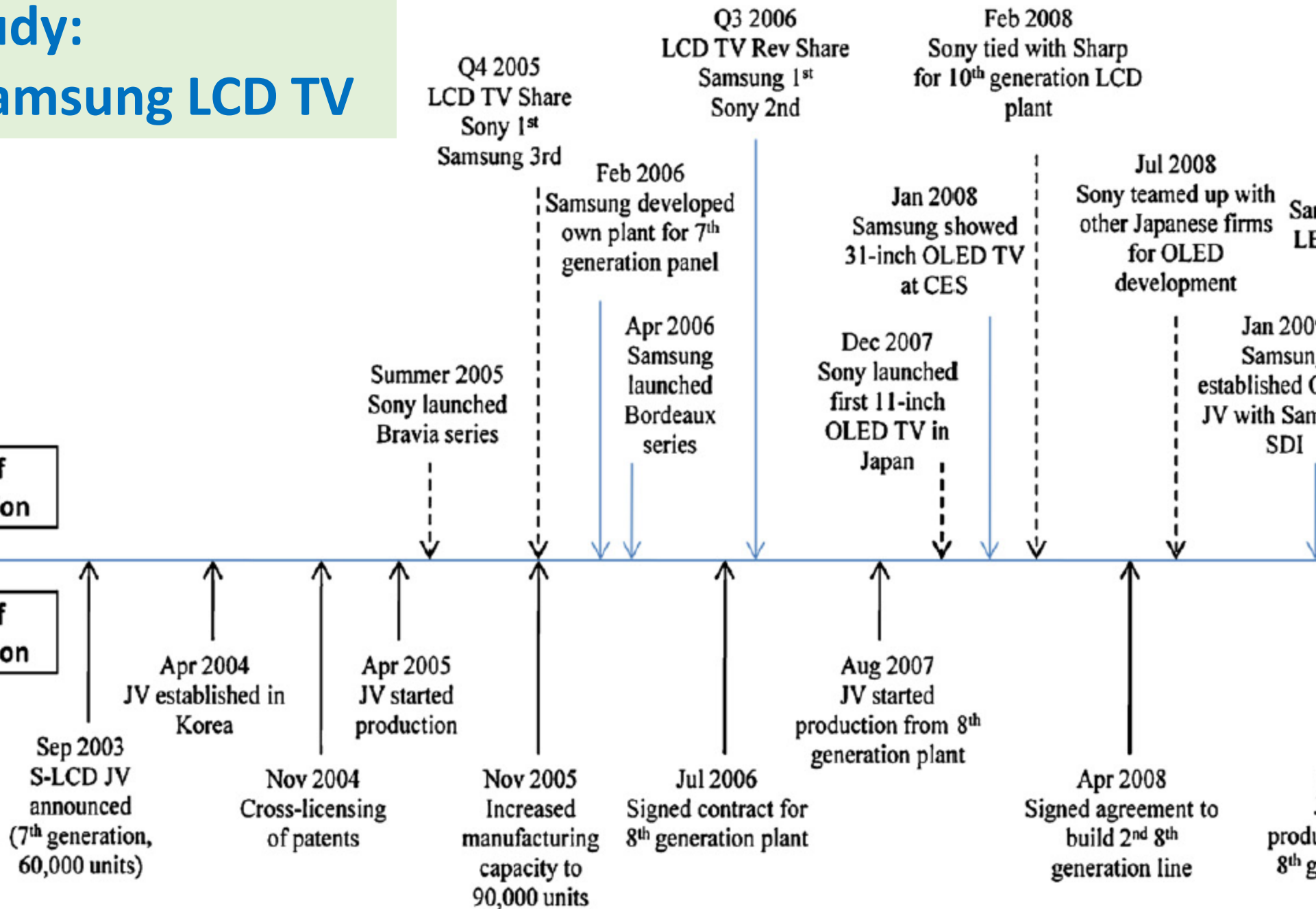
- Boeing 787 Dreamliner is 20 per cent more fuel efficient than similar aircraft and to achieve this Boeing planned to make the aircraft significantly lighter, necessitating extensive use of lightweight composite materials.
- With more than 50 per cent of the aircraft constructed from carbon fibre , the Dreamliner was highly innovative.
- Boeing looked to innovation offshoring to foreign aerospace companies for access to knowledge and expertise in composite manufacturing.
- Although assembled in the US, the airliner's composite wing was designed and developed in Japan by Mitsubishi, while Italy's Alenia along with Kawasaki of Japan developed the bulk of the fuselage, some 65% of this iconic American airliner has been developed overseas .



Case Study: & Samsung LCD TV

Areas of
Competition

Areas of
Cooperation



Access Important Information from Patent Databases

Google Patents

patent handbag



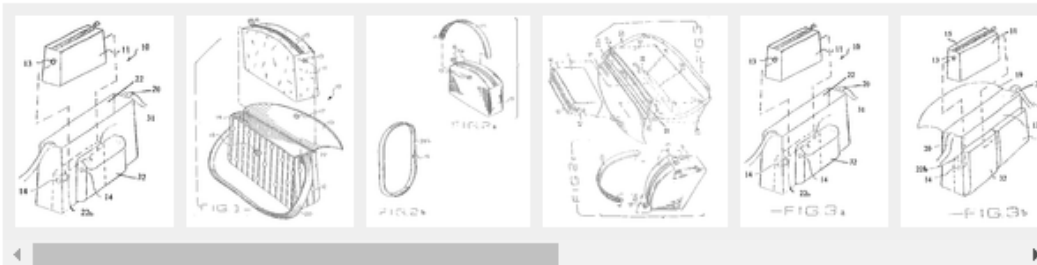
← Back to results patent handbag;

Carry bag with pouch insert and cover

Abstract

A carry bag with a pouch insert and magnetic fasteners is provided. The carry bag comprises a three-dimensional, generally rectangular pouch having a zipper or drawstring closure in the top face and a detachable carrying strap. One portion of each magnetic fastener is secured to each end face of the front and back faces of the pouch, and an opposite portion of each magnetic fastener is secured to the carrying strap.

Images (11)



US6446688B1

United States

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Inventor: [Alice B. Sutton](#)

Worldwide applications

1999 • [US](#) 2000 • [US](#) 2002 • [US](#)

Application US09/666,069 events

1999-01-26 • Priority to US09/237,755

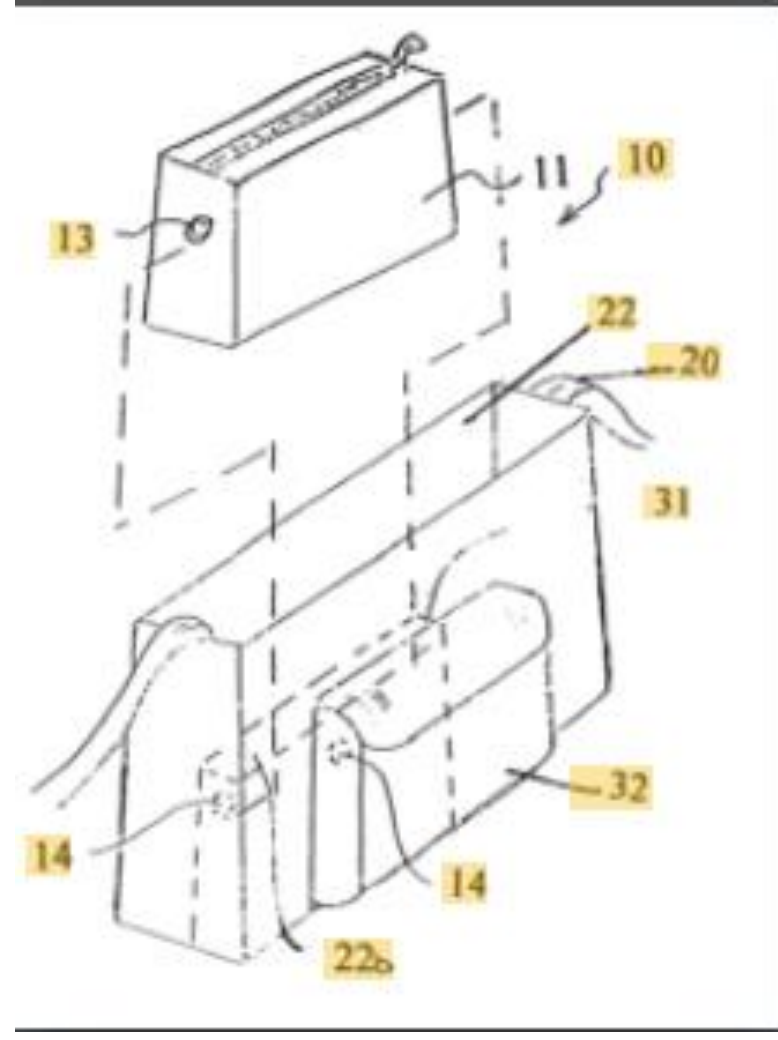
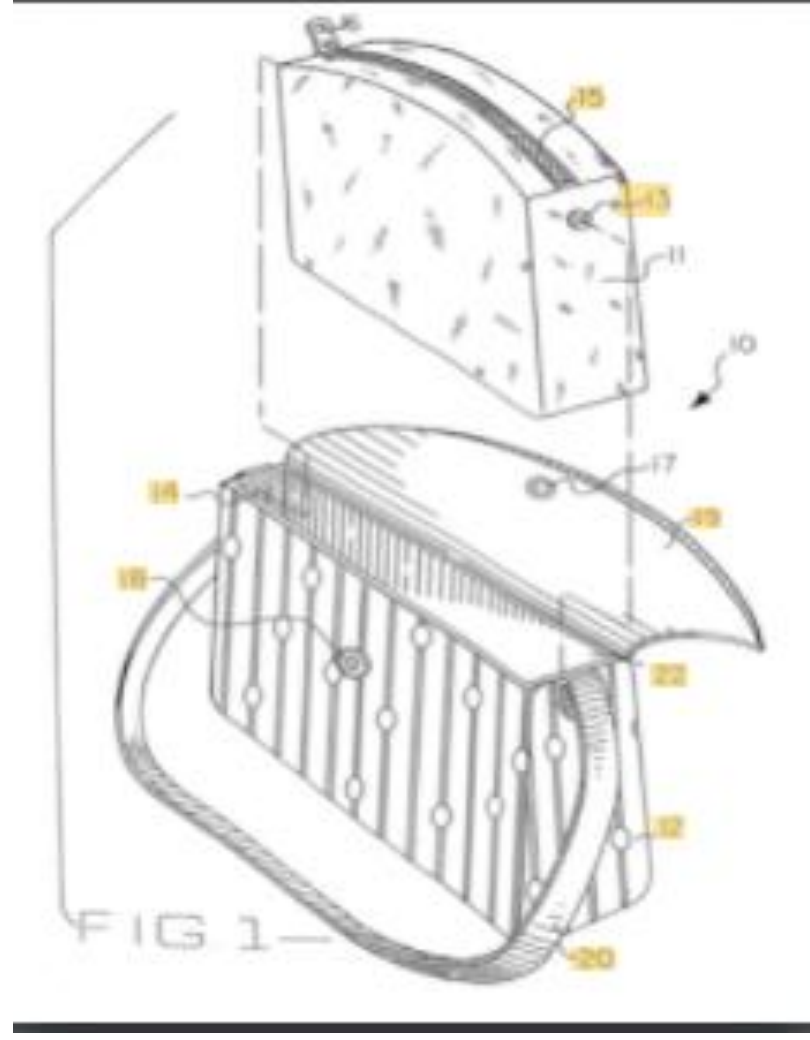
2000-12-06 • Application filed by Alice B. Sutton

2000-12-06 • Priority to US09/666,069

What is claimed is:

1. A carry bag for storing personal articles, the carry bag comprising:
 - a. a closable, free-standing pouch comprising a central storage area and two end faces, which are substantially parallel and opposite to each other;
 - b. a closure mechanism in the top face of the pouch;
 - c. at least two magnetic fastener, each comprised of at least two corresponding portions; at least one of the magnetic fastener portions being secured to the pouch;
 - d. a detachable, carrying strap having two opposite ends with a corresponding magnetic fastener portion secured to each end of the carrying strap; or a detachable circular strap with a corresponding magnetic fastener portion secured to the circular strap, a bottom section of the circular strap being adapted to support the pouch, and an opposite, top section of the circular strap being adapted for carrying the carry bag over a user's shoulder or hand; and
 - e. a cover comprising an open cover compartment comprising an interior partition dividing the cover compartment into a pouch compartment and another compartment, the pouch compartment comprising two compartment ends, which are opposite and generally parallel to each other, the interior partition comprising one of the pouch compartment ends; and

wherein a magnetic fastener portion is secured to a cover compartment end panel; and wherein the carry bag further comprises a storage compartment



Further reading

- Lewin, Massini & Peteers (2008) Why Are Companies Offshoring Innovation? The Emerging Global Race for Talent, Journal of International Business Studies ; 40(8)
- Ernst (2006) Innovation Offshoring. Asia's emerging role in GIN

Managing Innovation

Chapter 11

Topics to be covered

During the session, you will be able to:

1. Discuss the concept of the development funnel in innovation;
2. Evaluate the scope of management activities relating to innovation;
3. Discuss the contribution the four management functions (planning, organising, leading, and controlling) make to the innovation process;
4. Debate the importance of a project approach in the innovation process
5. Analyze the four components of Innovation Management
6. Describe the 70-20-10 rule to allocate resources to the innovation process

Advanced Organizer

- Given the complexity and uncertainty associated with Innovation, as well as the fact that innovation involves creativity and thinking out of the box, managing the Innovation process is a challenging task. In this session, you will learn key concepts relating to innovation management.

Generate a lot of Ideas/Insights

- As a start, it is recommended to generate several Ideas/Insights as potential solutions to an identified problem/issue
- This enhances the probability of identifying a winning idea/insight
- The ideas may then be discussed over a brainstorming session for their potential in resolving the problem
- Finally, the alternative solutions are screened and evaluated according to some established criteria, and the optimal idea/insight may be chosen
- The process of screening uses what is called a Development funnel



Development funnel (قمع) (Smith, 2015)

- Filtering comprises of a process where ideas are screened, initially to demonstrate that the technical principles work effectively, then for marketability and then for manufacturability. Those ideas which are not technically feasible, too expensive to manufacture or lacking commercial potential are successively screened out.
- The outcome is a much smaller number of ideas selected as acceptable concepts which may then be pushed forward into a rapid new product development process. The aim of the funnel process is not simply to back only those concepts which offer the highest pay-off, but rather, by employing the planning dimension, to use the funnel to build a portfolio of development projects that meet the needs of the business.
- This will typically mean ensuring a steady stream of new products coming on to the market, and this is something that clearly results from careful planning.

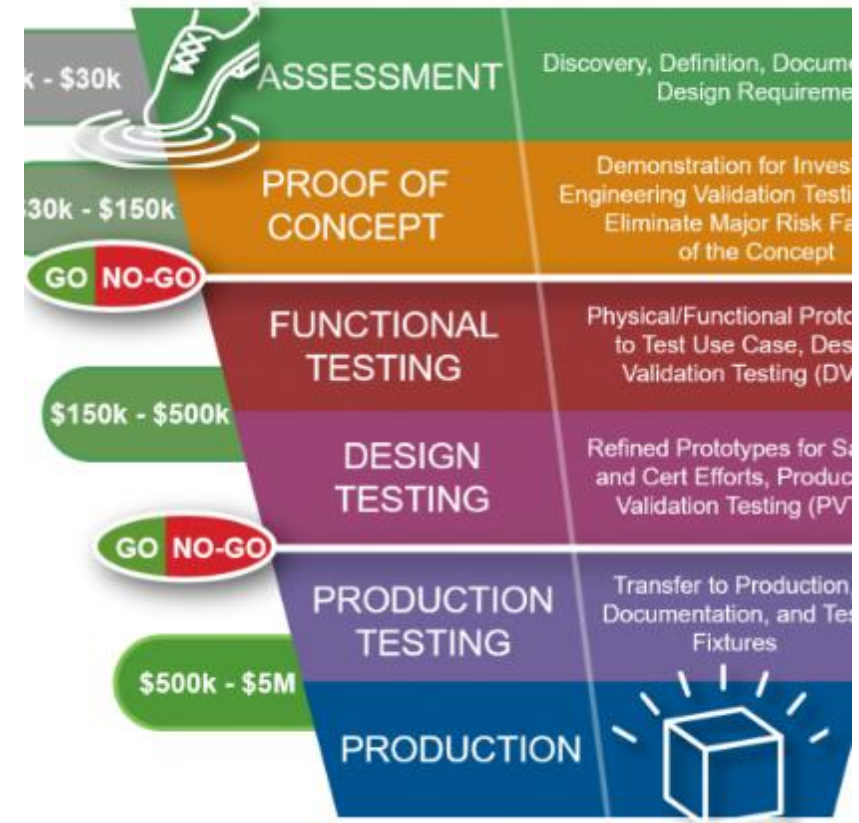
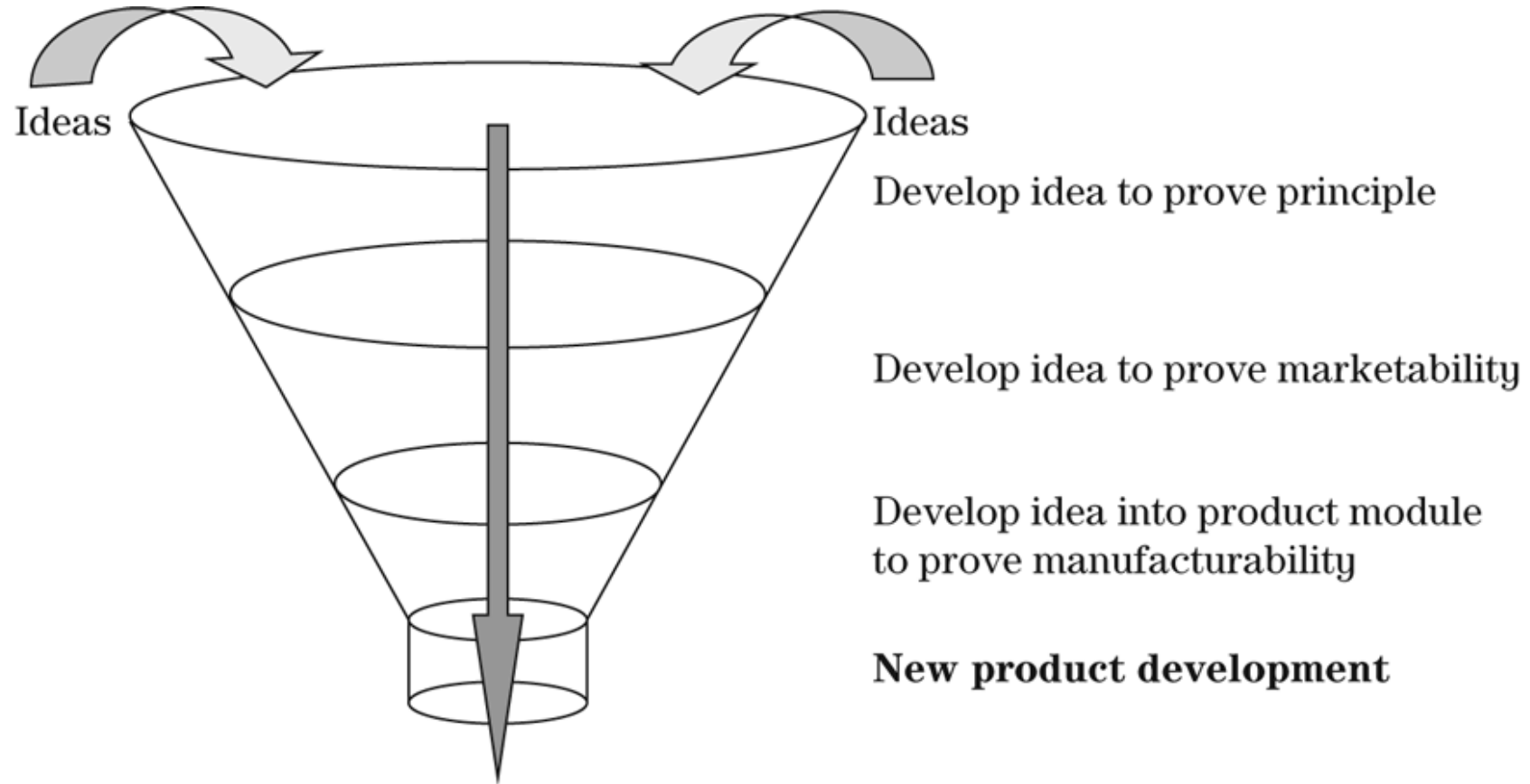


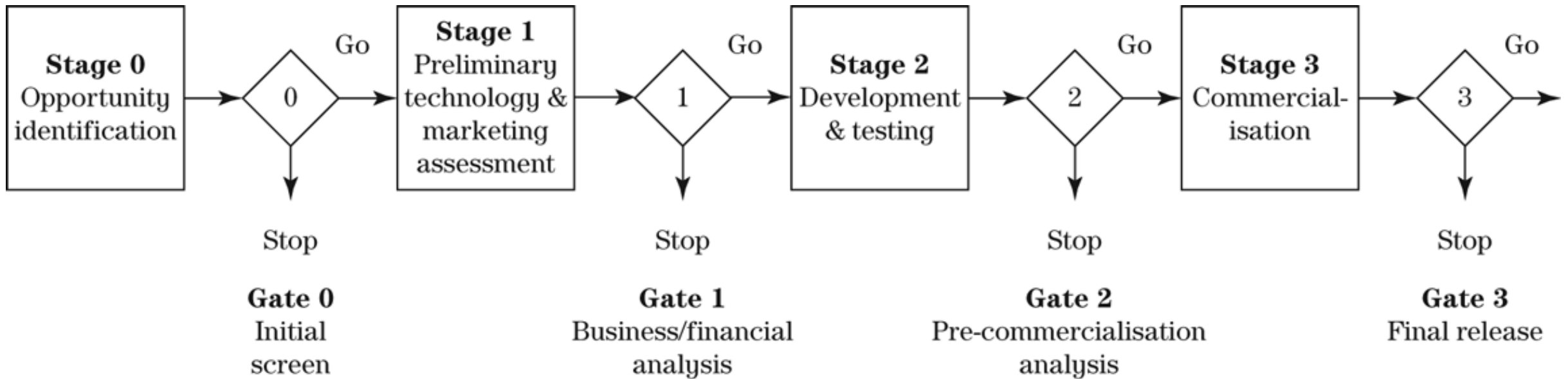
Figure: Development Funnel (قمع) (Source: Smith, 2015)



Screening Ideas-Development funnel (Smith, 2015)

- The development funnel is a 'structured' approach to getting from an idea or ideas to the successful launch of a new product or service.
- The principle starts with many ideas and uses a systematic method of filtering them so that only a few are actually developed and launched as commercial products or services.
- Ideally there should be an input of many ideas with the mouth of the funnel deliberately kept wide to encourage a wide range of potential new products and services. This increases the likelihood that the idea being retained finally would have the characteristics to be successful.
- The narrowing of the funnel represents the filtering processes as the large number of ideas is reduced to a much smaller number. As Figure below shows, the funnel may comprise multiple filtering points.

Srccreening: The stage-gate process (Source: Smith, 2015)



Stage-gate process

- There are a minimum of three possible gates in the course of any project, although it is common practice to have more. These are:
 1. initial screening – decision to commit resources
 2. business/finance analysis – aimed at deciding whether to go to full-scale development
 3. pre-commercialisation business analysis – decide whether to move to full-scale production and market launch.
- Each of these gates represents a key decision point in the new product development process where, depending on the outcome of the decision made at the gate, major resources in terms of finance, people and equipment can potentially be committed to the project.
- Cooper (1988) suggests that at each gate it is necessary to pose three questions:
 1. does the project continue to make economic/business sense?
 2. have the essential steps (necessary to pass that particular gate) been completed?
 3. is the project on time and budget and have the milestones been hit?
- The answers to these questions determine what happens, in terms of the three possible decision outcomes at the gate in question. The outcomes will be arrived at on the following basis:
 1. GO – if the answers to questions 1 to 3 are positive
 2. HOLD – if the answer to question 2 is negative
 3. KILL – if the answer to question 1 is negative

Basic Concepts of Management

A Brief Recap of Management Functions





Four functions of Management

Good managers are considered to perform four tasks with the collaboration and support of their colleagues and subordinates

1.Planning

2.Organizing

3.Leading

4.Controlling

Functions of Management (Source: Smith, 2015)

- Planning is a decision-making activity, involving looking into the future and making decisions that will enable a particular outcome to be achieved. Appropriate goals need to be set to achieve the desired objectives. Then action plans are to be developed to achieve these goals along appropriate resources being allocated to enable the activities to be undertaken.
- Organizing, in contrast, tends to be a more people-centred activity, involving specifying and structuring working relationships between individuals. This enables the planned activities to be carried out by ensuring that the required human resource expertise is available.
- Leadership involves conveying a sense of purpose to the whole organization and individuals. Leadership thus involves providing a vision of where the organization is going, and motivating people to make the necessary effort so that the vision is implemented.
- Finally, control involves monitoring and evaluating the progress of activities to ensure that they are going according to plan.

Functions of Management

- This raises the question: Why does innovation have to be managed? Although innovation may appear to be a spontaneous and creative activity that is difficult to manage, effective management of innovation is nonetheless essential.
- The contextual factors associated with innovation making them difficult to manage are:
 - Uncertainty;
 - complexity;
 - messiness/untidiness;
 - disruptiveness; and creativity



1. Planning

- Planning is determining organizational objectives and formulating approaches to achieve them.
- It constitutes a key skill as “A job well planned is half done.”
- During planning one needs to ask oneself the following questions:
 1. What am I trying to accomplish, i.e. what is my objective?
 2. What resources do I have and do I need to accomplish the objective?
 3. What are the strategies to achieve the objectives?
 4. Is this the optimal approach?

Planning

- Two planning techniques commonly associated with innovation management are project management and the development funnel.
- Project management (Smith, 2015):
 - A project is any one-off, often unique, activity which is time constrained and has a specific goal.
 - Project management is a general purpose planning technique that is widely used to help in project management, so that they are completed on schedule, to required specifications, and within budget.
 - Since the development of a new product/service is typically a one-off experience, it shares some common characteristics with project management. Hence project management techniques may prove useful in managing innovation projects.

Planning

- As a management technique, Project management comprises five main steps (Smith, 2015):
 1. determine the goal(s)
 2. identify the activities or tasks to be undertaken.
 3. estimate the duration of the activities/tasks
 4. determine the sequence in which the activities/tasks have to be completed and link them into a model of the overall project
 5. develop a project plan that provides an overall schedule for the project.



Environmental Analysis: SWOT

- To plan the direction an organization, want to head to, a realistic evaluation of the current situation is required. It helps us understand where we are now, which in turn helps us to define how to get to our planned destination.
- A **SWOT** analysis is a tool used for this purpose. It analyzes the organization's environment in terms of:
 - Strengths,
 - Weaknesses,
 - Opportunities and
 - Threats.

Strategic Planning: Mission, Goals, Objectives, & Values

- **Mission** typically focuses on its present business purpose - “who we are and what we do”. It includes:
 - The organization’s self–concept
 - Its philosophy
 - Long–term survival needs
 - Customer needs
 - Social responsibility
 - Nature of the product or service
 - RTA’s Mission is:
“Prepare legislation and develop integrated solutions of road systems and land/marine transportation networks that are safe and in line with Dubai’s economic development plans with the highest international standards”
- **Goals** are the broad, long-term accomplishments an organization wishes to attain.
- **Objectives** -- Specific, short-term statements detailing how to achieve the organization’s goals.
- **Values**-- how to accomplish goals?
 - They are the beliefs, traits, and behavioral norms that employees are expected to display in conducting the firm’s business and in pursuing its strategic vision and mission. They become an integral part of the firm’s culture.

Strategic Planning: Vision & Mission statement

- At strategic level, planning consists of developing a Mission and Vision statements.
- *Vision* -- A **vision** statement describes where the organization aspires to go.
 - It sets out management's future aspirations for the business to its stakeholders; Provides direction—"where we are going" by using distinctive and specific language to set the firm apart from its rivals. Example: RTA (Dubai) describes its vision as:

"The world leader in Seamless and Sustainable Mobility"

- *Mission*-- A **mission** statement defines the organization's business, its objectives, and how it will achieve these objectives. Example: RTA (Dubai) describes its vision as:

"We provide seamless and safe travel with innovative, sustainable mobility solutions and services to make every journey in Dubai a world-class experience."

Organising (Smith, 2015)

- In managerial terms organising concerns the arrangement or structure of the internal shape of an organisation.
- This usually involves assigning people to groupings (i.e. departments or teams) to facilitate effective working. As far as innovation is concerned organizing relates to the choice of internal structure or conducive to innovation, that is a structure that facilitates the creativity and flexibility required for innovation.

Organising (Smith, 2015)

- The internal configuration or structure of an organisation is important to innovation as it facilitates the innovation process by affecting numerous aspects of work activities including:
 - communication channels (between individuals and groups of individuals)
 - flows of information
 - working relationships
 - working practices
 - work environment
 - corporate culture.
- Innovation requires that communication and information flows are effective in order to facilitate the knowledge transfer, a notion captured by the concept the theory of absorptive capacity. Similarly, working relationships and practices have to be flexible to cope with uncertainty and unpredictability



Leading

- Leadership is about setting a direction or vision for others to follow.
- Leadership is associated with the following attributes:
 - Intentional influence is exerted over other people to guide, structure and facilitate activities and relationships in an organization
 - It stimulates high performance by employees to achieve organizational objectives
 - Managers motivate people to contribute their ideas
 - They do so by giving assignments, clarifying policies and providing feedback on performance
- Leaders must:
 - Communicate a vision and rally others around that vision.
 - Establish corporate values.
 - Promote corporate ethics.
 - Embrace change.
 - Stress accountability and responsibility
- Leadership and management must go hand in hand. They are not the same thing. But they are necessarily linked, and complementary. The manager's job is to plan, organize and coordinate. The leader's job is to inspire and motivate.



Leaders and Managers

The major difference between managers and leaders is based on the approach to motivate people and teams to achieve organizational objectives

Leaders act as visionaries to spearhead their team. They set goals and new directions, challenge the status quo, and motivate their team to reach the new vision.

Managers tend to maintain the status quo, focusing on conformance to the standard. They manage their teams around the boundaries that have been set, focusing on organizing and directing to achieve the objectives.

Peter Drucker recognized that with the rise of the knowledge worker, “one does not ‘manage’ people, the task is to lead people. And the goal is to make productive the specific strengths and knowledge of every individual.”

Progressive leaders give employees the authority to make decisions on their own without consulting a manager. This results in various benefits

- Customer needs are handled quickly.
- Manager's role becomes less of a boss and more of a coach.
- Enabling -- Giving workers the education and tools they need to make decisions.

Leading

- Regarding innovation, the vision needs to highlight innovation as important and highly valued by the organisation.
- Leadership can be manifested in a variety of ways. It may create roles within an organisation, that individuals can assume and which they can then use in order to provide leadership to others.
- Another way is for those in leadership roles to create a conducive culture or climate within an organisation where innovation can thrive. In the workplace, there are a number of practices or schemes that companies have implemented, which are designed to support employees in exercising their creativity by developing new products/services.
- While such practices tend to be very specific, leadership is about creating the kind of environment within an organisation where employees feel motivated to exercise their creativity and engage in activities that can lead to innovation.

Controlling (Smith, 2015)

- Controlling performance and making needed changes to plan is a skill of managers . Managers do so by:
 1. Measuring performance against corporate objectives
 2. Monitoring performance against standards
 3. Taking corrective action if required
- Control may appear to be antithesis of innovation. While excessive control may seriously deter innovation, nonetheless it does have a part to play in the effective management of innovation projects.

Controlling (Smith, 2015)

- Since innovation is associated with a high degree of uncertainty it makes control an important consideration. A lot of uncertainty, combined with the multidisciplinary nature of development teams and the unique nature of most innovations, means that control is necessary to manage the progress of an innovation project (Smith, 2015).
- There exists a very close link between planning and control techniques that can be used for managing innovation projects. The development funnel serves as both an aid to planning and a way of controlling and structuring the flow of new ideas and discoveries within an organization.
- Similarly, the stage-gate process outlined in figure below supports both planning and control. However, unlike the development funnel its primary function is to provide a means of exercising a degree of control over both the innovation process and the multidisciplinary teams working within the organization.

More on Planning: Project Approach

A project is defined as an effort to create or modify a specific product or service. Projects are temporary work efforts with a clear beginning and end. Projects can be completely contained within a specific unit or department, or include other organizations and vendors.

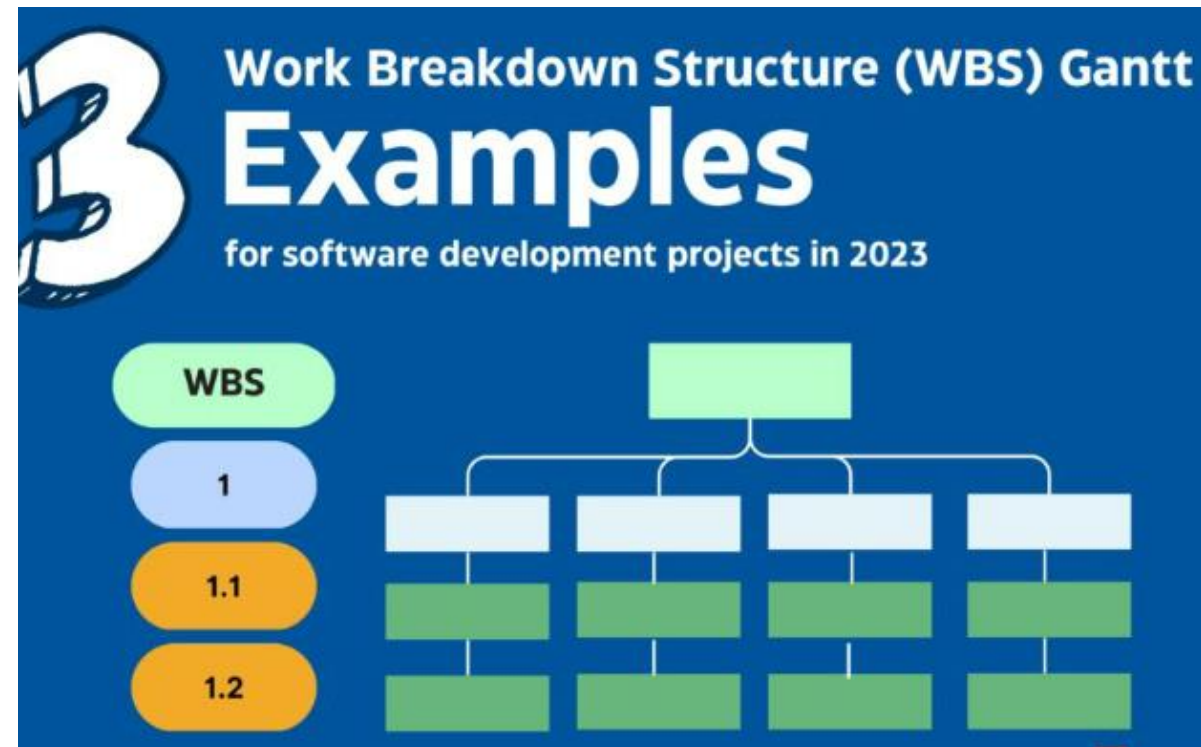
The project approach involves the selection of project management practices, means, and methods that the project management team should perform based on the specific, high-level project characteristics gathered from the project charter (Project Management Institute).

Project Management (Smith, 2015)

- Innovation involves the development of a new product or service. The firm will ideally aim to launch it in the market at a suitable time when it is meeting a market need, ideally before competitors can launch rival products/services.
- The activities/tasks associated in Innovation will typically include
 - design,
 - testing,
 - commissioning., etc.

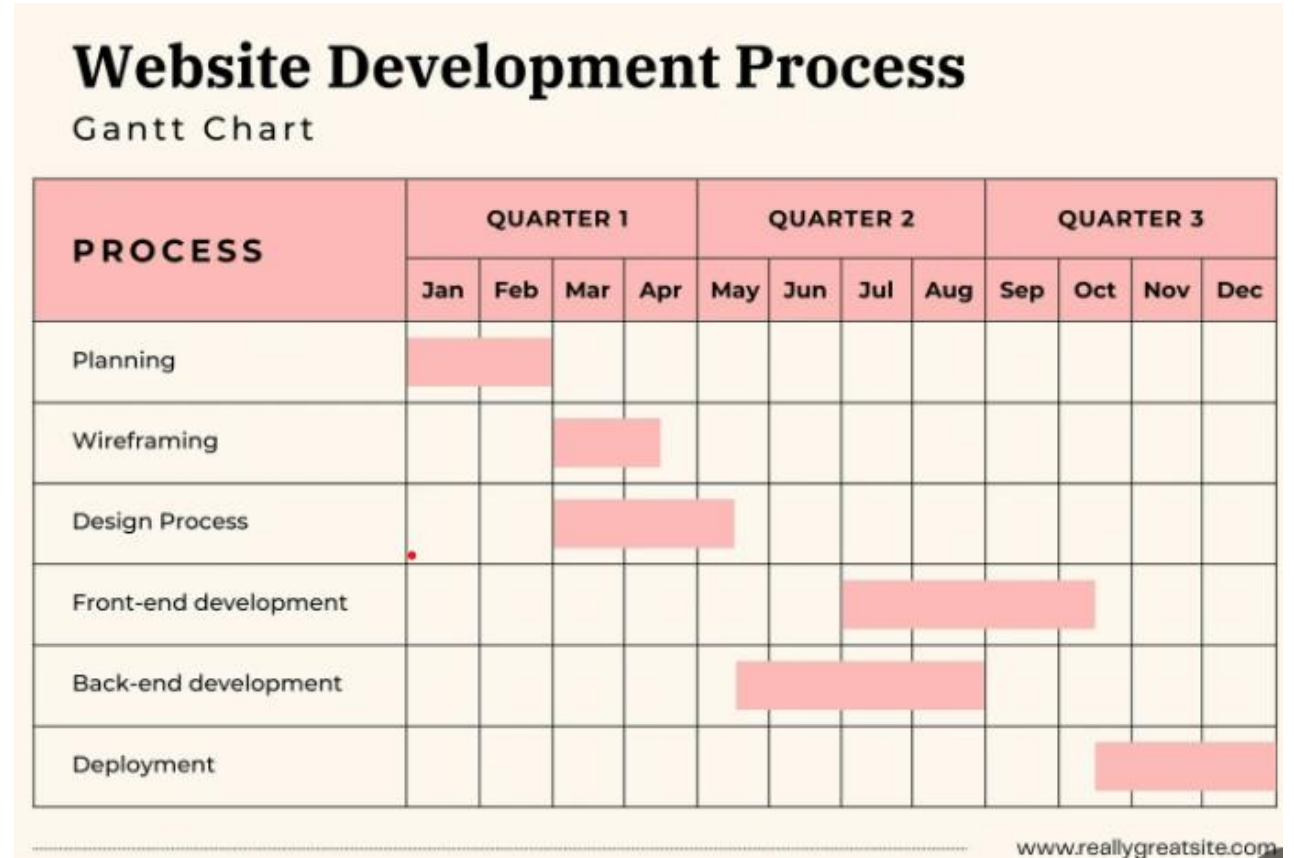
Project Management (Smith, 2015)

- Identifying the tasks to be undertaken is important as it forces those project personnel to think, not just about the immediate tasks, but also about future tasks.
- Further, while estimating the duration of these activities/tasks may be difficult, having to provide an estimate means that those involved have again to think carefully about the future course of the project.
- Similarly, project managers have to define the sequence or order in which the tasks will have to be completed. Once the sequencing has been defined the activities/tasks can be linked together in a plan or model of the project.

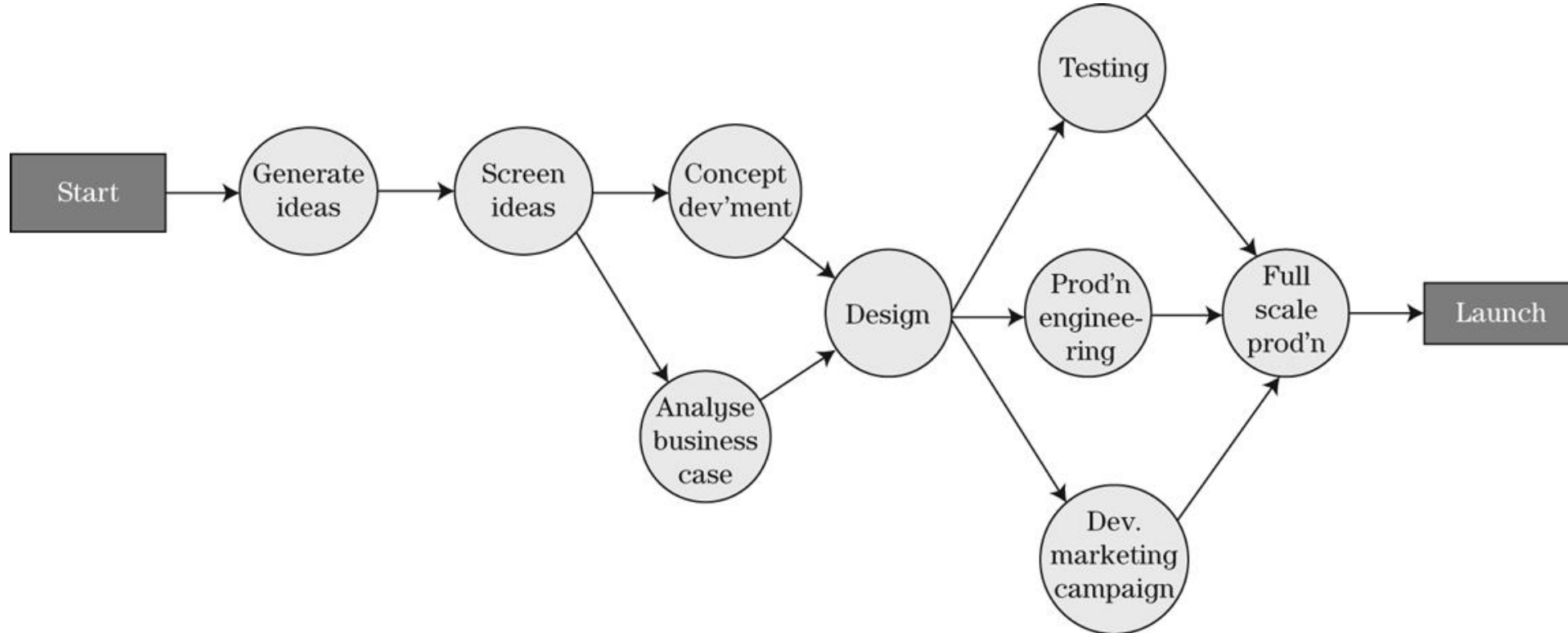


Project Management (Smith, 2015)

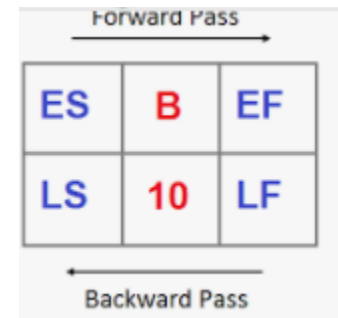
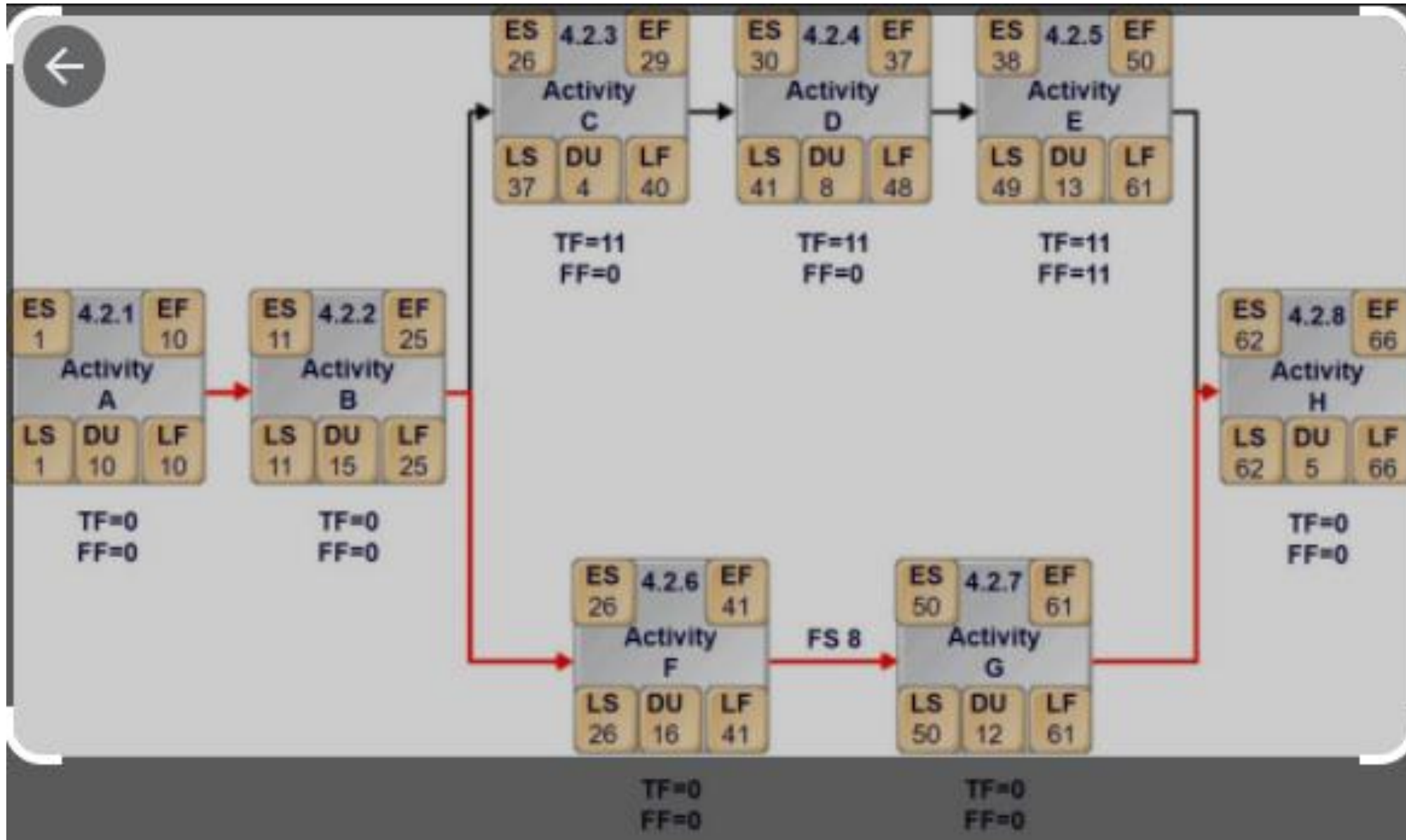
- Constructing a project plan is where plans are generated. Project managers typically used two major types plans.
 - Gantt or bar chart where the activities are listed on the vertical axis and the duration of them is recorded on the horizontal axis. With activities/tasks shown as strips the chart will show the cumulative time taken to complete the project.
 - Network diagram (Figure below). In this the activities/tasks are typically denoted as nodes and arrows showing how they are linked together. In this way the project is built up as a series of arrows and nodes beginning with the start of the project on the left-hand side and ending with completion of the project on the right-hand side



Project Management Network Diagram (Source: Smith, 2015)



Project Management Network Diagram





Project Planning

- Whichever type of chart is used the net result is a map of the whole project, forming a plan that shows what is to be undertaken and when (Smith, 2015).
- With innovation projects the level of uncertainty may often make it difficult to follow precisely the original plan, and it may be necessary to divert from the plan or to modify it extensively.
- Despite the uncertainty and the need to revise plans, project planning is essential in innovation. It enables innovation managers and those working on the innovation to think about the future, it also requires them to talk about and discuss the future.
- This involves not only a technical dimension but also a commercial one. Hence it is necessary to think about the innovation in much broader terms. This is what planning is really all about.



Organizing

- Organizing typically refers to coordinating the various resources to achieve organizational objectives. It involves:
 - Attracting people to the organization
 - Defining job responsibilities
 - Recruiting and developing employees
 - Developing an organizational structure
 - Grouping jobs into work units for efficiency

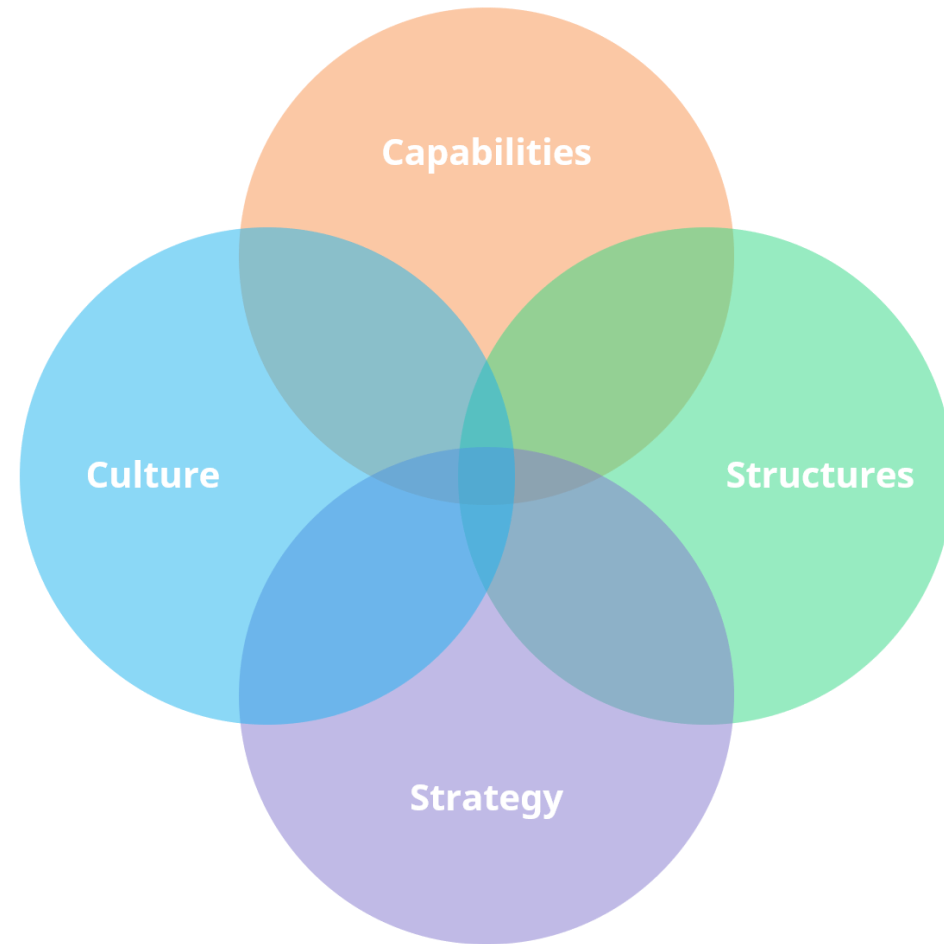
Innovation management

- What actually is innovation management?
- Innovation management is subject to much debate.
 - The very definition of innovation means that it cannot be managed
 - Yet others are stout believers in building systems and processes for the purpose of creating more innovation.
- The reality is probably somewhere in between

Components of Innovation Management

A simple approach to understand Innovation is to break it down and discuss each of its underlying components separately. The diagram illustrates its four components

[Source: <https://www.viima.com/blog/innovation-management>]



1. Capabilities

- The different abilities and resources the organization has for creating and managing innovation.
- It primarily involves people, as innovation relies on the abilities of both individuals and teams collectively.
- It refers first and foremost to the **abilities, unique insights, know-how and practical skills of the people** working for the organization.
- It also includes the **information capital** and **tacit knowledge** of the organization, as well as their **other resources** and available **financial capital**, all of which might be required to create innovation.

2. Structures

- The difference between structures and capabilities is that structures enable the institution to make effective use of its capabilities.
- This refers to the organizational structure, processes, and infrastructure of the organization, which may support the institution to operate and innovate much more effectively. This includes communication channels, processes for making decisions, and infrastructure for implementing ideas. Applications such as innovation management software may make a difference.
- Teams working on innovation projects need to be able to operate fast and adapt to their environment, while making decisions independent of the traditional approaches of the organization. Do NOT adopt the same rules and processes for everyone in an Institution. Economics of scale simply do not work for innovation.
- Google's 80-20 Formula. The ITO (Innovation Time out) policy encourages Google employees to spend 80% of their time on core projects, and roughly 20% (or one day per week) on “innovation” activities that speak to their personal interests and passions.

3. Strategy

- Strategy refers to a plan the organization has develop to achieve long-term success.
- It is ultimately about making a deliberate choice between a number of feasible options to have the best chance of success and implementing this choice.
- Innovation is one of the means to achieving your strategic goals, and is adopted by a number of successful companies (e.g. Google, Apple)
- In practice, “chance” innovation may reveal unrivalled opportunities that might be large enough to justify changing your strategy entirely. However, these are quite rare and practically impossible to foresee.
- The key is to align innovation activities with strategy, although it works out to be difficult to achieve in practice.
- You need to provide the institution with enough freedom to actually innovate, but also account for practical constraints (e.g. focus, available resources and your capabilities).

4. Culture

- A culture of innovation helps companies move through the innovation cycle. When multiple people, teams, and departments are working toward creative solutions, it's easier to consistently ideate and move ideas with potential to the next step of the innovation process.
- If structures allow the effective use of capabilities, culture is what enables the organization to acquire the capabilities related to people.
- The right pro-innovation culture supports the organization in recruiting and retaining the right people.
- It encourages innovation-friendly behavior and discourages the wrong kind. Culture can make a major difference for the innovativeness of an organization.

Culture: Common Traits

- Some commonly accepted traits for an innovative culture:
 - Emphasizes the need to always think of ways to get better
 - Values speed, learning and experiments
 - Considers failure as just a normal part of the process for creating anything new
 - Provides enough freedom and responsibility and is led primarily with vision and culture instead of a chain-of-command approach

Corporate culture and creativity (Smith, 2015)

- Each individual organization possesses its own distinctive corporate culture. However, corporate culture isn't easily defined, being much more elusive than resources like labour or capital. It is a uniquely human product that develops slowly, is tacit in nature and isn't easily transported across firms. Corporate culture is typically defined as, 'the way we do things around here' (Deal and Kennedy, 1982), or in terms of the values, norms and meanings that define an organisation's policies and practices.
- A key factor is that innovation typically relies on creativity. Creativity is a vital ingredient in developing innovations, and not just in providing the inspiration for new products and services. Creativity and innovation are key drivers of competitive advantage in today's fast-paced market-driven economy (Parahoo, Mumtaz; & Salem, 2017).
- Although the constructs of creativity and innovation are often used interchangeably (Martins and Terblanche, 2003), a distinction may be made between them (McLean, 2005). Creativity is generally conceptualized as the nature of the thought processes and intellectual activity used to generate insights or solutions to problems (Martins, and Terblanche, 2003), while Innovation refers to the development and implementation of new ideas, processes, products, or services (McLean, 2005). Smith defines 4 dimensions of creativity:
 - Inspiration: coming up with *novel* ideas
 - The ability to solve problems in a *unique* way
 - The ability to make connections that others *would not make*
 - The ability to see existing situations in an entirely *new way*

Corporate culture and creativity (Smith, 2015)

- Innovation is the implementation of creativity results, making creativity a part of the innovation process (McLean, 2005). McLean (2005) adds that all although employees may creatively generate brilliant ideas, innovation requires that the idea graduates from concept to market, overcoming various obstacles such as acceptability, securing scarce resources for development and funding..
- Research showed firms' internal corporate culture is a powerful driver of innovation (Parahoo, Mumtaz; Ayyagari, & AlKhazraji, 2017). Consequently, organisations keen to promote innovation and sustain innovative capacity over time need to give careful thought to developing an appropriate corporate culture (Smith, 2015). This in turn has implications for management practice as the dominant values, assumptions and beliefs within a work setting (i.e. the corporate culture) will be mirrored in the managerial practices of a firm (Smith, 2015).
- Smith discussed that the features of a creative corporate culture to support innovation include:
 - Focus on idea generation
 - Support continuous learning
 - Encourage risk taking
 - Tolerate mistakes/failure
 - Support change
 - Accommodate conflict

Corporate culture and creativity

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Investigating actors and enablers of innovation using a service ecosystem lens: a modelling approach

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Abstract

The study developed a conceptual model of service innovation viewed through a service ecosystem lens, and empirically validated the model using data gathered from 334 managers in the tourism industry of the United Arab Emirates. A mixed methods approach was used. Three significant factors emerged as direct antecedents of innovation, namely dyadic interactions between: *customers and employees*, *the firm and its partners*, and *the firm and its competitors*. Further, two 'enablers', technological and organisational support, reinforced these interactions. This study contributes to the emerging literature in two ways - operationalising the service ecosystem construct and examining the impact of dyadic interactions between actors within the ecosystem on service innovation. The findings and discussions provide managers in the UAE tourism industry with a holistic view of the innovation trajectory, helping them to identify critical enablers and manage key actor interactions in the ecosystem.

Rules for Fostering an Innovative Organization

Rule #1 – Unreasonable Expectations

- Only when people subscribe to unreasonable goals will they start searching for breakthrough ideas.
- There are no mature industries, only mature managers who unthinkingly accept someone else's definition of what is possible.

Rule #2 – Elastic Business Definition

- Too many companies define themselves by what they do rather than by what they know (core competencies) and what they own (strategic assets).

Rule #3 – A Cause, Not a Business

- Revolutionaries draw much of their strength from their allegiance to a cause that goes beyond growth, profits, or even personal wealth accumulation.
- The courage to leave some of oneself behind and strike out for parts unknown comes not from some assurance that "change is good" but from a devotion to a wholly worthwhile cause.

Rule #4 – New Voices

- Let the youth be heard.
- Listen to the periphery.
- Let newcomers have their say.

Rule #5 – An Open Market for Ideas

- Create a market for entrepreneurial ideas inside your company.
- New ideas are the currency of the realm.

Rule #6 – Create an Open Market for Capital

- Within a corporation, why set the hurdle for accessing a small investment for the purpose of funding an unconventional idea, building a prototype, or designing a market trial at the same difficulty as obtaining a large investment in an irreversible, existing business?

Rule #7 – Open a Market for Talent

- “A” people work on “A” opportunities.
- Provide incentives for employees who are willing to take a “risk” on something out of the ordinary.

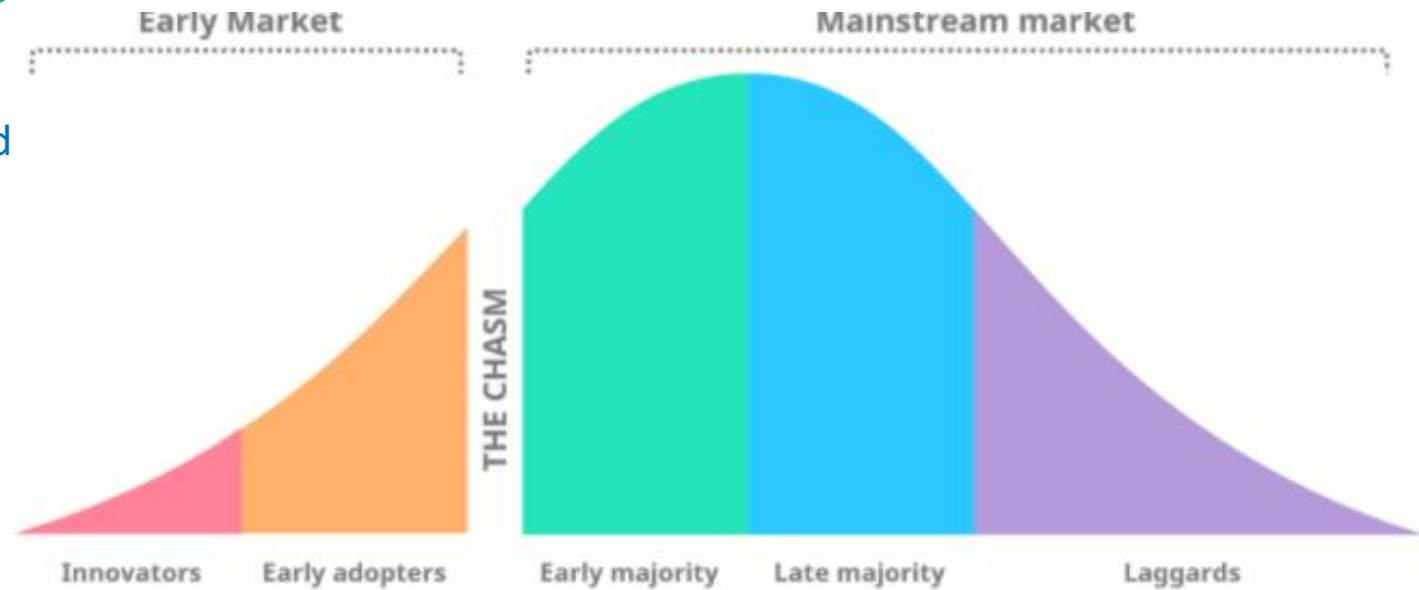
Rule #8 – Low-Risk Experimentation

- Being revolutionary does not mean being a high-risk taker.
- False dichotomy: Cautious follower vs. high-risk taker. Neither is likely to pay off in the age of revolution.

SOURCE: Adapted from Gary Hamel, *Leading the Revolution*, Boston, MA: Harvard Business School Press, 2000.

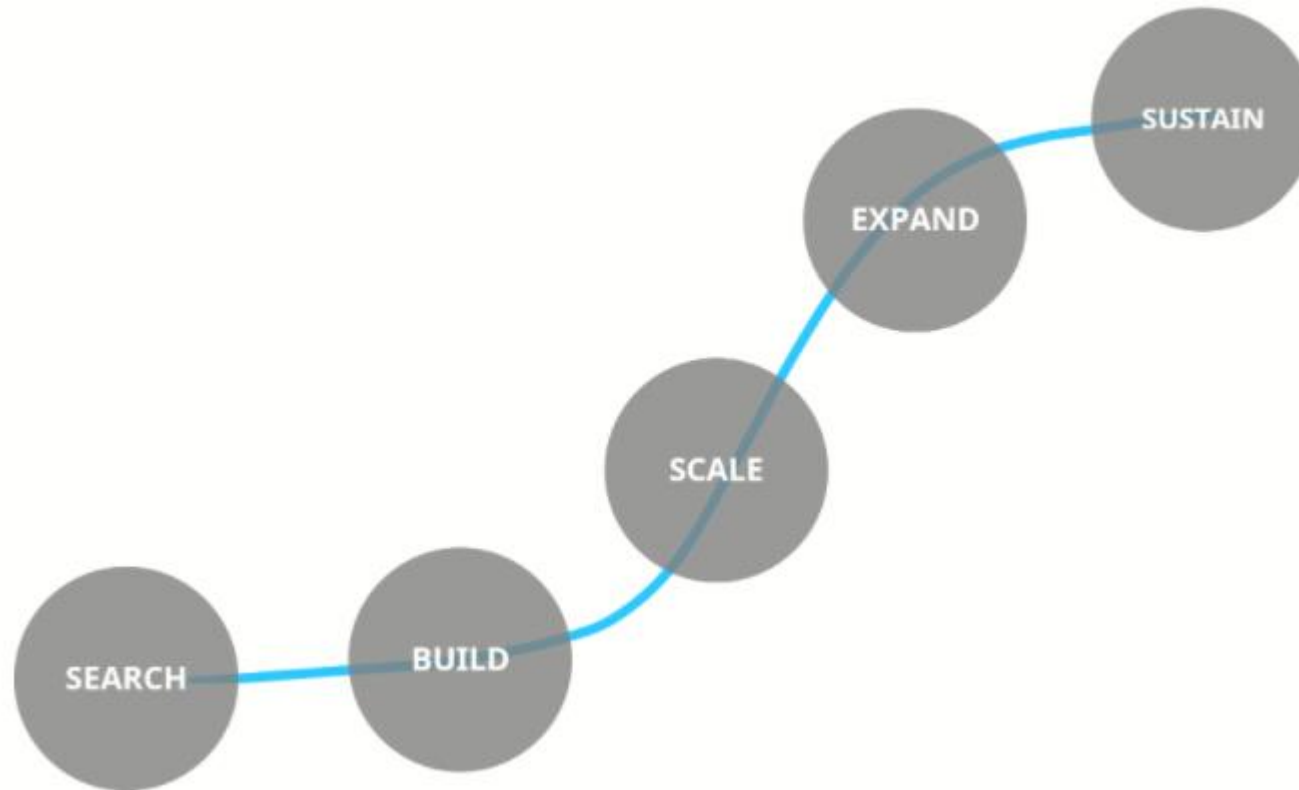
The Technology adoption life cycle

- The Technology adoption life cycle was first introduced by Geoffrey Moore in his 1991 book Crossing the Chasm.
- It predicts the path taken during the diffusion of innovations and explains why companies with disruptively innovative products (and/or technology) often have a hard time reaching success with the mainstream market.
- The basic idea is that the entire market can be represented with a bell curve that can be divided into segments based on how eager the customers are to adopt new technology with each segment having their own sets of expectations and desires.



The lifecycle of an innovation

- Every innovation, whether incremental or disruptive, passes through a number of essentially similar phases in their lifecycle:
- **Search – Build – Scale – Expand – Sustain**
- These phases cover everything from the innovation starting from a simple idea, to building and bringing it to market, as well as building a business out of it, and then sustaining it for as long as it makes sense. The timeframe may vary from months to years or even decades, depending on the type of innovation involved.



Innovation Management Success Factors

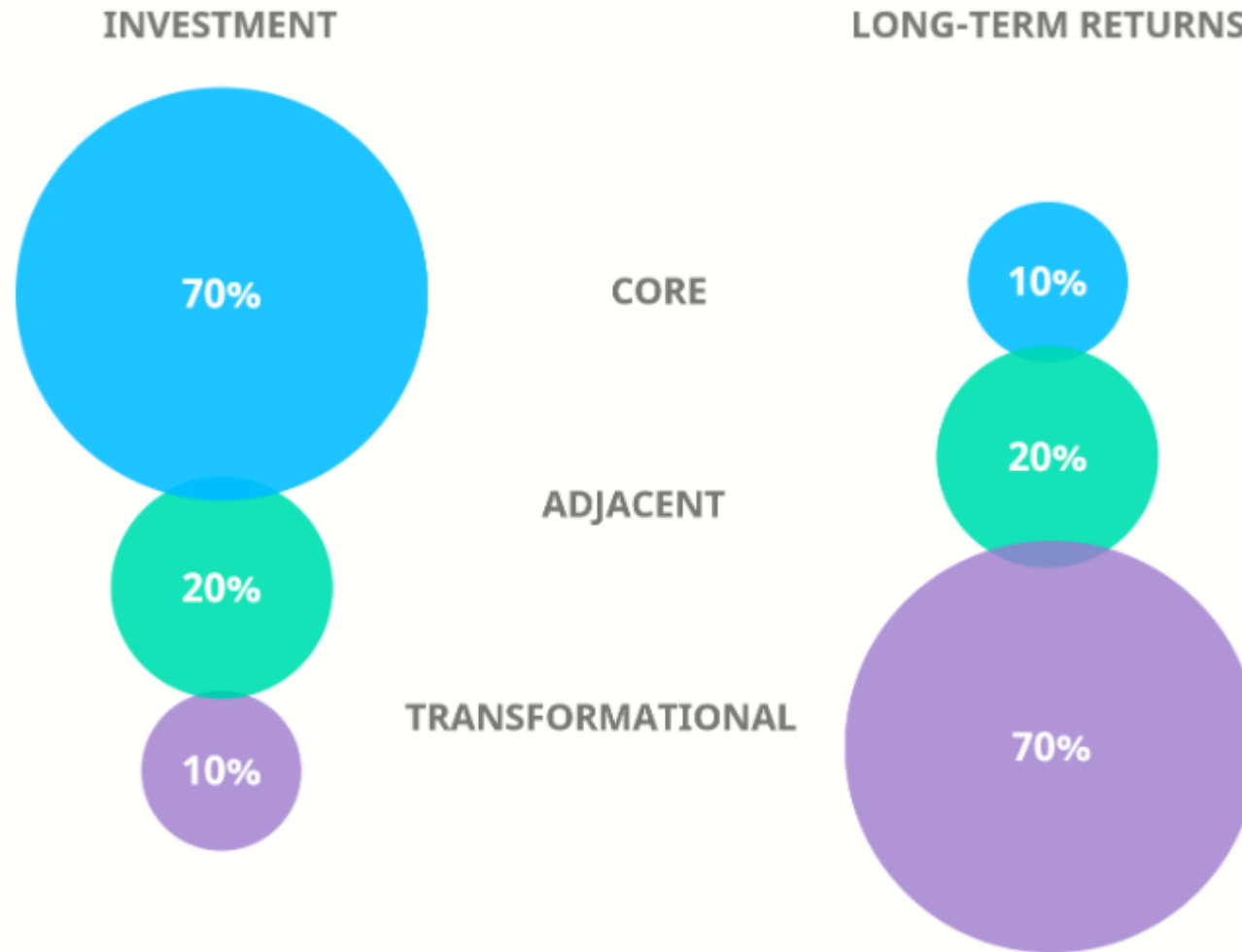
- Empower people but hold them accountable for their work and their decisions
- Have talented people on all levels of the organization
- Have tools and structures in place for making sure things happen and the right things move forward
- Have a clear vision and a strategy that everyone understands
- Be focused on implementing ideas as opposed to just talking about them

The 70-20-10 rule

- Introduced by Google CEO Eric Schmidt, the 70-20-10 rule is a thumb rule to allocate resources between “the core”, “the adjacent” and “the innovative stuff” (or “the transformational”).
- The core refers to all the activities that make up the majority of existing business of the organization
- The adjacent means new improvements or business areas that are logical extensions for the current business, such as opening new geographical markets for the existing services or products.
- The transformational may mean basically anything that is brand new for the organization, such as certain types of innovations.
- Research evidence suggests that companies that allocate their resources in this manner typically outperform their peers by a margin of 10-20% (as measured with their P/E ratio).

The 70-20-10 rule

Research also supports that the long-term returns for each type of investments are actually the inverse of the resources invested.



Contextual factors associated with innovation (Smith, 2015)

- Uncertainty may relate to how a new technology will perform, or how consumers will react to a new product. Similarly there will often be uncertainty about the level of resources or the time required to get an innovation ready for the market.
- Complexity forms an integral part of innovation because it involves the development of something new, which invariably requires a variety of expertise, including technical expertise to get the technology right, and commercial expertise (e.g. marketing, finance, etc.) to get the product or service ready for market. The involvement of staff from different disciplines inevitably leads to complexity, whether it is related to communication channels, the transfer of information or just different approaches to working. Further, innovation very often requires expertise from outside the organization which adds to complexity.
- Innovation does also tend to be messy and untidy. Innovation rarely proceeds in a smooth and straightforward way. Instead, innovation frequently comprises false starts when avenues of exploration prove to be blind alleys, new directions due to external influences and chance events leading to new insights.
- As innovation always involves uncertainty, risk is more than just something to minimize. For example, early-stage startups in general have very little to lose, hence they are prepared to bet 100% of their resources on working on a single ambitious project with a very high likelihood of failure hoping to achieve a large return.

Contextual factors associated with innovation (Smith, 2015)

- The fact that innovation is novel tends to mean that it is disruptive. When the innovation is radical, it can be very disruptive, requiring entirely new approaches. Even an incremental innovation may still be disruptive, leading to significant changes for the organization. Further, innovation almost invariably requires a degree of creativity. While management cannot directly enhance creativity in staff, it may use appropriate leadership to provide the an environment wherein which creativity can flourish.
- Faced with the uncertain, complex, messy, disruptive and creative nature of innovation, this session presents a range of tools and techniques that managers can employ to help and assist them with the difficult task of managing innovation. They will not guarantee success for an innovation, as there is nearly always far too much uncertainty linked to every innovation to ensure success. However, the techniques may help to make the process of innovation management more effective, and increase the likelihood that an innovation will succeed.

Findings of the PDMA Best Innovation Practices Study

In one of the major benchmarking studies of corporate innovation in American companies, the Product Development Management Association (PDMA) surveyed 189 large companies that are active product innovators. Below are some of their more notable findings.

1. Over 76 percent of the responding companies now use multidisciplinary teams to develop new products.
2. Only 56.4 percent of the companies had a specific new product strategy; only 54.5 percent had a well-defined new product development process; 32.8 percent had neither.
3. It takes the average company in the study 2.95 years to develop innovative types of new products.
4. Formal financial criteria to measure the performance of new products are developed by 76 percent of the companies.
5. Having insufficient resources is the most frequently mentioned obstacle to successful product development.
6. Companies are developing one successful new product for every 11 new product ideas or concepts they consider.
7. Over a recent five-year period, the companies introduced an average of 37.5 new products, whereas the median was 12.
8. The companies achieved a success rate of 58 percent of the products they introduced during the recent five-year period.
9. The companies spent 52 percent of their new product expenditures on new products that were financially successful.
10. Thirty-two percent of company sales came from new products introduced during the previous five years.

SOURCE: A. L. Page, "Assessing New Product Development Practices and Performance: Establishing Crucial Norms," *Journal of Product Innovation Management*, 10 (1993): 273–290. Reprinted with permission of John Wiley and Sons.

Supporting Materials & Extra Reading

Read Smith (2015) Exploring Innovation textbook, Managing Innovation Chapter, p 239-64

Read: Nickels, W., McHugh, J., & McHugh, S. (2016), Understanding Business, McGraw-Hill Higher Education- USA, ISBN-13: 978-0078023163; ISBN-10: 0078023165. Please read pages 187-189 of the Textbook, alternatively the corresponding pages 4/25-6/25 of the e-book

Activity: Decision Making Game: to apply your problem solving skills



Please complete the Game “Missionary and Cannibals” available from HBMSU LCMS repository by clicking here (Slide 3/11): <http://goo.gl/3IBI5C>
Your Instructor will provide you with the solution if you require support.

Activity: Decision Making Game: to apply your problem solving skills

Watch the following Video:

John Bessant on Innovation; Available at: https://www.youtube.com/watch?v=GRznoD_l4ZQ

Formative Assessment

1. Which of the following is NOT a contextual factor explaining why Innovation has to be managed :

- A. Uncertainty;
- B. Obviousness
- C. complexity;
- D. messiness/untidiness;
- E. disruptiveness; and creativity

Feedback: B

2. Which of the following is NOT a function of management?

- A. Planning
- B. Organizing
- C. Leading
- D. Financing
- E. Controlling

Feedback: D

3. Which of the following is Not a step in Project Management?

- A. Determine the goal (s) and Identify the tasks or activities to be undertaken
- B. Estimate the duration of the activities/tasks
- C. Determine the sequence in which the activities/tasks have to be completed and link them into the overall model of the project
- D. Raise funds to undertake the tasks
- E. Develop a project plan that provides an overall schedule for the project

Feedback: D

Formative Assessment

State if the following statement is True or False:

4. Development funnel is a planning techniques used in innovation management.

Feedback : True

Formative Assessment

6. Distinguish between Mission and Vision statements

▪ **At strategic level, planning consists of developing a Mission and Vision statements. Vision --** *More than a goal, it's a broad explanation of why the organization exists and where it's trying to go. It*

- Delineates management's future aspirations for the business to its stakeholders.
- Provides direction—"where we are going."
- Sets out the compelling rationale (strategic soundness) for the firm's direction.
- Uses distinctive and specific language to set the firm apart from its rivals
- **Mission** typically focuses on its present business purpose - "who we are and what we do". It includes:
 - The organization's self-concept
 - Its philosophy
 - Long-term survival needs
 - Customer needs
 - Social responsibility
 - Nature of the product or service
 - RTA's Mission is:

"Prepare legislation and develop integrated solutions of road systems and land/marine transportation networks that are safe and in line with Dubai's economic development plans with the highest international standards"

Formative Assessment

7. Distinguish between Creativity and Innovation?

Feedback:

Creativity is a vital ingredient in developing innovations, and not just in providing the inspiration for new products and services. Creativity and innovation are key drivers of competitive advantage in today's fast-paced market-driven economy (Parahoo, Mumtaz; & Salem, 2017).

Although the constructs of creativity and innovation are often used interchangeably (Martins and Terblanche, 2003), a distinction may be made between them (McLean, 2005). Creativity is generally conceptualized as the nature of the thought processes and intellectual activity used to generate insights or solutions to problems (Martins, and Terblanche, 2003), while Innovation refers to the development and implementation of new ideas, processes, products, or services (McLean, 2005). Smith defines 4 dimensions of creativity:

1. Inspiration: coming up with *novel* ideas
2. The ability to solve problems in a *unique* way
3. The ability to make connections that others *would not make*
4. The ability to see existing situations in an entirely *new way*

Innovation is the implementation of creativity results, making creativity a part of the innovation process (McLean, 2005). McLean (2005) adds that all although employees may creatively generate brilliant ideas, innovation requires that the idea graduates from concept to market, overcoming various obstacles such as acceptability, securing scarce resources for development and funding.

8. Why is corporate culture important for Innovation?

Feedback:

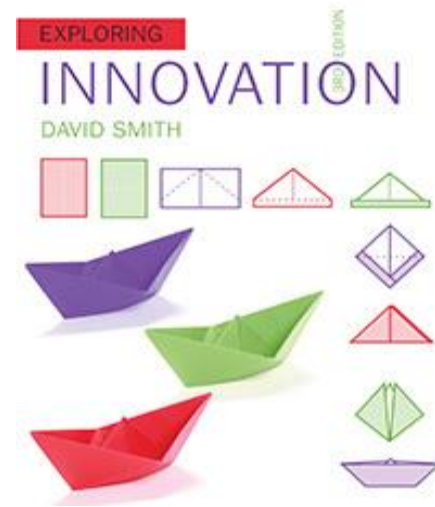
Research shows that firms' internal corporate culture is a powerful driver of innovation (Parahoo, Mumtaz; Ayyagari, & AlKhazraji). Consequently, organisations keen to promote innovation and sustain innovative capacity over time need to give careful thought to developing an appropriate corporate culture (Smith, 2015). This in turn has implications for management practice as the dominant values, assumptions and beliefs within a work setting (i.e. the corporate culture) will be mirrored in the managerial practices of a firm (Smith, 2015).

Summary

During the self-paced session, you will be given an Activity to analyse the tools that may be used to successfully manage innovation.

Chapter 10

Funding





Objectives

When you have completed this chapter you will be able to:

- Appreciate and understand the funding problems associated with innovation
- Analyse the means employed by innovators to reduce and minimise capital requirements
- Distinguish the different sources of capital available for innovation
- Differentiate the various forms of capital available for innovation evaluate the circumstances when particular forms of capital are most likely to be appropriate
- Identify some of the agencies available to assist in handling the funding innovation



Innovation cash flow & the cash flow gap (Smith, 2015)

- Invention and innovation are not the same thing. While inventors may be able to fund the process of invention from their own resources, innovation will normally require additional injection of funds.
 - Few innovators are likely to have the financial resources to fund the whole of the innovation process (Smith, 2015).
 - The need for cash would explain why successful innovators /technical entrepreneurs, such as James Dyson and Ron Hickman, initially tried to license their technology to large companies with substantial financial resources, rather than fund manufacturing and distribution themselves (Smith, 2015).
 - Since innovation involves exploiting ideas and inventions to turn them into commercial products, the exploitation part of the process has to be funded and it is expensive (Smith, 2015).
- 



Funding Innovation

- To fund their innovation, most innovators will start by making use of personal savings funds, usually supplemented by financial inputs from family and friends (F, F, & F).
- These funds are likely to be supplemented by financial bootstrapping – quite literally finding ways of acquiring resources either without having to pay for them or paying a lot less for them. An example of bootstrapping is to use a garage as office or to use personal car for business purposes.
- As the Innovator proceeds with the development of prototypes, setting up manufacturing/ service facilities and launching a product/service, the innovators will require ‘seed capital’ to fund further development (see Figure below).
- The inventors may approach ‘informal investors’ willing to provide either loan or equity capital. These business angels are high net-worth individuals willing to contribute some financial resources into new ventures.



Funding Innovation

- They do this not to obtain an income from dividends, but in the hope of capital gain at some later stage (Smith, 2015). There is no typical profile of a business angel but they have a common trait- they invest in people as much as in business propositions. A good business plan is necessary but never sufficient to clinch a deal. The chemistry between the investor and the management has to be right for both sides.
- Eventually, even this source of funds may not be adequate. As the innovation becomes ready for the market, venture capitalists may need to be approached. They provide larger sums than informal investors but operate on the same basis – that is to say, they invest in the expectation of capital gain.
- Finally, depending on the size of the venture, the innovator may turn to the equity market and seek additional capital through an initial public offering (IPO), perhaps on a junior market such as the Alternative Investment Market (AIM).

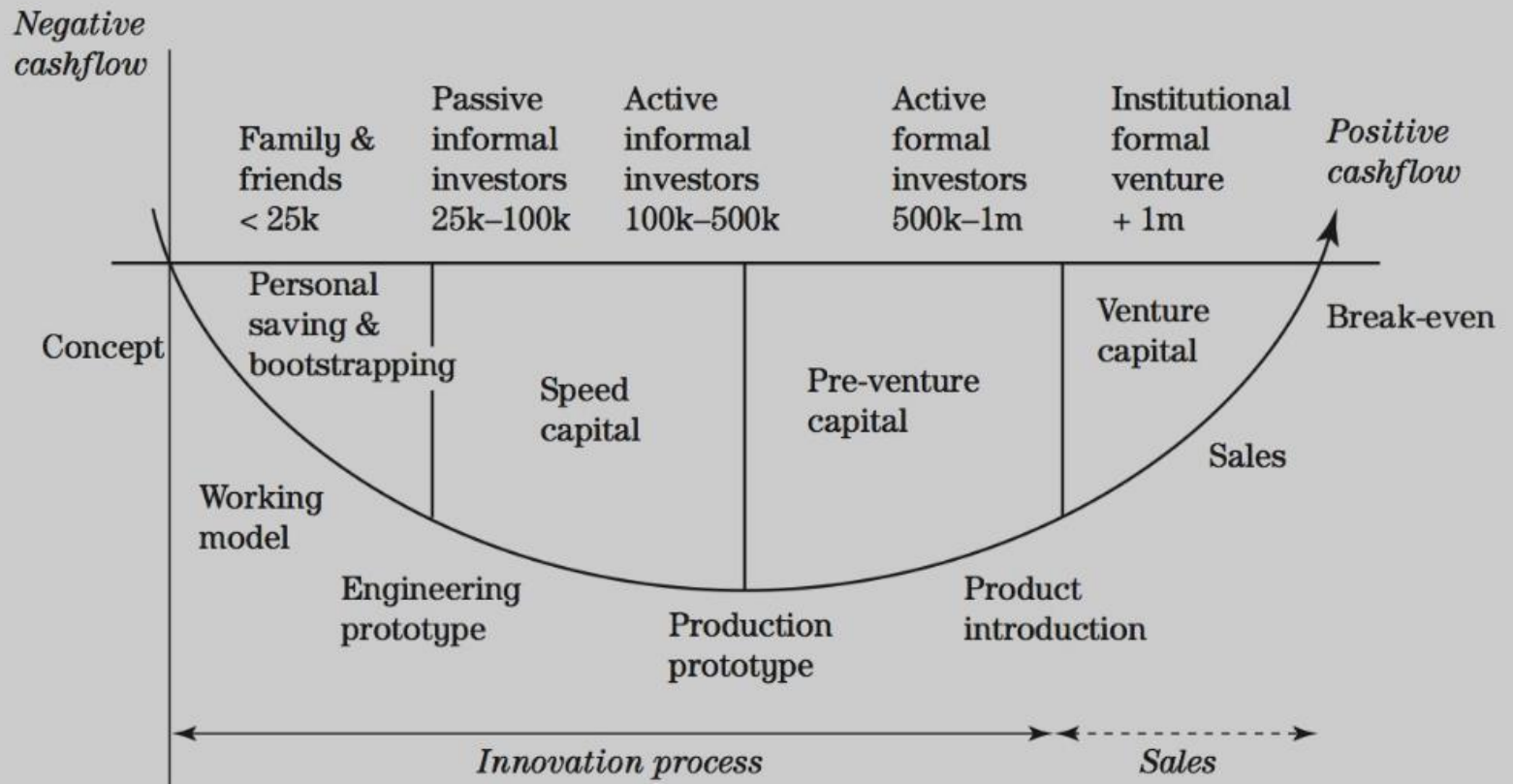


Early stages of innovation

- Negative cashflow
- Uncertainty about extent of negative cashflow
- Uncertainty about when the project will breakeven
- Lack of track record against which to borrow
- Unwillingness of financial institutions to lend

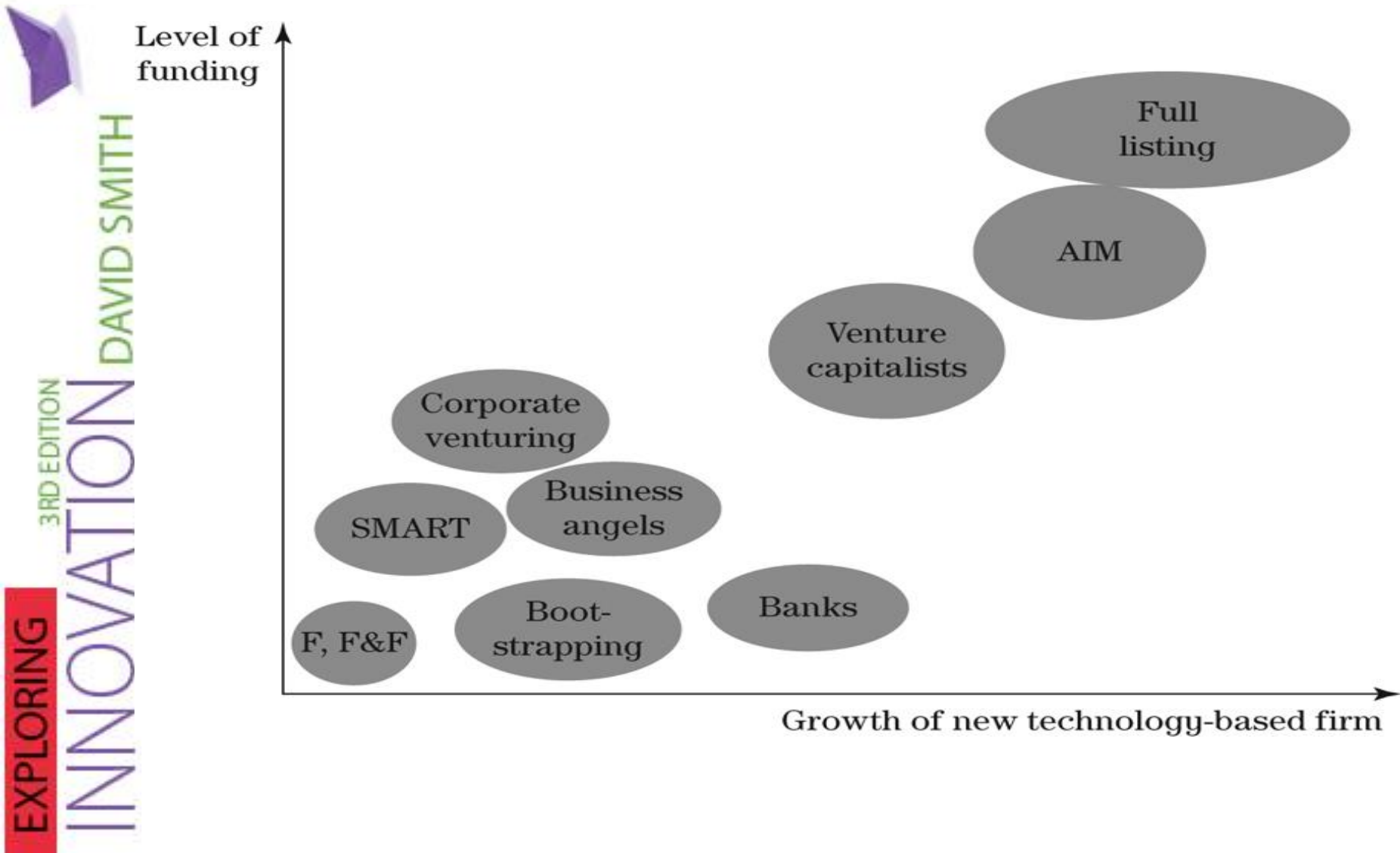
Innovation Cashflow

Figure 10.1 Innovation cashflow



Source: Adapted from Rorke et al. (1991: 19).

Sources of Funding (Source: Smith, 2015)





Founder, Family & Friends

- **Founder:** Personal savings, mortgaging the home, etc.
- **Family:** Guaranteeing loans, loans, equity stakes, legacies; e.g. Jeff Bezos of Amazon borrowed \$250,000 from his parent
- **Friends:** An instance of social capital i.e. a network of friends derived from friendship, sporting activities, school, leisure, e.g. James Dyson: Dyson Appliances; Martin Wood: Oxford Instruments

Financial Bootstrapping

Definition

“using creative ways of acquiring finance without recourse to banks or raising equity from traditional sources”

“Minimising or eliminating the need for finance by securing resources at little or no cost”

Steve Jobs and Steve Wozniak
Started Apple Computer in the garage of Jobs' parents' house.



Banks

Mike Lynch, the founder of the software company Autonomy described his experience of seeking funds from banks:

'I went to the bank first and met a very nice chap, very friendly. It turned out he was normally doing things like lending money to people to set up newspaper shops. He did not feel really qualified to comment on my adaptive non-linear pattern recognition technology. But he did give me a good piece of advice, which I carry with me even now, which is that people will always buy confectionery.'

Source: Doward (1999)

Micro-Financing Schemes: Grameem Bank



Prof Muhammad Yunus is a Bangladeshi social entrepreneur, banker, economist, and civil society leader. He was awarded the Nobel Peace Prize for founding the Grameen Bank and pioneering the concepts of microcredit and microfinance.



Innovation support Resources in UAE

- Listed below are two well-known innovation support resources in UAE:
 - 1. Mohammed Bin Rashid Innovation Fund**
(<https://www.mof.gov.ae/En/About/programsProjects/Pages/MohamedBinRashidInnovationBox.aspx>)
- An initiative of the Federal Government represented by the Ministry of Finance, His Highness Sheikh Mohammed bin Rashid Al Maktoum, Vice President and Prime Minister of UAE and Ruler of Dubai, launched "Sheikh Mohammed bin Rashid Al Maktoum Fund to Finance Innovation" which is worth AED 2 billion. The fund supports resident individuals and companies (of all sizes) registered in the UAE, provided that they offer unique and innovative ideas whether they be technologies, products, services and processes. Eligible applicants need to have registered their project as intellectual property of the respective individual or company, and the initial stage of the project must already be established. The fund's operations were launched officially in the second half of 2016.
- The objectives of the Fund are:
 - Supporting National Innovation Strategy, and achieving the UAE's developmental vision to be amongst the 10 most innovative countries internationally by 2021.
 - Providing financing solutions and the necessary guarantees to innovative entrepreneurs, in order to facilitate their access to business loans to finance their projects, and help them turn their ideas into actual projects.
 - Contributing to establishing a supportive environment for innovation in collaboration with financial institutions and funding entities, such as commercial banks, investment funds and family businesses, among others.



Innovation support Resources in UAE

- **Mohammed Bin Rashid Innovation Fund: Funding Conditions**
- The project idea must be unique and innovative, whether a technical project, processes, products or services that are registered as intellectual properties of their respective owner, and the initial stage of the project must already have been established.
- Applicants must be a resident, individual or a company registered in the UAE. Applying is not restricted to Emirati companies but also includes all companies in the region, provided that they are registered in the UAE.
- Introducing long-term business development plan supported by a feasibility study in order to qualify the project to acquire the appropriate funding.



Innovation support Resources in UAE

2. Khalifa Fund

(<https://www.mof.gov.ae/En/About/programsProjects/Pages/MohamedBinRashidInnovationBox.aspx>)

Khalifa Fund for Enterprise Development was launched on 3 June 2007 to help develop local enterprises in Abu Dhabi. The fund has a total capital investment of AED 2 billion and it aims to create a new generation of Emirati entrepreneurs by instilling and enriching the culture of entrepreneurship amongst young people, as well as supporting the development of small to medium investments in the UAE.

The Fund has also set up a system of support services for entrepreneurs, including training, development, data and consulting services, and marketing initiatives.



Finance for SMEs (Source: Khalifa Fund Report: SME Financing in the United Arab Emirates)

Overview of current activity

Banks constitute the principal source of funding for SMEs in UAE. The most commonly used financing mechanisms are letters of credit and overdrafts; relationship-based lending is not common.

- Despite the fact that the SME sector is regarded as a growth market, banks remain somewhat wary of financing SMEs. For instance, the rejection rate for SME lending in UAE ranges from 50% to 70%.
- In addition, interest rates are very high compared to typical SME lending. Given the importance of SMEs as a key contributor to economic growth and diversification, the UAE government has undertaken a number of initiatives to expand SME financing.
- Funding programs have been introduced by a number of entities including the Khalifa Fund; the Ministry of Economy; and the Mohammed Bin Rashid Establishment for Young Business Leaders among others.
- In 2009, the Ministry of Economy and the Central Bank launched a lending program focused on entrepreneurs.
- The scale of SME banking in the UAE has expanded significantly in recent years. Among the 52 banks that are authorized to operate in the UAE, 40 offer SME banking services. SMEs are regarded by most banks in the UAE as a major growth market; as a result, many of these banks have set up specialized banking units focused on this market. They have also introduced services that are specifically targeted to the unique needs of SMEs.

Sources of Finance (Source: Khalifa Fund Report: SME Financing in the United Arab Emirates)

Based on figures provided by UAE Central Bank, there are twenty-four national banks and twenty-eight foreign banks that are authorized (licensed) to operate in the UAE, forty of which actively compete for the SME segment (7 or 8 banks are dominant).

A list of the most prominent banks and other entities that provide funding to SMEs is provided in the table below.

Key Banks and Financing Programs Focusing on SMEs	
Standard Chartered Bank	One of the world's largest banks, worldwide SME strategy
RAK Bank	Rapidly growing, widely regarded as a leading innovator, aggressive in marketing to SMEs
Mashreq Bank	Especially popular with retailers
National Bank of Fujairah	Especially popular with Northern Emirates
ADCB	One of UAE's largest banks
Union National Bank	Popular especially with smaller businesses
Citibank	Small SME division, but regarded as innovative by other banks
Khalifa Fund (KF)	Supports SMEs in Abu Dhabi with a total capital investment of AED 2 billion
Tasne'a (KF)	Financing up to AED 10 million for greenfield industrial development projects
Microfinance (KF)	Funds raw materials for small home based businesses
Khutwa (KF)	Financial scheme program for micro-projects
Bedaya (KF)	Provides assistance to entrepreneurs by providing them seed capital
Zayada (KF)	Supports existing enterprises for their growth and expansion goals
Al Hasila (KF)	Supports fishermen who operate on a small scale

Sources: SME Lending in the UAE: 2008, Dun & Bradstreet Business Insight Series, 2008 and Khalifa Fund

Business Angels: Who are They?

Generally successful entrepreneurs themselves, or perhaps senior managers from industry, looking for the opportunity to invest in up and coming small businesses. There is no such thing as a typical business angel but one thing they have in common is that they invest in people as much as in business propositions. A good business plan is necessary but never sufficient to clinch a deal. The chemistry between the investor and the management has to be right for both sides.

Business Angels: Who needs them?

- Early stage and growing businesses with insufficient capital of their own to lever the right mix of other funding.
- Young companies requiring amounts of equity investment too small to interest mainstream venture capitalists, i.e. under £250K.
- Growing businesses who recognise the need to strengthen their board or management team to sustain successful development.

Business Angels

- “High net worth individuals seeking capital games over the life of their investment in a company”
- Sometimes called “informal investors”
- Approximately 18,000 business angels in the UK
- A heterogeneous group
- Aim to invest near start up and *harvest* as the business matures making a capital gain

Corporate Venturing

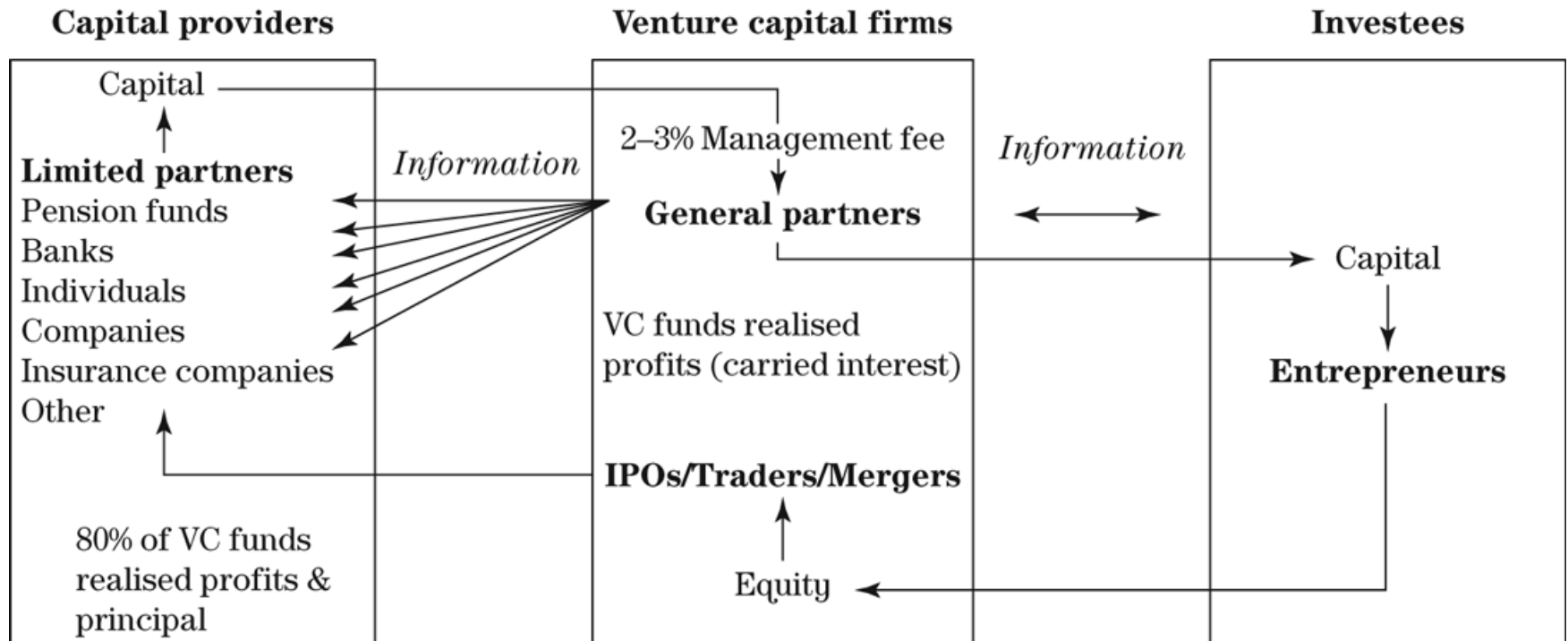
A recent report by the CBI (1999) describes corporate venturing as:

‘a formal, direct relationship, usually between a larger and an independent smaller company, in which both contribute financial, management or technical resources, sharing rewards equally for mutual growth.’

Venture Capitalists

- Borrow from institutional investors
- Aim for capital gain
- Act as intermediaries
- VCs expertise lies in evaluating the risks & growth potential of companies they back
- UK venture capital market is the largest in Europe

Venture Capital Funds Structure



Funding Trajectory: Pre-Seed, Seed, Series A, B, & C

(Source: <https://www.investopedia.com/>)

Background

- A startup based on a distinctive idea/insight will try to capture value using a great business model. In the initial stage of growth, the founder will usually try to leverage the contributions of friends, family and the founders' own financial resources to steadily grow in the market.
- Over time, the founder will hope to grow the customer base, and the business will soon begin to expand its operations and its objectives, establishing itself as a valuable competitor in the market, and opening up the possibilities for future expansion (e.g. new offices, new equipment, additional employees, often leading to an initial public offering (IPO)).

The Different Funding Rounds for startups

(Source: <https://www.investopedia.com/>)

- Unfortunately, in practice, the early growth stages of the majority of businesses rarely proceed as smoothly as described above.
- In fact, the vast majority of successful startups endeavour to raise capital through rounds of external funding, which provide outside investors the opportunity to invest cash in the startup in exchange for equity, or partial ownership.
- The process of growing a business through outside investment are referred to as Series A, Series B, and Series C funding rounds.

The Different Funding Rounds for startups

(Source: <https://www.investopedia.com/>)

- Usually, startups first resort to what is known as "seed" funding or angel investor funding. This is referred to as “bootstrapping” (relying on the generosity of friends, family, and founder’s own funds) and seed funding.
- However, it is often not sufficient, requiring additional rounds, referred to as Series A, B and C funding.
- As the business becomes increasingly mature, it usually complements the seed round with A, B, and then C funding rounds.
- From the investor perspective, If the business grows and earns a profit, the investor will be rewarded in proportion to the investment made.

Funding Valuation of startups

(Source: <https://www.investopedia.com/>)

- Before any round of funding begins, analysts make a valuation of the startup, based on different factors, including management, proven track record, market size, and risk.
- One of the key distinctions between funding rounds relates to the valuation of the startup, its maturity level and its growth prospects.
- In turn, these factors impact the types of investors likely to get involved and the reasons why the company may be seeking new capital.

Business Plan

Pitch:Reality TV Show

Shark Tank/ Dragon's Den

The panelists of seasoned investors are judges, and they are as well famous angel investors and venture capitalists (VC).

The show is popular because there are many startup founders seeking funding and so the competition is very high.

However, a good pitch has helped many contestants to stand out. Additionally, many businesses want to invest in startups with a bright future and in the same industry that they have invested in before.



Business Plan Competitions (Dragon's Den / Shark Tanks)

These are internal competitions in which teams of peers work together to, in each case, identify a new opportunity for the business, develop a relatively complete and well thought-out business plan for commercializing it, and then pitch that plan to an internal venture board, generally in order to get funding for it. This is not unlike what entrepreneurs do on the outside when they pitch to Venture Capitalists to get funding for a new business, except that in these cases, it is all internally-focused – the individuals are intrapreneurs pitching to an internal venture board for internal funding to develop a concept for use internally by the business. Otherwise the vibe is very much the same.

There are usually cash prizes and other incentives for the winners and runner-ups. After the competition, the winning teams are typically given a Series A round of funding to further investigate, research, and otherwise prove out their concept and business plan. If that phase proves successful, they may be awarded a Series B round for additional proof-of-concept development / testing, and so forth.

If a concept proves strong enough, it may ultimately make its way all the way through to commercialization (implementation) within the Core Innovation Team and/or one of the business' business units. How this happens can vary widely from one business to the next. In almost all cases the endeavor will be handed off to a special team dedicated to this purpose, but where exactly this takes place will vary considerably. In some cases, it will happen early on. In other cases, it will happen later, relying on the original team to continue pursuing the Series A, Series B, etc. work (on top of their normal day jobs) until a certain milestone has been reached. Considering the amount of (volunteer) work that teams must invest in these competitions, companies typically host them only once or twice per year. In some cases, they are also used as the venue for a formal intrapreneurship program.

The Valuation

In the Reality show Dragon' Den or Shark Tank , you will see founders asking for \$100,000 in exchange for a 10% stake in the company valued at \$1 million.

With that in mind, the judges will then ask the selling price of one unit of the product plus the cost of making it. The goal is to determine the profit margin. Therefore, a product retailing at \$10 and whose production cost is \$5 has a 50% margin.

If the company generated \$100,000 as revenue in the previous year, out of it 50% or \$50,000 is profit. So, asking for \$100,000 means that this investment is 10 times the revenue that your company generated or 20 times its profit.

Based on these numbers, an angel investor who accepts a 10% stack in your company in exchange for \$100,000 funding will receive \$5,000 profits every year.

That means it will take them 20 years to recoup their investment back if everything remains constant.

However, this might shorten by 5 – 7 years or more if the business hopefully grows and the revenue continues to rise, with the support of the Angel Investors.

Often, these investors can't touch any business opportunity that takes more than 7 years to recover their entire investment. Similarly, venture capitalists and private equity prefer investments that take 3-4 years but not beyond 10 years.

Pre-Seed & Seed Funding

(Source: <https://www.investopedia.com/>)

- The earliest stage of funding a startup occurs so early in the process that it is not generally included among the rounds of funding. "Pre-seed" funding typically occurs when the founder of a startup is launching its operations, and the funders are the founders themselves, as well as close friends, supporters and family.
- **Seed Funding** represents the first official equity funding stage, and it typically refers to the first official funds that a startup raises. Some startups do not extend beyond seed funding into Series A rounds or beyond.
- Think of the "seed" funding as an analogy for planting a tree, and this early financial support is ideally the "seed" which will help to grow the business. Seed funding helps a startup to finance its first steps. These include market research, product development, and the target market.
- The potential investors in a seed funding situation include: founders, friends, family, incubators, venture capital companies, and "angel investors". Angel investors tend to appreciate riskier ventures (such as startups with little by way of a proven track record so far) and expect an equity stake in the company in exchange for their investment.
- Once the startup has developed a track record, developed a customer base, generated consistent revenue, it is ready to raise additional capital.

Series A Funding

(Source: <https://www.investopedia.com/>)

- The first round after the seed stage is Series A funding.
- It requires the startup to demonstrate it has a viable business model that will generate long-term profit.
- The investors are not only looking for great ideas, but also for an effective strategy to turn that idea into a successful, money-making business.
- The investors involved in the Series A round come from more traditional venture capital firms, e.g. Sequoia Capital, IDG Capital, Google Ventures, and Intel Capital.

Series B Funding

(Source: <https://www.investopedia.com/>)

- Series B rounds take the startup past the development stage, and the investors help startups by expanding market reach.
- Startups that have gone through seed and Series A funding rounds have already developed substantial user bases and have proven to investors that they are prepared for success on a larger scale. Series B funding is used to grow the company so that it can meet these levels of demand.
- Growing and sustaining a winning product/service, as well as a team requires business development, sales, advertising, tech, support, as well as quality talent acquisition, and this consolidation requires substantial funds.
- Series B appears similar to Series A in terms of the processes and key players, who are often the same as the earlier round. The difference with Series B is the addition of other venture capital firms that specialize in later-stage investing.

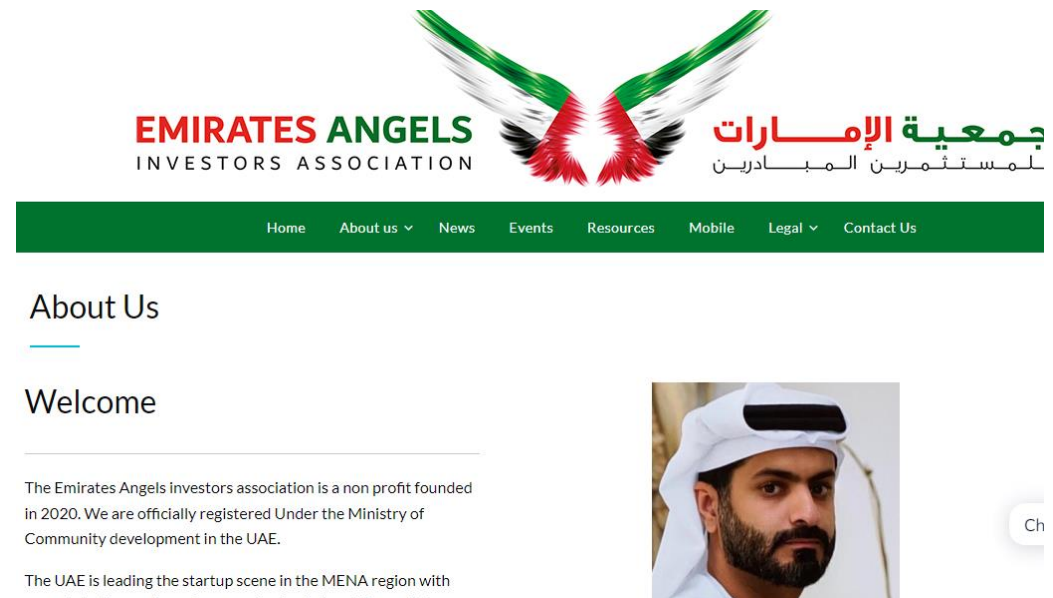
Series C Funding

(Source: <https://www.investopedia.com/>)

- Startups that raise a Series C funding are already quite successful and are looking for additional funding in order to help them develop new products, expand into new markets, or even to acquire other companies. Series C funding is focused on scaling up the company, growing as quickly and as successfully as possible.
- In Series C, the business has already established a successful business model, and it attracts groups such as hedge funds, investment banks, private equity firms, and large secondary market groups that are keen to accompany the other investors mentioned above.
- Many companies will complete an initial public offering (IPO) after their Series C funding round.

Funding Innovation: Angel Investors

- Angel investors fund young small businesses with a promising future.
- Contacting a potential investor can be a daunting task because they don't respond to unsolicited emails.
- UAE Angels Investors Association



Crowdfunding (Common in the West)

Crowdfunding the process of raising money to fund what is typically a project or business venture through many donors using an online platform, such as Kickstarter, Indiegogo and Crowdfunder.

Crowdfunding is about persuading numerous people to each give you a small donation – AED 50, AED 100, AED 200, AED 500, maybe more. Once you get a large number of donors, you are able to raise a substantial amount of hand. Crowdfunding has become possible in recent past due to a proliferation of websites that allow nonprofits, artists, and businesses -to raise money. This is the social media version of fundraising. There are over 600 crowdfunding platforms around the world, with fundraising reaching billions of dollars annually. . In 2015, it was estimated that worldwide overUS\$34 billion was raised this way. Crowdfunding is thus a form of crowdsourcing and of alternative finance

The fundraising window is usually finite - 90 days, for instance - and the fees and rules vary across platforms.

This modern crowdfunding model is generally based on three types of actors: the project initiator who proposes the idea and/or project to be funded, individuals or groups who support the idea, and a moderating organization (the "platform") that brings the parties together to launch the idea.

Crowdfunding has been used to fund a wide range for-profit entrepreneurial ventures such as artistic and creative projects, medical expenses, travel, or community-oriented social entrepreneurship projects.



Crowdfunding (Source: GInI Innovation Master Book)

Crowdfunding

In addition to online crowdsourced campaigns, several large-scale innovators have in recent years turned to running campaigns on **crowdfunding sites**... platforms traditionally reserved for cash-strapped entrepreneurs attempting to raise funding to launch very modest businesses and other projects. Examples of this include **GE** using **Kickstarter** and **OurCrowd**, **P&G** using **CircleUp**, and **Whirlpool Corporation** using **Indiegogo**.

The reason these large Fortune 100 businesses use such platforms is two-fold, neither of which involve a need for funding. First, it allows them to create very **grassroots word-of-mouth marketing** for some of the more radical innovation concepts they are considering. Second, and more importantly, it affords them yet one more opportunity to **market test** concepts. Those concepts that see a lot of traction on these platforms should, in theory, also be able to generate good traction within the general market. Conversely, poor traction on these platforms (which, after all, tend to be populated with “Innovator” and “Lead Adopter” type adopters) is generally taken as a sign of low demand for the idea, and thus an indicator that it possibly will not do well in the general market (and should therefore be nixed).

Characteristic examples of these platforms (there are estimated to be over 400 of them) include:

- Kickstarter www.kickstarter.com
- Indiegogo www.indiegogo.com
- CircleUp www.circleup.com

Crowd Funding

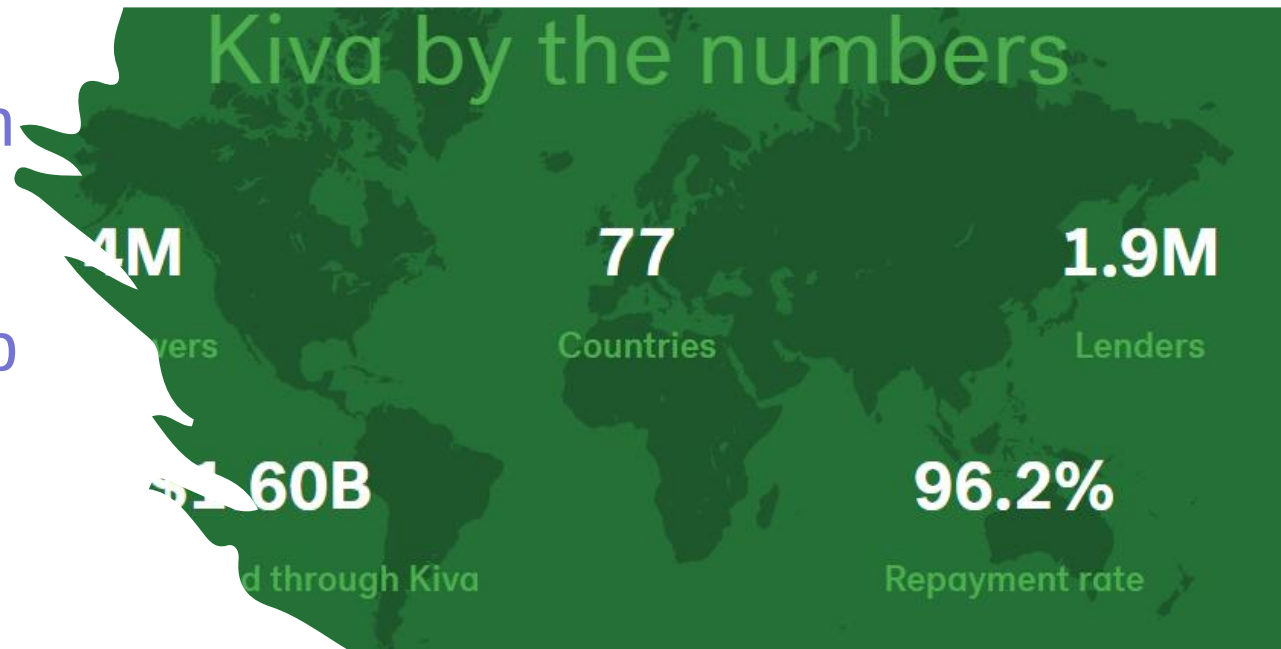
(Source: <https://www.investopedia.com/best-crowdfunding-platforms-5079933>)

The 6 Best Crowdfunding Platforms of 2024

- Best Overall: [Indiegogo](#)
- Best for Nonprofits: [Mightycause](#)
- Best for Investing: [StartEngine](#)
- Best for Individuals: [GoFundMe](#)
- Best for Creative Professionals: [Patreon](#)

Micro-Financing Schemes: Kiva.org

- More than 1.7 billion people around the world are unbanked and cannot access the financial services they need. Kiva is an international nonprofit, founded in 2005 in San Francisco, with a mission to expand financial access to help underserved communities thrive.



Micro-Financing Schemes: Kiva.org

- We do this by crowdfunding loans and unlocking capital for the underserved, improving the quality and cost of financial services, and addressing the underlying barriers to financial access around the world.
- Through Kiva's work, students can pay for tuition, women can start businesses, farmers are able to invest in equipment and families can afford needed emergency care.
- By lending as little as \$25 on Kiva, you can be part of the solution and make a real difference in someone's life. 100% of every dollar you lend on Kiva goes to funding loans.



Sandra

Nicaragua

A loan of \$875 helps to buy commercial inventory for her small retail store. [Read more](#)



MENA Perspective: Study by DSOA and Google (2014)

Dubai Silicon Oasis Authority (DSOA) and the Middle East and North Africa (MENA) Office of Google conducted a study in 2014 to better understand the emerging technology businesses (SMEs and start-ups) in the region. The study focused on technological entrepreneurs in the following countries: UAE; KSA; Egypt; Jordan; Lebanon; and Morocco.

The objectives of the study were to determine:

- a. The challenges technology SMEs and Start-ups face in current times
- b. Entrepreneur needs and the main issues faced when setting up businesses
- c. Whether government regulations were perceived to be inflexible/ ambiguous?
- d. What were the SMEs and Start-Ups perspectives on trade restrictions/customs? Their experiences and thoughts towards gateway payment systems or online payment security. Whether there is a talent pool to support businesses at the technical level and to what extent does the quality of shipping service impact their business?
- e. Is there a problem of too many SME options or will some be able to find a niche for themselves?

Sources of Finance

(Source: Khalifa Fund Report: SME Financing in the United Arab Emirates)

Based on figures provided by UAE Central Bank, there are twenty-four national banks and twenty-eight foreign banks that are authorized (licensed) to operate in the UAE, forty of which actively compete for the SME segment (7 or 8 banks are dominant).

A list of the most prominent banks and other entities that provide funding to SMEs is provided in the table below.

Key Banks and Financing Programs Focusing on SMEs	
Standard Chartered Bank	One of the world's largest banks, worldwide SME strategy
RAK Bank	Rapidly growing, widely regarded as a leading innovator, aggressive in marketing to SMEs
Mashreq Bank	Especially popular with retailers
National Bank of Fujairah	Especially popular with Northern Emirates
ADCB	One of UAE's largest banks
Union National Bank	Popular especially with smaller businesses
Citibank	Small SME division, but regarded as innovative by other banks
Khalifa Fund (KF)	Supports SMEs in Abu Dhabi with a total capital investment of AED 2 billion
Tasne'a (KF)	Financing up to AED 10 million for greenfield industrial development projects
Microfinance (KF)	Funds raw materials for small home based businesses
Khutwa (KF)	Financial scheme program for micro-projects
Bedaya (KF)	Provides assistance to entrepreneurs by providing them seed capital
Zeyada (KF)	Supports existing enterprises for their growth and expansion goals
Al Hasila (KF)	Supports fishermen who operate on a small scale

Sources: SME Lending in the UAE: 2008, Dun & Bradstreet Business Insight Series, 2008 and Khalifa Fund



Sources of Finance

(Source: Khalifa Fund Report: SME Financing in the United Arab Emirates)

Given the importance of SMEs and the reluctance of commercial banks to lend to SMEs, the government has taken initiatives to support the financial needs of entrepreneurs.

The Ministry of Economy along with the UAE Central Bank has announced that it will launch an SME start-up loan program targeting entrepreneurs.

The Khalifa Fund provides start-up and expansion loans as well as training for entrepreneurs in Abu Dhabi. It offers the following programs:

- Khutwa: a microfinance program that offers flexible loans up to AED 250,000 to support micro businesses. The program is social in nature, allowing the creation and growth of income streams for specific target groups, specially divorced women, widows and retirees.
- Bedaya: a start-up loan program, designed to support new SMEs with flexible loans of up to AED 3 million.
- Zeyada: an expansion and development loan program that provides existing, early-stage SMEs with flexible loans of up to AED 5 million.
- Tasnea: a program to finance greenfield industrial projects up to AED 10 Million.

In addition, the Mohammed Bin Rashid Establishment for Young Business Leaders provides start-up capital for aspiring UAE national businessmen in Dubai. The Mohammed Bin Rashid Establishment for SME Development offers a 5% Government Procurement Program, which requires that Dubai Government departments set aside at least 5% of their procurement budget to companies that are registered with the Establishment.

MENA Perspective

- Study by DSOA and Google (2014)

Study by DSOA and Google (2014)

The top two predominant financial challenges are:
Access to on-going/secondary capital and a lack of financial support from banks

Entrepreneurs face minimal difficulties with seed investments for the set-up; however, they are having issues obtaining sufficient secondary funding for their ongoing operations.

Nearly 70% of those surveyed face major to moderate challenges due to non-availability of support from banks over business accounts as well as a distinct lack of access to capital

- Availability of support from banks over business accounts poses a hindrance particularly for almost three quarters of North African entrepreneurs. Access to capital for start-up is claimed as a challenge for three quarters of respondents in the Levant and a majority of those in the GCC
- Inability to afford the initial investment for business set-up is cited as a challenge by around 58% (with 63% of GCC respondents citing a daunting challenge)

Challenges:

- Obtaining the initial seed investment to set up the business is perceived as fairly smooth, with a majority using private funds, working in pairs/trios (with one-two co-founders/stakeholders who also assisted with financing) as well as assistance from informal third parties (family, friends, ex-colleagues)
- Many mention they prefer to rely on internal revenue generation as challenges arise with fund raising, reasons for this include:

Other Challenges

- The technology field being 'off the radar' for investors in the region
- Government resources/subsidies seen as unavailable nor a viable option
- Suppliers being hesitant about adopting an online business model and carrying concerns that it will overshadow their traditional outlets

Upon their pursuit of secondary funding and raising capital, most entrepreneurs claim having faced numerous hurdles, including:

- Banks making it unnecessarily bureaucratic and costly to obtain funding: Bank views about technology Start-Ups laced with skepticism, thus making it difficult to obtain a business account, requirements of unwarranted paperwork and funding, and excessive lethargy (slow procedures)
- Nonexistent/ inadequate Angel investors
- A certain amount of difficulty in finding the right investor is expressed: Lack of commitment, understanding and cooperation from potential investors, such as dropping out mid-way, failing to meet promises, placing unrealistic expectations, and/or inefficiently allocating funds
- Intensive bureaucratic paper-work and/or procedures that is either a turn-off for potential investors and/or delays in the receipt of finances (sometimes up to 6 months after agreement)
- Predominance of the 'trader' profit-driven business mentality that does not comprehend the technology Start-Up business model requiring long-term investment
- Lengthy and time consuming process for financing and de-blocking funds

Resolutions:

- To decrease their dependency on banks, entrepreneurs have to increase their own down payment/funding from their side
- To attract investors on their second round of fundraising, some entrepreneurs use their own connections or crowd funding, others mention having to reassure potential and existing investors that they are in fact technically educated/aware, in an effort to enhance credibility and reassure investors about the future success of the business
- Some also mention that having an external/ international investor helps attract local investors - increasing their trust and confidence
- A few mention a lack of incubators as an addition challenge however others manage to collaborate with incubators who provide direct funding and/or connections to 3rd party investors, such as Oasis500, Groupon, iMena, Saudi Telecom, and Middle East Venture Partners



Chart 5.2.3: Financial challenges

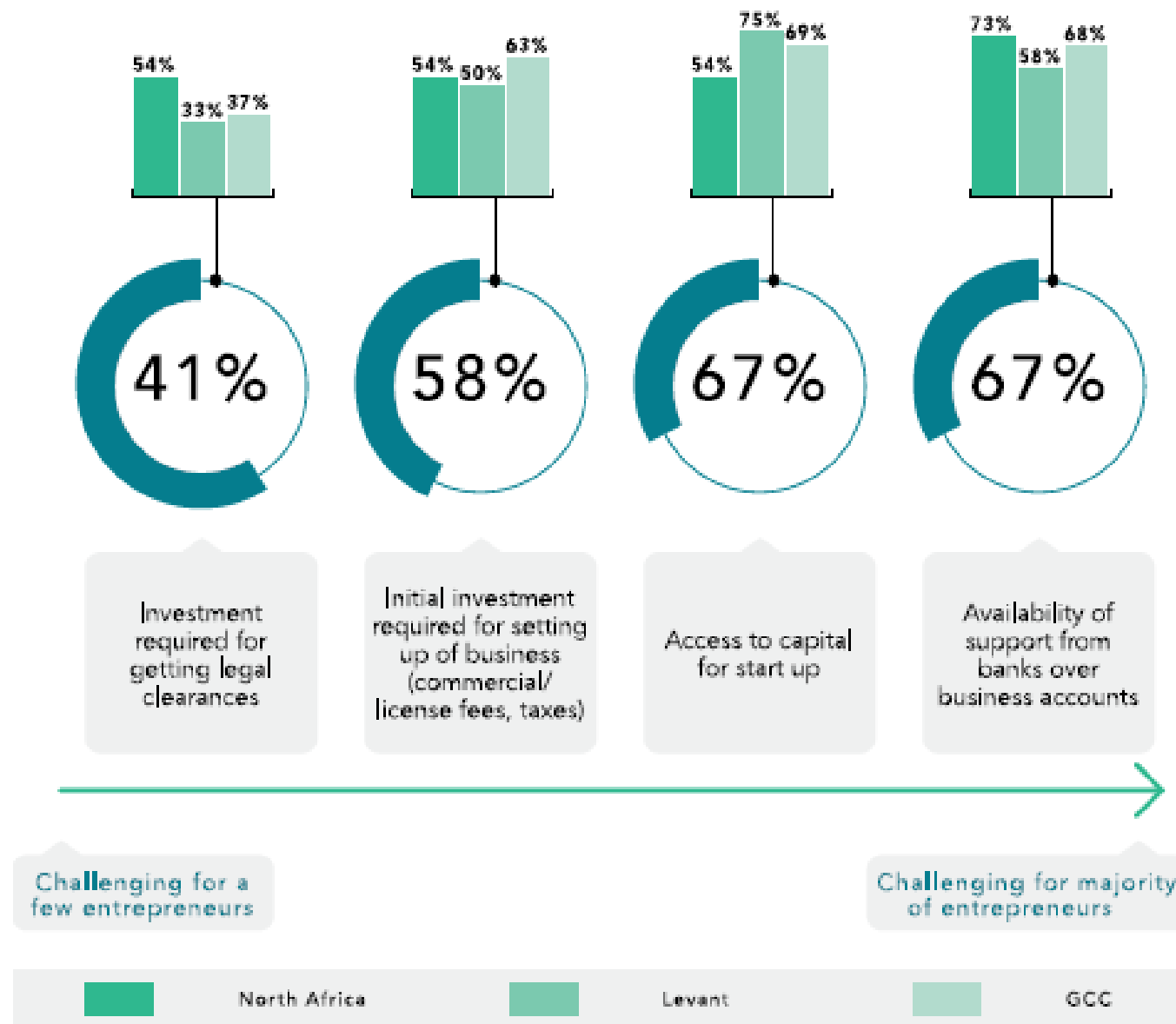


Chart 5.2.4: Cost incurred for start-up

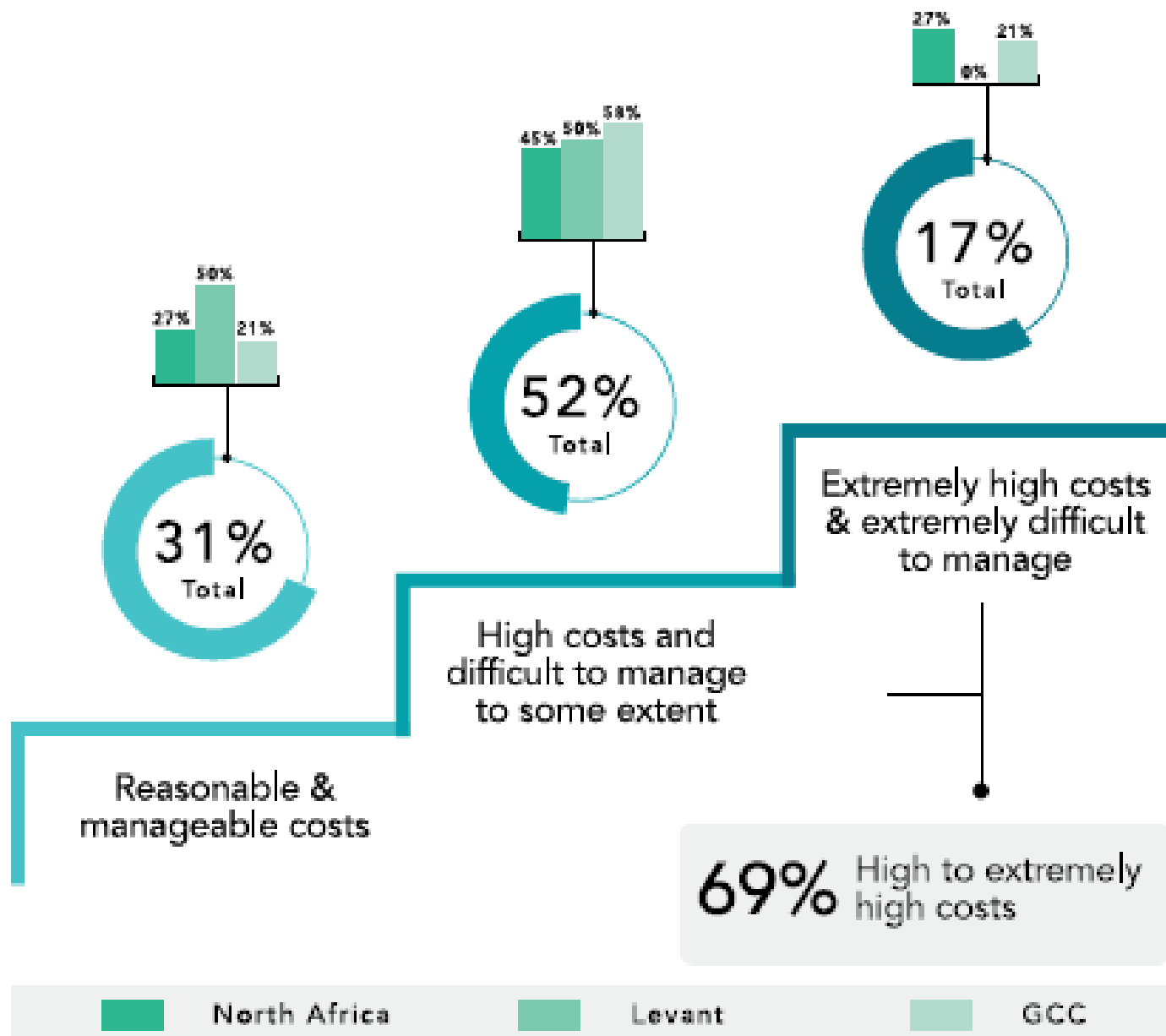
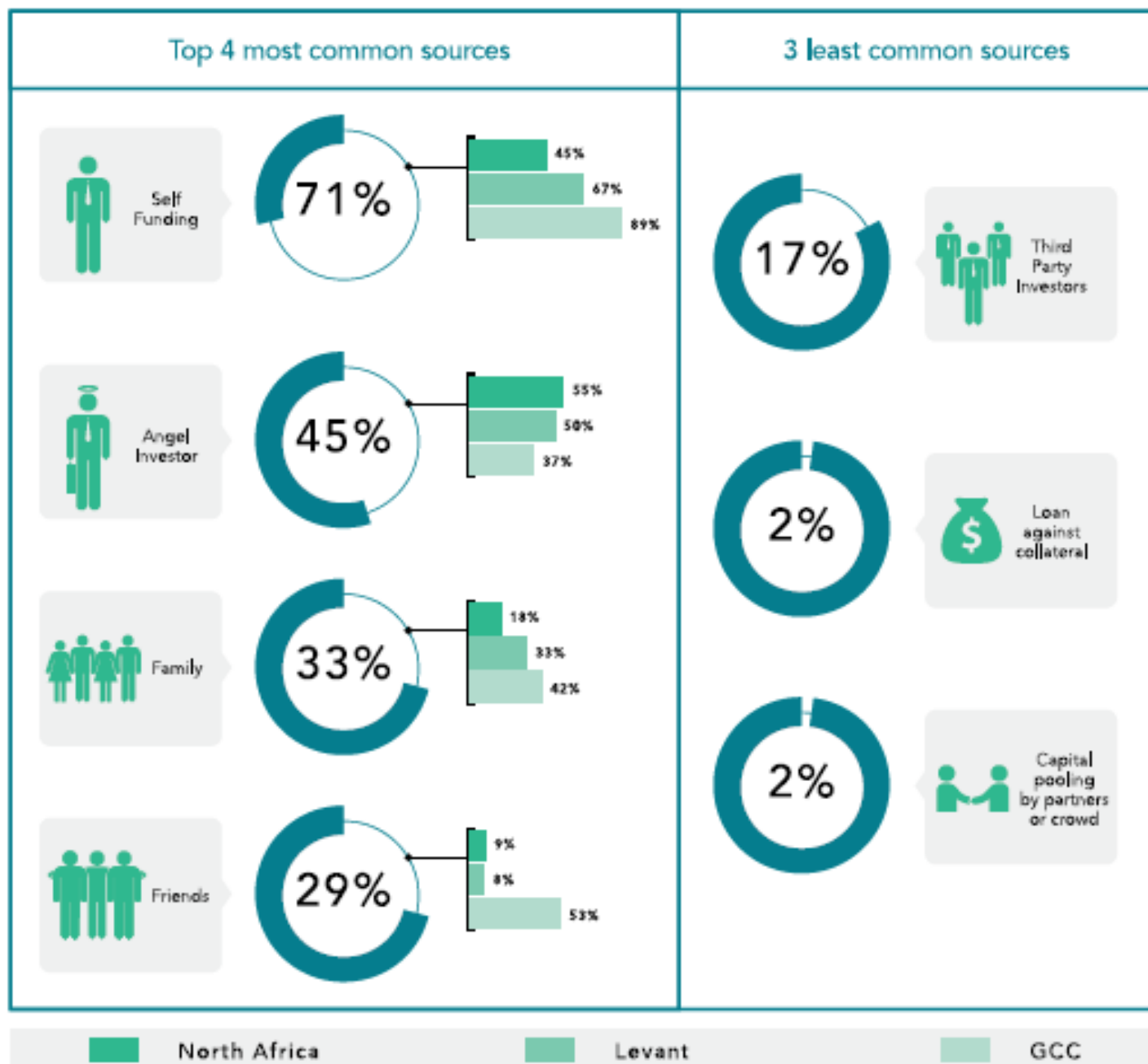


Chart 5.2.5: Sources of funds



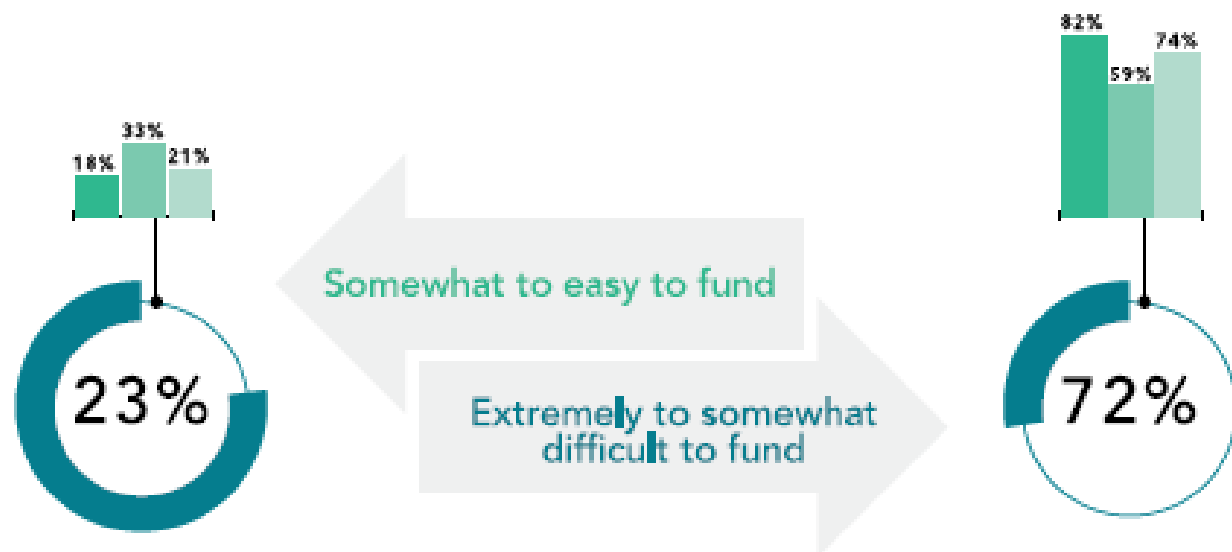
2- Funds sourcing: Keeping it within immediate social circle –across regions, around 70% claim self-funding while one in three rely on family

- Entrepreneurs in the GCC tend to rely more heavily on family (42%) and friends (53%) – more so than the Levant and North Africa
- Interestingly, slightly less than 50% have been able to find an **Angel investor** (especially in the Levant and North Africa)
- A sixth acknowledge receiving funds from **third party investors** (around a quarter of those in North Africa -27% and Levant – 25% while only 5% for the GCC)

3- Initial funding

- A majority (72%) claim that they found it difficult (28% extremely difficult and 44% somewhat difficult) to fund the business
- Less than a quarter claimed that funding was easy for them (to varying degrees)

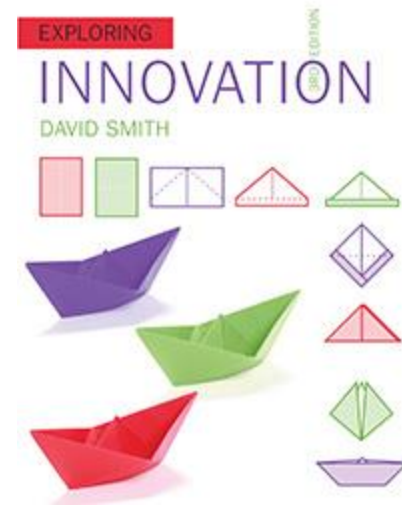
Chart 5.2.6: Ease or difficulty in initial funding



Thank you

Technological Entrepreneurs

Chapter 9



Objectives

At the end of this class you will be able to:

- Explain the nature of entrepreneurship;
- Distinguish different categories of entrepreneur;
- Identify the key characteristics of technological entrepreneurs;
- Categorize technological entrepreneurs;
- Analyse the factors that lead to the growth and development of technological entrepreneurs.

Entrepreneur

“a person who attempts to profit by risk and initiative”

Oxford English Dictionary

Entrepreneurs are not managers nor are they inventors.

True entrepreneurs need to know about management and they very often have to manage, but the terms ‘entrepreneur’ and ‘manager’ are not synonymous.

There is much more to being an entrepreneur than simply managing an enterprise. To try and understand just what it is that makes the entrepreneur different, it can be helpful to look at some of the different perspectives on entrepreneurship.

Entrepreneurs: what do they do?

- “organiser of production”: J.B.Say (1755)
- “risk taker”: J. Knight (1920s)
- “creator new products”: J. Schumpeter (1930s)
- “possession of certain *personality traits*”: McClelland (1960s)
- “alert to business opportunities”: Kirzner (1970s)
- “entrepreneurship process”: Gartner (1980s)



Technological entrepreneurship defined (Smith, 2015)

Definitions of technological entrepreneurship tend to focus on the technological perspective at the expense of the entrepreneurial (Smith, 2015).

“a company which emphasises research and development or which places major emphasis on exploiting new technological knowledge . It is often founded by scientists and engineers, and usually includes a substantial percentage of professional technologically trained personnel”
Cooper (1971)

While this definition rightly highlights the importance of technology for this sort of enterprise and the connection to innovation, it tends to ignore entrepreneurship. Hence perhaps a better definition comes from Jones-Evans (1997)

“... small technologically-based firms (that) display a distinct form of entrepreneurship, mainly because of the dependence of the venture on the owner-manager’s high degree of a technological expertise, translated into new technologies, products and processes” Jones-Evans (1997)

You may conceptualize it as

A technical entrepreneur leverages technology for problem-solving thereby developing innovative solutions that deliver more value to customers.



Examples of Technological entrepreneurs

Founder/ entrepreneur	Country	Company	Business
<i>Larry Page/Sergey Brin</i>	US	Google	Search engine
<i>James Dyson</i>	UK	Dyson	Household appliances
<i>Elon Musk</i>	South Africa	Tesla Motors; Space X	Automotive, Space, ...
<i>Jeff Bezos</i>	US	Amazon	E-commerce
Steve Jobs	US	Apple	Computers & Smartphones
Mark Zuckerberg	US	Facebook	Social Media

Tech Entrepreneurs: Who are they?

- Technological entrepreneurs possess a number of distinctive features that set them apart from what one might term traditional entrepreneurs.
- Two of the most significant features are the industry sectors in which they are found and their backgrounds. Technological entrepreneurs are found in sectors that utilise advanced technology.
- Typically this means sectors like information technology, biotechnology, communications and alternative energies (Yaffe-Yanai et al., 2010) together with rapidly changing sectors like automotive, aerospace and rail.
- In terms of background, technological entrepreneurs are trained in science or engineering.

Features of Technical Entrepreneurs

1. The founders of the company have been affiliated with the source of technology before establishing the company
2. The business idea of the company is essentially based on exploiting advanced technological knowledge developed or acquired in a source of technology
3. The company is independent
4. The company is entrepreneurial, that is, it is controlled and managed by an entrepreneur or group of entrepreneurs

Source: Autio (1995)



Features of Technical Entrepreneurs

5. **Visionary Thinking**: Technical entrepreneurs have a strong sense of vision and the ability to think outside the box. They see opportunities where others see obstacles and are constantly searching for ways to disrupt the status quo. They anticipate future trends and develop innovative solutions that solve real-world problems.
6. **Resilience**: Launching a tech company is a risky venture, and technical entrepreneurs are motivated to weather the storms and display determination and resilience.
7. **Customer Focus**: Successful technical entrepreneurs display customer orientation, constantly focusing on developing products and services that meet customer needs and exceed their expectations.
8. **Agility**: They adapt to changing market conditions and customer needs and adjust their business strategies as needed. They are constantly evolving and refining their business models to stay ahead of the competition.

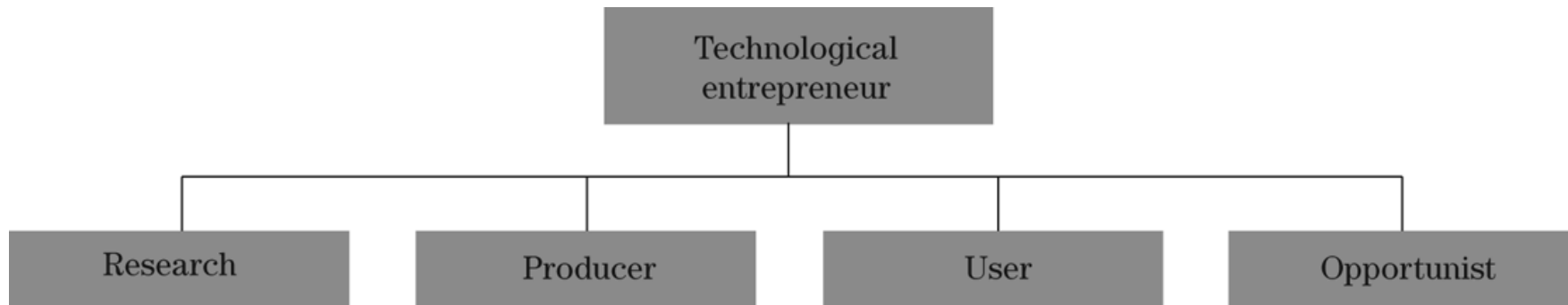
Technology entrepreneurs: Networking

Technological entrepreneurs typically make extensive use of networks comprising contact with universities, large companies and research institutes, as a source of both knowledge inputs and market opportunities. The mechanisms that help to maintain these contacts can take a variety of different forms:

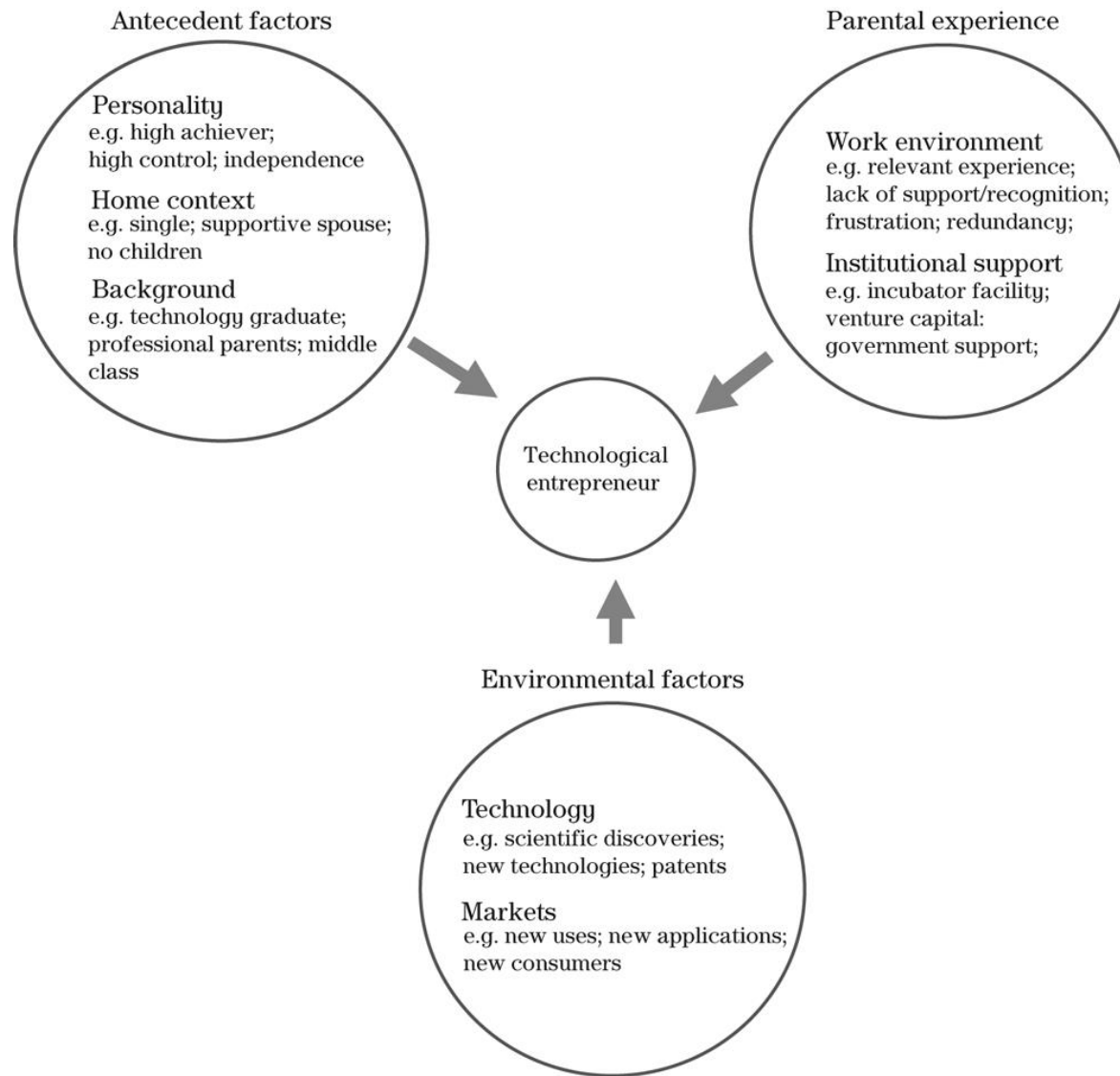
- Strategic alliances and
- R&D partnerships are often used as well as other forms of collaborative agreement.
- Open Innovation



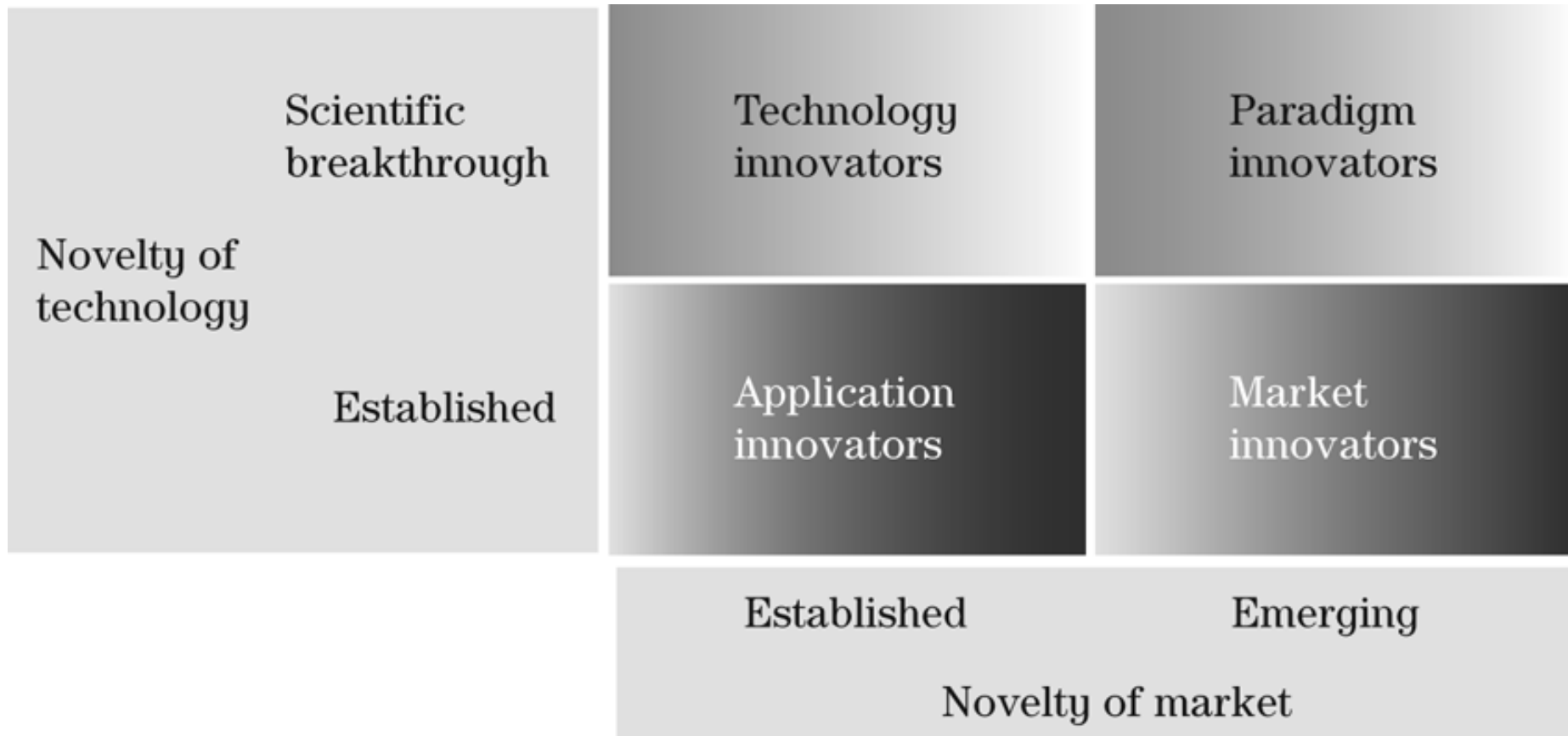
Typology of technological entrepreneurs



Factors affecting technological entrepreneurship



Technology/market taxonomy



Drivers of technological entrepreneurship

- Diffusion of knowledge-Easy access to information
- Staff churn-Spillover effects
- Improved institutional support-Innovation policies
- Availability and access to venture capital & funding opportunities
- Rise of open innovation paradigms
- The nature of new technologies- More Accessible
- Role models-Success Stories



Emergence of Technological entrepreneurship (Smith, 2015)

The closed innovation model began to lose its appeal at the end last century due to various factors: the greater mobility of skilled staff, the greater mobility of knowledge, and the increased availability of venture capital willing to back technology start-ups (Smith, 2015).

The technological sector experienced a combination of these factors and this led to what Chesbrough (2006) termed 'open innovation'. In this model, companies increasingly bring in new technologies from outside (i.e. via licensing). The trend towards systems of open innovation supported a powerful drive towards technological entrepreneurship, where technology informed individuals set up new technology-based companies (i.e. technology start-ups).

Although the concept of technological entrepreneurship predates open innovation with clusters of small technology-based firms in Silicon Valley or Boston's Route 128 being established in the 1960s and 1970s, it is the increasing popularity of open innovation has really promoted the era of technological entrepreneurship.



Technology Entrepreneurship occurs in Industrial sectors (Smith, 2015)

- The term 'technological entrepreneur' relates to what Oakey (2012) terms 'high-technology small firms' (HTSFs) or, as they are sometimes known, new technology-based firms (Smith, 2015).
- These are small businesses that are dependent on a high level of technological knowledge and expertise. The term is often used to describe a technology-based 'spin-off' business, formed by scientists and engineers leaving their current employment in a university, research-based institution or an industrial company and going it alone by setting up their own independent company.
- This sort of enterprise is typically to be found in industries such as electronics, computer services or biotechnology. In these sectors there is scope for individuals who have acquired technological knowledge and expertise through their work in a large organisation to set up on their own.



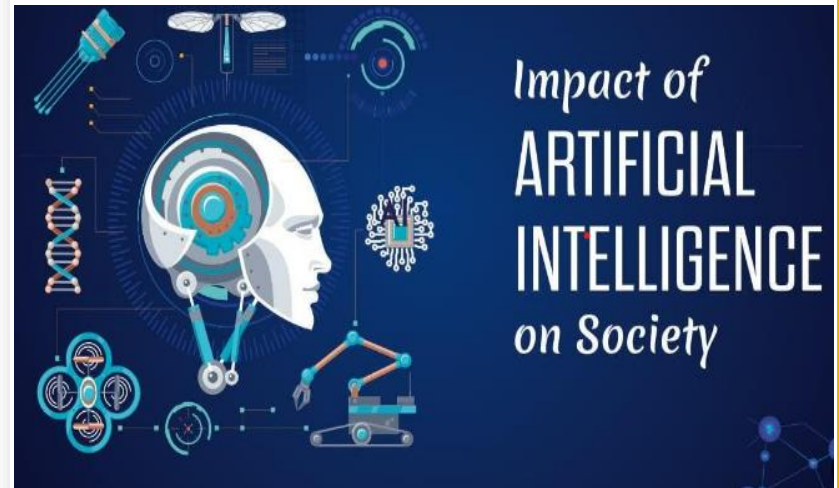


Technology Entrepreneurship occurs in Industrial sectors (Smith, 2015)

- As an independent firm they can then specialize in the supply of specialist components or specialist services.
- Nevertheless, they involve a particular form of entrepreneurship, one that relies heavily on technology and the creation of new products and services. As a result this is the form of entrepreneurship that is probably most closely connected to innovation.
- Technological startups are often acquired by large organizations to benefit from their unique Intellectual technology and reduce time to market.

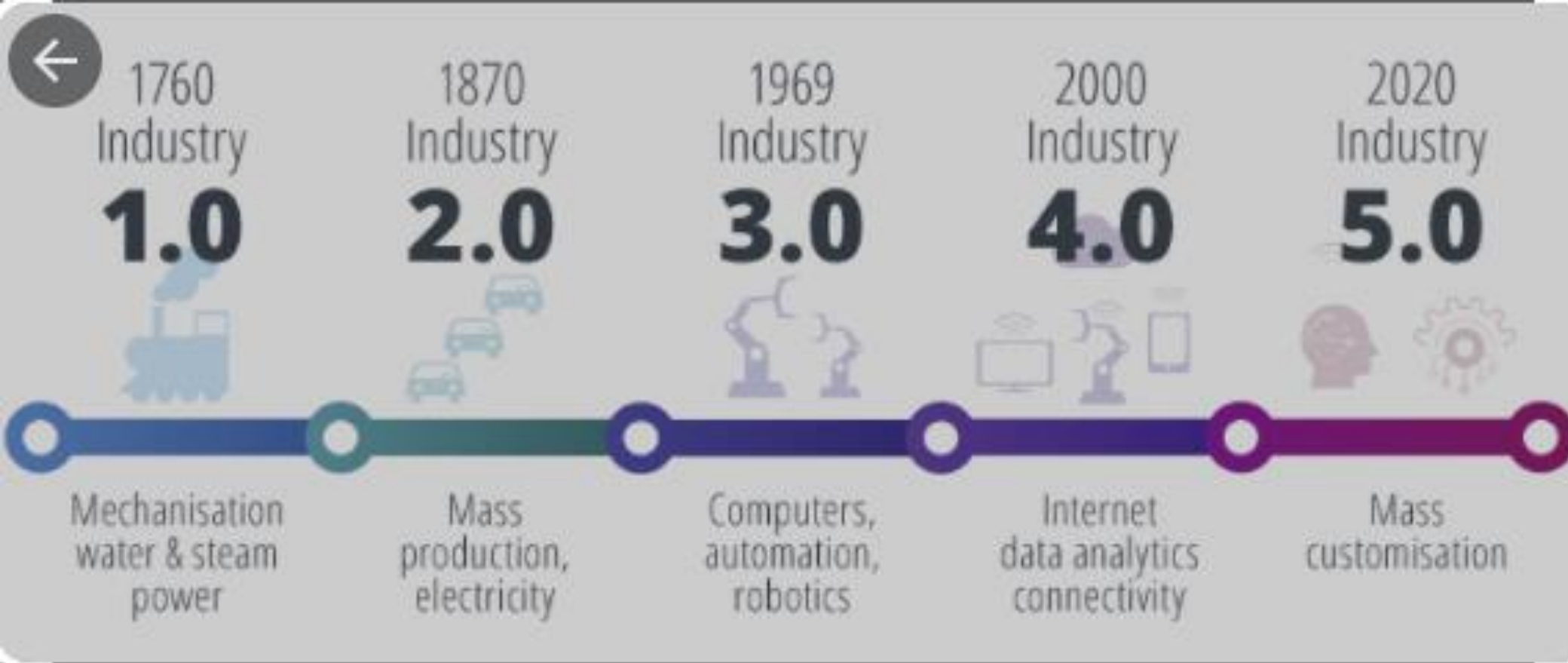
What Technological era are we in?

- We live in an information age, which is a period in human history characterized by the shift from industrial production to one based on information and computerization.
- The Digital Era is characterized by technology which increases the speed and breadth of knowledge turnover within the economy and society. Evolutionary theory, as an explanation of the system we live in, states that sustainability relies on knowledge turnover.
- Simultaneously, there have been breakthroughs in Artificial intelligence (AI) that are predicted to change the workforce and affect several sectors such as Healthcare, Education, manufacturing, agriculture, etc.



Industrial Revolution

- Throughout history, people have used technology to enhance their lifestyles.
- People adopted the technology they had available to help make their lives easier.
- At the same time they improved the application of the technology pushing its limits, until the next technology became available.
- This gave birth to the concept of industrial revolution began.
- Right now, we are going through the fourth industrial revolution, known as Industry 4.0.



Industry 5.0 seeks to foster a more balanced working relationship between increasingly smart technologies and humans.

Industry 5.0 seeks to foster a more balanced working relationship between increasingly smart technologies and humans.

Humans are now envisioned to collaborate with Robots, providing humans with greater opportunities to use their creative flair.

Space Travel

Entrepreneur Richard Branson went into space aboard his own winged rocket ship Sunday in his boldest adventure yet, ahead of fellow billionaire Jeff Bezos.

The 71-year-old Branson and five crewmates from his Virgin Galactic space-tourism company reached an altitude of about 53 miles (88 kilometers) over the New Mexico desert enough to experience three to four minutes of weightlessness and see the curvature of the Earth and then safely glided to a runway landing.

“The whole thing, it was just magical,” a jubilant Branson said after the trip home aboard the sleek, white space plane, named Unity.

Billionaire Richard Branson in his own ship

“The whole thing, it was just magical,” a jubilant Branson said after the white space plane, named Unity.

By AP New Mexico: Truth Or Consequences
Updated: July 14, 2021 2:00 a.m.



Billionaire Richard Branson makes a statement as crew members Seth Moses and Shrinia Gendia float in zero gravity on board Virgin Galactic's passenger rocket plane VSS Unity after reaching the edge of space above Spaceport America near Truth or Consequences, New Mexico, U.S. July 14, 2021. In a still image from video, Virgin Galactic/Handout via REUTERS.





Self-Paced



Wed, 07 Jul 2021



8:05 PM - 10:05 PM



<https://www.government.ae/en/about-the-uae/strategies-initiatives-and-awards/federal-governments-strategies-and-plans/uae-centennial-2071>

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<http://>

EXPLORE
IN

Technology Entrepreneur: Elon Musk
Tesla Roadster was the brainchild of Martin Heberhard, Marc Tarpenning and Elon Musk.

Musk is no stranger to high technology start-ups. South African-born Musk was a self-taught programmer who sold his first computer game at the age of 12, prior to moving to Silicon Valley. He initially made his mark as one of the co-founders of the electronic payment system PayPal, in 1999.

He sold it three years later to eBay for \$1.5 billion.

In 2002 he set up Space Exploration Technologies, or Space X as it is better known, to develop reusable technology for launching satellites into space.

Two years later he set about transforming the automotive industry by becoming one of the co-founders of Tesla Motors in early 2004.





- OnTime Group has history of 15 years, offering services aimed at simplifying the lives of customers in the UAE and beyond.
- Chief Executive Officer & Founder at OnTime Group: Mr. Waleid Abdulkareim
- First private Company in the Middle East to offer Government Services. Now serving over 1 million customers annually.
- Earned the trust and confidence of Its partners and customers.
- Expert knowledge and planning, hands-on experience, speed of service delivery, international standards, and true professionalism.

Technology start-ups in the MENA region

- Source: Study by DSOA and Google (2014)
- Tech start-ups increased 45% across MENA region
- More than double world average

TECHNOLOGY SMEs AND START-UPS

Business owners - the nationality of respondents was not a criteria for screening as in countries like KSA and UAE there are numerous expatriate business owners.



GEOGRAPHIC COVERAGE:

KSA, UAE, Jordan, Egypt, Lebanon and Morocco



- UAE leads the technology Start-Up revolution with more than 60% of total GCC sales (estimated at \$2 billion in 2010).
- KSA's sales were estimated at \$520 million and expected to grow significantly fueled by a \$66.7 billion retail sector.
- Although technology Start-Ups are claimed to account for only 0.3% of retail spending in Egypt, the rising Internet and smartphone penetration bodes well for e-commerce sites (41% have used their smartphone to make a purchase according to Google)
- In Jordan, 24% of Internet users are Technology Start-up users and a report shows they spent \$370 million on products/ services in 2011.

Profile of MENA Tech Entrepreneurs

- Young entrepreneurs, 88% below 35 years old
- Over 67% have a technology background prior to setting up their business
- Some had experience in working in technology companies
- The Entrepreneurs have a clear passion for their trade, they engage in online self-learning and sometimes attend external lectures at universities or other organizations
- Over 91% of respondents have technical skills

Profile of MENA Tech Entrepreneurs /...

- Most entrepreneurs had prior experience in the private sector
- 1/5 was associated with their own or family business prior to venturing into technology SME startup
- 1/3 owned other businesses



Previous Background

5.2.2. Financial challenges

The top two predominant financial challenges are:
Access to on-going/secondary capital and a lack of financial support from banks

Entrepreneurs face minimal difficulties with seed investments for the set-up; however, they are having issues obtaining sufficient secondary funding for their ongoing operations.

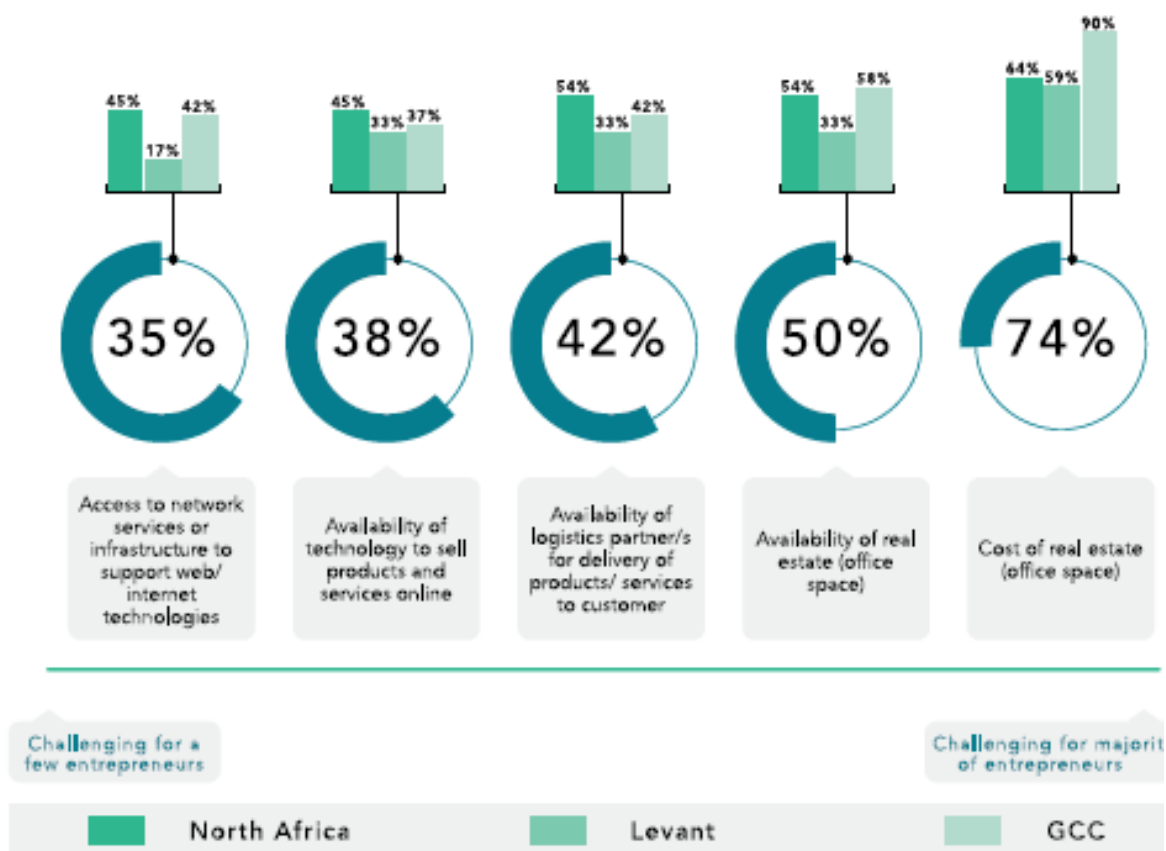
Nearly 70% of those surveyed face major to moderate challenges due to **non-availability of support from banks over business accounts as well as a distinct lack of access to capital**

- **Availability of support from banks over business accounts** poses a hindrance particularly for almost three quarters of North African entrepreneurs. Access to capital for start-up is claimed as a challenge for three quarters of respondents in the Levant and a majority of those in the GCC
- **Inability to afford the initial investment for business set-up** is cited as a challenge by around 58% (with 63% of GCC respondents citing a daunting challenge)

Infrastructure Challenges

- Cost of real estate as a significant challenge for an overwhelming 90% of GC entrepreneurs
- Lack of availability of real estate appears to be another hurdle faced by half of those surveyed (more so for the GCC and North Africa than for the Levant)
- Logistics partners come in second to real estate with just under half (42%) of respondents citing it as a challenge (particularly more than half in North Africa)

Chart 5.2.7: Infrastructure challenges



5.10.2. Factors that can help improve the technology environment

The key to promoting a better environment for tech start-ups lies with flexibility in government regulation & policies

- Reducing registration / licensing fee followed by provision of subsidies also feature amongst the top three factors for improving the environment for start-up business

Table 5.10.2: Factors that help improve environment for start-up business

Factors	Weighted rank	Weighted score
Increase flexibility in government regulations and policies	1	3.5
Reduce registration / licensing fees levied by government	2	3.3
Provide government subsidies to increase technology Start-ups	3	3.1
Reduce customs duties levied on products being sold	4	2.6
Increase transparency in customs and trade policies	5	2.4



Clusters

Definition: A number of similar things growing or grouped closely together

The idea of cluster was noticed as early as 1890 by Alfred Marshall “*An industry concentrated in certain localities is commonly, though perhaps not quite accurately, described as a **localized industry***”. Porter (1998) defines cluster as “a concentration of interconnected companies and institutions in a particular field”.

Probably some of the most famous clusters are Silicon Valley or Hollywood.

The success of Silicon Valley with computer and related industries during 1990s coupled with the theoretical explanation given by Michael Porter (1990s) re-ignited the interests of many countries and regions to adopt “economic clusters” as means for economic growth and internalization.





Clusters

- Porter (1998) further explained that clusters include a group of related companies and other institutions which provided inputs to a firm operating in the cluster, and supported it in competing effectively.
 - These inputs emanated from suppliers of specialized components, machinery, and services, and providers of specialized infrastructure. Porter took a holistic view of the value chain, and beyond and argued that a cluster extended downstream to the distribution network and customers, and horizontally to manufacturers of complementary products, companies in industries related by skills, and technology, public support institutions, and Research and Developments centers.
 - Today, many examples of successful clusters can be identified in Asia itself; such as, the automobile production in Kyushu cluster in Japan; the emergence of Hsinchu Science Park as an IT Cluster in; IT cluster in Bangalore, India; ICT in Singapore; and ICT cluster in Seoul, Internet city in Dubai, Silicon Oasis in Dubai.
- 

Clusters and Technological Entrepreneurship (Smith, 2015)

- Clusters comprising firms created by technological entrepreneurs generally take the form of Marshallian Industrial Districts (see Figure below). These are clusters that comprise large numbers of small, locally owned firms that are co-located (Markusen, 1996).
- Technological entrepreneurs sometimes locate in clusters of the state-anchored variety, including government laboratories or hospitals. Although governments have attempted to motivate cluster formation, in the USA the more successful ones, such as Silicon Valley and Silicon are generally ones that formed spontaneously rather than being planned (Smith, 2015).

Such clusters do tend to form around distinctive institutional facilities, Silicon Fen in UK has grown up around Cambridge University and Motor Sport Valley has grown up around the Silverstone Grand Prix circuit in Northamptonshire (Smith, 2015).

There are other factors that play an important part in attracting technological entrepreneurs to locate in clusters. These include: a local pool of technically skilled labour, local investors (i.e. business angels/venture capitalists) and the presence of specialised suppliers and services (Smith, 2015).

These factors thus provide an 'innovative milieu', a complex eco-system of science and technology based companies which constitute a 'knowledge community'. This provides a wide range of specialist services that supports, encourages and stimulates technological entrepreneurship.



Collaboration with Universities

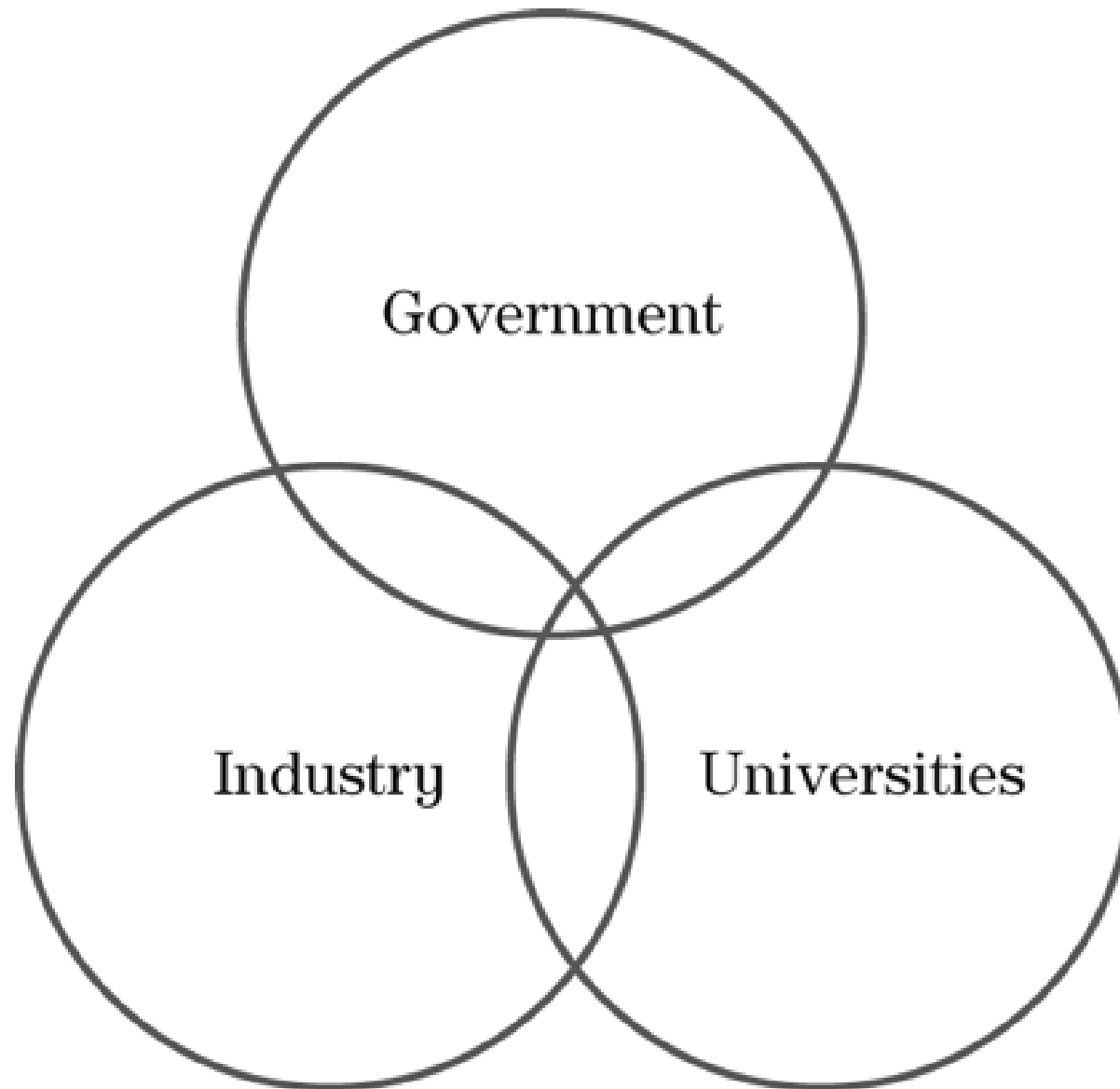
Another feature of technological entrepreneurship is that it may involve close links with universities. The potential for universities to contribute to technological entrepreneurship has long been recognised (Oakey, 2012).

As Etzkowitz and Leydesdorff (1997) put it, ‘the development of academic capacities carries within itself the seeds of future economic and social development in the form of human capital, tacit knowledge and intellectual property’.

In the case of Silicon Valley, for example, many of its most noted technological entrepreneurs, such as Bill Hewlett and Dave Packard, were alumni of Stanford University in Palo Alto in the heart of the valley and retained close links with the university.



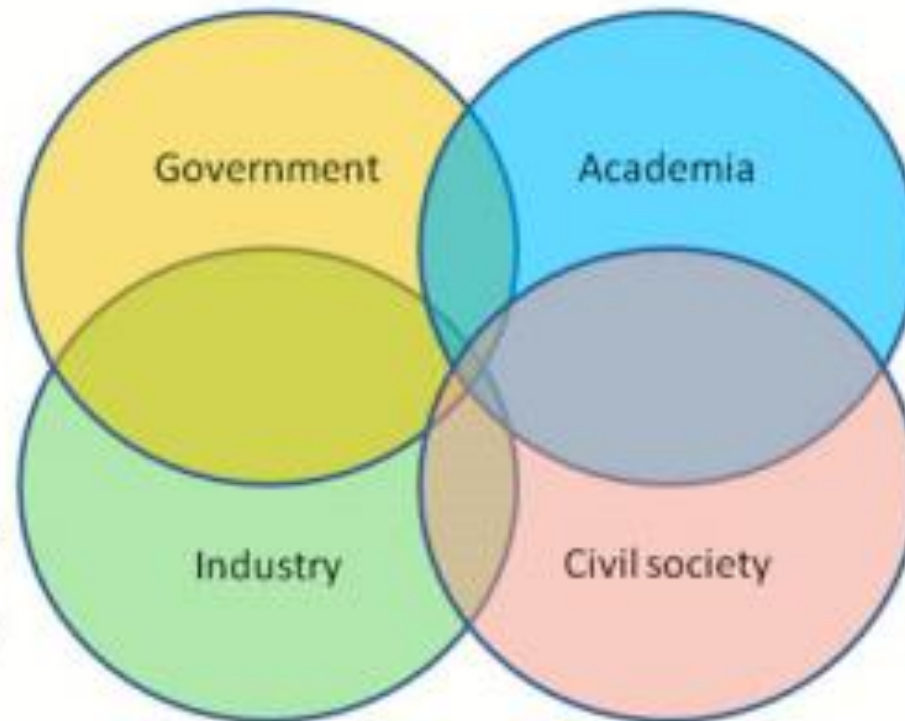
Triple Helix



Quadruple Helix

Stockholm County Administrative Board (SE), Municipality of Norrtälje (SE), Västabollands stad (FI), Destination Roslagen (SE), intermunicipal collaborative organization.

"E-teams"; Regional teams of entrepreneurs collaborating in EE, FI and SE. Local subcontractors and technology providers in EE, FI and SE.



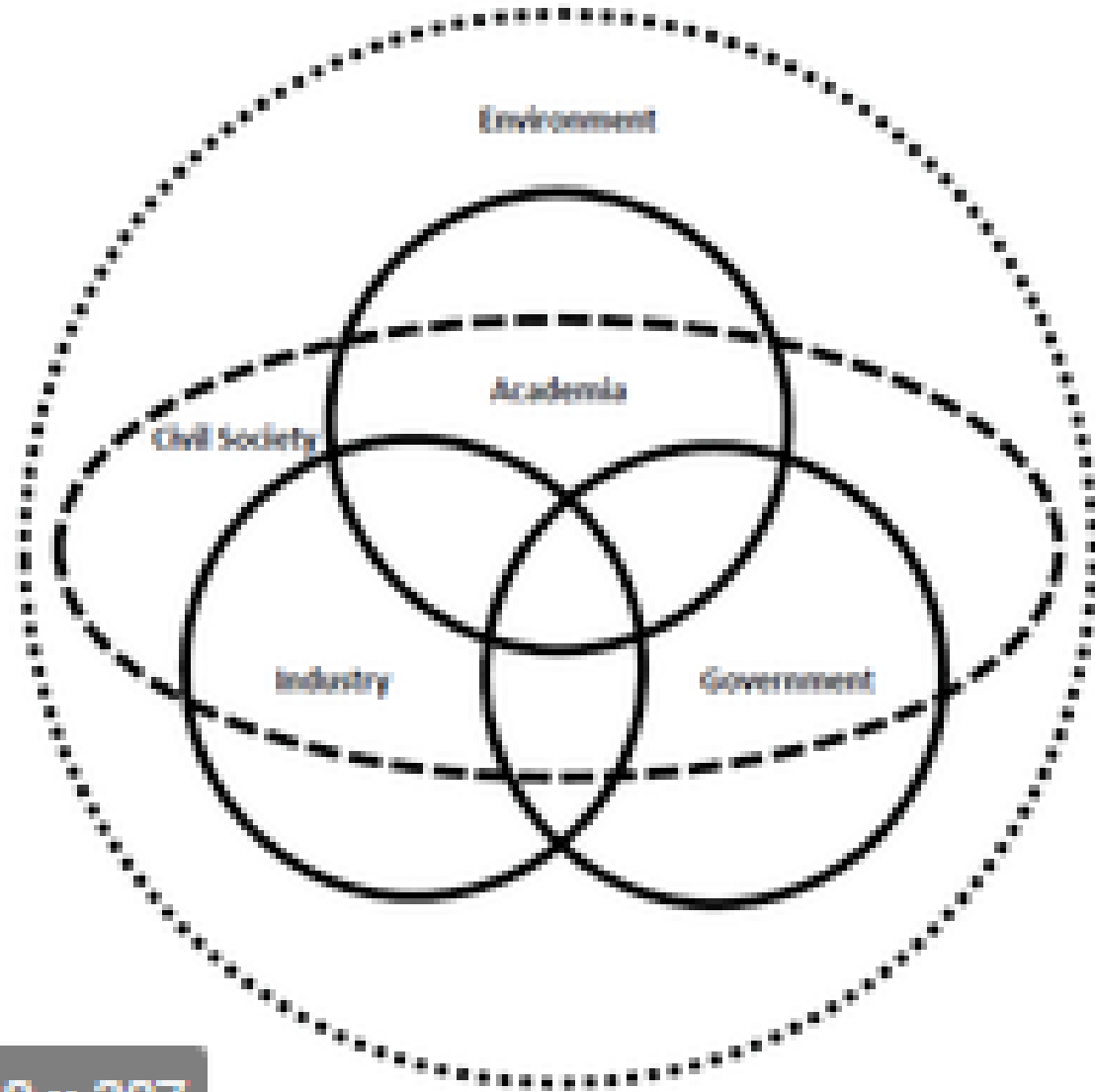
Saremaa University Centre (EE), entrepreneurship research.
KTH – Royal Institute of Technology (SE), innovation and gender research.
Åbo Akademi University (FI), gender and ICT research.

BalticFem (SE), NGO organizing local female entrepreneurs.
Eurohouse (EE), NGO organizing local female entrepreneurs.
Läänemaa Tourism (EE), local tourism association.
WINNET Europe (EU).

Fig. 2 The Quadruple Helix Central Baltic project. Formal project partners in *bold*. *EE, FI and SE* are the official European Union acronyms for Estonia, Finland and Sweden



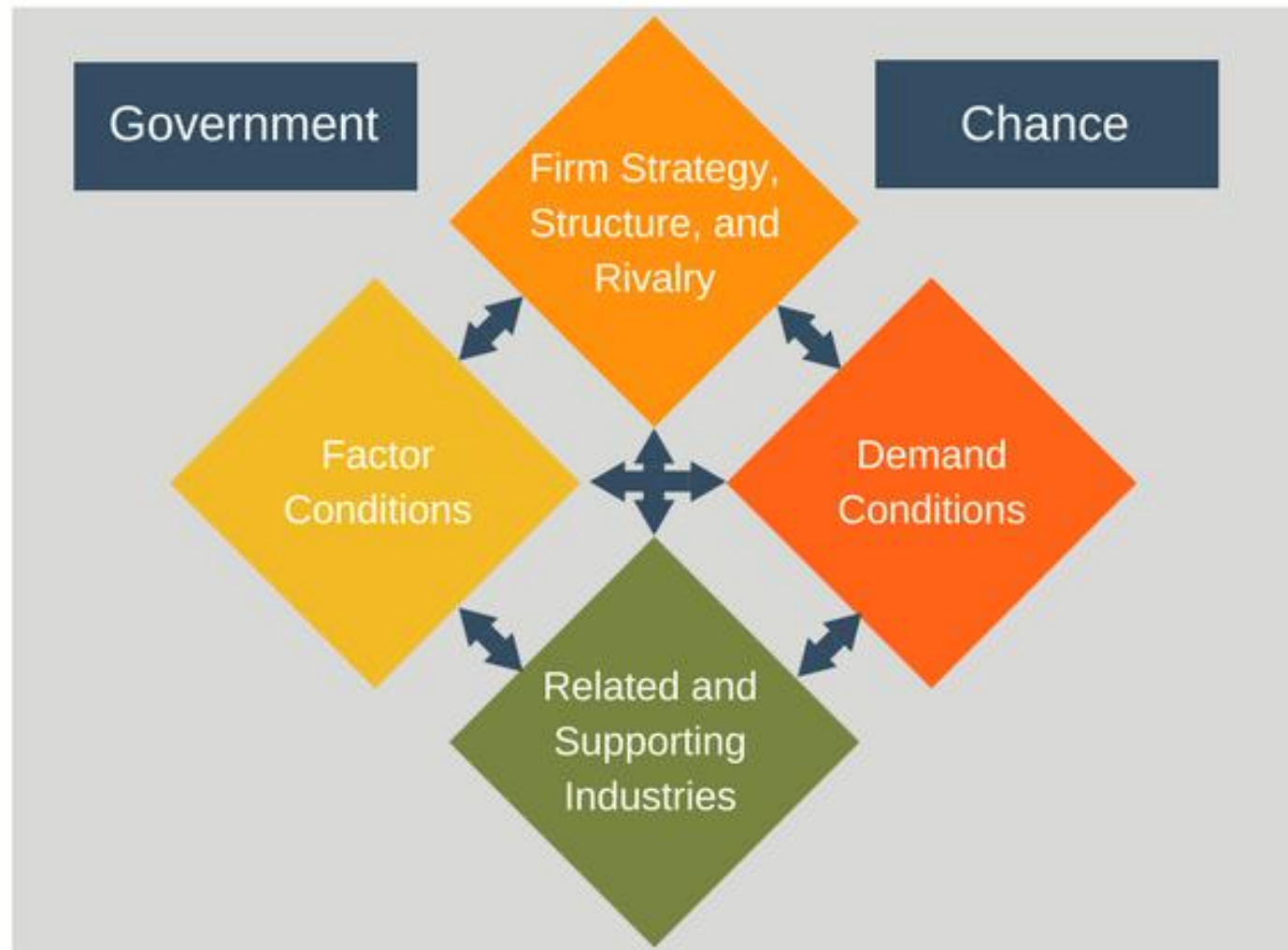
Quintuple Helix



Clusters and Technical Entrepreneurs

- In understanding how these benefits are leveraged, Porter (1990) explained the effects of the local business environment on competition through four interrelated factors in his “Diamond Model”.

Porter's Diamond Model





Porter's Diamond

- In understanding how these benefits are leveraged, Porter (1990) explained the effects of the local business environment on competition through four interrelated factors in his “Diamond Model”. The four influences in the Diamond Model are:
- **Factor conditions** representing the cost and quality of inputs such as natural resources, skilled labor, infrastructure and technology-related advancements on which local firms seek to compete.
- **Demand conditions** or the sophistication of local consumers in terms of demand for the industry's products or services.
- **Firm strategy and rivalry** which include the nature and intensity of local competition, which affects the way firms do business with each other, their employees and the government.
- **Related and supported industries** which represents the local extent and sophistication of suppliers and internationally successful competitors that stimulate cooperation and competitive rivalry?



Clusters as National Strategy for Economic Development in Dubai

- Dubai's economy is heavily focused towards the service industries.
- The service sector of Dubai accounts for three quarters of the economy while the primary sector is worth less than 1% and the rest is the secondary sector (mining and quarrying, manufacturing, utilities, construction).
- As the economy has moved towards high value activities and the development of the service sector, the secondary sector has declined in importance. This is reflective of economies that become more advanced and seek to become knowledge based economies rather than simple production centers.

Clusters in UAE

- One strategic approach used by Dubai to support its economic development is the development of a range of clusters to allow for organic growth and networking opportunities.
- Dubai has been experiencing considerable amount of success in developing clusters and attracting businesses within it.

Article | Publisher preview available

The Role of Place Branding and Image in the Development of Sectoral Clusters: The Case of Dubai

 Springer

July 2016 · *Journal of Brand Management* 23(4):383-402
DOI: [10.1057/bm.2016.18](https://doi.org/10.1057/bm.2016.18)
Project: [place branding](#)

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The role of place branding and image in the development of sectoral clusters: The case of Dubai

Original Article | Published: 08 July 2016

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Clusters as National Strategy for Economic Development in Dubai

- Over the last decade, Dubai's economy has diversified considerably. This is assisted by the emergence of new sectors such as financial services that have more than doubled (from 4.5% to 10.3% share of the economy).
- Real estate has gone through a rise and then a fall peaking in 2007 with 18% share of the economy.
- Logistics has also been a strong performer with a value of 14% of the economy up from 11% at the start of the millennium. However, the most dominant sector is still “trading”, accounting for about a third of the economy.

UAE Clusters

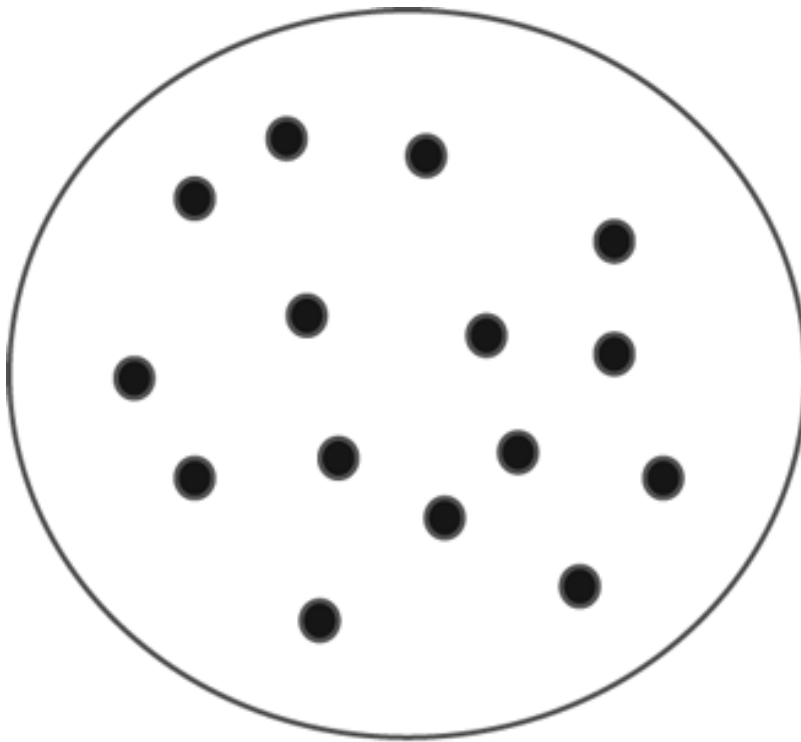
In UAE, we have examples such as Dubai Silicon Oasis, Media City, Healthcare City, Sharjah University city, Masdar Institute in Khalifa City, etc.



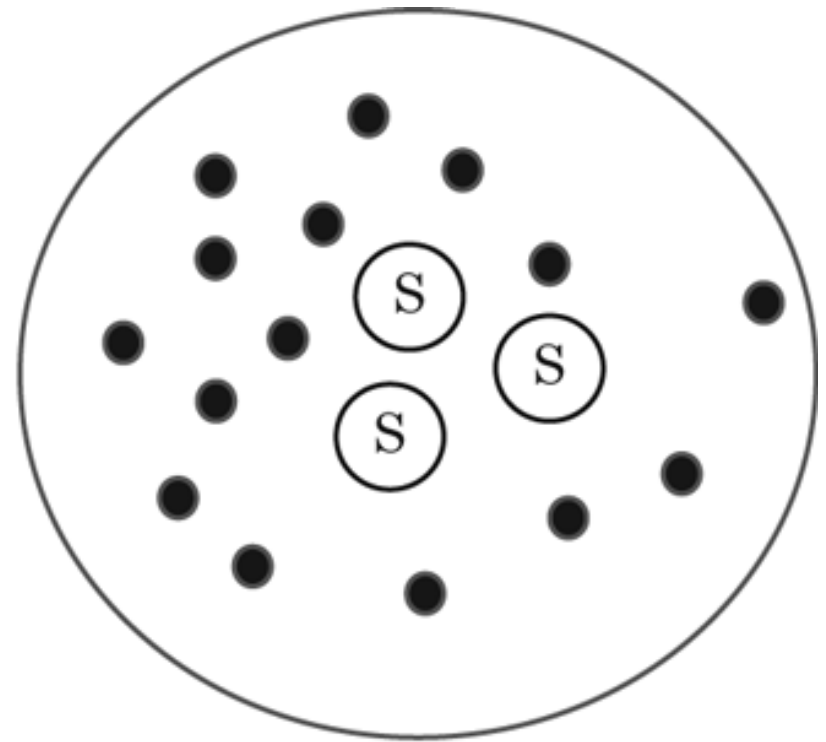
Types of clusters

- Clusters comprising firms created by technological entrepreneurs generally take the form of Marshallian Industrial Districts (see Figure next slide). These are clusters that comprise large numbers of small, locally owned firms that are co-located (Markusen, 1996).
- Alternatively, Technological entrepreneurs are sometimes drawn to locate in clusters of the state-anchored variety.

2 types of cluster



Marshallian ID cluster



State-anchored cluster

Class Exercise

Week 9: Sessions 1 and 2



ChatGPT is an artificial intelligence chatbot developed by OpenAI and launched in November 2022. It is built on top of OpenAI's GPT-3 family of large language models and has been fine-tuned using both supervised and reinforcement learning techniques.

Your niece Maysoon is an avid entrepreneur. She visits you at your as you come back from work. Maysoon tells you she wants to use ChatGPT to launch innovative services. As her elder cousin, please reflect on how Chat GPT may be used to improve existing products/services/processes.

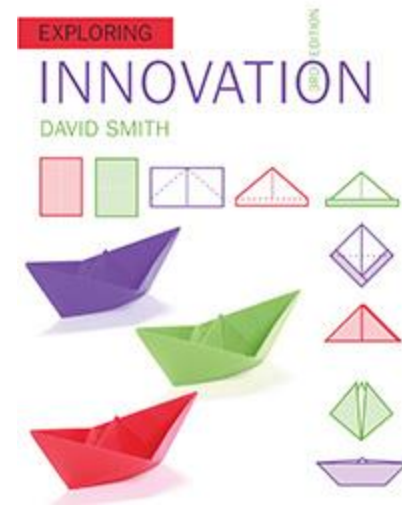
Maysoon is interested in making some money to fund her next holiday travel. Advise her how an entrepreneur may use CHAT GPT in their business model to earn revenue.

You will work in your own assigned Group for 15 minutes and you will then make a short presentation of 2-3 minutes per group.

Shukran

Intellectual Property Rights

Chapter 7



Objectives

At the end of this class you will be able to:

- Appreciate the rationale behind the various legal rights associated with intellectual property;
- Identify the various types of intellectual property right (IPR);
- Distinguish the benefits conferred by intellectual property rights;
- Differentiate the remedies available to those whose intellectual property rights have been infringed;
- Show how intellectual property rights can be used to create value for their creator.

Advanced Organizer

- An Inventor usually considers considerable time and resources to develop a business model for a new invention. In order to safeguard her efforts in developing the invention by preventing others to copy the invention, and to enable her to derive optimal revenue from it, the inventor may leverage the Intellectual Property Rights (IPR) regime. This session discusses how IPR may be used to create value for their inventor.

Intellectual Property

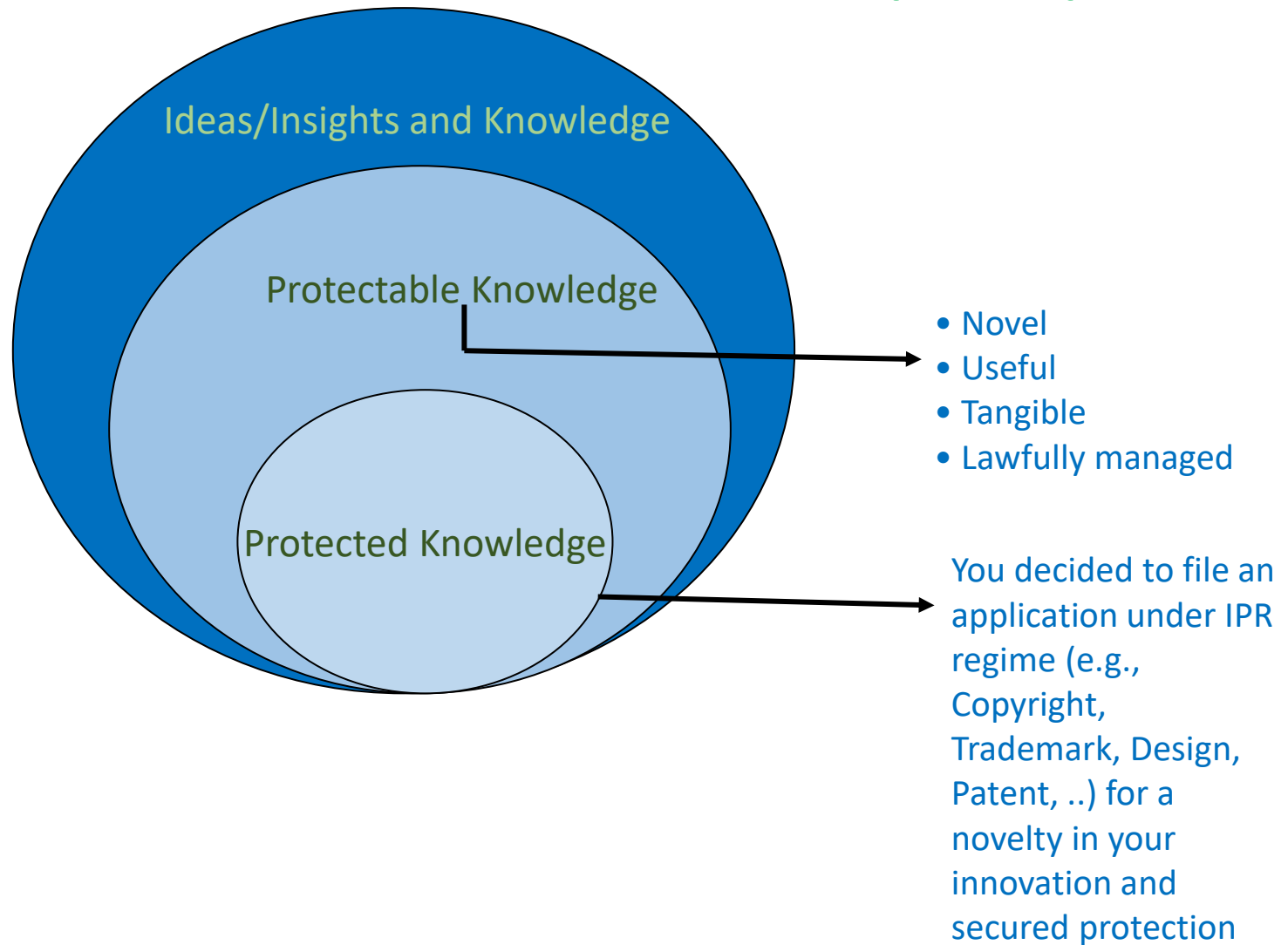
- Is any product of human intellect that is intangible but has value in the marketplace.
- It is called “intellectual” property because it is the product of human imagination, creativity, and inventiveness.
- Traditionally, businesses have thought of their physical assets, such as land, buildings, and equipment as the most important.
- Increasingly, however, a company’s intellectual assets are the most important.

Intellectual Property

Intellectual property refers to the subset of ideas that are:

- Novel (New)
- Useful (To improve our lives)
- Reduced to practice in a tangible form (can be implemented to improve our lives)
- Managed according to the law (Defined in legal IPR provisions)

Ideas and Intellectual Property



Intellectual Property In Practice

- Manifest in a variety of things and in various forms
 - New products and services
 - Designs and computer programmes
 - Literary and artistic works

Determining What Intellectual Property to Legally Protect

Criteria 1

Determine whether the intellectual property in question is directly related to the firm's competitive advantage.

Criteria 2

Decide whether the intellectual property in question has value in the marketplace.

Common Mistakes Organizations make

Not properly identifying
all of their
intellectual property.

Not fully recognizing
the value of their
intellectual property.

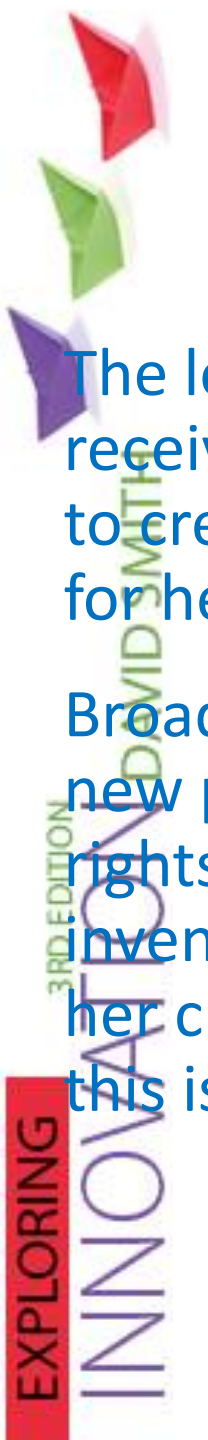
Not legally protecting the
intellectual property
that needs protecting.

Not using their
intellectual property as
part of their overall
plan for success.

Intellectual Property Rights

- To deter unlawful copying of an Invention, the state confers Intellectual Property Rights (IPR) certificates, which certifies that the inventor is the owner of the rights, just like for a physical property (e.g. A Car)





IPR owned by an Innovator

The logical follow up question is to ask: Once an innovator has received legal protection for the rights, how can she leverage her IPR to create value for her product/service and derive financial rewards for her efforts?

Broadly, the IPR certificate gives the innovator legal rights over the new product/service and she may hence (like for any other property rights) determine how she wishes to use it for commercializing her invention. She may as well as prevent other from copying or using her creation without her formal approval. Let us therefore discuss this issue in more depth.

Intellectual Property Rights

- The certificate give legal recognition to the ownership of new ideas, designs, etc.
- They give the owner of IPR the right to stop others exploiting his/her property
- They create for the innovator a system by which he/she can benefit from his/her ingenuity
- For example, the owner of the IPR may *license* it to others provided certain conditions are met





IPR owned by an Innovator

It was discussed above that there are various types of IPR that are offered protection under the legal regime, some of the common ones are:

- Trademarks;
- Copyright and Related Rights;
- Patents;
- Industrial Designs
- + *Try to Leverage: Traditional Knowledge; Geographical Origin*



Copyright

- An intangible right that grants to the owner of a work of authorship the legal right to determine how the work is used and to obtain the economic benefits from the work.
- Springs into life when work is created, and it belongs to the person creating the work
- A work does not have to have artistic merit to be eligible for copyright protection.
- If the author is an employee it goes to the employer
- Ownership can be assigned & licensed



What is Protected by Copyright?

- Literary works
- Dramatic; Pantomimes and choreographic works
- Musical Compositions & artistic works;
- Films; sound recordings; broadcasts; etc.
- Computer Software
- Pictorial, graphic, and sculptural works
- Operating manuals and sales brochures are eligible for copyright protection.

Copyright

Copyright provides exclusive rights to the right holder enabling her to prevent anybody to perform the following activities with her work without her formal permission:

- Copying or reproducing
 - Adapting
 - Distributing
 - Issuing and renting
 - Public performance
 - Broadcasting
- Viewed in another way, the rights enable the author to derive financial rewards by enabling others to use the assigned rights. For example, when a movie is based on the storyline of a book, the movie producers need to pay an agreed sum to the author for the right to use the story in their movie.



United Arab Emirates

Federal Law No. 7 of 2002 Concerning Copyrights and Neighboring Rights

Section One Scope of Protection

2. Authors of the works and the holders of the neighboring rights, shall enjoy the protection of this law in case that an aggression against their rights occurs in the State, namely in the following works:

1. Books, pamphlets, essays and other written works.
2. Computer programmes and applications, databases and works analogous to them as defined by a Minister's decision.
3. Lectures, addresses, sermons and other works of similar nature.
4. Dramatic, dramatico-musical works and dumb shows.
5. Music compositions with or without words.
6. Sound and audiovisual works.
7. Architecture works, engineering plans and layouts.

United Arab Emirates

Federal Law No. 7 of 2002 Concerning Copyrights and Neighboring Rights

8. Works of drawing, painting, sculpture and lithography (fabric, metal, stones, wood) and engrave or similar works in the scope of fine arts.

9. Photographic works and works analogous to photography.

10. Works of applied and plastic art.

11. Illustrations, geographical maps, sketches, three-dimensional works relative to geography, topography and architecture designs etc.

12. Derived works without prejudice to the protection stipulated for the works been derived from. The protection includes the title of the work if innovated and the innovated theme written for a broadcasting programme.

3. Protection does not extend to mere ideas, procedures, methods of work, mathematical understandings, principles, and abstract facts, but extend to creative expression in any of them.

Protection does not extend also to the following:

(1) The official documents, whatever is their original language or the language translated to them, such as the texts of laws, regulations, decisions, international agreements, judgments, arbitrators' awards and the decisions of the administrative committees having judicial competence.

(2) The news, events and the current facts, which constitute merely media news.

(3) The works transferred to public property.

Nevertheless, the items in paragraph 1, 2 and 3 of this article if their compilation or arrangement or any effort thereof is characterized by creative manner shall enjoy protection.

Copyright

- In other applications, Copyright may help an Innovator protect the content on its brochures and web site, preventing others from freely copying it, which would dilute its brand.
- Copyright will also protect the communication messages of the innovator, and protect his slogan, tagline and other brand assets.
- Customers associate the slogan “Just do it” with Nike and “Impossible is nothing” with Adidas. Copyright and trademark may jointly protect this important tagline and hence avoid copying by competitors.



Copyright

- Depending on the area of the innovation, copyright may also help an Innovator to derive substantial revenue from his innovation.
- For example, in 2012, author J K Rowling was estimated to have a personal wealth of about \$1 billion, with most of her money coming from her books and the "Harry Potter" film franchise and merchandise products (e.g. mugs, t-shirts, bags, etc.) .



Copyright & Related Rights timescales



Literary, musical, artistic & dramatic works	➡	Author lifetime 70 years n USA; 50 Years in many other countries
Films, TV, radio & cable	➡	Date of 1 st broadcast + 50 years
Sound recordings (incl. performers rights)	➡	Date of recording + 50 years OR if published + 70 years
Publisher's right (layout)	➡	Date of publication + 25 years

Section Four

The Term of Protection and the License to Use the Works

20.—1. The economic rights of the author stipulated in this law shall be protected through his lifetime and fifty years beginning from the first day of the calendar year following the death.

2. The economic rights of the joint authors shall be protected through their lifetime and fifty years beginning from the first day of the calendar year following the death of the last surviving author.

3. The economic rights of the authors of collective works except the authors of applied arts shall be protected for a period of fifty years beginning from the first day of the next calendar year of the first publication in case that the author is a legal person but in case that the author is a natural person, the period will be calculated according to the rule stipulated in (1) and (2) of this article.

The economic rights of the works published for the first time after the death of their author expire after fifty years starting the first day of the next calendar year of its first publications.

4. The economic rights of the works published anonymously or pseudonymously shall be protected for fifty years from the first day of the next calendar year of the first publication if the author is specified and known, or has disclosed his identity, the period of protection will be calculated according to the rule stipulated in paragraph 1 of this article.

5. The economic rights of the authors of the applied art works shall expire after twenty-five years of its first publication, starting the first day of the next calendar year.

6. In the events the term of protection is calculated from the date of first publication, the date of the first publication will be the beginning for calculating the term irrespective of the re-publication unless the author has entered in his work after revision essential changes that make it look as new work.. If the work was composed of many parts or volumes published separately in many terms each part or volume shall be considered independent work at the time of calculating the term of protection.

7. The economic rights of the performers shall be protected for a period of fifty years to be calculated from the beginning of the next calendar year in which the performance was

Copyright Law: A Case Study



Copyright Law: A Case Study

- In 1939, a Zulu migrant worker and entertainer, Solomon Linda stood before a microphone in Johannesburg's first recording studio, improvising vocal lines.
- He named the song *Mbube*, Zulu for lion. Linda came up with melodious notes that became a famous tune from Africa. This became the song *The Lion Sleeps Tonight*.
- There are versions in several languages. More than 150 different performers have recorded it and it is used in over 15 movies and musicals. It has earned over \$15 million in royalties.
- Linda's role in the song's creation is undisputed, but he died a poor man, leaving his family too poor even to afford a headstone for his grave. This is the story of the legal battle to claim back for Linda's children a share in the proceeds from their father's creation
- Read more at:
https://www.wipo.int/wipo_magazine/en/2006/02/article_0006.html





Exclusions From Copyright Protection

The main exclusion is that copyright laws cannot protect ideas.

- For example, an entrepreneur may have the idea to open a soccer-themed restaurant. The idea itself is not eligible for copyright protection. However, if the entrepreneur writes down specifically what his or her soccer-themed restaurant will look like and how it will operate, that description is copyrightable.
- The legal principle describing this concept is called the idea-expression dichotomy:
- An idea is not copyrightable, but the specific expression of an idea is.

Trademarks & Brands

- Although the term “brand” is sometimes used interchangeably for a “trademark”, a “brand” usually refers to a combination of tangible and intangible elements, such as a trademark, design, logo and tagline.
- Brands are generally perceived as a key asset for creating value for a business. Any organization can develop a brand image. However, many innovators devote little attention to developing brand value for their products/services.
- Often many entrepreneurs do not quite understand or are not fully convinced that a brand image can be a powerful tool for promoting and marketing product/services.



الاتحاد
ETIHAD
AIRWAYS



كارفور
Carrefour



تعاونية الاتحاد
UNION COOP



Trademarks & Brands

- A trademark is any word, name, symbol, or device used to identify the source or origin of products or services and to distinguish those product or services from others.
 - Many innovators may wish to devote their limited resources to other aspects of the business, that they might perceive as more important.
 - Further, many innovators may not be conversant with developing a brand image or may believe that brand development is only feasible by large businesses.
 - However, these are unhelpful perceptions, which may be counter productive for a business. In fact, the potential commercial value of a brand cannot be underestimated, as it remains the key vehicle for engagement between an enterprise and its consumers.
- 

Trademarks & Brands

- Innovators should therefore focus on developing the brand image in parallel to developing the business, and a branding strategy should form a central part of their business plan.
- The brand enables the communication of value inherent in the product/service to customers.
- The literature has firmly established how a brand image will attract and help retain consumers.



Trademarks & Brands (2022)

The Kantar BrandZ Top 10 Most Valuable Emirati Brands 2022

Rank 2022	Brand	Category	Brand Value 2022 (\$M USD)
1	Etisalat	Telecom Providers	11,847
2	FAB	Banks	7,719
3	ADNOC	Energy	3,208
4	Emirates	Travel Services	3,200
5	Emaar	Real Estate	3,043
6	Du	Telecom Providers	2,128
7	ADCB	Banks	1,046
8	Emirates NBD	Banks	904
9	Aldar	Real Estate	725
10	Dubai Islamic Bank	Banks	619

STC/AL RAJHI BANK/
SAUDI NATIONAL BANK/
ETISALAT BY E&/FAB/
EMIRATES/ADNOC
DISTRIBUTION/EMAAR/
ALMARAI/MOBILY/
JARIR BOOKSTORE/
NAHDI/RIYAD BANK/
BUPA ARABIA/DU/
ALINMA BANK/BANK
ALBILAD/SAB/ADCB/
AL OTHAIM/EMIRATES
NBD/ALDAR/JAHEZ/
ARAB NATIONAL BANK/
TAWUNIYA/DUBAI
ISLAMIC BANK/
DAR AL ARKAN/EXTRA/
PANDA/SAUDIA

KANTAR BRANDZ
2023 MOST VALUABLE
EMIRATI AND SAUDI BRANDS



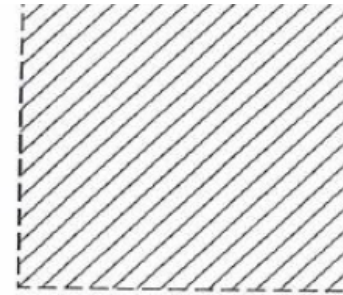
Trademarks & Brands

- At the corporate level, the value of brands is strongly recognized, and innovators should similarly enhance the brand image of their start-ups.
- There are also strong financial incentives to do so. Imagine that an innovator wants to franchise or license its innovation. The value of its brand would be an important asset. The same logic would apply if the innovator wishes to sell her innovation. For example, a few years ago, it was reported that Facebook acquired WhatsApp for \$19 billion.

Trade marks

- Trade marks have been in use for hundreds of years
- Legal protection for trade marks since 1875
- Previously covered under ‘passing off’
- Trade marks have become more important as brands and branding have grown
- Recent legislation substantially extended what could be registered *e.g. shape, Sound (Nokia ring tone)*

UPS brown



Trademark:	Design Only (Color)
Reg. No.:	2901090, 09 Nov 2004
Status:	Registered
Goods and Services:	Transportation and delivery of personal property by air and motor vehicle (1920)
Registrant:	United Parcel Service of America, Inc.
Design Phrase:	The drawing is lined for the color brown. The mark consists of the color chocolate brown, which is the approximate equivalent of pantone matching system 462c, as applied to the entire surface of vehicles and uniforms. The mark consists of the color brown alone. The broken lines indicate the position of the mark and do not form part of the mark.



Ups Color Brown

- © 2016 United Parcel Service of America, Inc. UPS, the UPS brandmark, and the color brown are trademarks of United Parcel Service of America, Inc. All rights reserved.



Tiffany Blue

Tiffany Blue was first associated with the upscale jeweler in 1845, when Charles Lewis Tiffany chose the robin's egg shade for the cover of the company's first catalog, or "Blue Book."

According to the company, he may have selected the color because turquoise was a popular gemstone at the time.

Today the color is not only trademarked (it has been since 1998), it also has its own custom Pantone number: 1837, the year the company was founded.





Tiffany blue



Trademark:

Design Only (Color)

Reg. No.:

2359351, 20 Jun 2000

Goods and Services: Jewelry, precious metals (1939)

Leather products (1939)

Registrant:

Tiffany and Company

Design Phrase:

The mark consists of a shade of blue often referred to as robin's-egg blue which is used on boxes. The matter shown in broken lines represents boxes of various sizes and serves to show positioning of the mark. No claim is made to shape of the boxes.



The Process of Obtaining a Trademark





Some Well Known Trade Marks

Barbie

Virgin



Rectangula



Google



Trademarks

Trade Marks are used to distinguish products and services

Any sign which is capable of distinguishing the goods and services of one trader from those of another

Trade Mark can be

- Word
- Logo-
- Sound- Check Nokia tune <https://www.youtube.com/watch?v=yq0EmbY3Xyl>
- Smell- See http://www.wipo.int/wipo_magazine/en/2009/01/article_0003.html



Requirements of a Trade mark

- Should satisfy the requirements of the Trade Marks Act 1994
 - Now even includes sounds, smells and containers
- Should be distinctive
 - *Can't be a "laudatory" term*
- Must not be deceptive
- Must not cause confusion with previous trademarks
- Initially for 10 years but renewable
- Can be revoked

The Coca-Cola logo, featuring the brand name in its iconic red script font.

Examples of Trademarks

- Coca-Cola bottle
- Chanel perfume bottle
- Bach's 'Air on a G-string'
- The colour green
- The sound of dog barking
- The slogan 'exceedingly good cakes'



Passing Off

- A common law right
- No requirement for registration
- Is a form of mis-representation
- Based on the principle that a trader must not sell goods under the pretence that they are the goods of another
- In an action for passing off one has to show not only that misrepresentation has occurred but that goodwill or reputation has been damaged.

Patents

3 Requirements for a Patent

Useful

It must have utility.

Novel

It must be different from what has come before (i.e., not in the "prior art").

Not Obvious

It must be not obvious to a person of ordinary skill in the field.



Patents

- A patent is a grant from the federal government conferring the rights to exclude others from making, selling, or using an invention for the term of the patent.
- Since Patent #1 was granted in 1790, the U.S. Patent & Trademark Office has granted over six million patents.
- The patent office is strained. It now takes an average of 29.1 months from the date of first filing to receive a U.S. patent.



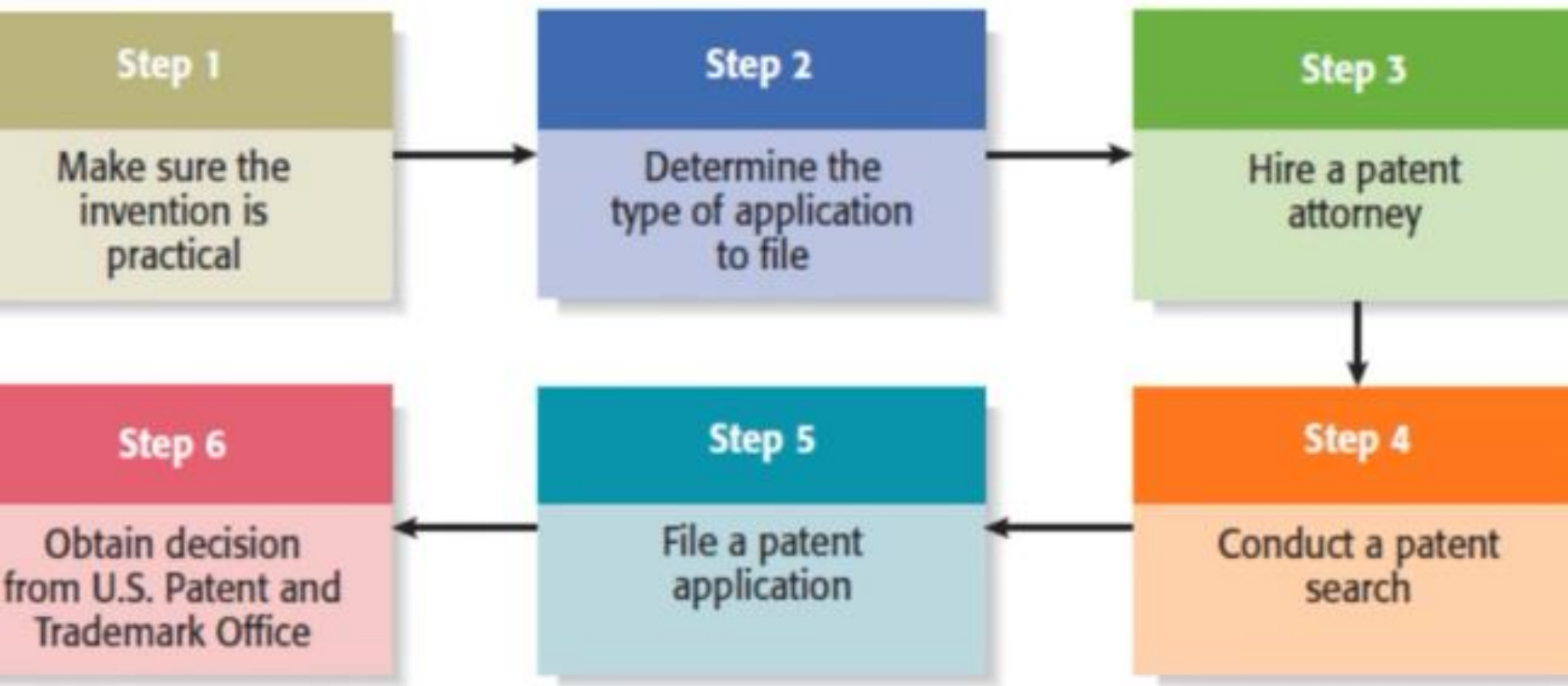
Patent rights owned by an Innovator

If the IPR associated with an invention has been legally established through a patent, the rights holder (inventor) may then permit someone else to produce the invention in return for a fee. Such an arrangement is usually known as a 'licensing agreement'. It was discussed in an earlier session how Procter & Gamble license many of their inventions to be manufactured and launched into different global markets.

A patent will therefore provide the innovator a time frame of 20 years to exploit her invention or to license it to others. During this time, she is expected to derive sufficient financial rewards to cover her efforts and expenses. After 20 years, anybody may freely reproduce the invention without paying the rights holder.

Therefore many innovations in the technological arena would find that securing a patent is a useful strategy to enable them to trade the right through licensing or otherwise prevent others from infringing it.

The Process of Obtaining a Patent





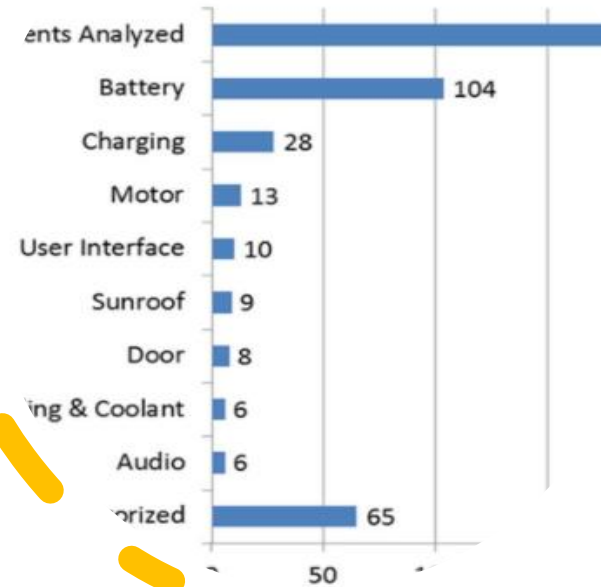
Patent Infringement

- Takes place when one party engages in the unauthorized use of another party's patent.
- The tough part (particularly from a small entrepreneurial firm's point of view) is that patent infringement cases are costly to litigate.
- A typical patent infringement case costs each side at least \$500,000 to litigate.

Patent rights owned by an Innovator

- IBM used an aggressive intellectual-property enforcement effort that boosted its annual patent-licensing royalties from \$30 million in 1990 to nearly \$1 billion ten years later.
- Many business leaders are therefore focusing on the management of intellectual property as a core competence for the success of their firms.
- These leaders recognize that the knowledge economy establishes a new environment of competition in which intellectual rather than physical assets have become the principal indicator of shareholder wealth and competitive advantage.
- Innovators must therefore strive to leverage the power of patents if their invention is protectable by such a regime.

TESLA'S PATENTS BY



Patent Database

<https://patentscope.wipo.int/search/en/search.jsf>

WHAT IS THE WIPO PATENTSCOPE SEARCH SYSTEM?

The PATENTSCOPE search system is the FREE OF CHARGE patent search system provided by the World Intellectual Property Organization (WIPO) that allows you to access millions of patent documents.

- You're a patent attorney and need to find a specific patent document...
- You're an inventor and want to see whether your latest invention has already been patented...
- You're a researcher and are interested in seeing which technologies have been developed in your field...
- You're an entrepreneur and want to find out who your competitors are and what they're up to... The PATENTSCOPE search system is the right tool for you!

Google Patent Database

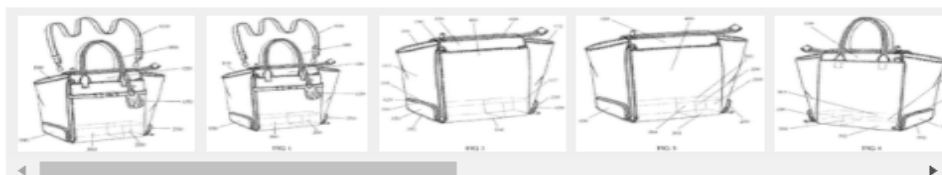
<https://patents.google.com/>

Carry-all bag with fixed insulated and non-insulated compartments

Abstract

The present invention is directed to a soft-sided carry-all bag containing fixed insulated and non-insulated compartments for extensive storage of various items needed for everyday use including, among other things, food, footwear, laptop computers, cellular phones, books and notebooks, writing or eating utensils, water and drinks bottles, eyeglasses, coins, wallets, credit cards, cosmetics, toiletries, and clothing. The bag is meant to provide an individual with a means for consolidating such various items of everyday use without sacrificing aesthetic features, and generally includes a flexible main compartment for the storage of larger items, a rigid insulated compartment for the storage of perishable items, and a rigid non-insulated compartment for the storage of smaller items, each with separate access. The rigid insulated and non-insulated compartments comprise the base of the bag, and function to both provide support for the flexible main compartment and prevent crush damage to the contents within each lower compartment.

Images (11)



Classifications

■ **A45C13/02** Interior fittings; Means, e.g. inserts, for holding and packing articles

[View 6 more classifications](#)

Description

PRIORITY NOTICE

The present application is a non-provisional utility application and makes a claim of priority under 35 U.S.C. § 119(e) to prior provisional application 62/480,034 filed in the United States Patent and Trademark Office on Mar. 31, 2017.

CROSS REFERENCE TO RELATED PATENT APPLICATIONS

The present application makes no reference to any other related filed patent applications.

STATEMENT REGARDING FEDERAL SPONSORSHIP

No part of this invention was a result of any federally sponsored research.

TECHNICAL FIELD OF THE INVENTION

The present invention relates in general to utility bags, and more specifically to soft-sided carry-all bags containing fixed insulated and non-insulated compartments for extensive storage of such various items needed for everyday use including, among other things, food, footwear, laptop computers, cellular phones, books and notebooks, writing or eating utensils, water and drinks bottles, eyeglasses, coins, wallets, credit cards, cosmetics, toiletries, and clothing.

US10602821B2

United States

[Download PDF](#) [Find Prior Art](#) [Similar](#)

Inventor: Ta'Loria Da'Shea Young, Eunice Damaris Martinez-Gonzalez

Current Assignee : Martinez Gonzalez Eunice Damaris , Young Taloria Dashea

Worldwide applications

2018 [US](#)

Application **US15/910,841** events [⌄](#)

2017-03-31 [Priority to US201762480034P](#)

2018-03-02 [Application filed by Ta'Loria Da'Shea Young, Eunice Damaris Martinez-Gonzalez](#)

2018-03-02 [Priority to US15/910,841](#)

2018-10-04 [Publication of US20180279733A1](#)

2020-03-31 [Application granted](#)

2020-03-31 [Publication of US10602821B2](#)

Status [Active](#)

2038-04-02 [Adjusted expiration](#)

Info: [Patent citations \(13\)](#), [Cited by \(14\)](#), [Legal events](#), [Similar documents](#), [Priority and Related Applications](#)

External links: [USPTO](#), [USPTO Assignment](#), [Espacenet](#), [Global Dossier](#), [Discuss](#)

Claims (8)

[Hide Dependent](#) [^](#)

We claim:

1. A carry-all bag combination, comprising:
 - a main compartment;
 - an insulated compartment;
 - a non-insulated compartment; and
 - a rigid compartment assembly;wherein said main compartment comprises a flexible compartment;
- wherein said insulated compartment comprises a rigid compartment;
- wherein said non-insulated compartment comprises a rigid compartment;
- wherein said rigid compartment assembly comprises said insulated compartment and said non-insulated compartment;
- wherein said insulated compartment and said non-insulated compartment are affixed to one another;

Industrial Design

- In many instances, a new product may be offered additional protection against illegal copying by using industrial design protection. Let us imagine that the innovator has created a new functional piece of furniture. Design protection is a useful means for protection here.
- Similarly, if the innovator is designing fashion items such as jewellery, watches, belts, hats, handbags, accessories and clothing, protection as industrial design is worth considering.
- Design registration protects the overall aesthetic appearance of the design. As an illustration, Louis Vuitton owns a design patent for a handbag, while Tag Heuer owns a design patent for a watch. In a competitive market, many products are thus now differentiating themselves by using design to allow important features relating to ease of use or creative shapes to be protected.
- Having an IPR relating to Industrial Design will therefore allow the innovator to derive benefits from his or her creation, while preventing copying by others.



Traditional knowledge

Middle East Traditional medicine



China Traditional medicine



Peru Traditional crafts



Arabic Traditional embroidery
(Palestine, Tareez)



Turmeric

Traditional knowledge

Hoodia Cactus-Southern Africa



Turmeric



Turmeric &
Cranberry Seed...

AED 183.53
Kiehl's Official

3RD EDITION

INNOVATION

DAVID SMITH

EXPLORING

Traditional knowledge

- **Traditional knowledge** refers to the **knowledge**, innovations and practices of **indigenous** and local communities around the world.
- Developed from experience gained over the centuries and adapted to the local culture and environment, **traditional knowledge** is transmitted orally from generation to generation.

Geographical Indication

In order to function as a GI, a sign must identify a product as originating in a given place.

A **geographical indication (GI)** is a sign used on products that have a specific **geographical** origin and possess qualities or a reputation that are due to that origin.



Roquefort cheese

The unique blue cheese from France is aged in natural caves around the Roquefort-sur-Soulzon region and protected using geographical indications.



Darjeeling Tea

BASMATI, GEOGRAPHICALLY

The Intellectual Property Appellate Board will next week take a call on granting the 'geographical indicator' tag to Basmati, the world-famous long grain aromatic variety of rice grown in India and Pakistan.





Supporting Materials & Extra Reading

Read Smith (2015) Exploring Innovation textbook, IPR Chapter, p 149-159

Read Graham, S. J., Merges, R. P., Samuelson, P., & Sichelman, T. (2009). High technology entrepreneurs and the patent system: Results of the 2008 Berkeley patent survey. *Berkeley Technology Law Journal*, 1255-1327.

Read various relevant Articles by World Intellectual Property Organization on: Small and Medium-Sized Enterprises

Available at: <http://www.wipo.int/sme/en>

Read Case Study

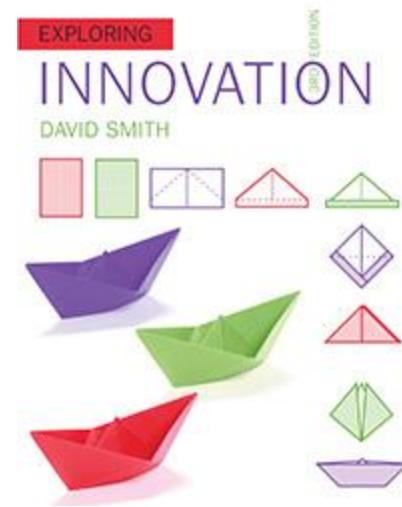
All for the Love of Chocolate, which describes how Patchi, Lebanon has leveraged IPR regime to further its growth.

Available at: <http://www.wipo.int/ipadvantage/en/details.jsp?id=2848>





Thank you



Innovation Strategy

Chapter 8

Objectives

At the end of this class you will be able to:

- Understand the nature of Innovation strategies;
- Explain the nature and importance of innovation strategy;
- Recognise the various innovation strategies that are available as a means of exploiting innovations;
- Discuss practical examples of innovation strategies.
- Discuss Innovation Maturity and its Measurement

Advanced Organizer

- The exploitation/commercialization phase of Innovation involves taking the right business decisions. Many innovations have failed to make it due precisely to poor strategic decisions. Innovators need to develop a sound understanding of the nature of Innovation Strategies. This session discusses strategic management and its core concepts.

Innovation Virtuous Circle

- Companies like Apple or P&G are known to launch regular Innovations. Yet, this is NOT linked solely to their resources.
- It is because they follow a disciplined innovation process Morris (2006). Therefore, it is important to develop an institutional Innovation strategy.
- However, such as strategy will only be as good as the institution's ability to execute it, and to a great extent on the mindset.



Figure 1
Innovation's Virtuous Cycle



What is an Innovation Strategy?

- An innovation strategy in business is defined as a commitment to a common innovation mission and a structured set of activities designed to support the future growth of an organization. No two innovation strategies are the same.
 - It is the roadmap for innovative ideas.... The plan that guides how you will implement those ideas and make them thrive. It is the playbook that turns creativity into success.
 - Role of innovation strategy as the captain steering a ship. Without a clear strategy, you might end up sailing in circles. But with it, you're navigating towards the path of success.
 - When Apple launched the iPhone, it was not just about a cool gadget; it was a result of a meticulous innovation strategy. Apple did not just create a phone; they created a unique customer experience, and this is what made it a game-changer.
 - It is a common innovation mission and a detailed plan that aims to create added value which presents customers with a desirable benefit, and for which the customers are willing to pay more than the current offering. It includes a set of policies or behaviors geared toward achieving future organizational growth.
- 



Benefits of an Innovation Strategy

- Leading Innovative businesses recognize that big, new ideas do not happen in a vacuum. Instead, innovations emanate from ecosystems (environments and procedures) that are deliberately created to facilitate their development.
 - Developing an innovation strategy is important as it may (Alex Elkins, 2021):
 - **Clarify priorities and goals:** An innovation strategy outlines the goals of the organization's innovation activities and helps focus efforts on reaching those goals.
 - **Foster alignment:** With a plan in place, diverse groups within an organization will all be working toward common goals rather than pursuing their own individual priorities.
 - **Keep a business from resting on its laurels:** Even businesses that start out as innovators must continue to innovate in a strategic way, as copycats and innovative competitors are likely to take market share over time.
 - **Help a business achieve long-term success:** Without ongoing innovation, a company is unlikely to gain (or maintain) competitive advantage or keep customers engaged over the long term.
- 



Innovation and 'Creative Destruction'

- A few decades back most large companies were protected from outside impacts by their size that prevented new entrants from entering the market, hence shielding them from many strategic challenges. The large organizations outspent new entrants on marketing, logistics, followed their competitors' successful innovations, tied up distributors with exclusivity agreements, sold their products/services at low prices, and leveraged economies of scale to out-muscle their competitors.
 - However, with globalization and the emergence of the knowledge economy, these tactics are no longer effective, so that the strategic vulnerabilities of all organizations, large and small, is increasing.
 - Such a situation is not new. In 1942, Schumpeter coined the term "creative destruction," whereby radical innovations result not from undefined external factors, but from within organizations, due to the very nature of competition. Creative destruction led to every major shift in how businesses and consumers operate. For example, Netflix and streaming replaced Blockbuster and video rental stores, Amazon and Souq.com replaced used the convenience of e-commerce to forced shopping malls to adapt what they offer to shoppers or go out of business. And the cycle perpetuates itself.
 - Change, Schumpeter observed, is the common condition, not stability.
- 



Key Components of an Innovation Strategy

An innovation strategy is an organizational plan to foster innovation within the organization. It involves recognizing opportunities for innovation, setting objectives, assigning resources, and implementing activities to promote creativity and exploration.

An effective innovation strategy should support an organization's business strategy. The key components of an innovation strategy include:

1. **Vision:** A clear and convincing vision that defines the direction for innovation efforts and inspires employees to pursue new ideas or insights.
2. **Goals:** The strategy should define specific goals and objectives for innovation, such as launching new products/Services/processes/business models, entering new markets.
3. **Resource allocation:** The strategy should allocate resources, such as funding and personnel, to support innovation efforts and ensure that they are given the priority they deserve.
4. **Processes and structures:** The strategy should define processes and structures for managing innovation, such as idea generation, evaluation, and implementation, and ensure that they are aligned with the company's existing structures.
5. **Culture:** The strategy should promote a culture of innovation within the organization, encouraging employees to share ideas, take risks, and experiment with new approaches.



Role of an Innovation Strategy

- Without an innovation strategy, innovation improvement efforts can easily become a mix of best practices: R&D across decentralized autonomous teams, launching internal entrepreneurial ventures, launching up corporate venture-capital arms, pursuing external alliances, embracing open innovation and crowdsourcing, collaborating with customers, and implementing rapid prototyping, ...
- There is nothing wrong with these initiatives. However, the problem is that an organization's capacity for innovation derives from an innovation system: a coherent set of interdependent processes and structures that defines how the company searches for solutions to problems, determine which projects get funded, and develops a business model.
- An innovative strategy guides decisions on how resources are to be used to meet a business's objectives for innovation, deliver value and build competitive advantage. Strategies should include: an analysis of a business's competitive and technological environment.
- Individual best practices involve trade-offs. And adopting a specific practice generally requires a host of complementary changes to the rest of the organization's innovation system. A company without an innovation strategy won't be able to make trade-off decisions and choose all the elements of the innovation system.
- In the absence of an innovation strategy, different departments may easily end up pursuing conflicting priorities—even if there is a clear business strategy.

Why Strategic planning skills are important for Innovators?

- Strategy consists of ***competitive moves*** and ***business approaches*** used by managers to run the company
- It represents management's "***action plan***" to
 - Grow the business
 - Attract and satisfy customers
 - Compete successfully
 - Conduct operations
 - Achieve the targeted levels of organizational performance
- Hence Strategic planning is a key skill for Innovators.



Innovation Strategy: Practical Considerations

Developing and Driving Strategy

- Create, build, strengthen, and maintain your competitive advantage.

Successfully Managing Innovation Ecosystems

- Explore the concept of innovation ecosystems and set up the right network of partnerships to contribute to innovation and success.

Balancing Exploitation and Exploration

- Assess how your organization can evolve its capabilities to remain successful and relevant in a rapidly changing environment.

Open Innovation: How Organizations Source Ideas

- Manage successful open-innovation efforts, and benchmark on leading global organizations, and learn how they managed the process.

Partnering to Commercialize Technology

- Discover and engage in technology partnerships to drive mutual value.

Organizational Innovation and Societal Influence

- Understand how a large-scale initiative can have implications not only for your organization but also for partners and society.



Innovation Strategy: Examples (Alex Elkins, 2021)

You may think of innovation strategy as the heartbeat of an organization's ecosystem. It's not just about launching products; it's about creating a culture where everyone is thinking: "How can we do this better?"

Let us illustrate by some practical examples:

- **Apple**
 - Apple builds innovations by acquiring startups and partnering with other companies to develop new products, such as a partnership with Volkswagen to produce driverless shuttle vans for employees. The company also has an internal accelerator that leverages Apple employees to push forward new research projects, such as Titan, an electric car project.
- **Amazon**
 - In addition to an innovation lab that creates new hardware devices such as the Kindle and Echo, Amazon also encourages all employees to pitch innovative ideas. For example, an employee spontaneously pitched the idea for Amazon Prime in a memo, and leaders provided internal funding to make it happen.
- **Airbnb**
 - They did not just luckily develop a new way to book a room; it was a result of a well-thought-out innovation strategy that disrupted the entire hospitality industry.



Innovation Strategy: Examples

- Google

- Google offers innovation classes for employees and runs an R&D facility that has launched products like self-driving cars and Google Glass. But the company also looks to bring in innovations from external sources, with GV, its own venture capital fund for innovative startups, and an incubator for promising entrepreneurs.

- Zara

- What makes them stand out? Their innovation strategy is not just about clothing; it is about speed. Zara disrupted the fashion industry by introducing fast fashion – getting the latest trends from runway to store shelves in record time. It is a strategy that keeps customers excited and competitors on their toes.

- Tesla

- Elon Musk did not just want to build electric cars; he aimed to make them desirable. Tesla's innovation strategy goes beyond vehicles – it is about creating an entire sustainable energy ecosystem. They are not just driving cars; they are steering the future.

Strategic decision making is sometimes classified according to three levels of strategy based on which part of the organization they relate to. In ascending order of hierarchical organizational level, they may be classified as:

Corporate strategies

- Many organizations diversify their operations as they grow and often acquire or develop a portfolio of more or less related businesses. Corporate strategies are concerned with the broader issues of what industries the organization wants to compete in, involving considerations such as mergers and acquisitions and allocation of resources between Strategic Business Units (SBUs) (Henry, 2008). These strategies are therefore not concerned with targeting users in individual markets, but rather in establishing appropriate networks and synergies between its constituent business units and sharing resources between them to achieve efficiency and effectiveness in operations.

Business Strategies

- Such strategies are sometimes called competitive strategy and they define how the organization will compete within a particular industry or market, and establish a competitive advantage over its rivals. They involve:
 - Identifying which segments the organization should target, the services it should offer them, as well as differentiation and pricing strategies.
 - Obtaining inputs through an effective supply chain and using the organization's resources effectively within a value chain for efficiency and effectiveness.
 - Developing good internal communications so that the value chain may function effectively and for organizational learning to take place.

Functional strategies

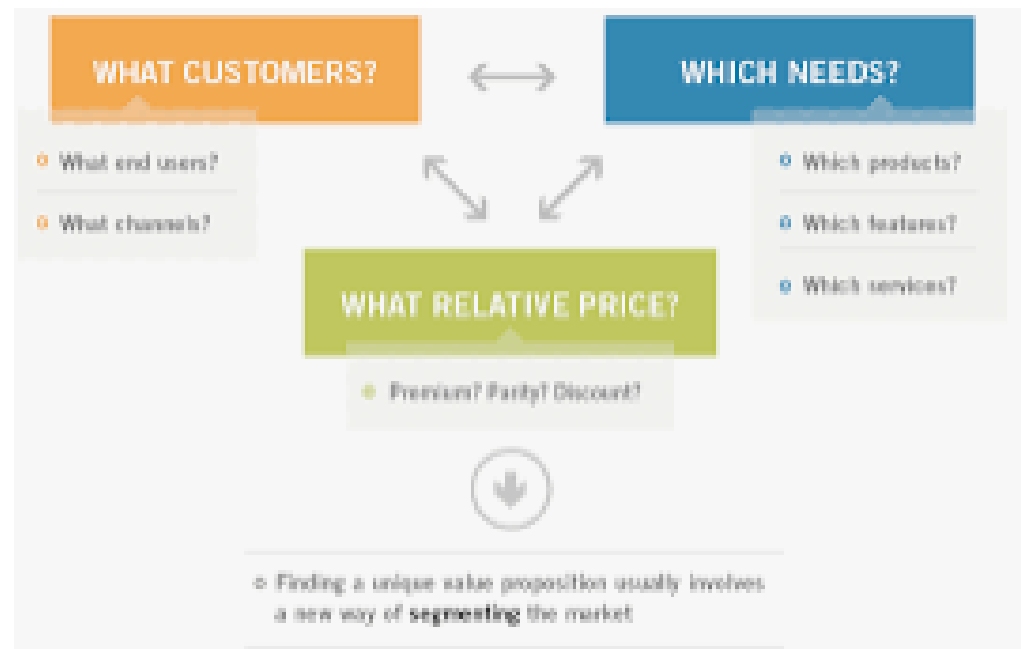
- Each of an organization's individual functions will have its own strategy. For example, Research and Development and Marketing may have their own strategies that are in support of the overall organization's strategy. Such strategies are not of particularly great magnitude and are more likely to be short-term.

LEVELS OF STRATEGY



Major Elements of a Successful Strategy

- Developing a successful strategy *hinges* on making *competitive moves* aimed at
 - Appealing to buyers in ways to set the company apart from rivals and
 - Carving out its own market position
- Involves *developing* a *distinctive* element to
 - Attract customers *and* retain them
 - Provide a competitive edge





Alignment Between Innovation and Business Strategies

- Strategic Alignment is like the GPS for your business strategy, ensuring that your innovative efforts are on the right track to success.
- Imagine you're embarking on a cross-country road trip. You have got a destination in mind, but how do you ensure you're taking the right turns and staying on the best route?
- Strategic Alignment is about making sure our innovation strategy aligns seamlessly with the broader business goals and objectives. It is not just about having cool ideas; it is about making sure those ideas contribute to the big picture success of the entire company.
- For example-if your destination is profitability, your innovation strategy is the roadmap. If you take detours or choose the wrong roads, you might get lost or delayed. So, aligning your strategy with the broader business objectives is like having a GPS that keeps you on track towards success.

Some Key Innovation and Strategic Frameworks



ISO 56000 Standard

- ISO 56000:2025 defines the fundamental concepts, principles, and vocabulary for innovation management systems (IMS).
- It provides a standardized framework to help any organization—regardless of size or sector—systematically manage innovation, covering products, services, processes, and business models. The standard fosters a shared understanding to improve innovation
- The **ISO 56002** framework is an international guideline for establishing an **Innovation Management System (IMS)**. While ISO 56002 itself provides the "how-to" for managing innovation, its sister document, **ISO 56004**, specifically provides the guidance for the assessment of those capabilities



ISO 56000 Standard: The 8 Guiding Principles

ISO 56000 assessment evaluates an organization against eight core principles:

1. **Realization of Value**

Creating financial or social value, not just "newness."

2. **Future-focused Leaders**

Leaders who prioritize long-term vision over short-term gains.

3. **Strategic Direction**

Innovation activities must align with the organization's core goals.

4. **Culture**

Promoting a climate of curiosity, risk-taking, and collaboration.

5. **Exploiting Insights**

Using data and "customer empathy" to drive decisions.

6. **Managing Uncertainty**

Managing risk through experimentation and pivoting.

7. **Adaptability**

The ability to change internal structures as the market shifts.

8. **Systems Approach**

Treating innovation as a set of interconnected processes.



Innovation Maturity

- **Innovation Maturity** is a measure of an organization's systemic capacity to consistently generate, develop, and commercialize value-added ideas.
- Rather than viewing innovation as a series of spontaneous "eureka" moments, maturity models evaluate the extent to which innovation is embedded into the institutional DNA through formal processes, strategic alignment, and a supportive cultural climate.
- It is a critical concept for three primary reasons:
 - **Predictability and Scalability**

It shifts the organization from "hero-based" innovation (dependent on specific talented individuals) to "process-based" innovation, ensuring repeatable success.
 - **Risk Mitigation**

High maturity levels allow firms to balance their portfolio through disciplined experimentation, reducing the sunk costs associated with unsuccessful projects that fail to align with corporate strategy.
 - **Dynamic Capability**

It fosters the organizational agility required to sense and respond to market disruptions, ensuring long-term survivability in increasingly volatile competitive landscapes.



Innovation Maturity Models

There are several Proprietary Frameworks to Assess Innovation Maturity. Generally, the Maturity models generally follow a five-level progression:

1. Level 1 (Ad-hoc/Reactive)

Innovation is unplanned, dependent on individuals, and rarely succeeds.

2. Level 2 (Repeatable/Novice)

Some successes, but processes are inconsistent and localized.

3. Level 3 (Defined/Managed)

Structured, company-wide processes and strategies are in place.

4. Level 4 (Predictable/Experienced)

Data-driven,, with measurable outcomes and ROI on innovation.

5. Level 5 (Optimized/Leader)

Continuous improvement of innovation capabilities, deeply embedded in the culture.



Global Innovation Institute – GInI Framework for Innovation Maturity



Level 1 (★★★) – Initiating

The first level, indicating a desire to be innovative and the start of a journey towards it.

Level 2 (★★★★) – Developing

Demonstrates a strong desire to be innovative, with a formal program and structure in place.

Level 3 (★★★★★) Achieving

Strong progress in innovation strategy and systems, with clear impacts from innovation efforts.

Level 4 (★★★★★★) Leading

Excellent progress in strategy and systems, with demonstratable business impact.

Level 5 (★★★★★★★) Sustaining

(Implied top level in the framework, often reflecting

Measuring Corporate Entrepreneurship

Business Horizons (2014) 57, 37–47



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Diagnosing a firm's internal environment for corporate entrepreneurship

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Measuring Corporate Entrepreneurship

The corporate entrepreneurship assessment instrument (CEAI)

- 1: Management support for corporate entr. (19 Items)
- 2: Work discretion (10 Items)
- 3: Rewards/Reinforcement (6 Items)
- 4: Time availability (6 Items)
- 5: Organizational boundaries (6 Items)

CEAI-Some measures

Section 3: Rewards/Reinforcement

- 30. My manager helps me get my work done by removing obstacles and roadblocks.
- 31. The rewards I receive are dependent upon my innovation on the job.
- 32. My supervisor will increase my job responsibilities if I am performing well in my job.
- 33. My supervisor will give me special recognition if my work performance is especially good.
- 34. My manager would tell his/her boss if my work was outstanding.
- 35. There is a lot of challenge in my job.

Section 4: Time availability

- 36. During the past three months, my workload kept me from spending time on developing new ideas.
- 37. I always seem to have plenty of time to get everything done.
- 38. I have just the right amount of time and workload to do everything well.
- 39. My job is structured so that I have very little time to think about wider organizational problems.
- 40. I feel that I am always working with time constraints on my job.
- 41. My co-workers and I always find time for long-term problem solving.

Section 5: Organizational boundaries

- 42. In the past three months, I have always followed standard operating procedures or practices to do my major tasks.
- 43. There are many written rules and procedures that exist for doing my major tasks.
- 44. On my job I have no doubt of what is expected of me.
- 45. There is little uncertainty in my job.
- 46. During the past year, my immediate supervisor discussed my work performance with me frequently.
- 47. My job description clearly specifies the standards of performance on which my job is evaluated.
- 48. I clearly know what level of work performance is expected from me in terms of amount, quality, and timelines of output.

Mechanistic vs. Organic Organizational Structures

Mechanistic Structure

- The mechanistic structure is traditional and it has a hierarchy with leaders delegating work to those at lower level. Teams are generally led by managers rather a team approach. The mechanistic structure tends to be more stable, and also more rigid in delegating leadership roles.
- In a centralized leadership approach, the structure endeavors to have specific job descriptions for all staff who work on individual tasks, with the managers controlling the work. Written communication is intensively the means used to organize work; this is a heavy-reporting organizational structure.
- The structure requires leaders to build loyalty among workers who report directly to them. Workers must have explicit trust that the directions they are given will move the company closer to its goals.



Mechanistic vs. Organic Organizational Structures

Organic Structure

- The organic organization structure is flat, with no typical pyramid of leadership. Many employees work in horizontal clusters, rather than in vertical clusters. In terms of organizational structure, the objective is to have a more homogeneous culture of employees via decentralized management roles.
- Employees are encouraged to work together and to have teams working on tasks together or coordinating the flow of information and duties. Teams develop with horizontal communication, rather than vertically as in a traditional hierarchy.
- Many organizations adopting an organic structure implement open floor plan workspace, where verbal communication is adopted instead of written communication. A team member's role can alter, as the needs of the organization change over time.



Linking Strategy with Ethics

1 The Strategic Tension: "Move Fast" vs. "Move Right"

The primary ethical challenge in innovation management is:

- **Early Stage:** Impact is easy to influence, but consequences are hard to predict.
- **Late Stage:** Consequences are clear, but the technology is so embedded that change is expensive and difficult.

2 To link these two domains, adopt three lenses:

A. Triple Bottom Line (TBL)

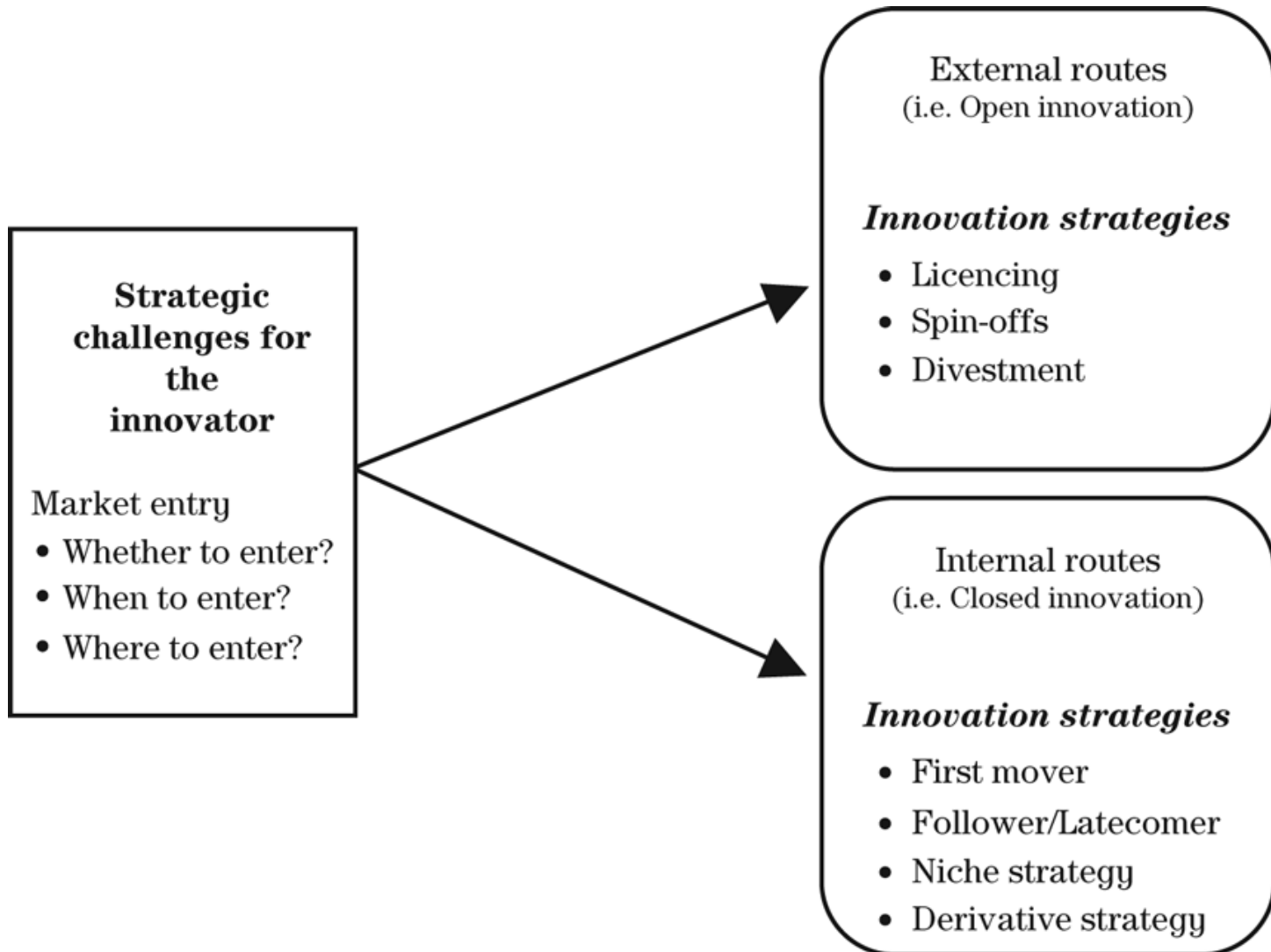
B. The 4 Dimensions of Responsible Innovation (Stilgoe et al.)

- **Anticipation:** Asking "What if?" to envision potential side effects of a new technology.
- **Reflexivity:** Recognizing the limitations of one's own knowledge and the biases inherent in the design team.
- **Inclusion:** Engaging diverse stakeholders (not just shareholders) early in the innovation cycle.
- **Responsiveness:** Having the organizational agility to pivot or kill a project if ethical risks emerge.

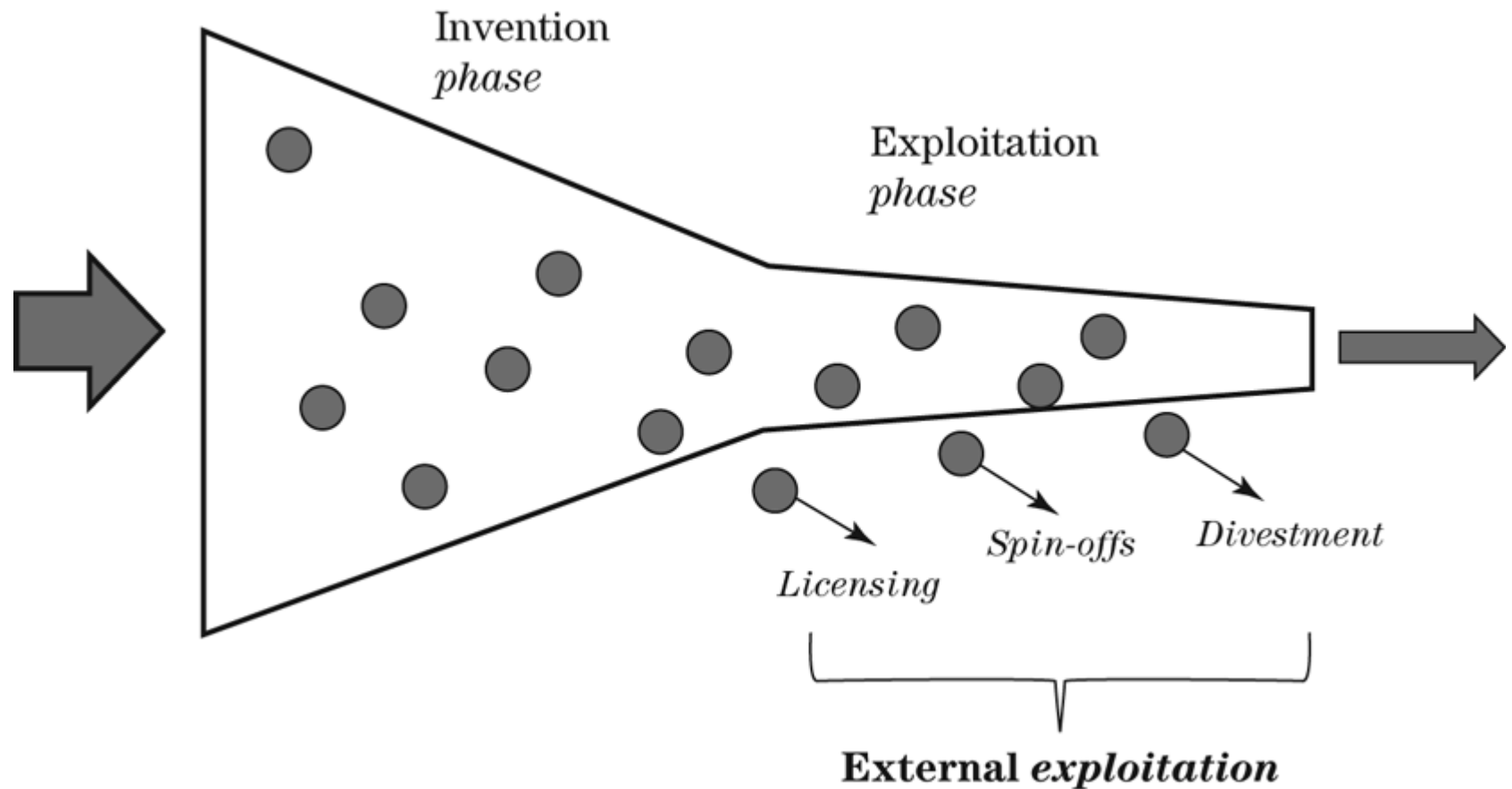
C. Distributive Justice in Innovation

- Ethics in strategy asks: *Who benefits from this innovation, and who bears the risk?*
- An ethical strategy ensures data representativeness as a core technical requirement.

Innovation Strategies (external & internal)



Exploitation via 3 external routes



Factors leading to use of external routes to Innovation

- Lack of resources
- Lack of knowledge
- Poor fit with company strategy
- Lack of reach

Cesaroni's 3 factors for *Licensing*

- Lack of complementary assets in production & marketing
- Transaction costs of licencing are lower than purchasing complementary assets
- Intense competition in the final product market

How Google encourages innovation among its employees

Google's Objective

- Organize the world's information and make it universally accessible and useful. When they introduced innovations like Google Maps or Google Drive, it wasn't just about cool technology. These innovations were aligned with their overarching mission, making information accessible and useful to everyone.





How Google encourages innovation among its employees

1. 20 Percent Time

- One day per week for employees to work on their own ideas

2. The architecture of innovation

- Google has always had trendy offices. With plenty of fun features (such as vertical ladder chutes between floors). The Office design ensures that employees are never more than 50 meters from food. This “encourages employees to snack constantly as they bump into coworkers from different teams within the company.” This kind of collaboration is at the heart of much of the innovation at Google.

3. Outside inspiration

- Over the past 11 years, there have been more than 4 000 Talks at Google events. It started with talks from Googlers themselves, then expanded to authors and experts from all around the world with different backgrounds. The talks are hosted by Google volunteers in offices around the world, with about 12 talks happening each week

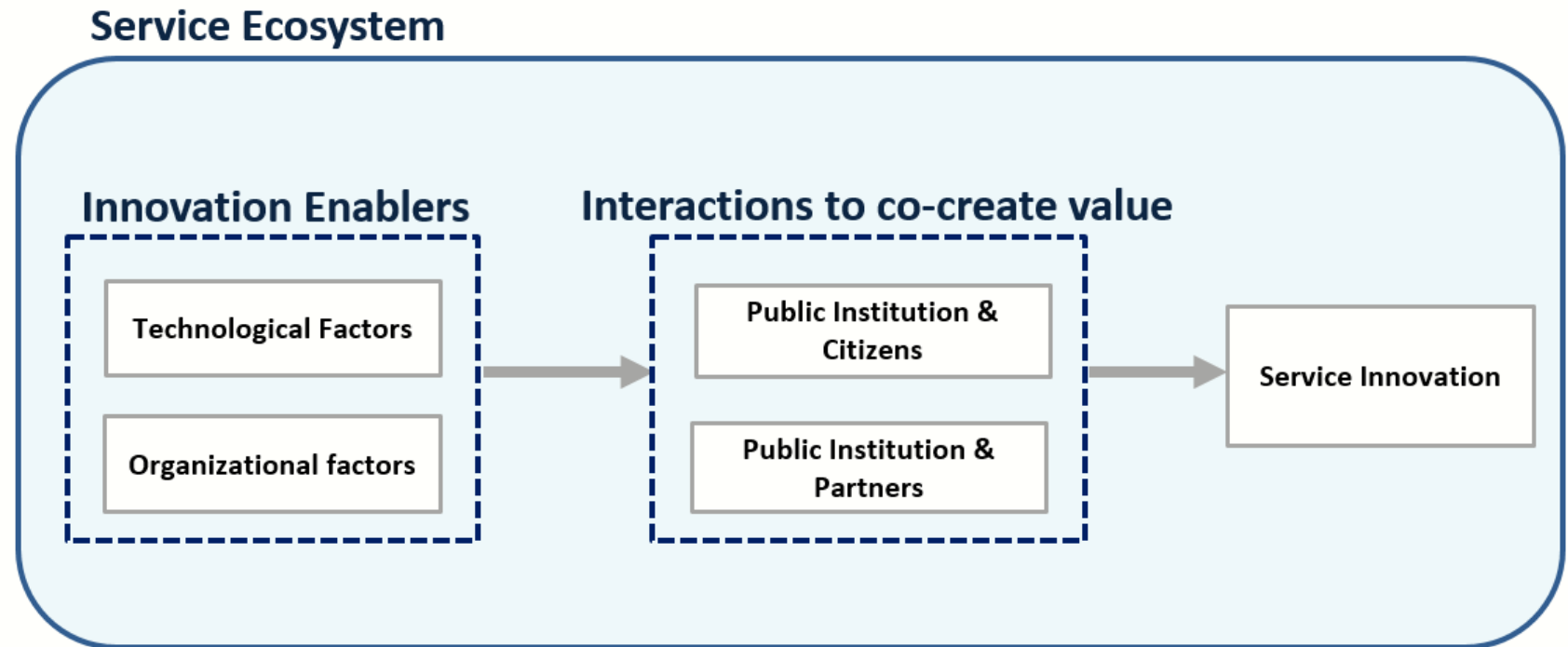
4. Empowerment

- Perhaps most importantly though, Google does a lot of work around empowering its employees across the board.
- 



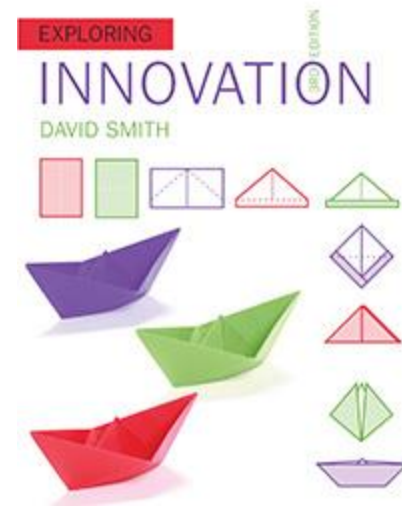
- The pre-dominant leadership or management style in Procter & Gamble is that of participatory, delegating, and empowerment. Management has decentralized decision making process in such a manner that middle level management most at times do not have to wait for headquarters approval and funding; in order to embark on certain key innovative projects.
- Because of the empowerment given to mid and senior level management within this multinational corporation, it is much easier for management to customize products and customer services internally. It is abundantly clear that, the success of this giant corporation can be closely tied to its management and leadership style.
- This is a clear customer focus leadership style to strengthen employee morale and to refocus employees' attention to providing the needs and wants of customers in this ever-changing global market. P&G management has recognized that in order to be a leader in the industry, they have to

Innovation Ecosystem in Public Sector



Parahoo, Sanjai & AlNakeeb, Ahmed (2019)

Thank You



Theories of Innovation

Chapter 3

Objectives

When you have completed this class you will be able to:

- Distinguish and analyse the concepts of technological paradigms and technological change;
- Analyse the long wave theory of innovation;
- Identify and analyse innovation theories;
- Evaluate the most appropriate theories for the analysis of innovations;
- Demonstrate practical benefits that can be derived from the application of innovation.

Science

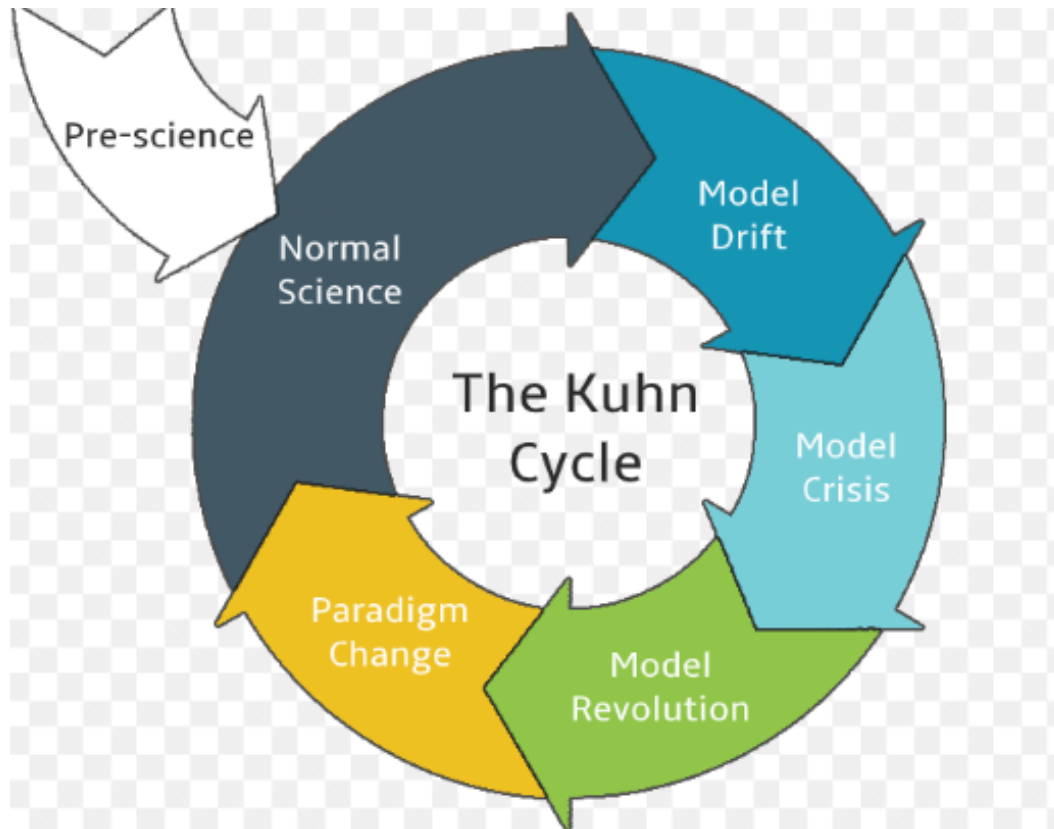
- *Science, is all about understanding the natural world (i.e., fauna and flora).*
- *Science involves the use of systematic rigorous methods of enquiry in order to develop logical, self-consistent explanations of natural phenomena (Littler, 1988).*
- *A critical aspect is the application of the scientific method involving observation, the development of hypotheses and the systematic collection of data in order to arrive at explanations.*
- *The rigorous application of scientific method results in knowledge about the natural world being recorded and codified as a formal body of knowledge that can be relatively easily communicated between individuals through books and papers*

Kuhn's scientific paradigm

A paradigm is a way of organizing and condensing sensory information . Like learning in general, paradigms help in the study of physical science by helping us to organize information and understand our world.

According to Kuhn scientific advances proceed through a series of periodic revolutions (see Figure), such as those initiated by the work of Copernicus, Newton or Einstein that led to paradigm shifts resulting in new schools of thought replacing existing ones.

Kuhn's scientific paradigm



According to Kuhn scientific advances proceed through a series of periodic revolutions (see Figure), such as those initiated by the work of Copernicus, Newton or Einstein that led to paradigm shifts resulting in new schools of thought replacing existing ones.

Definition of technology

1. **Merriam-Webster** defines technology as: "the practical application of knowledge especially in a particular area."
2. According to the **Cambridge Dictionary**, technology is: "the study and knowledge of the practical, especially industrial, use of scientific discoveries."
3. The **National Academy of Sciences** defines technology as: "the purposeful application of information in the design, production, and utilization of goods and services, and in the organization of human activities."
4. The **United States National Aeronautics and Space Administration (NASA)** defines technology as: "the purposeful application of information in the design, production, and utilization of goods and services, and in the organization of human activities."

Technological paradigm

Building on Kuhn's seminal contribution, paradigms proposed in many scientific fields.

A technological paradigm is different from scientific cognition and its related paradigm due to the different objective of science: research in technology is goal-oriented. We are trying to make human lives easier, operational process more efficient, or achieve sustainable development, for example.

Therefore, a technological paradigm represents:

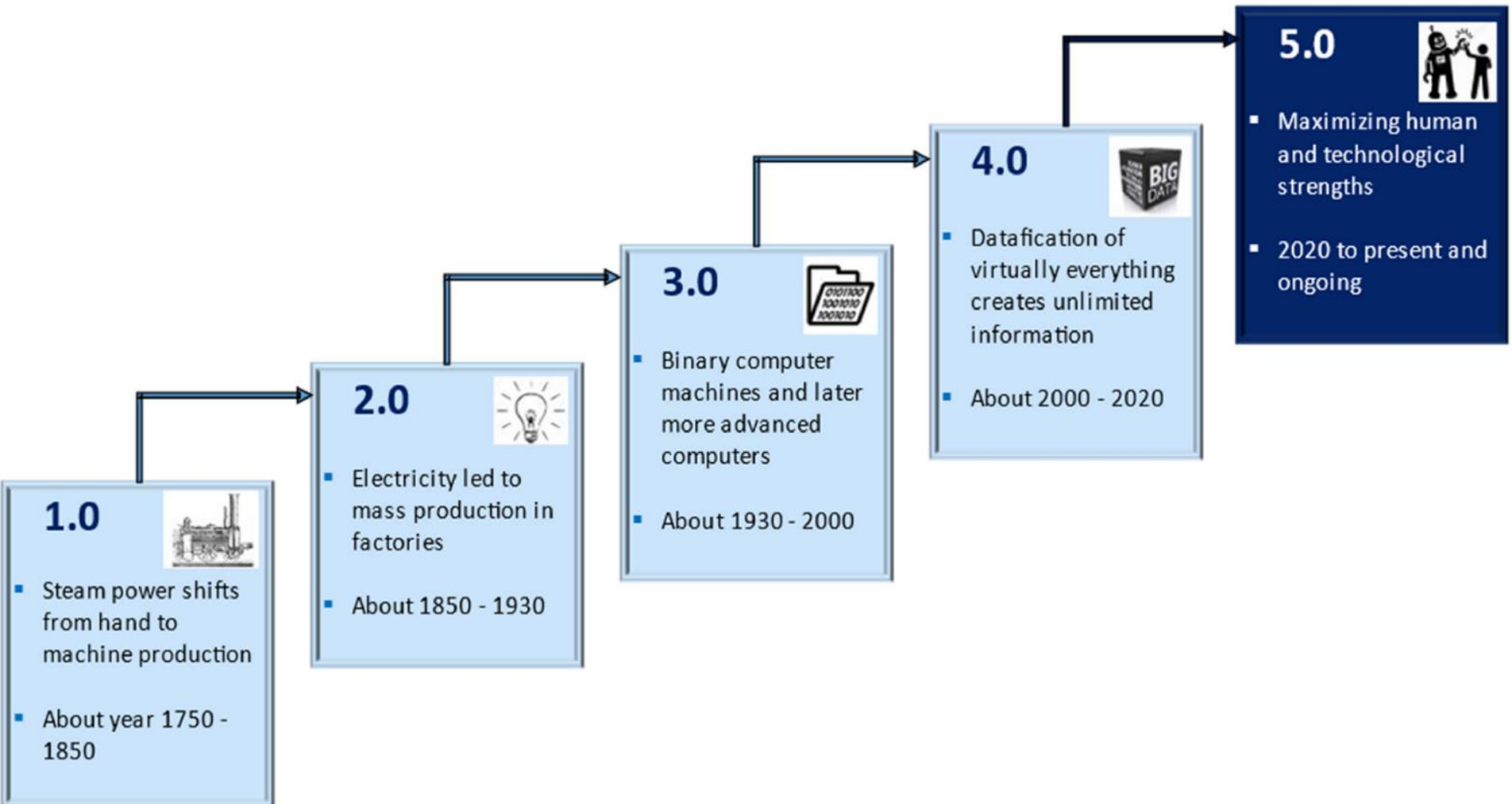
“the technological domain” within which technological advances take place.”

Von Tunzelmann (1995)

“a strong interrelated constellation of technological innovations”

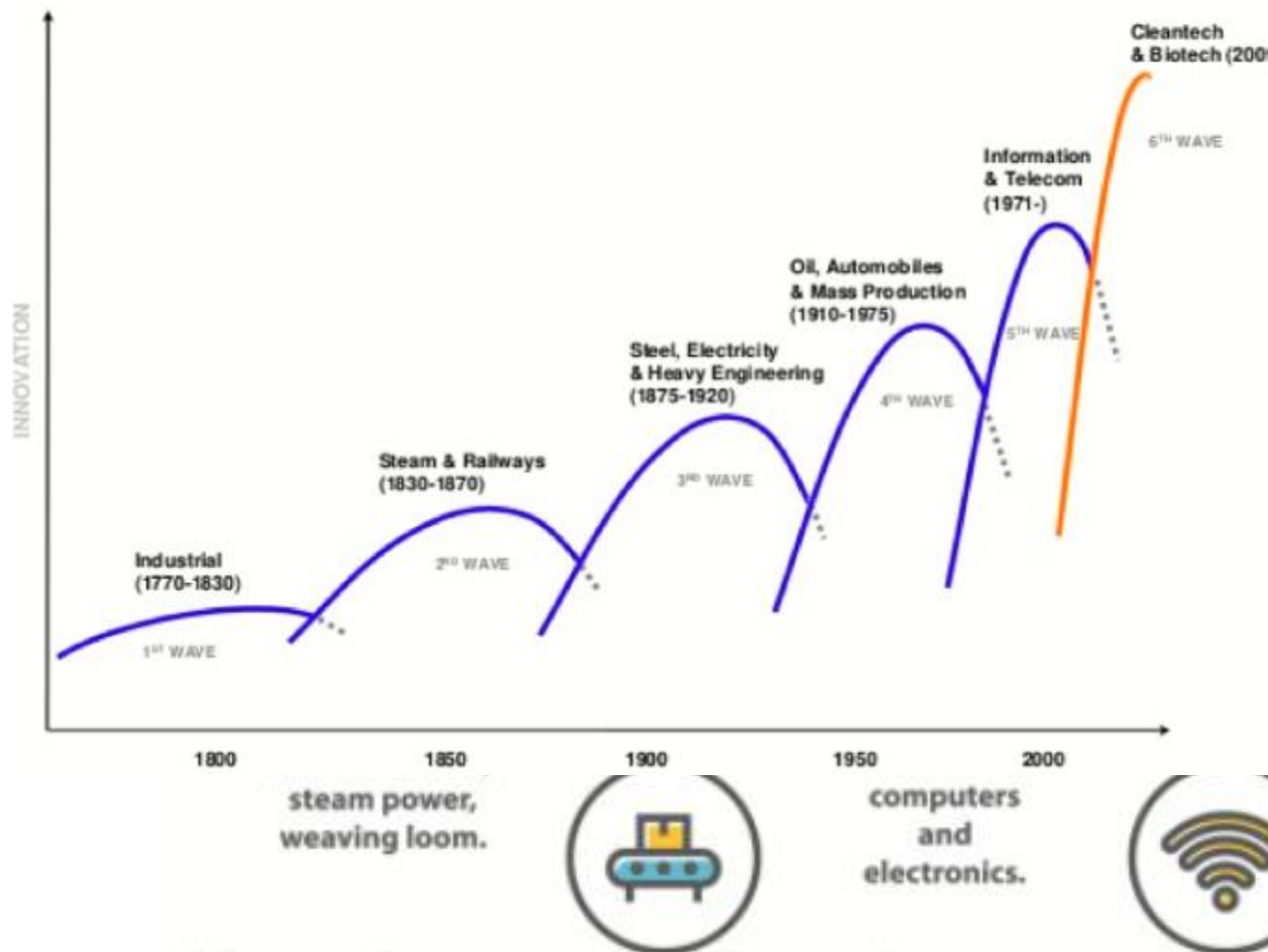
Perez (2002)

Paradigm shifts: Industrial Revolution (Noble, Mende, Grewal, & Parasuraman, 2022)



Industrial Revolution

Waves of innovation



Fifth Industrial revolution

The Fifth Industrial Revolution, or 5IR, encompasses the notion of harmonious human–machine collaborations, with a specific focus on the well-being of the multiple stakeholders (i.e., society, companies, employees, customers).

It thus paves the way for a (r)evolution in thinking about and leveraging human–machine collaborations for greater societal well-being.

We consider how the 5IR relates to retailing and services (and the broader field of marketing) specifically, to establish an overarching framework of the (r)evolution involved in embracing human–machine collaborations (Noble et al, 2022).



Practical benefits from the application of technological paradigms

The application of technological paradigms can lead to a wide range of practical benefits across various sectors. In the current technology-enabled era, some practical benefits derived in different domains:

1. Healthcare:

1. **Telemedicine:** Technology enables remote medical consultations, increasing access to healthcare for patients in remote areas.
2. **Electronic Health Records (EHR):** EHR systems improve patient care coordination, reduce errors, and provide quick access to medical information.
3. **Medical Imaging:** Advanced imaging technologies like MRI and CT scans improve disease diagnosis and treatment planning.

2. Education:

1. **E-Learning:** Online platforms and educational apps facilitate learning from anywhere, making education more accessible.
2. **Personalized Learning:** Technology allows for tailored instruction and adaptive learning, catering to individual student needs.
3. **Global Access:** Online courses and resources make education accessible to a global audience.

3. Manufacturing and Industry:

1. **Automation:** Robotics and automation technologies enhance productivity, reduce human error, and streamline manufacturing processes.
2. **3D Printing:** Additive manufacturing allows for rapid prototyping, customized products, and reduced waste.
3. **Internet of Things (IoT):** IoT sensors optimize maintenance and improve efficiency by providing real-time data on equipment and processes.

4. Agriculture:

1. **Precision Agriculture:** Technology like GPS, drones, and sensors enable precise crop management, reducing waste and increasing yields.
2. **Genetic Engineering:** Biotechnology improves crop resistance to pests, diseases, and environmental conditions.
3. **Data Analytics:** Data-driven insights help farmers make informed decisions about planting, harvesting, and resource allocation.

Practical benefits from the application of technological paradigms /...

5. Transportation:

1. **Electric Vehicles (EVs):** EVs reduce greenhouse gas emissions and dependence on fossil fuels, leading to a more sustainable transportation system.
2. **Autonomous Vehicles:** Self-driving cars have the potential to enhance road safety and increase transportation efficiency.
3. **Ride-Sharing Apps:** Services like Uber and Lyft provide convenient, cost-effective, and environmentally friendly transportation options.

6. Communication:

1. **Smartphones and Messaging Apps:** These technologies enable instant global communication and have transformed the way people connect and share information.
2. **Social Media:** Social platforms facilitate networking, information sharing, and online communities.
3. **Video Conferencing:** Tools like Zoom and Microsoft Teams have become crucial for remote work and collaboration.

7. Environmental Conservation:

1. **Renewable Energy:** Solar, wind, and hydroelectric technologies offer cleaner energy sources, reducing greenhouse gas emissions.
2. **Waste Reduction:** Recycling technologies and waste management systems contribute to a more sustainable environment.
3. **Environmental Monitoring:** Remote sensing and data analysis help track and address environmental issues.

8. Finance:

5. **Online Banking and Payments:** Technology provides convenient and secure financial transactions, reducing the need for physical banks.
6. **Cryptocurrency:** Digital currencies offer alternative financial systems and secure, decentralized transactions.
7. **Algorithmic Trading:** Automated trading systems optimize investment strategies and increase market efficiency.

Practical benefits from the application of technological paradigms /...

9. Entertainment:

1. **Streaming Services:** Online streaming platforms like Netflix and Spotify offer a vast array of entertainment content on-demand.
2. **Virtual Reality (VR) and Augmented Reality (AR):** These technologies enhance gaming, immersive experiences, and interactive storytelling.

10. Space Exploration:

1. **Satellite Technology:** Satellites aid in navigation, weather forecasting, and global communication.
2. **Exploration Robotics:** Robots like the Mars rovers enable scientific research and the potential for human exploration of other planets.

These examples highlight how technological paradigms have the potential to improve efficiency, accessibility, sustainability, and quality of life across various fields. The practical benefits of technology are continually evolving as new innovations and paradigms emerge.

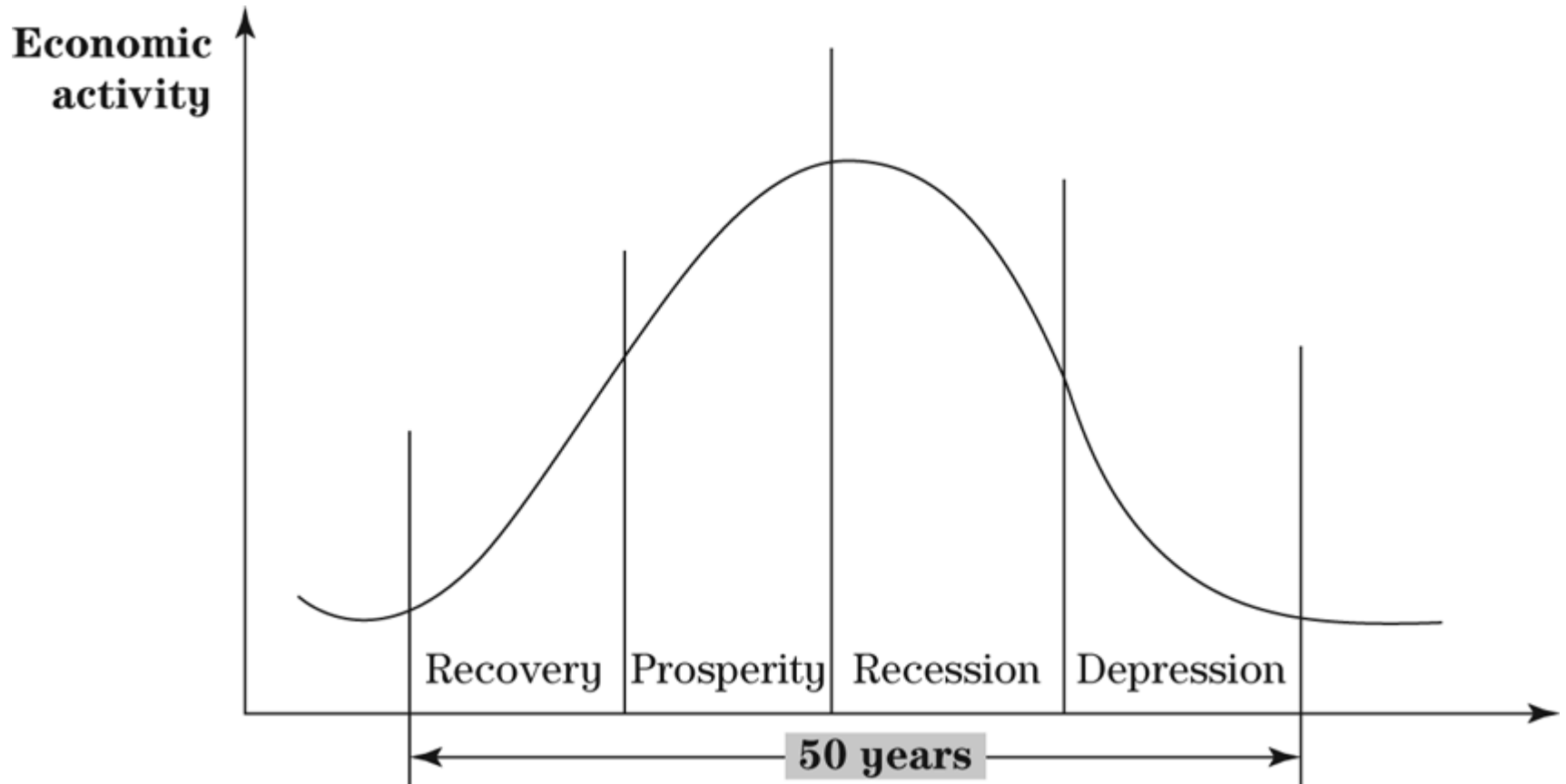
Creative destruction

- Introduced by Joseph Schumpeter
- Dramatic changes associated with a new technology
 - i.e. a paradigm shift
- Re-configuration of the economy
- New industries arise/emerge
- Old industries decline
- Examples: steamship, car, aeroplane, computer

Mini Case: Creative Destruction in Photography

Kodak was once the Google of its day. Founded in 1888, in the early years it was a pioneer of the new technology of photography. In the 1930s it pioneered colour pictures with its 'Kodachrome' film. It also made use of innovative marketing techniques. 'You press the button, we do the rest', was its strap line in 1888.

The Long Wave cycle



4 Phases of the Long Wave Cycle

- **Recovery**
 - Increased innovation; new products/services; competing designs; high novelty value; premium prices
- **Prosperity**
 - Diffusion stage - spread of innovations; new technology spreads to other industries; new industries emerge; standards emerge; expansion & prosperity
- **Recession**
 - Risky investments; spectacular collapses
- **Depression**
 - Consolidation; new industries mature; intense price competition; inventions & discoveries of next phase

Joseph Schumpeter

The idea of a long wave cycle was taken up by Joseph Schumpeter, who came across Kondratiev's work in Germany before he moved to Harvard University in the US. Schumpeter used the long wave cycle as a feature of his work on business cycles. Schumpeter (1939) put forward the idea that each long wave represented the application of a new group of technologies, each of which had a very powerful transforming effect on the economy, effectively bringing about another industrial revolution. In Schumpeter's analysis each revolution was based on a major technological change that, just like the first industrial revolution, brought major shifts in productivity, consumption and the organisation of productive activity. The technological change of the first long wave centred on the development of iron smelting using coal, pioneered by Abraham Darby, the application of water power and above all the mechanisation of the textile industry especially cotton manufacture in Lancashire. Associated with these new technologies was an organisational innovation – the development of the factory system, initiated by Richard Arkwright with his mill at Cromford in Derbyshire.

Long Wave Cycle and Technological change

One has only to reflect on the course of technological change to appreciate that it is not constant, being sometimes rapid, as in the early years of the Internet, and sometimes relatively sluggish. Nor are the Internet boom years of the late 1990s different from other periods. The 1870s and 1880s saw a flurry of innovations surrounding electricity. This same period also saw innovations surrounding the internal combustion engine. Further evidence of the cyclical nature of innovation and the technological change that accompanies it can be found in the cyclical nature of academic work. It is no coincidence that the rise of the Internet in the early 1990s was accompanied by a renewed interest on the part of academics in technological change, manifested in an increase in academic output of journals and books.

Kondratiev Cycle

As a result, he is particularly associated with the notion of a 50-year cycle of economic activity stretching from a phase of depression through recovery and boom back to depression again. Such was his contribution that this long cycle, or “long wave” as it is sometimes termed, is often referred to as the “Kondratiev cycle”.

Kondratiev's long wave cycles

Date	Cycle/wave	Leading Sector/Technology
1780–1830	First	Cotton, iron, water power
1830–1880	Second	Railways, steam power, steamship
1880–1930	Third	Electricity, chemicals, steel
1930–1980	Fourth	Cars, electronics, oil, aerospace
1980–2030	Fifth	Computers, telecommunications and the Internet
2030-onwards	Sixth	Nanotechnology, biotechnology, Robotics ?



Kontradiiev Long Wave Theory

- The Kondratiev Long Wave Theory, also known as the Kondratiev Wave or K-Wave, is an economic theory proposed by Russian economist Nikolai Kondratiev in the 1920s. This theory suggests that capitalist economies go through long-term cycles of expansion and contraction, each lasting several decades, driven by technological innovations. Here's an analysis of the Kondratiev Long Wave Theory of innovation:

1. Wave Theory Overview:

- Kondratiev identified long economic cycles lasting approximately 50-60 years, characterized by alternating periods of economic growth and decline.
- He argued that these cycles were primarily driven by technological innovations that create major new industries and economic opportunities.

• 2. Phases of the Kondratiev Wave:

- **Spring Phase:** This is the upward phase of the cycle characterized by innovation, technological advancements, and economic growth. New industries and technologies emerge during this phase.
- **Summer Phase:** The upswing continues, and the economy experiences sustained growth, as these new industries mature and expand.
- **Autumn Phase:** This is the downward phase where economic growth slows, and financial speculation increases. Existing industries become overextended.
- **Winter Phase:** The economy experiences a period of recession and depression as overextension and speculative bubbles burst.

• 3. Role of Innovation:

- Technological innovation is at the core of Kondratiev's theory. Each wave begins with a significant technological breakthrough that drives economic growth.
- Innovations can include new technologies, such as the steam engine during the First Industrial Revolution, or new industries, like the information technology and internet sectors in the late 20th century.

Kontradiiev Long Wave Theory

■ 4. Criticisms and Controversies:

- The Kondratiev Wave theory has faced criticism for its generality and the difficulty of predicting the timing of these waves.
- Some argue that economic cycles are influenced by a wide range of factors, including geopolitical events, monetary policy, and social factors, not just technological innovations.
- The concept of a 50-60 year cycle has also been questioned, as the timing of these cycles has been inconsistent.

■ 5. Contemporary Relevance:

- While the Kondratiev Long Wave Theory is not universally accepted by economists, it continues to be a topic of interest and debate.
- Some scholars and analysts use the theory to help explain historical economic trends and to make long-term forecasts.
- The theory's emphasis on the importance of technological innovation aligns with the role of innovation in shaping modern economies and industries.

In summary, the Kondratiev Long Wave Theory of innovation suggests that economies undergo long-term cycles driven by major technological innovations. While the theory has faced criticism and skepticism, it highlights the significant impact of innovation on economic growth and is part of the broader discussion of long-term economic cycles and innovation's role in shaping the trajectory of economies

Theories of innovation (Source: Smith, 2015)

- This section focuses on specific innovations and discusses factors that influence the nature and the success or failure of particular innovations. Within this perspective, a number of theories are presented.
- The value of understanding theories of Innovation is that they may be used as analytical tools to understand and make sense of innovation. The theories of Innovation thus have considerable explanatory power in explaining and even predicting the outcome of the innovation process.
- Another use of Theories of Innovation is to help categorise and classify innovations. At the same time, these theories can help in identifying some of the factors that bring about successful innovation.

Theories of innovation





Theories of innovation

A Theory may be summarized as a system of linked ideas intended to explain something, it represents an idea used to account for a situation or justify a course of action

- There are many theories associated with innovation, and four are covered here. This is not meant to imply that other theories are not relevant, but the selection is for the sake of parsimony. The four theories are:
 - Technology S-curve
 - Punctuated equilibrium
 - Dominant design
 - Absorptive capacity.
- These theories are associated mainly with technological innovation. Nonetheless, they all provide adequate scope for analysing innovations in general

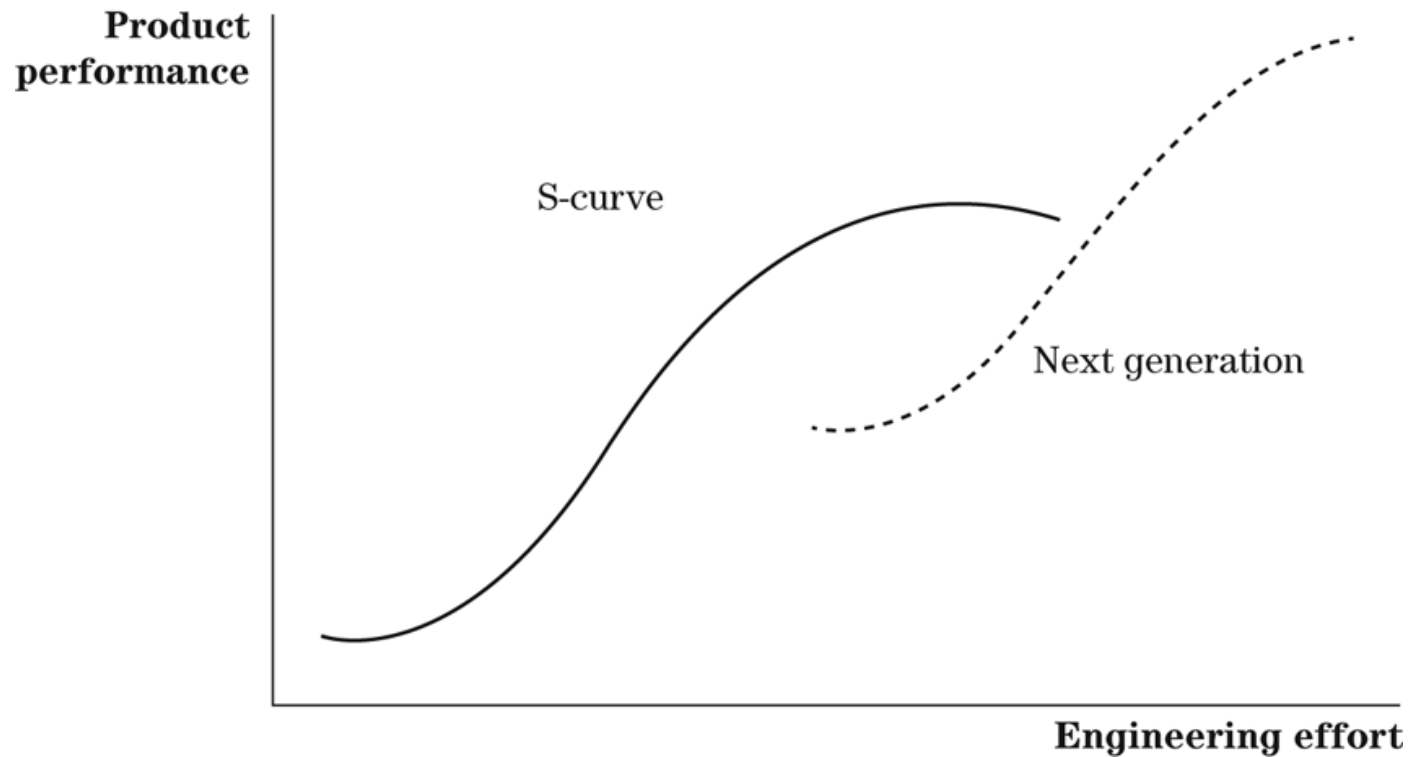
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Technology S-curve (Source: Smith, 2015)



Technology S-curve

The S-Curve theory of innovation is a conceptual framework used to describe and analyze the life cycle of a technology or product and its adoption in the market. This theory visualizes innovation and adoption as an "S" shaped curve, which represents the typical pattern of progress and growth for new technologies. Here's a description of the stage in the S-Curve theory of innovation:

1. Introduction

- The S-Curve begins at the bottom, indicating the initial stages of a new technology's introduction into the market. At this point, the technology is novel and not widely adopted.

2. Slow Adoption and Learning Phase:

- In the early stages, the technology experiences slow adoption. This phase is marked by limited awareness, skepticism, and relatively low growth in usage.
- Innovators and early adopters typically drive the initial growth, experimenting with the technology and providing feedback.

3. Accelerating Growth:

- As the technology becomes more refined, reliable, and user-friendly, it enters the phase of accelerating growth. This is the steep incline of the "S" curve.
- Early adopters and a broader segment of the market start embracing the technology, leading to rapid adoption and increased market penetration.

Technology S-curve

Maturity and Market Saturation:

- The curve begins to level off as the technology reaches maturity. Most of the target market has adopted the technology.
- Competition often intensifies during this phase as more companies enter the market.

5. Plateau and Decline:

- After reaching maturity, the technology may eventually plateau as the market becomes saturated. Growth in adoption slows down, and the technology may face competition from newer innovations.
- In some cases, the technology may enter a phase of decline if it becomes obsolete or is replaced by superior alternatives.

6. Replacement or Evolution:

- Depending on the industry and technology, the S-Curve may signify the need for either a replacement technology or a new iteration of the existing technology to stimulate growth.
- Companies must continue innovating to stay competitive and extend the life of the technology.

Technology S-curve

7. Examples:

- The S-Curve theory can be applied to various technologies and products, such as personal computers, smartphones, and even more complex technologies like artificial intelligence or renewable energy.
- For instance, the adoption of smartphones followed an S-Curve, starting with slow adoption in the early 2000s, accelerating growth in the mid-2000s, reaching saturation by the 2010s, and continuing to evolve with new features and applications.

The S-Curve theory of innovation is a valuable tool for understanding how technologies and products evolve over time, from initial introduction to eventual saturation or replacement. It helps businesses, policymakers, and analysts anticipate and plan for the different phases of a technology's life cycle, enabling more informed decision-making and innovation strategies.

In the product sector, one may cite the example of information storage devices which moved from floppy disks, CDs, USB drives, external drives, and more recently cloud storage. Similarly, in services, online commerce is threatening the market shares of many brick-and-mortar companies. Amazon is a typical example. Similarly, online sale of music has affected traditional music shops.



Technology S-curve (Smith, 2015)

A key idea behind the theory of the technology S-curve is the concept of a technology life cycle. This implies that over time the ability of a technology to deliver enhanced performance will vary.

Early in the life cycle the ability of a certain level of investment in engineering effort to deliver enhanced performance will be high. However, successive amounts of additional engineering effort produce ever greater improvements in performance. This is commonly referred to as 'learning curve' effect, whereby when a new technology is better understood, and better managed by a firm, the latter is able to deliver better quality at a cheaper cost, which then in turn supports the wider diffusion of the product/service. Eventually the rate of quality/performance improvement per unit engineering effort will begin to lessen until a stage is reached where increasing engineering effort produces diminishing returns in terms of performance improvement. This implies that a given technology eventually reaches some kind of 'natural limit' as it matures.

Foster (1986) argues that each new technology simply has a physical limits. He cites the technology of sailing ships whose speed was limited by the physics of wind and water (Smith, 2015).

Technology S-curve (Smith, 2015)

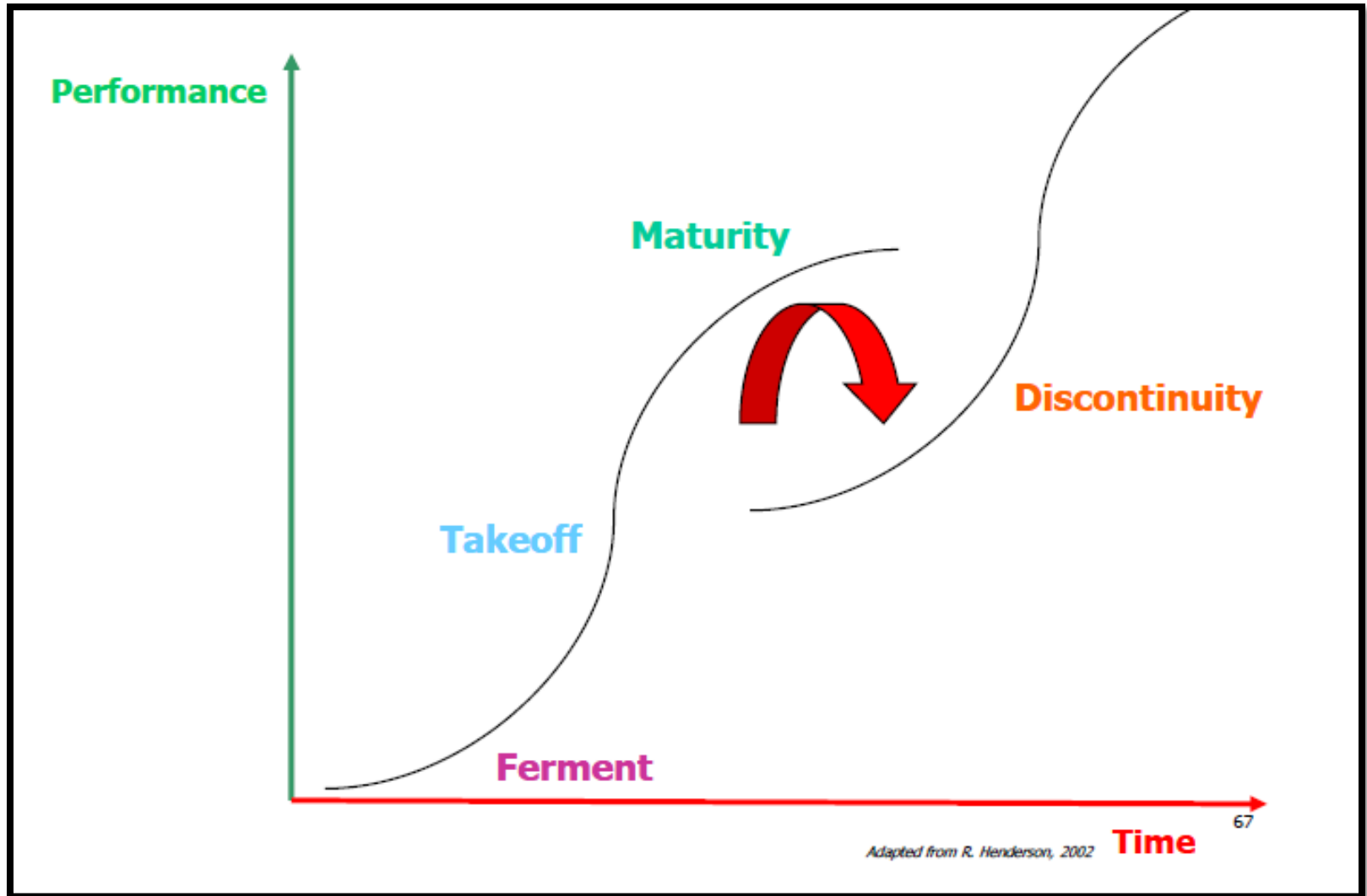
Christensen (1997: 34) suggests that the technology S-curve can be useful not only in terms of its descriptive value (i.e. the extent to which it reflects what happens in real life) but also in terms of its predictive value. In other words, the technology S-curve may be used to predict the course or developmental path of innovation. The point of inflection is where the existing technology has reached maturity and is embarking into relative decline. Although the technology may still make an effective contribution, the point has been reached where it is time to find a successor technology (i.e. another technology S-curve) to arise. The inability to anticipate new technologies is often the reason why incumbent firms often fail with new entrants entering the market with new technologies.

In the product sector, one may cite the example of information storage devices which moved from floppy disks, CDs, USB drives, external drives, and more recently cloud storage. Similarly, in services, online commerce is threatening the market shares of many brick-and-mortar companies. Amazon is a typical example. Similarly, online sale of music has affected traditional music shops.

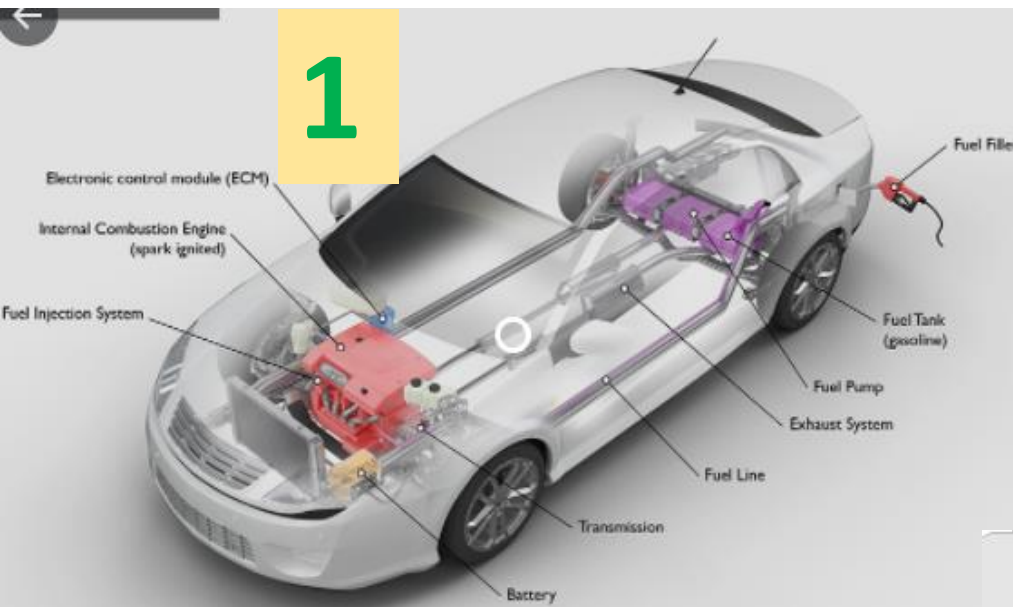
Stages in Technology S-curve

- Ferment: the introduction of the newly implemented service or product
- Take off: the rapid growth of the innovation in to the market achieving acceptance
- Maturity: the innovation is accepted by the public and it was become a success
- Discontinuity: An innovation has fulfilled its purpose and a new technology or development may risk this technology.

Technology S-curve

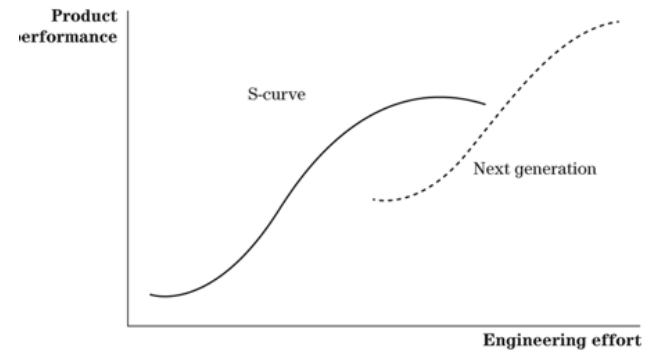


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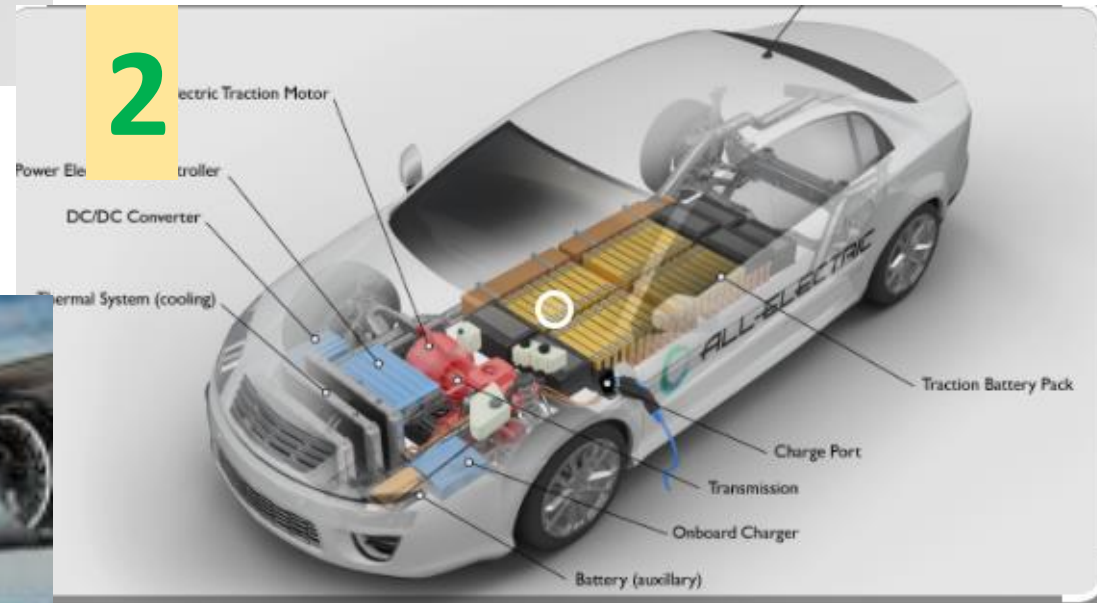


Internal Combustion engine

Technology S-curve



2



Electric Motor

3



Hyperloop

Technology S-curve (GInI Handbook)

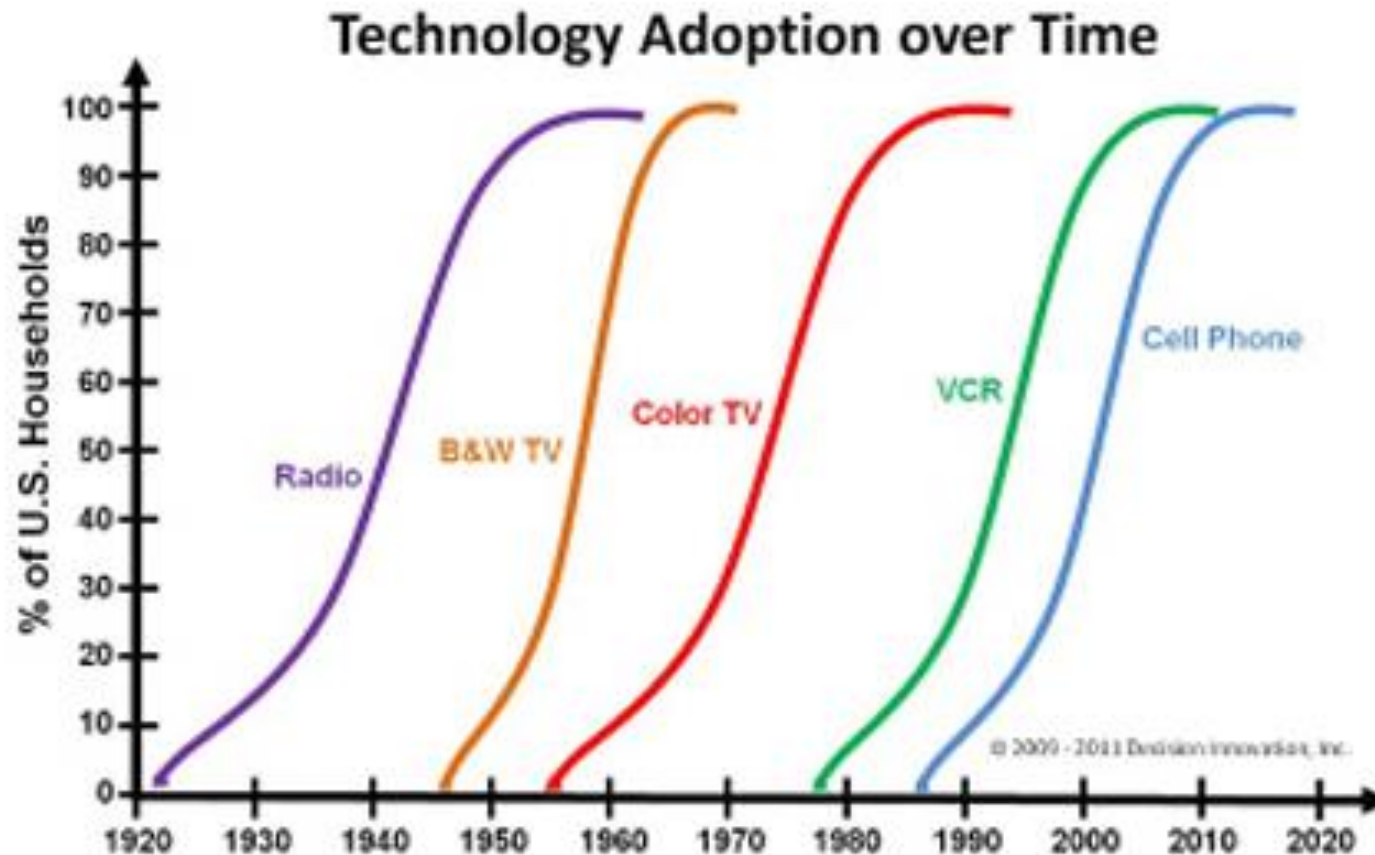


Figure 4-2: Characteristic Historical Adoption Curves (S Curves) for Select Technologies.

Punctuated equilibrium (Smith, 2015)

Abernathy and Utterback (1978) used the distinction between incremental and radical innovation to show how established industries go through periods of stability with changes confined to incremental innovations.

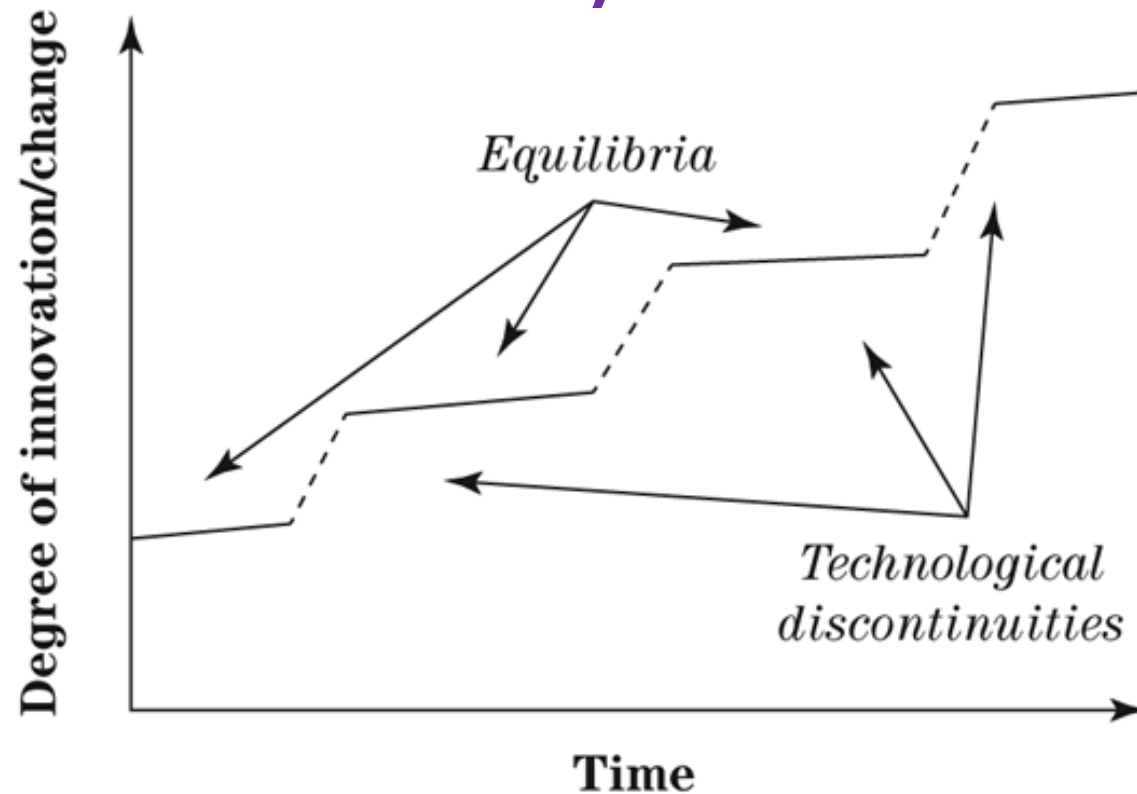
Gradualism suggests that change is gradual with the accumulation of unique morphological adaptation. Punctuated Equilibrium suggests that rapid change.

After some time, the stability is broken by a radical innovation that is highly disruptive, bringing the period of stability and equilibrium to an end. This idea of an interrupted, stepped pattern of change comes from biology as a theory of evolution in the early 1970s.

Tushman and Anderson (1986) applied the concept to describe a pattern of innovation that progresses through a succession of fits and starts. In this paradigm, major technological breakthroughs that form technological discontinuities are relatively rare with most innovations being incremental.

The discontinuities that punctuate periods of equilibrium are associated with major technological advances. These step changes are very significant, so that increases in scale, efficiency or design are unable to enable older technologies to compete with the new technologies. In other words, any amount of incremental change in the existing technology will still not make it competitive.

Punctuated equilibrium (Smith, 2015)





Punctuated
equilibrium
??

Punctuated equilibrium



Storage Formats for Music

The music industry has used different storage formats: Vinyl (1950s), Cassette (1960s & 1970s), CD (1982-1997), and mp3 (1997-date). Apply the theory of punctuated equilibrium by drawing a diagram (graph) showing the innovations in music recording over 60 years.



Punctuated Equilibrium (Smith, 2015)

- A key feature of technological discontinuities is that they require new skills, competencies, capabilities, and new knowledge in both the development and the manufacture of the product.
 - As a result the new innovations may render competencies acquired by existing firms irrelevant. In other words, these firms are unable to use the knowledge and experience they have accumulated during the period of equilibrium.
 - Given that the existing knowledge represents a big investment made over a long period of time, they would like to use this knowledge. Hence they are more likely to opt for incremental innovations as opposed to radical ones. Nor is it just that 'incumbent' firms possess knowledge and expertise linked to the old technology that gives rise to inertia; other factors are also involved.
- 



Punctuated Equilibrium (Smith, 2015)

In addition to knowledge and expertise linked to the old technology, other factors also give rise to inertia:

- Traditions
 - sunk costs
 - internal political constraints
 - commitment to outmoded technology.
- Sometimes working practices have changed very little in a long time, becoming so widely accepted and so deeply ingrained in the organization, that they have become traditions.
 - As traditions, the rationale and reasoning behind the working practices may no longer be valid, but since this is the way things have been done for a long time, traditions are very hard to break. Traditions are not limited to working practices, but they may relate to other aspects of a business as well.
 - Sunk costs are costs associated with previous investments. These could cover machinery, infrastructure, systems or even training. Firms make investments all the time, but where the investment is in technology, the cost of the investment becomes a sunk cost. The key issue is that the investment cannot be transferred to the new technology – it is forever linked to the old technology.



Punctuated equilibrium

The Punctuated Equilibrium Theory of Innovation is a concept that offers a unique perspective on how innovation occurs over time. It was introduced by organizational theorists Michael Tushman and Romanelli in the late 1980s. This theory challenges the conventional idea that innovation is a gradual and continuous process and suggests that it often follows a pattern of long periods of stability interrupted by sudden, significant breakthroughs. The key concepts include:

1. Stasis and Transformation:

- Punctuated equilibrium in the context of innovation refers to the long periods of stasis or stability in organizations or industries where there is little change.
- These stable periods are punctuated or interrupted by brief episodes of significant transformation, which can be disruptive innovations or major changes.

2. Triggering Events:

- According to the theory, these episodes of transformation are often triggered by external or internal events. These triggers can include technological advancements, changes in market conditions, regulatory shifts, or even leadership changes within organizations.
- Such events act as a catalyst, pushing organizations to adapt and innovate more rapidly.

3. Rethinking Innovation:

- The Punctuated Equilibrium Theory challenges the traditional view of innovation as a gradual, linear process. Instead, it emphasizes the importance of recognizing and responding to these periods of stasis and transformation.
- Organizations must adapt to rapid changes and harness the opportunities presented by transformative events to stay competitive.



Punctuated equilibrium

- **Some Examples of the application of the theory in Practice:**
 - This theory can be observed in various contexts. For instance, consider the smartphone industry. There were relatively stable periods with incremental innovations in mobile phones, followed by a punctuated transformation with the introduction of the iPhone, which revolutionized the industry.
 - Another example is the pharmaceutical industry, where periods of drug discovery stasis are punctuated by groundbreaking discoveries and innovations.
- **Implications for Innovation Management:**
 - **Anticipating Change:** The theory suggests that organizations should be vigilant in anticipating triggering events and be prepared to adapt quickly when they occur.
 - **Balancing Stability and Innovation:** It's crucial to strike a balance between maintaining stability and fostering an innovative culture. Organizations need to be able to transition from stasis to transformation effectively.
 - **Leveraging Disruption:** Innovation managers should recognize that periods of transformation can provide unique opportunities to gain a competitive edge. Being prepared to embrace disruptive innovations can lead to significant advantages.
- **In conclusion:**
 - The Punctuated Equilibrium Theory of Innovation challenges the traditional view of gradual innovation by emphasizing long periods of stability punctuated by rapid transformation. Understanding this theory can help innovation leaders and organizations navigate the dynamic landscape of innovation more effectively by anticipating and capitalizing on transformative events.



Punctuated Equilibrium (Smith, 2015)

Internal political constraints occur for different reasons.

For example, managers sometimes develop a strong commitment to the old technology due to their familiarity with it, and they may hence resist a new technology they know little about. This resistance to change may even lead the managers to be constraint to pursuing innovation.

These four factors constrain or limit the responsiveness of existing firms, and this is a cause for the occurrence of periods of equilibrium.

During this time, the existing firms may confine themselves to incremental innovations, thereby extending the period of equilibrium.

Eventually, radical innovations typically brought about by a major technological breakthrough will lead to discontinuities and punctuate the equilibrium.

A period of 'ferment' then follows. Usually, new firms, or outsider firms on the fringes of the industry, are the first to apply the new technology and develop radical innovations. These new firms are able to respond more effectively to the new conditions as they are free from the obstacles of an old technology, traditional ways of doing things and an outdated view of the world.



Dominant design (Smith, 2015)

It represents the design or product configuration that comprises that wins the favor of the marketplace, the one that competitors and innovators must adopt to command a significant market following (Nordström and Biström, 2002: 713).

In fact, all or most firms will eventually adopt the dominant design. The theory of a 'dominant design' is associated with ideas about the evolutionary development (Teece, 1986) of science.

In science ideas are constantly evolving, and the evolution process comprises two stages:

- a pre-paradigmatic phase, when many ideas are circulating and no one explanation of a phenomenon holds sway, and
- a paradigmatic phase, when a single explanation or theory becomes widely accepted.

The paradigmatic phase with the emergence of a dominant paradigm indicates scientific maturity. This paradigm becomes the accepted view, until it is eventually overturned by another paradigm.

Smith (2015) mentions that in the seventeenth century Copernicus's theories of astronomy overturned those of Ptolemy. The early phases relating to the introduction of new technologies innovation are often very similar. They tend to be associated with an era of 'ferment'), where designs and configurations are fluid.



Dominant design (Smith, 2015)

Abernathy and Utterback (1978) suggest different factors may give rise to dominant designs.

- Consumer preference, where a particular design mix present in preferred by consumers as they perceive it to better meet their requirements.
- It is important to recognize that these factors will not all be technical, as dominant designs are rarely technically superior to rival designs. An illustration is the touch screen phone introduced by iPhone in 2007.
- The market power of a dominant producer may also help to establish a dominant design.
- Finally, regulation, by either government or an industry body, may be instrumental in a dominant design appearing.

It is important to recognize that the theory of dominant design highlights the importance of the user, who may be less interested in technical features and more interested in usability. It also emphasizes the importance of standards, specifically where compatibility is an issue for the user. Finally, it highlights the role of business strategy within innovation.

In the Blu-ray case, for instance, Toshiba lost out to Sony because its business strategy was geared to being first to market rather than customer acceptance. Please read this case to understand the role of Business strategy (Smith (2015), p83).

Dominant design (Smith, 2015)

Initially it is simply an issue of new designs being substituted for old ones. Following substitution there is a phase (often overlapping) of intense design competition (see Figure below).

Competitors soon enter the market, with rivalry based on competing designs, each of which is significantly different. In this pre-paradigmatic phase no one design or configuration stands out.

Typically several design alternatives emerge, each incorporating the fundamental breakthrough technology, but applied in a different way to give a different product architecture.

Smith (2015) mentions the evolution of the bicycle (Rosen, 2002), which was associated with numerous competing designs in the late nineteenth century, as a typical example of the pre-paradigmatic period of ferment. Ultimately the dominant design emerges during the paradigmatic phase.

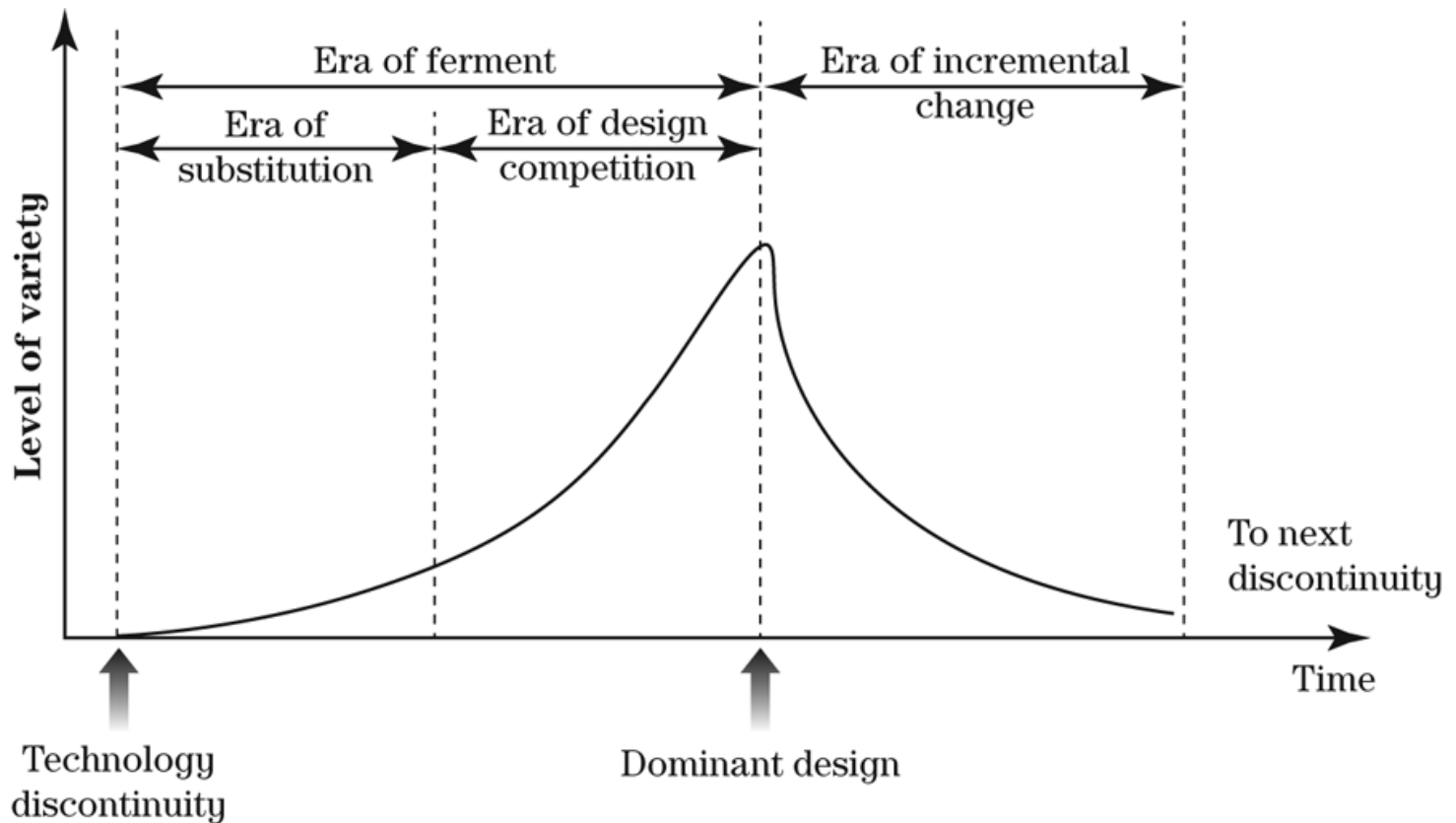
A process of technological evolution involving variation, selection and retention leads over time to the emergence of a single design that forms the accepted market standard (see Figure below, Smith, 2015)

Dominant design in Smartphones

A dominant design is the one that wins the loyalty of customers, the one that competitors and innovators must adhere to if they hope to command significant market following.



Dominant design



Different eras or phases of Dominant Design Theory

When discussing the concept of Dominant Design in the context of technological innovations, it's helpful to break it down into different eras or phases. These eras represent the evolution of technology and design within an industry. Here's an elaboration on the different eras of Dominant Design:

Era 1: Fluid Phase

- **Characteristics:** The initial phase is characterized by a lack of a dominant design. There are multiple competing designs, approaches, and technologies within the industry.
- **Innovation:** Innovations during this phase are often radical and exploratory. There is a high degree of experimentation, and companies are exploring different paths.
- **Competition:** Competition primarily revolves around the creation and promotion of alternative designs. Standards are not well-established, and companies seek to gain a foothold with their unique innovations.

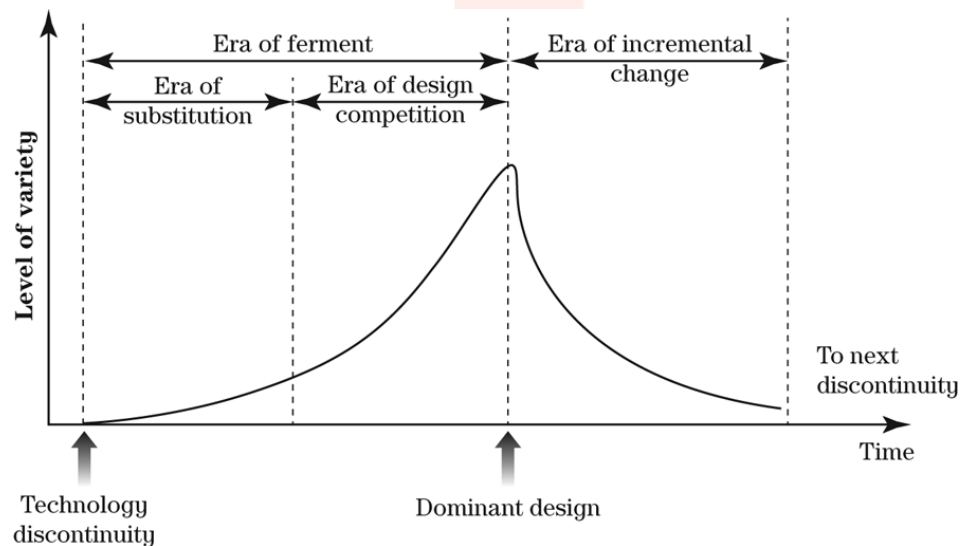
Era 2: Transitional Phase

- **Characteristics:** In this phase, a dominant design begins to emerge. It may not be universally accepted, but it gains significant recognition.
- **Innovation:** Innovations become more focused on refining the emerging dominant design. Companies begin to standardize around the dominant design.
- **Competition:** Competition shifts from design exploration to improving the features and efficiency of the emerging dominant design. Companies compete to establish themselves as leaders in this design.

Different eras or phases of Dominant Design Theory

- **Era 3: Specific Phase**

- **Characteristics:** The specific phase is marked by the widespread acceptance of a dominant design. A single design or standard becomes the industry norm.
- **Innovation:** Innovations in this phase are incremental and focus on improving the established dominant design. Companies seek to optimize processes, reduce costs, and add incremental features.
- **Competition:** Competition shifts to efficiency, cost management, and customer experience within the constraints of the established dominant design.



Smith (2015) refers to the three phases as Substitution, Design Competition, and incremental change (see Figure)

Different eras or phases of Dominant Design Theory

- **Era 3: Specific Phase**

- **Characteristics:** The specific phase is marked by the widespread acceptance of a dominant design. A single design or standard becomes the industry norm.
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- **Competition:** Competition shifts to efficiency, cost management, and customer experience within the constraints of the established dominant design.

- **Transition and Innovation Trajectories:**

- **Transition between Phases:** The transition from one era to the next can be triggered by external events, market demands, or technological breakthroughs. For example, the introduction of a game-changing technology or a disruptive innovation can accelerate this transition.
- **Innovation Trajectories:** Each era is associated with specific innovation trajectories. In the fluid phase, you have a diversity of exploratory innovations. In the transitional phase, the focus shifts to incremental improvements on the emerging dominant design. In the specific phase, innovation concentrates on efficiency and optimization.

Implications for Innovation Management

- **Timing:** Understanding which era an industry is in is critical for innovation management. It informs the timing of innovation efforts and strategy. During the fluid phase, focus may be on experimentation, while in the specific phase, it's on refinement and cost reduction.
- **Competitive Advantage:** Companies that recognize the emerging dominant design and align their innovations with it can gain a significant competitive advantage. They can also influence the standardization process by actively participating in setting industry standards.
- Understanding the different eras in Dominant Design is crucial for innovation managers. It allows them to navigate the changing landscape of technology and design within their industries, make informed strategic decisions, and capitalize on emerging dominant designs.



Absorptive Capacity

The theory of absorptive capacity is different from previous theories as it integrates both the external dimension of innovation (relating to the evolution of technology), and the internal dimension (relating to learning and the knowledge transfer process within the innovating organization).

The emphasis is on learning, making Absorptive Capacity as a very different analytical framework from the S-curve or punctuated equilibrium.

Absorptive Capacity deals with the ability of a firm to recognise the value of new, external information, integrate it and use it for commercial purposes.

Figure below provides a schema illustrates the processes of recognition, assimilation and application at work. In this schema the external environment where technological evolution takes place is at the top and the internal environment of the firm is at the bottom (Smith, 2015).

The process of recognising external trends and technological opportunities is represented by the arrow linked to the box on the left. This box represents the part of an organisation's absorptive capacity that focuses on assimilation. Hence the link from the external environment to this box represents a conduit or channel through which external ideas and opportunities are fed into the organization (Smith, 2015). The capacity of the conduit is a crucial feature of organisations with a strong absorptive capacity. If it is effective, the organisation will be good at recognising external ideas.



Absorptive Capacity

Although 'external influences' are vital for innovation, recognition of such influences is only one feature of absorptive capacity (Smith, 2015).

For effective innovation an organization must also be capable to assimilate ideas within the organisation. As Figure below shows, assimilation depends on the ability to bring external ideas in and to absorb such ideas within an organisation.

This process is called 'knowledge diffusion', and this has to extend right across an organisation and to all its levels. The 'silo' mentality has no place within the theory of absorptive capacity (Smith, 2015).

For effective innovation, once the ideas are absorbed within an organization, they have to be applied. An organisation's capability to apply ideas it has absorbed is represented by the box in the centre of Figure below (Smith, 2015).

Smith (2015) discussed that there is a tension between looking inward or outwards in an organization, and he argues that effective absorptive capacity requires the organisation to maintain a balance between information absorbed from internal and external environments for effective innovation (Smith, 2015).



Absorptive Capacity (Smith, 2015).

Absorptive capacity is more sophisticated than some of the other theories of innovation (Smith, 2015).

It demonstrates the importance of external knowledge as a critical element in innovation.

It further helps to explain why some organisations, despite being exposed to external knowledge are still poor at innovation, as they fail to absorb and make use of the knowledge.

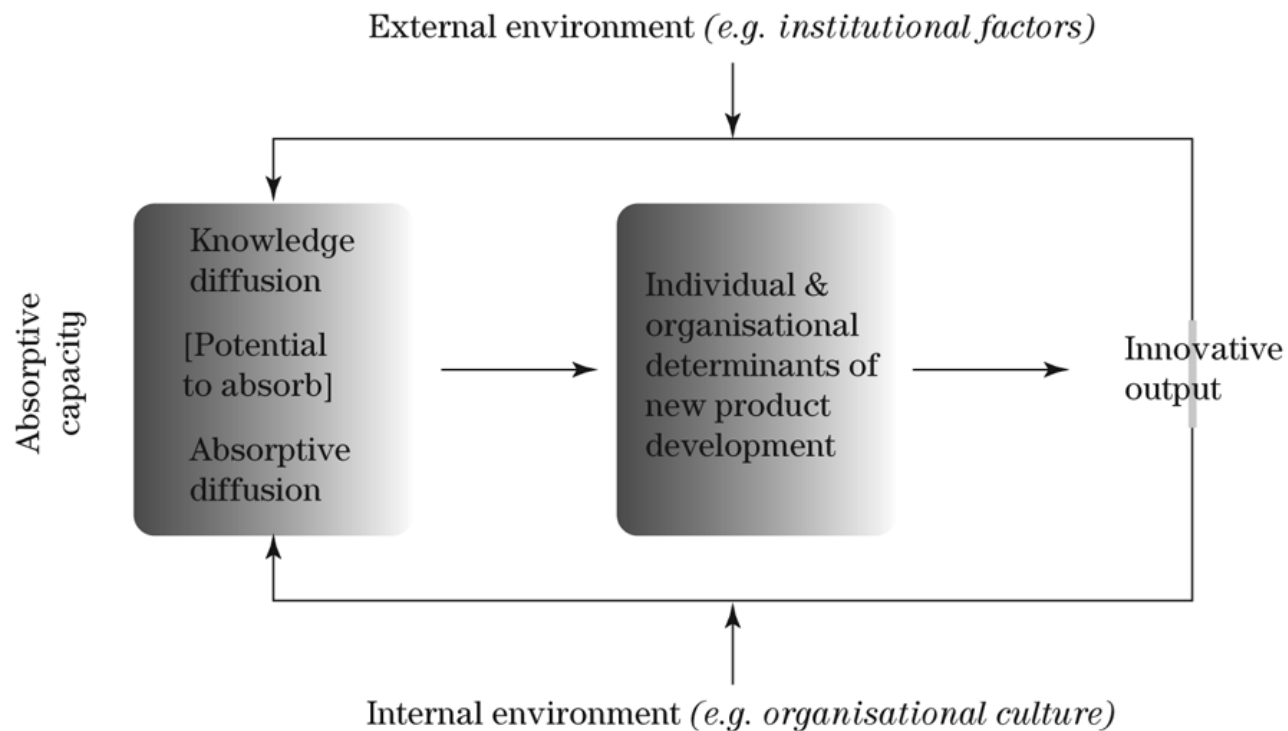
Absorptive capacity demonstrates why networks and networking is so important to innovation. However, the reason why Absorptive capacity has been widely used lies in the fact that it integrates a number of ideas. These include technological evolution, the learning process and networking. Absorptive capacity thus offers a synthesis that draws these different strands together. In the process it offers a powerful tool for analysing innovation.



Absorptive capacity

Potential to absorb
innovative inputs

Ability to generate
innovative outputs



Absorptive Capacity (Smith, 2015).

The Absorptive Capacity Theory of Innovation emphasizes the importance of the learning process and networking in an organization's ability to effectively absorb and utilize external knowledge. These elements play a critical role in innovation, and understanding their significance is essential for postgraduate students in an Innovation course:

1. Learning Process:

Continuous Learning: In the context of absorptive capacity, the learning process is ongoing. Organizations must have mechanisms in place to continually acquire and update their knowledge base. This involves staying up-to-date with the latest research, technologies, and market trends.

Exploration and Exploitation: The learning process should balance exploration (seeking new knowledge) and exploitation (using existing knowledge). While exploration is crucial for staying innovative, exploitation ensures that the organization effectively applies the knowledge it has absorbed.

Internal Knowledge Management: Organizing and managing the knowledge internally is vital. This includes categorizing, storing, and disseminating knowledge within the organization. Effective internal knowledge management ensures that absorbed knowledge is readily available for innovation initiatives.

Absorptive Capacity

2. Networking:

External Knowledge Sources: Networking serves as a gateway to external knowledge sources. Organizations that actively engage in networks, partnerships, and collaborations with universities, research institutions, industry peers, and customers can access a wider pool of knowledge.

Open Innovation: Networking is a foundation for the open innovation model, where organizations seek external knowledge intentionally. Collaborations and partnerships with external entities allow for the exchange of ideas, expertise, and technologies, fueling innovation.

Cross-Pollination of Ideas: Effective networking promotes the cross-pollination of ideas. Interactions with diverse stakeholders can lead to serendipitous discoveries and creative problem-solving that might not have occurred in isolation.

Peer Learning: Interactions with peers in the industry or innovation community provide opportunities for peer learning. Exchanging experiences and insights with others can accelerate the learning process and inform innovation strategies.

Absorptive Capacity: Implications for Innovation Management:

Strategic Alliances: Innovation managers should proactively seek out strategic alliances and partnerships. These can provide access to complementary knowledge and resources, accelerating the innovation process.

Innovation Ecosystems: Consider the broader innovation ecosystem within which the organization operates. Engaging with academic institutions, government agencies, and industry clusters can expand networking opportunities and knowledge resources.

Cultural Considerations: Fostering a culture of continuous learning and openness to external knowledge is crucial. Encouraging employees to network, attend conferences, and collaborate with external partners can enhance absorptive capacity.

Monitoring and Assessment: It's important to continually assess the effectiveness of the learning process and networking initiatives. This includes tracking the impact of acquired knowledge on innovation outcomes.

In conclusion, the Absorptive Capacity Theory of Innovation highlights that learning processes and networking are not just supportive factors but core components of an organization's innovation strategy. Embracing continuous learning and building a robust network of external knowledge sources can significantly enhance an organization's absorptive capacity, driving innovation and competitiveness. Postgraduate students in an Innovation course should recognize the strategic importance of these elements in their future roles as innovation leaders and managers.

Change comes from the outside ...

- The **digital watch** did not come from the established watch industry.
- **Video games** did not come from board game companies.
- The **ballpoint pen** did not come from fountain pen manufacturers.
- **On-line insurance** did not come from the insurance industry.
- **Amazon** did not come established book sellers.
- **iTunes** did not come from the music industry.
- **Wikipedia** did not come from Encyclopaedia companies.
- **Budget airlines** did not come from the established carriers.
- **VOIP** did not come from the telephone industry.

Revolutionary new products and services come from outside an industry where there are no mental, financial and infrastructure investment legacies.

Thank you